MANUAL · 10-INCH VERSION · IN PREPARATION



Samba Solar Track

The controller that keeps your solar module pointed at the sun and brings it to safety in a storm.

Waveshare ESP32-P4-WIFI6-Touch-LCD-10.1 with the RearCircuitBoard_10 board
The main variant of Samba Solar Track. This manual describes the assembly, the circuit board, the connections, the battery, flashing the firmware for the first time and testing the hardware. The firmware for this board has been started and flashed onto a test board (it boots and Wi-Fi works), but has otherwise not yet been tested on the device; the circuit board exists as draft 0.12 and has been neither manufactured nor measured.

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Samba Solar Track

Manual · 10-inch version

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In preparation: please do not build this yet

Everything in this manual is a draft. The RearCircuitBoard_10 board (draft 0.12, status 9 November 2026) has only been checked on the computer: the KiCad rule checks report no errors, and the manufacturing data for JLCPCB is available. Nothing has been manufactured, assembled or measured; all values are calculated or taken from datasheets. The dimensions of the rear panel are **assumptions** until the device has been measured. Before ordering, measurements on the delivered board are required (Appendix 11 E). Firmware that lets the 10.1-inch board control a tracker does not exist yet; work on it has started (Chapter 8).

1 What is Samba Solar Track?

A solar module delivers the most power when the sun shines perpendicularly onto it. Samba Solar Track turns and tilts the module throughout the day so that this is the case as often as possible, and lays it flat in a storm.

Follows the sun

A sensor with four light sensors detects where the sun is. Two drives align the module accordingly.

Protects in a storm

An anemometer monitors the wind. In a storm, the module lies flat and stays there until it is calm.

Large display

The 10.1-inch touch display is the main device. A built-in battery keeps it running if the 24 V supply fails.

Protects the mechanics

Limit switches, pauses when changing direction and run-time monitoring prevent damage.

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The parts of the system

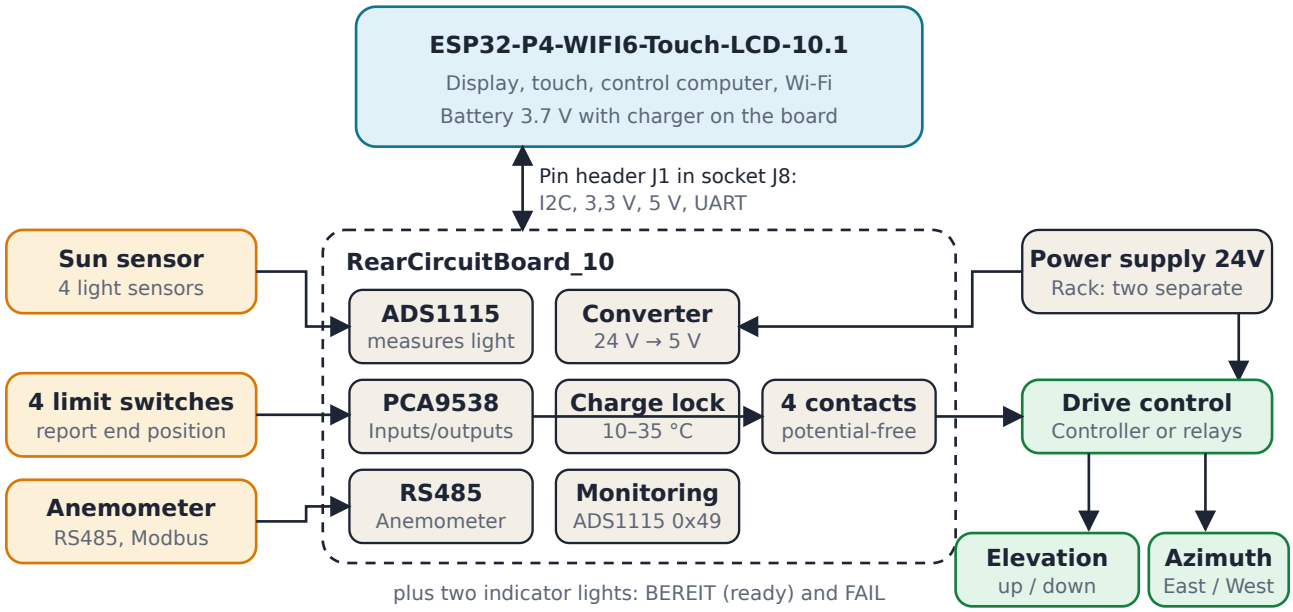


Fig. 1 – System overview of the 10-inch version. Yellow: sensors. Blue: the control computer. Green: everything that moves the module. On the rack, two separate 24 V power supplies feed the drive control (HLG-600H-24 on the rack next to the control box, also for the HallTapBoard) and the RearCircuitBoard_10 board (small power supply next to the tablet); the circuit board supplies the board and its battery with 5 V. A shared power supply remains permitted as a variant (Chapter 6, Fig. 6).

The two axes

The module moves in two directions. The **elevation** is the tilt: in the morning and evening the sun is low, at midday it is high. The **azimuth** is the rotation from east to west, following the path of the sun across the sky.

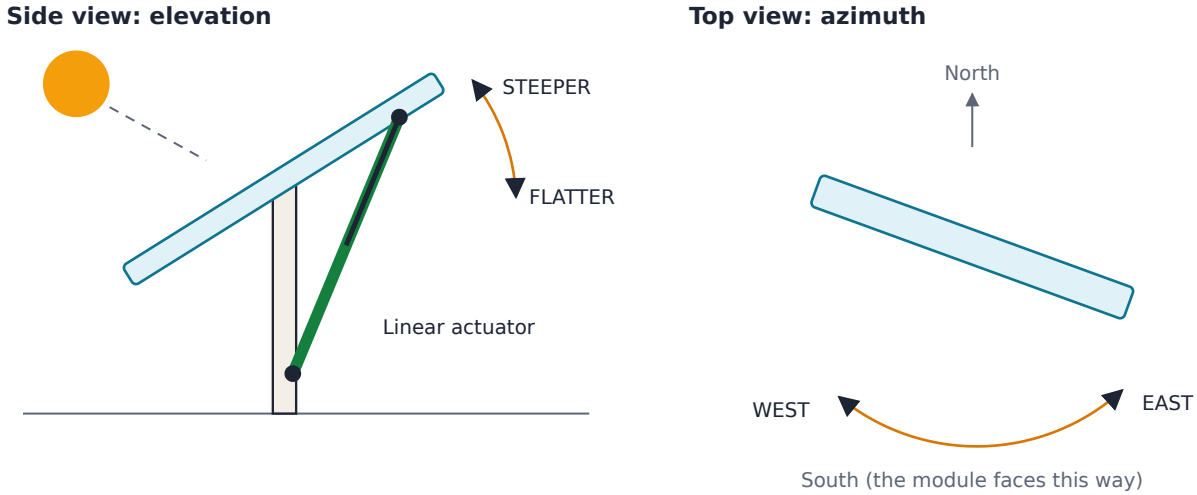


Fig. 2 – Schematic sketch of the two axes. Your mechanics may look different. The only important thing is: one linear actuator per axis that can move in two directions.

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ℹ️ **One or two axes?**
Samba Solar Track comes in two operating modes: **vertical only** (one axis, the elevation) or as a **full tracker** with elevation and azimuth. You choose which one applies when you switch on the display for the first time (in preparation, see Chapter 8).

2 Safety first

The controller moves heavy parts with powerful drives and has a battery. Read these points before you connect anything.

⛔ **Risk of crushing**
A linear actuator pushes with several hundred newtons and does not stop if a hand is in the way. Only work on the mechanics when the 24 V power supply is switched off (with two power supplies, the one for the drives). The automatic mode can start moving on its own at any time.

ℹ️ **What the circuit board is intended for**
The RearCircuitBoard_10 board is an expansion board for Samba Solar Track. On its own it has no function. It requires a compatible Waveshare board with display (for the RearCircuitBoard_10 board, the ESP32-P4-WIFI6-Touch-LCD-10.1), an external 24 V DC supply as well as sensors and drives. These are not part of the circuit board (not included in delivery). The circuit board itself is supplied only with 24 V. The drives may run on 12 or 24 V, as long as no switching contact carries more than 30 V and 1 A (Chapter 6).

⛔ **Safety extra-low voltage only, no mains voltage**
The RearCircuitBoard_10 board is intended exclusively for safety extra-low voltage (SELV): none of its connections may carry more than 30 V DC. The controller receives 24 V DC, the drives typically run on 12 or 24 V. Mains voltage (230 V) must never be connected to the circuit board, its terminals or its switching contacts. The switching contacts (PhotoMOS) are not mains relays: they switch control signals and relay coils up to 30 V and 1 A, but no motor and no device on the mains. The power supply that turns 230 V into 24 V is a ready-made, purchased device with an output approved for SELV (check the datasheet); with two power supplies this applies to both. Work on 230 V belongs in the hands of a qualified electrician.

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⚠️ **Soldering and connecting yourself: note on liability**

In the “smd” assembly variant you solder the terminals, the pin headers J16 and J21 and the pin header J1 yourself; in the “voll” (full) assembly variant only J1 (Appendix 11 B). After that you connect the power supply, sensors and drives yourself. No liability is accepted for damage caused by improper soldering, reverse polarity or incorrect connection of external components, to the extent permitted by law. Liability that cannot be excluded by law (for example in the case of intent, gross negligence or under product liability law) remains unaffected. This is a notice, not legal advice.

⚠️ **Install a fuse**

Put a fuse that suits the drive into the positive line of each drive (the value is in its datasheet). A blocked motor draws many times its normal current. With a shared power supply, each line gets its own fuse directly at the power supply: the drives 25 A, the controller 2 A (Chapter 6).

⚠️ **Storm protection is no substitute for a sturdy construction**

The controller moves the module flat, but it cannot compensate for a power failure. It detects an anemometer that no longer responds at all. It does not detect one that delivers wrong values (for example a stuck cup wheel). The mechanics must survive a storm even in an unfavourable position.

💡 **What the controller safeguards for you**

With every fault, all switching contacts open and the drives stop. A broken limit switch cable acts like a reached end position. If the anemometer no longer responds, the automatic mode lays the module flat as a precaution. Storm has priority over everything, even over your fingers on the display. The firmware for the 10.1-inch board that implements this is in preparation (Chapter 8).

⛔ **Battery only with a firmware that checks the lock – at your own risk**

The 10.1-inch board has a battery (3.7 V). Its charger on the board does not check the temperature. The RearCircuitBoard_10 board therefore blocks charging itself when it is colder than 10 °C or warmer than 35 °C at the rear panel (Chapter 7). Because some rare soldering faults can defeat this lock unnoticed, since 3 Nov 2026 the firmware cross-checks the lock continuously and warns in the header (Chapter 8). With this firmware you may connect the battery (decision of 3 Nov 2026); however, the lock in the hardware has not yet been measured on a manufactured device, and the firmware cannot prevent charging itself. The risk is yours. Do the cold and heat test from Chapter 9 first.

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⚠️ **Power supply 24 V, never permanently above 30 V**

Use a power supply with 24 V DC and at most 5 % deviation (i.e. 22.8 to 25.2 V); set an adjustable power supply to 24.0 V. The circuit board absorbs short voltage spikes, but not a permanent voltage above 30 V: then the protection diode D6 conducts and is destroyed above about 33 V, usually as a short circuit; in this way it protects the rest of the board. The fuse F1 does not reliably trip on a current-limited power supply. D6 must be replaced afterwards.

⚠️ **Installation site dry, not above 3050 m**

Only install the controller in a dry place where no condensation can form on the circuit board, and not higher than 3050 m above sea level. The board itself is specified for 0 to 60 °C. The cover of the housing must be flame-retardant (UL94 V-0).

3 What you need

The electronics consist of the ready-made display board and the custom RearCircuitBoard_10 board, which sits on its rear panel.

PART	PURPOSE	NOTE
Waveshare ESP32-P4-WIFI6-Touch-LCD-10.1	Display, touch, control computer, Wi-Fi	Exactly this model (1280 × 800, in metal housing, with 3.7 V battery). Nothing is modified on the board or the battery.
RearCircuitBoard_10	Power supply, measurement, switching outputs, anemometer connection, charge lock	Custom circuit board, manufactured and assembled by JLCPCB (Chapter 5, bill of materials in the Appendix).
Sun sensor	Detects where the sun is	Mandatory: a ready-made, purchased solar tracker detector with four analogue outputs and a 3 m cable. No home-made sensor (Chapter 4).
Anemometer with RS485 output	Measures the wind	Mandatory: RS485 (Modbus RTU), wind speed only or wind speed with wind direction. Other outputs (pulse, 0–10 V, 4–20 mA) do not work.
Linear actuators	Move the module	With a controller and push-button input or with two power relays per axis (Chapter 6). Mandatory: separate limit switches in the motor circuit.
4 limit switches	Report the end positions	With normally closed contact (NC)

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PART	PURPOSE	NOTE
Power supplies 24 V DC	Sole power supply of the controller; on the rack a second one for the drives	24 V ± 5 %. On the rack two separate power supplies: Mean Well HLG-600H-24 (IP67, 600 W / 25 A) on the rack for the sync controller, drives and HallTapBoard, and a small power supply (SELV, 2 A or more) next to the tablet only for the controller (Chapter 6). The circuit board turns this into 5 V for the board and battery; you do not need a USB power supply. One shared power supply for everything remains allowed as a variant.
Cables and fuses	From the power supply to the drives, controller and HallTapBoard	One separate cable with its own fuse at the power supply per outgoing line: drives from 4 mm² with 25 A, controller 2 × 1.5 mm² up to 15 m with 2 A.
2 lights for the housing (optional)	Show from outside whether everything is running	One green LED and one RGB LED with common anode, both **without** a built-in series resistor.
Rear housing from the 3D printer	Covers the circuit board	Custom design in the project folder `cover/`, not larger than the display, 50 mm or 40 mm deep (your choice), with 6 holes for PG 11 cable glands on the underside.
Mechanics	Mounting and heat path	4 M2.5 × 12 + 6 standoffs with male and female thread and one nut each, 4 M2.5 × 6 screws (M2.5 × 5 if the thread is too shallow), 4 M2.5 washers (DIN 433, 5 mm outer diameter) under the screw heads and up to 8 more for the standoffs, only if needed. For the heat path an M3 × 12 or M3 × 13 brass spacer and thermal pads of 0.5, 1.0 and 1.5 mm. Which of these are needed is shown by the check during the first assembly (Chapter 5, Chapter 7).

4 The sun sensor

The sensor is the eye of the tracker. It is bought ready-made and reports how bright it is in each of the four compass directions.

How it works

Inside the sensor there are four light sensors, one each for east, west, south and north. If the sun is exactly in front of the sensor, all four are equally bright. If it moves to the side, the sensor on its side becomes brighter and the opposite one darker. The controller always compares two opposite sensors and moves towards the brighter one until both are equal again.

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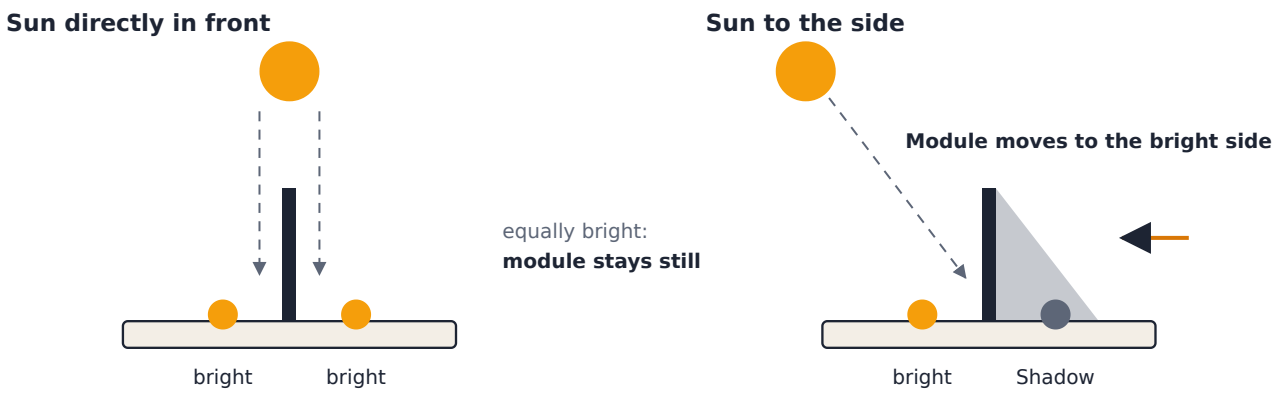


Fig. 3 – The principle, shown simplified with two opposite sensors and a shade between them. The manufacturer does not say exactly how the purchased sensor is built inside.

COMPARISON	MEANING	THE CONTROLLER MOVES
North brighter than south	The sun is higher than the module is facing	flatter (factory setting: output UP)
South brighter than north	The sun is lower	steeper (factory setting: output DOWN)
East brighter than west	The sun is further east	to EAST (only with two axes)
West brighter than east	The sun is further west	to WEST (only with two axes)

Connection and wire colours

The sensor has a three-metre cable with five wires: one for the supply and four for the signals. There is no separate ground wire. The circuit is closed via four small resistors on the RearCircuitBoard_10 board. The supply runs through a current limiter so that a short circuit in the long cable does not disturb the controller.

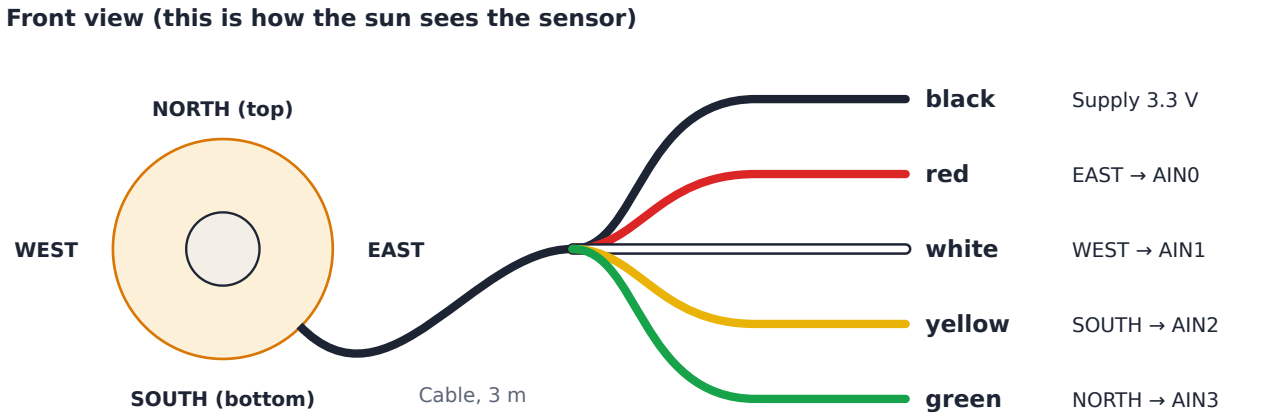


Fig. 4 – The sun sensor and its five wires. AIN0 to AIN3 are the inputs of the measuring chip on the RearCircuitBoard_10 board.

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Mounting

1. **Attach to the module**
   The sensor is firmly attached to the module and faces exactly the same direction as the module surface. So it moves together with the module.
2. **Align**
   The NORTH side points to the top edge of the module, the EAST side to the east.
3. **Clear view**
   Nothing may cast a shadow on the sensor, not even the frame of the module or the anemometer.
4. **Route and connect the cable**
   Run the cable to the controller and connect the five wires to the terminals of the RearCircuitBoard_10 board as shown in Fig. 4.

Holder for a 40 × 40 profile

For frames made of aluminium profile 40 × 40 there is a matching holder for the sun sensor. It is not part of Samba Solar Track; it will later be available in the shop in several versions, soon as a finished holder (without sensor) for €14.99. On the Samba Solar Rack the sun sensor sits on the sensor carrier P15 above the centre beam P11, in the middle of the tilting frame; position and cable route are shown in the wiring manual.

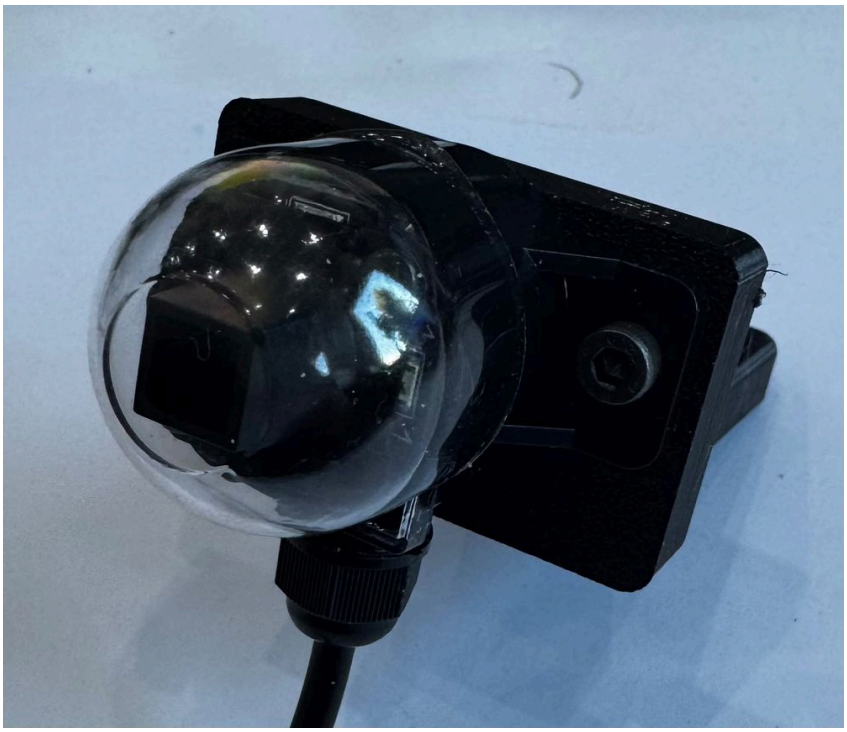


Fig. 4a – Photo: the sun sensor on the holder.

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🧩 **Still to be checked on the device**

The manufacturer only supplies a connection sketch. Three things are therefore assumptions: that the output of a direction is highest when the sun is in that direction; that NORTH belongs to the top edge of the module; and how high the signals are in full sun. The commissioning checks all three. If a direction is wrong, it can be swapped in the settings without rewiring.

ℹ️ **Intended by the manufacturer for its own controller**

The manufacturer points out that the sensor is mainly suited to its own controller. For other systems it specifies exactly the circuit that Samba Solar Track uses: 3.3 V supply and 200 Ω to ground per output.

ℹ️ **Shield of the sensor cable**

On the RearCircuitBoard_10 board, terminal J2 has a sixth pole for the shield of the extended sensor cable. Connect the shield only there at the controller, not at the sensor.

5 The RearCircuitBoard_10 board

The 10.1-inch board sits in a metal housing. The RearCircuitBoard_10 board sits on the outside of its rear panel and is connected to the board via a long pin header.

The 40-pin header J1 is on the underside of the circuit board. It goes through a slot in the rear panel into the socket header J8 of the board. It carries the data (I2C), the 3.3 V from the board, the 5 V to the board and the connection to the anemometer. There are no cables between the circuit board and the board.

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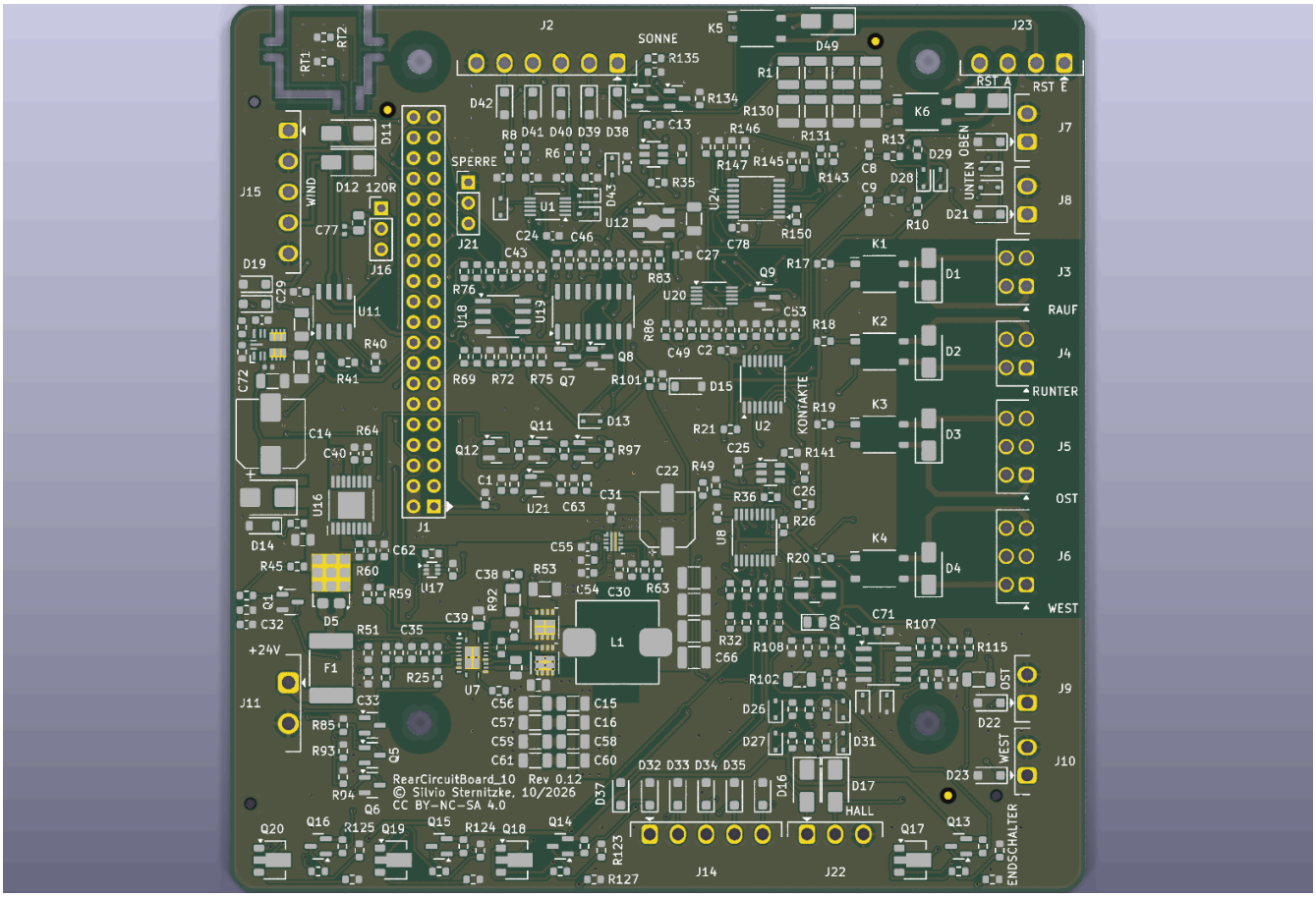


Fig. 5 – Draft 0.12 of the RearCircuitBoard_10 board from above (image from the KiCad project, without component bodies). All terminals sit at the edge, the cables go outwards: on the right the limit switches of the elevation, the four switching outputs and below them the limit switches of the azimuth (since 3.11.2026 no longer at the top), at the top the sun sensor, at the bottom the lights and to the right of them the HALL pulse input (J22, prepared), on the left the anemometer and 24 V. Since 3.11.2026 the board measures 106 × 110 mm (previously 100 × 102 mm): the additional margin on the right and at the bottom carries the protection of the limit switches and lights against 24 V; below the sun sensor sit its protection diodes, at the top right the eight measuring resistors. Since draft 0.12 (9.11.2026), terminal J23 for the reset contacts of the sync controller (“RST E”, “RST A”) sits at the top right next to hole H2. Next to each terminal except J14, the printing says what it is for.

What is on the circuit board

MODULE	WHAT IT DOES
24 V input	Terminal J11, self-resetting fuse F1, reverse polarity protection D5, overvoltage protection D6 and an electronic fuse U16. It switches off after 4 ms in the event of overload and tries again after about half a second. A temperature switch U17 switches it off above 110 °C.
5 V converter	A buck converter (LM5146 with two transistors) turns 24 V into about 5.05 V. It delivers 4 A continuously, enough for the board and for charging the battery. In the event of a short circuit, it switches off and on again in cycles (hiccup).
5 V fuse	The electronic fuse U15 protects the 5 V to the board (limit 4.4 to 5.3 A) and lets no current flow backwards into the converter.
Overvoltage shutdown	If the 5 V rises above about 6 V due to a defect, a latch switches off the 24 V input and keeps it off until you switch off the 24 V for at least 30 seconds.

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MODULE	WHAT IT DOES
Charge lock	Lowers the 5 V to about 4 V as long as charging the battery is not allowed (Chapter 7). Jumper J21: 2-3 = operation, 1-2 = force lock.
Light measurement	Measuring chip ADS1115 (address 0x48) with measuring resistors and overvoltage protection for the four outputs of the sun sensor; separate, current-limited 3.3 V supply for the sensor.
Monitoring	Second ADS1115 (address 0x49): temperature at the rear panel, the 5 V at the board and whether charging is enabled. Read-only, it switches nothing.
Switching outputs and limit switches	PCA9538 (0x70) with four potential-free PhotoMOS contacts and four limit switch inputs.
Indicator lights	Second PCA9538 (0x71) with the BEREIT (ready) and FAIL lights on the circuit board and at terminal J14; plus the status signals of the current limiters and the inputs “24 V present” and “5 V present”.
Anemometer	Terminal J15 for both wind sensors with all four wires: at pole 1 a separate, current-limited supply of about 11.5 V (regulator U23 with protection diodes D18 and D19), at poles 3 and 4 the RS485 chip THVD2410 with protection diodes; termination resistor via jumper J16.
HALL pulse input	Terminal J22 for two potential-free contacts (channel A and B), intended for the planned HallTapBoard. Per channel a pull-up to 5 V, a protection diode at the terminal, a filter and a comparator (U22) with hysteresis; it passes the signal on at 3.3 V to GPIO29 and GPIO30 of the board. The 10-inch firmware evaluates the input when “Hall tap present” is switched on under SETTINGS (Chapter 8; only built and tested on the computer).
Reset contacts (from draft 0.12)	Third PCA9538 (address 0x72) with two further potential-free PhotoMOS contacts K5 and K6 at terminal J23: RST elevation and RST azimuth. Built the same way as the switching outputs, with a protection diode across each contact. With them, the firmware resets the sync controller after a power failure (Chapter 8).

💡 **Safe by itself**

After switching on and after every restart, all pins of the PCA9538 are “open” (high-impedance) for a moment. Then all contacts are open (the four switching outputs and, from draft 0.12, the two reset contacts at J23) and the lights are dark until the firmware switches them. If the board is off via its slide switch, the contacts also stay open.

What the firmware expects from the circuit board

The 10-inch firmware (in preparation) needs this information. It is listed here so that circuit board and firmware match (rule H7).

CHIP	ADDRESS	EXPECTATION
ADS1115 (sun sensor)	0x48	AIN0 east, AIN1 west, AIN2 south, AIN3 north, each 200 Ω (2 × 100 Ω) to GND, 22 kΩ in series, protection diode at the terminal and clamp diode at the input.

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CHIP	ADDRESS	EXPECTATION
ADS1115 (monitoring)	0x49	Supplied by the circuit board's own 3.3 V supply; not reachable without 5 V (for example in battery operation), in which case the controller only issues a warning. AIN0 temperature sensor RT2, AIN1 half of 3.3 V, AIN2 half the voltage at the J8 socket of the board, AIN3 enable of the charge lock. The firmware measures the four inputs one after another, one pass every 0.8 seconds, and uses them to cross-check the charge lock (Chapter 8).
PCA9538 (switching outputs)	0x70	IO0–IO3 contacts UP, DOWN, EAST, WEST (LOW = contact closed), IO4–IO7 limit switches (LOW = closed = movement allowed). After a reset all pins are inputs.
PCA9538 (lights)	0x71	IO0 BEREIT (ready), IO1–IO3 FAIL red, green, blue (LOW = on); IO4/IO5 fault of the current limiters (LOW = fault); **IO6 “24 V present” (LOW = present)**, **IO7 “5 V present” (HIGH = present, also during the charge lock)**. Operate IO6 and IO7 only as inputs, never as output HIGH.
PCA9538 (reset contacts, from draft 0.12)	0x72	IO0 contact RST elevation (J23 pole 1–2), IO1 contact RST azimuth (J23 pole 3–4), **LOW = contact closed**. After a reset or switching on, all pins are inputs, so both contacts are open. IO2–IO7 are unused, each has 10 kΩ to GND on the board, and the firmware sets them as output LOW. The RESET input has its own 10 kΩ resistor to 3.3 V (R150). If 0x72 does not respond (board 0.11), only the reset via J23 is “not available”; this is not an error, everything else keeps running.
Pulse input J22 (U22)	– (no I2C)	GPIO29 = channel A (J1.19), GPIO30 = channel B (J1.21), as inputs of the pulse counter (PCNT) of the ESP32-P4. Its driver switches on the internal pull-up; this does no harm because there are 10 kΩ to 3.3 V on the circuit board and the comparator drives the line. Contact closed = HIGH (Hall signal HIGH), open or no plug = LOW; a constant level counts nothing. Uncritical after a reset. Every edge of both channels is counted (quadrature), noise filter 5 µs, expected 200 to 400, at most about 1,000 pulses per second and channel (**assumption**). Evaluated only when “Hall tap present” is switched on.
Remote access	–	The “Remote control locked” switch belongs to the firmware (Chapter 8); the circuit board does not change anything about it.

Mounting

⚠️ **Mount the copper island at the top left carefully**

The small copper island with the temperature sensor is separated from the rest of the circuit board by slots so that its heat hardly reaches it. It hangs on a finger (1.9 × 2.5 mm) and is supported on the left and right by a narrow bridge of PCB material each (implemented on 11.10.2026: two bridges). Nevertheless: hold the circuit board by the edge, lower it straight and do not press on the island. Press the thermal pads only lightly. If the island breaks off anyway, the sensor no longer measures and the battery charge lock does not work as intended; the circuit board must then not be operated with the battery connected.

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1

Replace the screws

Unscrew the four screws that hold the board's PCB from the outside through the rear panel one after the other and replace each with an M2.5 standoff with male and female thread. The male thread must not be longer than the old screw. With the M2.5 × 12 + 6 standoff, an M2.5 nut goes onto the male thread, but no washer: this way the standoff can be screwed in fully, and the nut rests on the rear panel. Then use the depth gauge of the caliper to measure how deep the female thread of a standoff is, and note the value (it determines the length of the screws in the last step). *Expected:* The four standoffs hold the PCB as firmly as the screws did before. From the outside of the rear panel to the top of the standoffs it is 13.9 mm (measured on the device, 05.11.2026).

2

Check during the first assembly: first fit J1 without washers

Do not solder in the pin header J1 yet. Insert it into the circuit board from below so that its plastic carrier rests against the underside (hold it at the top with adhesive tape, for example). Leave out the spacer and pads for this step. Place the circuit board on the standoffs without washers, pressing only directly above J1, and screw the screws in only lightly, each with a washer under the head. Then measure the distance g from the underside of the carrier to the top edge of J8; the pins go e = 15.5 mm − g deep into J8. *Expected:* The circuit board lies flat on all four standoffs, J1 sits straight and without tension in J8, and g is 12.3 to 12.5 mm, so the pins go 3.0 to 3.2 mm deep (calculation; 2.5 to 4.0 mm is permitted, i.e. g 11.5 to 13.0 mm).

3

Washers on the standoffs only if needed

If J1 hits the bottom of J8 (the circuit board then does not rest on the standoffs) or the pins go deeper than 3.7 mm, place one M2.5 washer to DIN 433 (0.5 mm thick, 5 mm outer diameter) on each standoff and repeat the previous step. Two washers per standoff are too many: the pins then go only 2.0 to 2.2 mm deep. The small DIN 433 form is chosen because it stays inside the copper-free ring around the hole (6.5 mm) even if it slips; an ordinary 6 mm washer could reach tracks on the underside. *Expected:* The pins go 2.5 to 4.0 mm deep; the same number of washers (none or one) lies on all standoffs.

4

Solder in J1

Only now remove the circuit board again and solder in J1 straight, with the carrier resting against the underside. *Expected:* All pins protrude equally far below the circuit board; the carrier rests without a gap.

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5 Place the thermal pad and spacer

Place a thermal pad (6 × 6 mm) on the rear panel under the copper island at the top left of the circuit board, stand the M3 brass spacer on it, and put the second pad on top. Which combination it will be is likewise decided during the first assembly. With 13.9 mm between circuit board and rear panel (50 mm housing, without washers on the standoffs), either the 12 mm spacer with one 1.0 mm and one 1.5 mm pad fits (the pads are compressed by about 24 %) or the 13 mm spacer with two 0.5 mm pads (about 10 %, at the lower limit). With one washer per standoff (14.4 mm), for example the 12 mm spacer with two 1.5 mm pads (20 %) or the 13 mm spacer with two 1.0 mm pads (30 %) fit. With 9 mm (40 mm housing), the 6 mm spacer and two 1.5 mm pads fit; with a different distance, the choice depends on the measured distance (Chapter 7). *Expected:* The spacer stands vertically under the four lines in the printing on the underside.

6 Hold the circuit board by the edge and align it

Hold the circuit board with the components facing outwards only by the edge, not in the area of the copper island at the top left. Hold it over the rear panel and check from the side that the spacer stands exactly under the four lines. *Expected:* No finger and no tool touches the island; the spacer stands under the lines.

7 Plug on the circuit board

Push the long pin header J1 straight and without tilting through the slot in the rear panel into the socket header J8; press only directly above J1, never on the island. The pads are only compressed slightly. *Expected:* The four holes of the circuit board are above the standoffs, the circuit board rests on the standoffs (or the washers) and on the pad; no washer has slipped.

8 Screw down

Screw the circuit board onto the standoffs with four M2.5 × 6 screws, each with a washer under the head, only moderately tight (the thread of the standoffs is brass). Without washers on the standoffs, the screw engages about 3.9 mm deep in the standoff, with one washer per standoff about 3.4 mm. This only works if the female thread is deep enough (measured in the first step: without washers at least about 4.2 mm, with one about 3.7 mm; **assumption**, the depth is not specified anywhere). If it is shallower, use M2.5 × 5 screws; without washers they engage about 2.9 mm deep. Screw head and washer may be at most 7 mm in diameter; here too the M2.5 washer to DIN 433 with 5 mm is used, on which the head (4.5 mm) rests completely. *Expected:* The circuit board lies flat on all four standoffs and does not wobble; the pin header sits in the J8 socket of the board without tension; the spacer sits firmly between the pads and does not slip.

⚠️ **From draft 0.12: hole H2 only with plug J23 removed**

Terminal J23 sits at the top right, close to hole H2; its tracks run just under 5 mm past the centre of the hole. Therefore only screw the screw in H2 in or out when the plug of J23 is removed.

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Measured, checked with a test fit and still open

The pins of the pin header J1 should go 2.5 to 4.0 mm deep into the socket header J8. **Measured** on the device (05.11.2026): standoffs with nut, without washer, give 13.9 mm from the outside of the rear panel to the underside of the circuit board. **Position of J1/J8 checked on the device with a test fit (06.11.2026):** a printed test piece of the circuit board sat on the standoffs without washers; its block above J1 stood cleanly above J8. **Calculated** from this (not measured with a caliper): J8 lies about 0.9 to 1.1 mm below the outside of the rear panel, accurate to about 0.2 mm (dimensional accuracy of the print). The pins stand 18 mm below the circuit board. **Assumptions:** the usable depth of J8 (about 4.2 mm according to the 3D model from Waveshare) and the depth of the female thread of the standoffs (measure before screwing down). Whether washers go on the standoffs and which heat path it will be is only decided during the first assembly on the manufactured circuit board. The four prepared holes of the rear panel (“reserved mounting holes”) lie under the circuit board and remain free.

WASHERS PER STANDOFF	DISTANCE REAR PANEL–CIRCUIT BOARD	PINS IN J8	RESERVE TO THE BOTTOM OF J8	M2.5 × 6 SCREW ENGAGES	ASSESSMENT
none	13.9 mm (measured)	3.0–3.2 mm	1.0–1.2 mm	3.9 mm	probably suitable according to the test fit
one	14.4 mm	2.5–2.7 mm	1.5–1.7 mm	3.4 mm	at the lower limit
two	14.9 mm	2.0–2.2 mm	2.0–2.2 mm	2.9 mm	too tight, do not use

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ℹ️ **Rear housing**

The housing for the rear side is a custom design for 3D printing (folder `cover/`): as large as the rear panel of the display (245.2 × 164 mm). There are two depths to choose from; each has its own files with `_50mm` or `_40mm` in the name:

- **50 mm deep:** The circuit board sits 13.9 to 14.9 mm above the rear panel of the display (pin header J1 21 mm, standoffs M2.5 × 12 + 6 with nut, washers on top only if needed); according to the calculation 9.6 to 10.6 mm of air remain to the bottom of the housing; the cables to the glands run in this gap. The glands sit in the middle of the side face.
- **40 mm deep:** The circuit board sits 9 mm above the rear panel (pin header J1 15 mm, C52461). Standoffs M2.5 × 6 + 6 with nut and washer on the male thread fit for this (measured by the user: 8.9 mm distance); a JLCPCB number for these standoffs is **open**. The glands sit in the middle of the wall height. With 13.9 to 14.9 mm distance, the circuit board does not fit into this housing (according to the calculation at most 0.6 mm of air remains to the bottom; at 14.9 mm J11 hits it by 0.4 mm).

Caution, do not swap the pin headers: The long 21 mm pin header (C51353) belongs only to the 13.9 to 14.9 mm distance (50 mm housing). At 9 mm distance its pins would go about 9 mm deep into the socket header J8 instead of 3 mm (calculated with a 1.5 mm rear panel, **assumption**) and would damage J8 of the board. For 9 mm use only the 15 mm pin header (C52461).
The cables come in at the bottom through six PG 11 cable glands (hole 18.9 mm). The housing is built so that the image is the right way up when the glands are at the bottom and the keyholes at the top (as of 28.10.2026; with the previous housing the image was upside down; **assumption** until installed in the new housing). It is fastened with M2.5 screws to the eight edge holes of the rear panel, and it hangs on the wall on two concealed keyholes (screw Ø 4 mm). It is printed with the bottom facing down. For the Creality K2 Plus with Hyper PETG-CF, ready-made print projects are included, for each depth one for the 0.4 mm and one for the 0.6 mm nozzle (for example `Rueckgehaeuse_ESP32-P4_10.1_50mm_K2Plus_HyperPETG-CF_0.4mm.3mf`, likewise `_40mm_…` and `…_0.6mm.3mf`): 5 walls, 6 top and 6 bottom layers each, gyroid infill, supports only from the print bed. The keyholes print without supports. The holes for the glands get no support and may sag slightly at the top; if so, ream them out. Check the preview in the slicer before printing (**assumption**: not checked in the slicer). According to the calculation, the circuit board fits inside; only a test fit will show whether the nuts of the glands have enough clearance inside to the connectors of the circuit board (**assumption**).

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6 Connecting and wiring

All terminals of the circuit board are pluggable: the wires go into a loose plug, and the plug onto the circuit board. This way the circuit board can be replaced without loosening any wires.

The complete wiring on the Samba Solar Rack with two drives, sync controller BST-KZQ-15A-2 and HallTapBoard is described step by step in the wiring manual (verkabelung.html, PDF `Samba-Solar-Track-Verkabelung-10Zoll.pdf`). This chapter explains the terminals of the circuit board and the circuits behind them.

> 🧩 **Settings on the display**
> Where this chapter mentions settings on the display (for example under SETTINGS), they are contained in the firmware for the 10.1-inch board, but have not yet been tested on the device (Chapter 8).

The terminals of the circuit board

TERMINAL	PRINTING	ASSIGNMENT (POSITION OF PIN 1 FOR THE VIEW OF THE BOARD AS IN FIG. 5)
J2	SONNE	Pin 1 on the right. 1 = 3.3 V (black), 2 = east (red), 3 = west (white), 4 = south (yellow), 5 = north (green), 6 = shield. Every wire withstands 12 V permanently; 24 V on a measuring wire (2 to 5) destroys two resistors of the board (warning below).
J3, J4	RAUF, RUNTER	Switching outputs of the elevation: two rows with 2 × 2 pins, one potential-free contact between two diagonally opposite pins (Fig. 12)
J5, J6	OST, WEST	Switching outputs of the azimuth: two rows with 2 × 3 pins, contact between the two outer, diagonally opposite pins (Fig. 12)
J7, J8	OBEN, UNTEN	Limit switches of the elevation (right edge, top): 1 = signal, 2 = GND. Since 3.11.2026 every wire withstands 24 V permanently.
J9, J10	OST, WEST	Limit switches of the azimuth (right edge, bottom, below the switching output WEST; until 3.11.2026 at the top next to J2): 1 = signal, 2 = GND. Every wire withstands 24 V permanently.
J11	+24V	Pin 1 at the top. 1 = +24 V, 2 = minus (larger plug, pitch 5.08 mm)
J14	(no printing; designator “J14”, to the left of HALL)	Pin 1 on the left. 1 = plus of both lights, 2 = red, 3 = green, 4 = blue (FAIL), 5 = minus of the BEREIT (ready) light. Since 3.11.2026 every pole withstands 24 V permanently; each light is switched by a current sink (7 to 8 mA, also for green, blue and white lights with up to 3.3 V forward voltage).
J15	WIND	Pin 1 at the top. Wind sensors (pitch 3.81 mm): 1 = +12 V (red, about 11.5 V from the circuit board), 2 = GND (black), 3 = A (yellow), 4 = B (green), 5 = shield
J16	120R	Termination resistor jumper: 1-2 = on, 2-3 = off
J21	SPERRE	Charge lock jumper: 2-3 = operation, 1-2 = force lock (for testing only)

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TERMINAL	PRINTING	ASSIGNMENT (POSITION OF PIN 1 FOR THE VIEW OF THE BOARD AS IN FIG. 5)
J22	HALL	Pin 1 on the left. Pulse input for the HallTapBoard (see below): 1 = channel A, 2 = channel B, 3 = GND and shield; potential-free contacts only, no supply
J23	RST E, RST A	From draft 0.12. Pole 1 on the right. 1–2 = contact RST elevation, 3–4 = contact RST azimuth; each potential-free, at most 30 V and 1 A. To the RST and COM terminals of the sync controller (see below). Only fit it if you connect the reset via J23.

⛔ **Plug connectors in meticulously correctly and not offset**

Every plug belongs on exactly its own terminal, flush, with all pins in the plug and not offset by one pin. Especially at the upper end of J7, at the end of J10 and at both ends of J14, a plug with more poles can stick out beyond the terminal without it being noticed. Then wires are on the wrong pins: a three-pole plug on J7 can put a limit switch input into an undefined state, and the six-pole sensor plug at the end of J14 puts a sensor wire onto the positive pole of the lights. The circuit board no longer has its own protection against this; only your care counts.

Therefore always do it like this: De-energise the drives before plugging in or pulling out. Label every plug exactly like its terminal. Plug in J2, J14 and J15 first, the plugs of the switching outputs last. Before switching on for the first time and after every re-plugging, carry out the test step in Chapter 9 (step 3).

ℹ️ **What the plugs are protected against nonetheless**

The plug for the 24 V is larger (pitch 5.08 mm) and fits no other terminal; terminal J11 has end walls, so it cannot be plugged in offset either. The plugs of the switching outputs have two rows and fit no single-row terminal. If a single-row plug sits in one row of J3 to J6, it touches at most one contact pin and switches nothing. Swapped plugs of RAUF and RUNTER or OST and WEST also switch nothing: the drive does not move. All of this is derived from the manufacturer's drawings and will be checked on the prototype.

⚠️ **These mix-ups remain possible**

The two-pole plugs of the limit switches (J7 to J10) are identical to each other: if they are swapped, a drive moves past the end position until the maximum run time stops it. Two-pole plugs also fit J2, J14 and J15, the anemometer plug also fits J2 and J14, the light plug also fits J2. Then the sun sensor delivers wrong values, the lights show wrong things or the anemometer is considered failed. Terminal J15 (WIND) is the only one with a pitch of 3.81 mm instead of 3.5 mm: its plug fits no other terminal, and no other plug of the controller fits J15. This way no sensor and no light accidentally receives the 11.5 V from J15.1 (**Assumption** until the plug test on the sample). Two-pole plugs also fit J22 (HALL).

J23 (from draft 0.12) is not protected against mix-ups: it is the only four-pole terminal, but its plug fits onto J2 and J14 with an overhang, and a two-pole limit switch plug fits onto J23. Therefore label the plug clearly with “J23 RST” and check it against the picture before switching on for the first time.

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⛔ **Sun sensor J2: never 24 V, route the cable separately**
Since 3.11.2026 the limit switches (J7 to J10), the lights (J14), the anemometer (J15) and the pulse input (J22) withstand 24 V permanently on every wire, for example from a swapped wire in the cable. **The sun sensor at J2 only partly:** every wire of J2 permanently withstands 12 V, for instance from a mix-up with the wind sensor supply at J15. 24 V on a measuring wire (pole 2 to 5) destroys the two measuring resistors of that wire on the board (R1 to R4 and R130 to R133), nothing else; that channel then delivers wrong values until the two resistors are replaced. 24 V on pole 1 (3.3 V) or pole 6 (shield) does no harm. Nevertheless, route the sensor cable separately from all 24 V cables (drives, power supply), never connect a 24 V wire to J2, and check every wire at J2 against the table above before switching on for the first time.

Power supply: two power supplies on the rack, a shared one as a variant

The controller needs only one power source: 24 V DC at terminal J11. The circuit board turns this into 5 V and passes it to the board via pin header J1. You do not need a cable to the board.

Since 3.11.2026 the following applies to the installation on the Samba Solar Rack: **two separate power supplies**. A Mean Well **HLG-600H-24** (IP67, fixed output 24 V ± 1 %) lies on the right foot beam of the rack, on the inside of the carriers of the control box (E1) and of box 2 (E1b), and feeds only the sync controller BST-KZQ-15A-2, the drives and the HallTapBoard. With two drives, the star point (busbar terminals), the fuses and the HallTapBoard sit in box 2 (E1b), the sync controller in the control box (E1) (assembly manual Rack 2-E Plus from V3.0.0). The cable to the HallTapBoard (L2) is protected in box 2 by a fuse of its own: fuse terminal block Phoenix Contact UK 6-FSI/C-LED24 with a 1 A automotive blade fuse in the positive wire directly at the star point (since V3.1.1, chapter 13.3). The self-resetting fuse F1 on the HallTapBoard only protects the board, not the cable. A **small 24 V power supply** (SELV, 2 A or more) next to the tablet feeds only terminal J11. The wind sensors are powered by the circuit board itself via J15 (12 V from regulator U23). **The negative poles of the two power supplies are never connected:** the Hall pulses come via optocouplers, the switching contacts are potential-free. The complete wiring on the Samba Solar Rack with two drives, the BST-KZQ-15A-2 sync controller and HallTapBoard is described step by step in the wiring manual (verkabelung.html, PDF `Samba-Solar-Track-Verkabelung-10Zoll.pdf`).

Variant with a shared power supply (decision of 12.10.2026, still permitted, Fig. 6): one powerful 24 V power supply near the drives, for the rack at the control box. From there, each consumer gets its **own cable with its own fuse** (star wiring): the drives a short, thick cable, the controller a thinner one up to 15 m long, the HallTapBoard a small one. This way the motor current never flows through the controller cable, and its voltage stays steady when the drives start. The 0.5 to 1 A fuse for the HallTapBoard belongs to this variant only.

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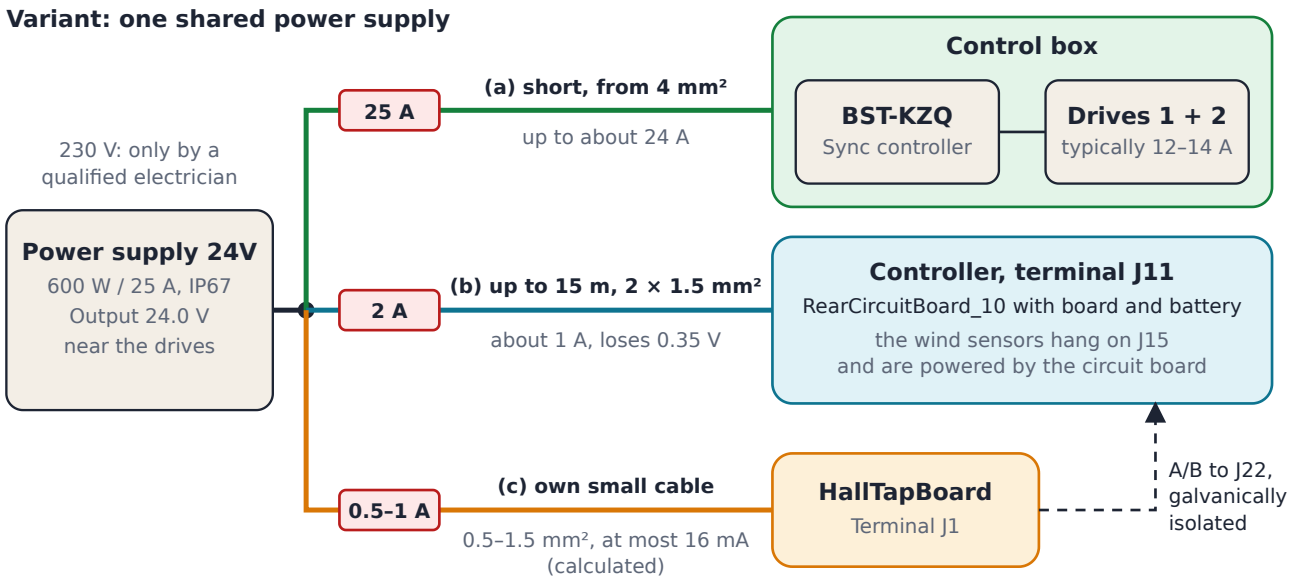


Fig. 6 – Variant with a shared power supply (decision of 12.10.2026, still permitted): one 24 V power supply near the drives, wired in a star. (a) Drives and sync controller via a short, thick cable, (b) the controller via its own cable up to 15 m, (c) the HallTapBoard via its own small cable with a 0.5 to 1 A fuse. The installation on the rack with two separate power supplies (since 3.11.2026) is drawn in the wiring manual.

1 Keep everything de-energised

The power supplies stay switched off until all the following steps are done. The 230 V side is connected by a qualified electrician. *Expected:* There is no voltage at the output of the power supply.

2 Set the output voltage

After connection by the electrician, switch on the power supply once without any load and measure the output voltage with the multimeter. If it has an adjuster, set it to **24.0 V**. Then switch it off again. *Expected:* 22.8 to 25.2 V; set an adjustable power supply to 24.0 V (a power supply with a fixed output such as the HLG-600H-24 delivers 24 V ± 1 % according to the datasheet; the circuit board is designed for 24 V ± 5 %).

3 Fit the fuses

On the rack (two power supplies): directly at the power supply of the drives, fit a 25 A fuse holder into the positive wire to the sync controller (for racks with one drive, the value from its assembly manual); the cable to the HallTapBoard gets its own 1 A fuse at the star point (fuse terminal block Phoenix Contact UK 6-FSI/C-LED24 with a 1 A automotive blade fuse, assembly manual Rack 2-E Plus V3.1.2, chapter 13.3; F1 on the HallTapBoard only protects the board, not the cable). On the Samba Solar Rack 2-E Plus, the 25 A fuse is also such a fuse terminal block in box 2 (E1b). In the variant with a shared power supply, three fuse holders in the positive wires: 25 A for the drives, 2 A for the controller, 0.5 to 1 A (slow-blow, **suggestion**) for the HallTapBoard. *Expected:* Each cable that needs a fuse has its own; no thin cable hangs behind the 25 A fuse.

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4 Lay the cable to the controller

Two wires with 1.5 mm² (0.75 mm² or more possible) from the power supply of the controller to terminal J11: on the rack from the small power supply next to the tablet (short), in the variant with a shared power supply up to 15 m. Label plus and minus at both ends. Do not branch off together with the drive cable at a shared terminal. *Expected:* The cable starts at the terminals of the power supply and ends at the plug of J11.

5 Wires into the J11 plug

Plus to contact 1, minus to contact 2. Screw the wires tight in the unplugged connector, and only then plug it in. *Expected:* Both wires sit firmly, no bare metal sticks out.

6 Plug in the connector

Push the plug all the way into J11. *Expected:* It sits firmly between the end walls.

ITEM	WHAT APPLIES
Power supplies	24 V DC, output approved for SELV. On the rack: for the drives the Mean Well **HLG-600H-24** (IP67, 600 W / 25 A, fixed output 24 V ± 1 %) on the rack; for the Samba Solar Rack 2.0-E Plus with the sync controller BST-KZQ-15A-2, the drives together typically need 12 to 14 A, at most about 24 A (overcurrent limit of the sync controller as delivered, ordering specification of the rack; up to 15 A per drive). For the controller a small power supply of 2 A or more next to the tablet; it needs about 1 A (with battery charging). In the variant with a shared power supply, one device with **600 W / 25 A** for everything.
Behaviour under overload	Variant with a shared power supply: A power supply with only 500 W is only an option if it limits the current under overload and keeps supplying (current limiting, CV+CC). A device that switches off and restarts under overload (hiccup) also cuts the power to the controller at every start-up peak; the board then runs on the battery.
Voltage	Fixed output 24 V ± 1 % (HLG-600H-24); set only an adjustable unit to 24.0 V. The circuit board is designed for 24 V ± 5 % (22.8 to 25.2 V); after 15 m of cable, about 23.65 V arrive at J11, without load the full 24.0 V. **Never permanently above 30 V.** At only 20 V, fuse F1 is not sufficient for the full current in the heat.
Sense lines	If the power supply has connections for sense lines (remote sense), they stay at the power supply as described in the datasheet. Never run them to the drives: the power supply would then compensate for the loss in the motor cable and drive the voltage at J11 too high.
Installation site	Shady. For the Samba Solar Rack 2-E Plus lying flat on the inside of the carriers of the control box (E1) and of box 2 (E1b) on the right foot beam, position and orientation according to its assembly manual (from V3.0.0); otherwise vertical, cables downwards. In the heat, a power supply delivers less (derating, curve in the datasheet).
Fuse F1	Resets itself. If it has tripped, switch off the 24 V, remove the cause and wait about one minute.
After an overvoltage shutdown	Switch off the 24 V for at least 30 seconds, then switch it on again. If the circuit board switches off again afterwards, it is defective.
PC at the “UART” socket	The PC also supplies the board via USB. Do not plug in a USB cable below 10 °C: the battery would then be charged despite the charge lock (Chapter 7).

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ITEM	WHAT APPLIES
Anemometer	Not connected to the power supply and not to a distribution terminal. Both wind sensors are connected with all four wires to J15 of the controller; the circuit board supplies them itself (Anemometer on RS485). Their current flows through the cable to the controller.
Drives	No motor current flows through the circuit board. The drives together with the sync controller get their own short cable from the power supply with the 25 A fuse: on the rack from the HLG-600H-24 (star point and 25 A fuse in box 2 (E1b)), in the variant with a shared power supply from that one.

Voltage loss on the drive cable (calculated with outgoing and return wire, copper). As a guide value, at most about 1 V should be lost at full current (**assumption**, the manufacturer gives no value). That is why the power supply is located at the drives: the long cable to the controller carries only about 1 A.

CROSS-SECTION AND LENGTH	AT 14 A	AT 20 A
4 mm², 2 m	0.25 V	0.35 V
4 mm², 3 m	0.37 V	0.53 V
4 mm², 5 m	0.61 V	0.88 V
6 mm², 5 m	0.41 V	0.58 V
6 mm², 10 m	0.82 V	1.17 V
4 mm², 15 m (for comparison: power supply far away)	1.84 V	2.62 V

ℹ️ **Two power supplies, two grounds – the HallTapBoard always stays isolated**
On the rack, the negative poles of the two power supplies are never connected to each other, not even via a cable shield: the Hall pulses come via optocouplers, the switching contacts are potential-free. In the variant with a shared power supply, the negative wires of all cables meet at the negative terminal of the power supply. There is then only a small difference between the ground of the controller and the ground of the sync controller: at most about 0.2 V on the negative wire of the controller and about 0.26 V on the negative wire of the drives (3 m, 4 mm², 20 A), calculated. However, it jumps every time the motors start. The HallTapBoard therefore keeps the two sides isolated with optocouplers. Do not create a second connection between the ground of the controller and the control box, for example via a cable shield.

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⚠️ **What the controller knows about the power supplies**

The “24 V present” input of the circuit board measures the voltage at J11. On the rack (two power supplies) it therefore reports only the power supply of the controller. Whether the power supply of the drives has failed, the fuse of the drives has tripped or the cable to the sync controller is interrupted, it does **not** show; the Hall tap is to detect this (HallTapBoard at J22, work in progress). In the variant with a shared power supply it also shows when the power supply fails (or the cable to the controller or its fuse). The firmware reads the input, shows a failure in the header (“24 V MISSING / CHECK POWER SUPPLY”, Chapter 8) and in the browser and reports it in the log; it has no effect on the control.

⚠️ **If a power supply fails**

If the power supply of the controller fails, the board keeps running on its battery (Chapter 7); the drives stand still when their power supply fails (in the variant with a shared power supply both at once). The switching contacts keep switching and move nothing. Without limit switches, the firmware may then remember a wrong end position and a wrong tilt; in a storm, on anemometer failure and when parking, it therefore moves flat again (Chapter 8). If the power supply fails during a storm, the module stays where it is. No controller can compensate for that.

Sun sensor at the ADS1115

Each of the four outputs of the sun sensor delivers a signal that increases with brightness. A 200 Ω resistor to ground, specified by the sensor manufacturer, turns it into a voltage. The ADS1115 measuring chip measures it: the brighter, the higher.

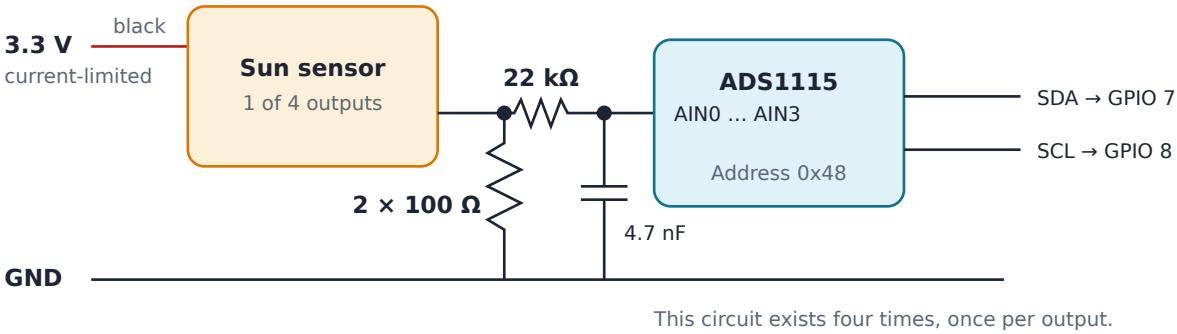


Fig. 7 – One output of the sun sensor at the measuring chip. The small capacitor smooths interference on the long cable. Fig. 4 shows which wire goes to which input.

The four switching outputs

The RearCircuitBoard_10 board issues movement commands via four **switching contacts**: RAUF (up), RUNTER (down), OST (east) and WEST. Each contact is potential-free, i.e. just a switch with

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two connections, without connection to the rest of the electronics. It tolerates up to 30 V and 1 A; above 33 V the protection diode across the contact conducts. That is enough for control signals and relay coils, not for a motor. The motor current therefore never flows through the circuit board.

How the contacts are connected depends on the drive. There are two variants, and you can choose **separately for each axis** which one you use:

🎛️

Variant A: controller with push-button input

The drive has its own control unit with connections for two push-buttons. The contacts of the circuit board replace the push-buttons. Example: sync controller for two drives (Fig. 10).

🔁

Variant B: external power relays

The drive has only two motor wires. The contacts of the circuit board switch two power relays that reverse the polarity of the motor (Fig. 8).

AXIS	CONTACTS OF THE CIRCUIT BOARD	VARIANT A	VARIANT B
Elevation	RAUF (IO0), RUNTER (IO1)	RAUF to UP and COM, RUNTER to DOWN and COM	RAUF switches relay 1, RUNTER switches relay 2
Azimuth	OST (IO2), WEST (IO3)	OST to UP and COM, WEST to DOWN and COM	OST switches relay 1, WEST switches relay 2

💡 **Nothing needs to be set in the software for this**

For the controller, both variants are the same: it closes a contact as long as it should move, and opens it to stop. You make the choice solely through the wiring. Elevation and azimuth may also be connected differently. The only things to adjust per axis are the maximum run time and whether limit switches are connected (Chapter 8).

Variant B: external power relays (polarity reversal)

A linear actuator moves in one direction when plus is on wire A, and in the other when plus is on wire B. Two power relays with changeover contacts swap the poles. They are located outside the RearCircuitBoard_10 board, for example as a ready-made relay module in the control cabinet. Their coils are switched on and off by the switching contacts of the circuit board.

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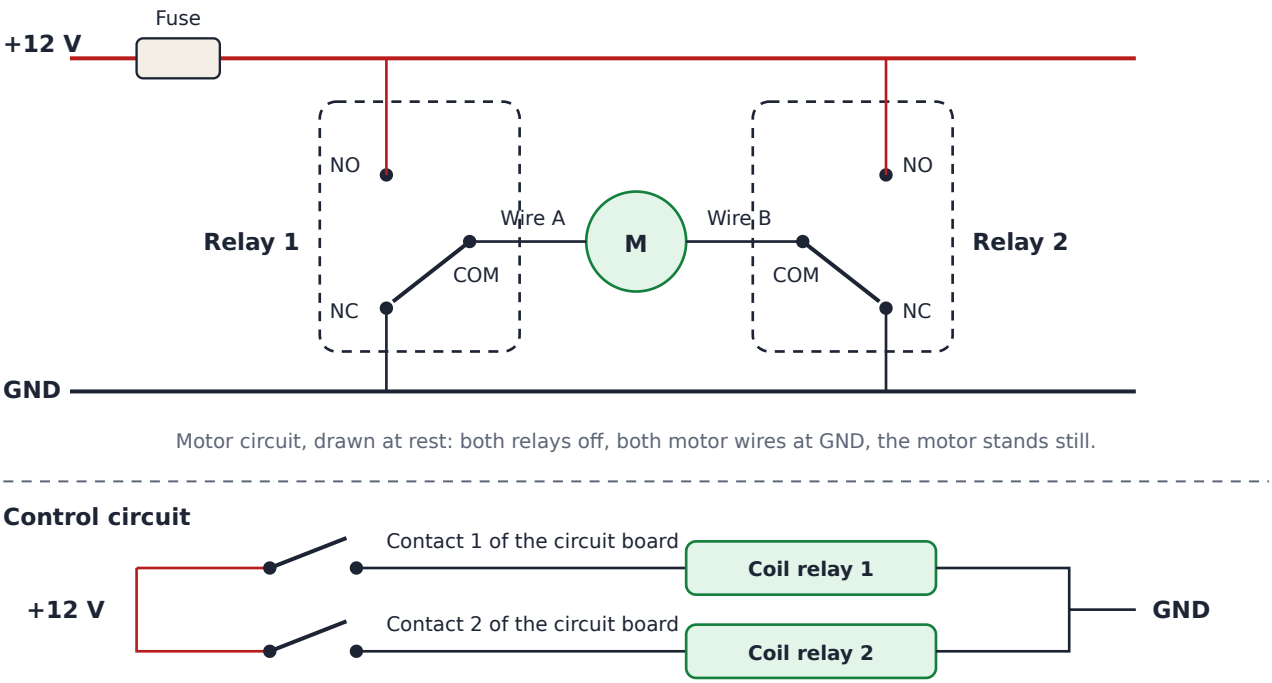


Fig. 8 – Polarity reversal circuit for one axis with two external power relays. At the top the motor circuit, at the bottom the control circuit: the two contacts of the circuit board (RAUF and RUNTER, or OST and WEST) switch the relay coils. NO = normally open, NC = normally closed, COM = common contact.

RELAY 1	RELAY 2	WIRE A	WIRE B	MOTOR
off	off	GND	GND	stands still (braked)
on	off	+12 V	GND	moves in direction 1
off	**on**	GND	+12 V	moves in direction 2
on	on	+12 V	+12 V	stands still. No short circuit, but never happens in the firmware.

💡 **Why exactly this circuit?**
Even if both relays were to pick up at the same time, there would be no short circuit of the power supply. The circuit is inherently safe, and the firmware additionally prevents this case.

⚠️ **Use relays with a flyback diode**
When a relay coil is switched off, a voltage spike occurs that can destroy the switching contact of the circuit board. Use relays or relay modules with a built-in flyback diode, or solder a diode in parallel to the coil. The coil voltage must be at most 30 V, for example 12 V or 24 V.

Assignment on the PCA9538

In “vertical only” operation, the azimuth outputs and limit switch inputs remain unused.

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PIN	FUNCTION	DIRECTION
IO0	Contact elevation RAUF (up)	Output
IO1	Contact elevation RUNTER (down)	Output
IO2	Contact azimuth OST (east)	Output
IO3	Contact azimuth WEST	Output
IO4	Limit switch elevation top	Input
IO5	Limit switch elevation bottom	Input
IO6	Limit switch azimuth east	Input
IO7	Limit switch azimuth west	Input

Limit switches

A limit switch sits at the end of the travel path and is pressed when the module arrives there. The **normally closed contact (NC)** is used: normally the circuit is closed, in the end position it opens.

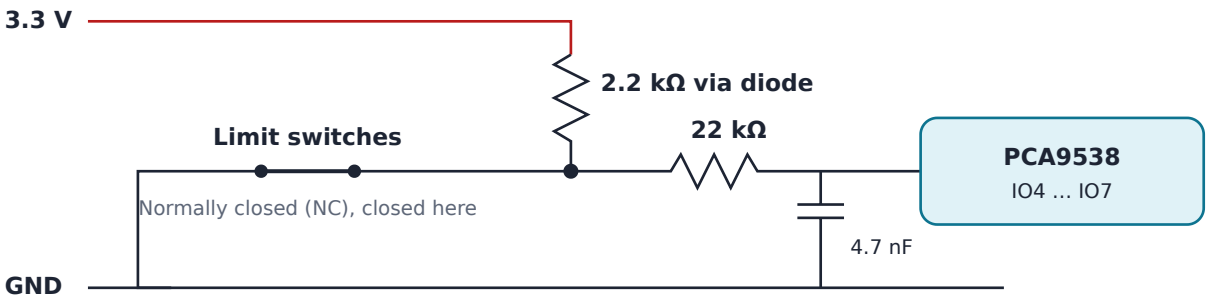


Fig. 9 – A limit switch input. When closed, the input is at GND (LOW): movement allowed. If the switch opens or the cable breaks, the resistor pulls the input to HIGH: movement blocked.

💡 **Interference pulses cannot cancel an end position**
On a long cable, an interference pulse can make the input look “closed” for a moment. The controller therefore reacts asymmetrically: if the switch opens, it blocks immediately. The direction is only released again once the switch has been continuously closed for a tenth of a second.

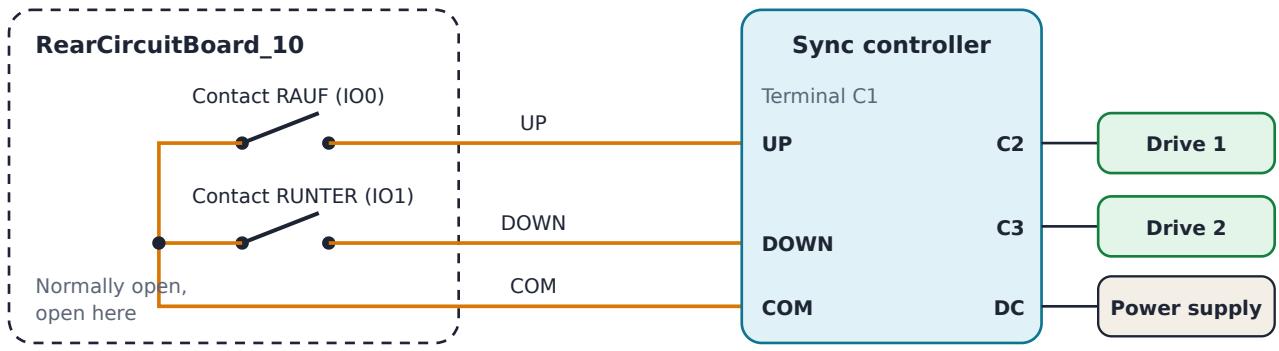
ℹ️ **Drive with built-in end-of-stroke cut-off?**
Many linear actuators switch off by themselves at the end of the stroke. In that case you can do without external limit switches and set “… limit switch fitted” for this axis to *NO* on the display under SETTINGS. The controller then assumes that the end position has been reached after the maximum run time has elapsed.

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Variant A: controller with push-button input

Some drives come with their own control unit to which two push-buttons can be connected. The most important example is the sync controller. A large module is often lifted by two linear actuators at the same time. So that it does not tilt, both must move at exactly the same speed. This is handled by a **sync controller**: it supplies both drives and continuously matches them. For the elevation, such a set with two 3500 N drives is presumably planned.

The controller has an input for two push-buttons, UP and DOWN (terminal C1 with the wires UP, DOWN and COM). Samba Solar Track “presses” these push-buttons with two switching contacts of the RearCircuitBoard_10 board. The motor current flows only through the controller.



Contact RAUF closed: both drives extend. Both contacts open: stop.

Fig. 10 – Connection to the sync controller. Two switching contacts of the circuit board replace the UP and DOWN push-buttons. They only switch a control signal.

The same wiring applies to the azimuth if a controller with push-button input also works there: then the contacts OST (IO2) and WEST (IO3) go to its push-button input.

⛔ **Order only in momentary mode**

The controller comes in two versions. For Samba Solar Track it must be **momentary mode (push-to-run)**: the drives only move as long as the contact is closed. In latching mode (self-holding), a short pulse is enough and the drives move all the way to the end. If the controller then fails, nobody stops them any more. The firmware does not support this version. This applies to every controller with push-button input, on the elevation as well as on the azimuth.

ITEM	WHAT TO DO
Contacts	The switching contacts of the circuit board are already potential-free normally open contacts. Contact RAUF (IO0) to UP and COM, contact RUNTER (IO1) to DOWN and COM.
Run time	Recalculate “Max. run time elevation” and set it under SETTINGS: stroke divided by speed, plus 20 %. Example: 300 mm stroke at 5 mm/s is 60 seconds; set 72 seconds.
Limit switches	The drives switch off by themselves at the end of the stroke. Without additional limit switches, set “Elevation limit switch fitted” to *NO* under SETTINGS.

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Direction of travel	If the module moves down on RAUF (up), swap the UP and DOWN wires.
Remote control	The radio remote control and the hand control of the controller bypass Samba Solar Track. Anyone moving with them can interfere with the storm protection. Best use them only for setting up.

🧩 **Still to be checked on the device**

Three things are not clear from the product description and are therefore assumptions: whether the controller needs a reset run via the hand control after a power failure, how it reacts to two contacts closed at the same time (the firmware never closes both) and what voltage is present at terminal C1.

⛔ **Mandatory: limit switches in the motor circuit**

A component at the switching output can age over the years so that the contact no longer opens completely. The controller does not notice this. Therefore the drives must have their own limit switches in the motor circuit that interrupt the motor current themselves at the end of the stroke. With variant B, a fuse also belongs in the coil circuit of the relays.

Reset of the sync controller: J23 or J5

The sync controller of the Samba Solar Racks (**BST-KZQ-15A-2**) has a control terminal with four connections: **COM · RST · DN · UP**. UP and DN act like the UP and DOWN buttons, RST like the Reset button. Each input is switched against COM, and only with potential-free contacts without external voltage. There are two ways for the reset:

WAY	WHEN	CONTACT AT RST AND COM	SETTING (CHAPTER 8)
J23	Board from draft 0.12; with “vertical only” and with “vertical + horizontal”	Elevation: J23 pole 1 (RST) and pole 2 (COM); azimuth: J23 pole 3 (RST) and pole 4 (COM) at the sync controller of the azimuth	Reset sync controller elevation or azimuth = J23
J5	Only with “vertical only”: the EAST output of the azimuth is then free. J23 does not have to be fitted.	The contact of J5 (EAST, the two outer, diagonally opposite pins, Fig. 12)	Reset sync controller elevation = J5

On the rack with four wires: The cable to the sync controller needs only four wires for moving and reset: UP, DN, RST and COM. COM is routed only once and bridged at the controller to J3, J4 and J23 pole 2; the three contacts are potential-free, so they may share one side. This way UP (J3) belongs to UP and COM, DOWN (J4) to DN and COM, RST E (J23 pole 1 and 2) to RST and COM. Without J23, with “vertical only”, RST goes to J5 instead of J23, and the COM bridge accordingly to J5. How the wires go into the plugs step by step is described in the wiring manual (verkabelung.html, Chapter 9).

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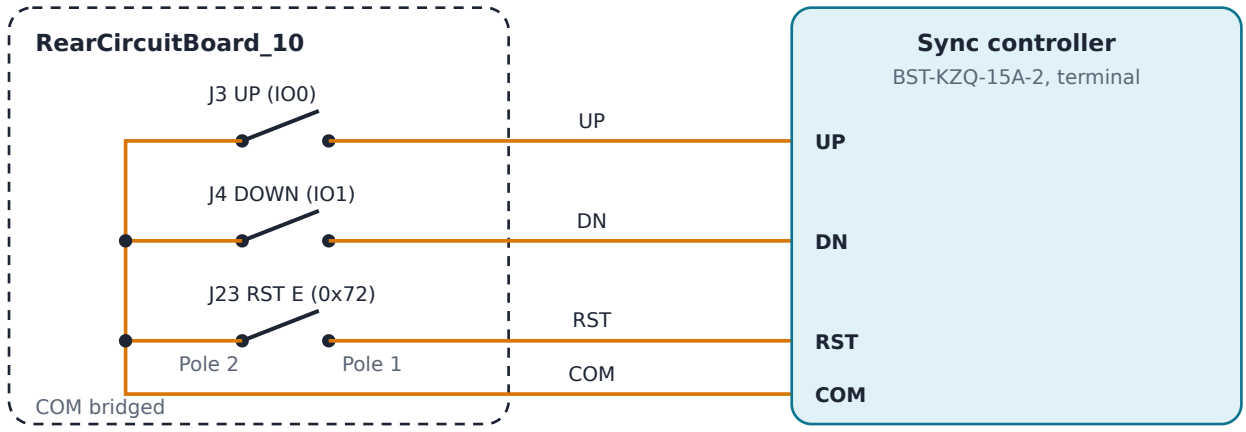


Fig. 10a – Moving and reset via four wires: J3 (UP) to UP, J4 (DOWN) to DN, J23 pole 1 (RST E) to RST; the second side of the three contacts (at J23 pole 2) is bridged at the controller and goes to COM as one wire. With “vertical only” without J23, contact J5 (EAST) takes over the role of J23.

⚠️ **Label the plug of J23**

J23 is not protected against mix-ups (see above). Label the plug with “J23 RST”. Never apply a voltage to an RST terminal: the sync controller expects a potential-free contact, and the contact on the board tolerates at most 30 V.

🧩 **Still to be checked on the device**

The manufacturer does not state how long RST must be connected to COM. The firmware closes the contact for 5 seconds (**assumption**, to be checked during the first installation on the rack). The board draft 0.12 has not been manufactured; J23 has only been checked on the computer.

Anemometer on RS485

The anemometer is connected with all four wires to terminal **J15** of the circuit board (1 = +12 V, 2 = GND, 3 = A, 4 = B, 5 = shield). The circuit board carries the RS485 chip for it; the board itself has none. The circuit board also supplies the sensors itself: J15.1 carries about 11.5 V from a separate, current-limited regulator (U23). You do not need a separate power supply or a branch at the power supply. The sensors from ACOVMELST are intended: the wind speed sensor (address 3 ex works) and, for the type with wind direction, additionally the wind direction sensor (address 4 ex works). Both are connected side by side to the same terminal.

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⛔ **Connect only when de-energised**

The manufacturer warns: if the anemometer is wired while live, it can burn out. First switch off the controller (unplug plug J11), then wire the plug of J15 and plug it in. According to the manufacturer, the sensors tolerate 5 to 12 V (5 to 24 V in its table); the circuit board supplies about 11.5 V.

ITEM	WHAT TO DO
Wires	According to the datasheet: red = plus, black = minus, yellow = A, green = B.
Supply	Red (plus) to J15.1, black (minus) to J15.2. The circuit board supplies about 11.5 V at J15.1, current-limited; together the sensors may draw at most 0.3 A continuously (two sensors need about 0.12 A, **Assumption**). A short circuit in the cable does no harm; the anemometer is then considered failed.
A and B	Yellow (A) to J15.3, green (B) to J15.4. For the type with wind direction, the wires of both sensors go in parallel to the same contacts: red to red, black to black, A to A and B to B. If swapped, nothing breaks; the anemometer simply does not respond. Even if 24 V or the 11.5 V from J15.1 in the cable gets onto A or B, the chip is not damaged.
Shield	Connect the shield of the cable to J15.5, only at the controller.
Wire the plug	Strip each wire by 7 to 8 mm (datasheet of the plug KF2EDGK-3.81-5P, pitch 3.81 mm: conductors AWG 28 to 16, up to 1.5 mm², screw M2 with at most 0.2 N·m). For the type with wind direction, two wires per contact go into the plug, preferably in a twin ferrule (**Assumption**, not tried out). *Expected:* Every wire sits firmly; no bare wire sticks out at any clamping point.
Termination resistor	Jumper J16 on 1-2 (on) if the controller sits at the end of the cable, otherwise 2-3. The other end of the cable gets a 120 Ω resistor between A and B at the last sensor (manufacturer's recommendation). On the board a capacitor sits in series with the resistor: if a 24 V wire accidentally ends up on A or B, the termination is not damaged. An ohmmeter therefore does not show 120 Ω between A and B.
Installation	The arrow on the base of the wind direction sensor points north.

Pendulum holder for the anemometer

An anemometer only measures correctly when it is vertical; on a tilting table, however, everything mounted on the module carrier moves along. For this there is a pendulum holder that keeps the anemometer vertical at any tilt of the table. It is not part of Samba Solar Track. Coming soon in the shop (not yet available): build instructions as PDF for self-assembly €19.99, set of printed parts €69.99, complete kit €129.99.

On the Samba Solar Rack, two wind sensor profiles (40 × 40, 410 mm) sit on the left side of the rack, 15 cm to the left and to the right of the left P5; they butt against the rear face of P6, are screwed there and flush at the top. On each profile a wind sensor hangs outwards in its pendulum; you choose which sensor goes on which profile. At 0° about 3 cm remain to the ground, at 60° the rack is about 2.2 m high, and behind the rack about 0.63 m more space is needed. The junction box E6 sits on the underside of the rear P6 on the tilting frame; E6 and all cables lie only on undersides. Each wind sensor cable must be at least 1.9 m long. The cable route is described in the wiring manual.

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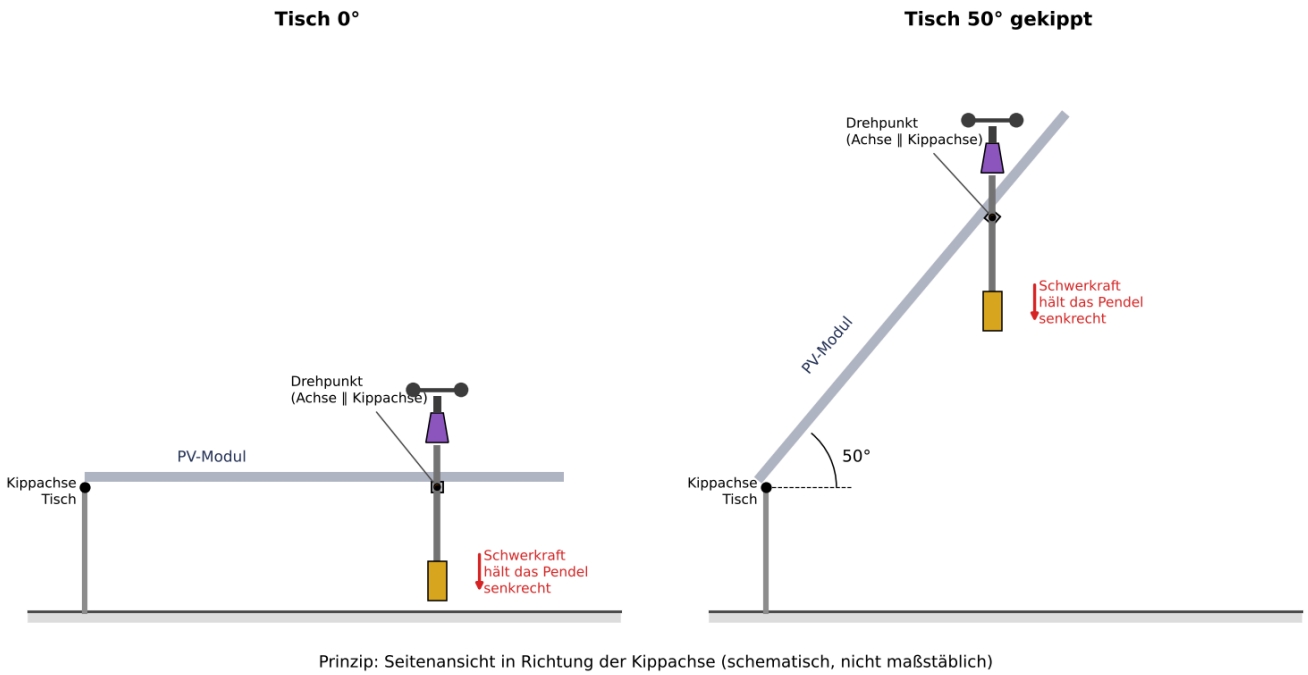


Fig. 10a – The principle, seen from the side in the direction of the tilt axis (schematic, not to scale). On the left the table at 0°, on the right tilted to 50°: in both cases the pendulum with the anemometer hangs vertically.

Lights in the housing

The circuit board carries the indicator lights BEREIT (ready, green) and FAIL (three-colour). For a second light on the outside of the housing, both are connected to the five-pole terminal J14: pole 1 is the common positive pole of both lights (easiest to solder together in the housing and run one wire to pole 1), poles 2 to 4 are red, green and blue of the FAIL light, pole 5 is the negative pole of the BEREIT light.

ITEM	WHAT TO DO
Which lights	Ordinary LEDs **without** a built-in series resistor; for FAIL an RGB LED with **common anode**. The series resistors are on the circuit board.
Cable length	Only short cables in the housing.
Short circuit	Does no harm: the supply of the lights is limited to about 135 mA.
During the charge lock	Only about 4 V are present; the lights may then be somewhat dimmer (**assumption**).

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HALL pulse input (J22)

🧩 **Draft, only checked on the computer**
The 10-inch firmware evaluates J22 when “Hall tap present” is switched on under SETTINGS (Tilt with Hall tap); this has only been built and tested on the computer. The HallTapBoard intended for it is only a draft so far (see below). Without a HallTapBoard, J22 stays free and the Hall tap off. The circuit has only been checked on the computer, not manufactured and not measured.

What is it? J22 is a three-pole plug-in terminal on the bottom edge of the circuit board, to the right of the terminal of the lights (J14, without printing), printed “HALL”. Two potential-free contacts are connected here: switches without their own voltage, for example the output of an optocoupler. The input is intended for the planned HallTapBoard. This small board is meant to tap the pulses of the Hall sensor of drive 1 at the sync controller (BST-KZQ-15A-2) and pass them on via optocouplers galvanically isolated as two contacts (channel A and B). The controller counts how often these contacts close and open and thereby detects whether and how far the drive moves (Chapter 8).

What the circuit board does for it: It applies a small test voltage (5 V via 4.7 kΩ) to each wire itself and detects whether the contact is open or closed. A protection diode at the terminal, a filter and a comparator with hysteresis make the input insensitive to interference on long cables. It is designed for 15 m of cable and up to 5,000 pulses per second and channel (**assumption**, the pulse rate of the drives has not yet been measured).

POLE OF J22	SIGNAL	GOES TO
1	Channel A (contact to pole 3)	GPIO29 of the board (J1 pin 19)
2	Channel B (contact to pole 3)	GPIO30 of the board (J1 pin 21)
3	GND: common return of both contacts and shield	Ground of the circuit board

⛔ **Never connect Hall sensors directly to J22**
Never connect the Hall signals of the drives directly from the sync controller to J22. The sensor cable of the drives is only intended for 4 m, its ground is on the negative wire of the drives and jumps with the motor current (ground offset, even with a shared power supply), and a wrong count could later cause the frame to twist. Connect only via the HallTapBoard with its optocouplers.

⚠️ **No voltage at J22**
No supply voltage and no sensor that expects power from here belongs at J22; the circuit board supplies no power there. If a wire with 24 V accidentally gets onto pole 1 or 2, the circuit board survives this even permanently, with the display board plugged in and switched on. At pole 3 (return), on the other hand, it short-circuits the external voltage source.

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This is how you connect the cable from the HallTapBoard to J22, once the HallTapBoard exists:

1. **De-energise the controller**
   Switch off the 24 V power supply (or pull the fuse of the cable to the controller). *Expected:* The BEREIT (ready) light is off, the display dark or in battery operation. Even in battery operation you can safely continue working, because at most 5 V are present at J22.

2. **Prepare the cable**
   Take a shielded cable with three wires (for example LiYCY 3 × 0.25 mm²), at most 15 m long. Strip the wires far enough that the bare end disappears completely into the clamping point of the plug (about 6 to 7 mm, **assumption**, check on the plug), and twist the shield. *Expected:* three bare wire ends and one twisted shield end, no wire nicked.

3. **Wire the plug**
   Plug KF2EDGK-3.5-3P: pole 1 = channel A, pole 2 = channel B, pole 3 = common return **and** shield. Pole 1 is marked with a triangle on the circuit board; in the view of Fig. 5 it is on the left, on the installed device (turned by 180°, cable glands at the bottom) seen from the back on the right. Tighten each screw firmly and label the plug “J22 HALL”. *Expected:* Each wire sits firmly, no copper strand sticks out.

4. **Connect the other end**
   On the HallTapBoard the same assignment: A, B and the common return. Do **not** connect the shield there. *Expected:* The shield is only connected at J22 pole 3.

5. **Plug into J22**
   Into the terminal printed “HALL”; it is the only three-pole terminal. *Expected:* The plug sits flush on the header, no pin is free.

6. **Rule out a short circuit**
   Still without voltage, measure with the multimeter (resistance) between pole 1 and pole 3 and between pole 2 and pole 3. *Expected:* no short circuit (no value near 0 Ω). An optocoupler only conducts when the HallTapBoard itself is supplied; without voltage, the multimeter therefore usually shows a high resistance.

7. **Switch on and measure**
   Switch the 24 V power supply back on, the drives stand still. Measure the DC voltage between pole 1 and pole 3, then between pole 2 and pole 3. *Expected:* about 5 V with the contact open, about 1 V with it closed (the HallTapBoard has 1 kΩ in series with the contact). Depending on the position of the drive, either value may occur. During a movement the value jumps too fast for the multimeter; it then shows some value in between.

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⚠️ **No test setup without the board with 24 V at the pulse input**
If the circuit board lies on the table without the board plugged in and a wire at J22 gets 24 V (for example a swapped connection at the HallTapBoard), current flows through the pull-ups R102/R103 into the board's 5 V rail. Without the board as a load it rises to about 11 V according to the calculation. This endangers the regulators U12, U5 and U9. Therefore only connect the pulse input to the HallTapBoard with the board plugged in.

ℹ️ **Route the cable separately**
Do not run the cable together with the motor cables in one cable where it can be separated. Connect the shield only at J22 pole 3.

🧩 **Still open**
The pulse rate of the drives, the switching time of the HallTapBoard optocouplers and the capacitance of the cable are **assumptions** until they have been measured. With the optocouplers of the HallTapBoard (draft 0.6), according to the calculation it is certainly sufficient for a few hundred pulses per second; 1,000 are only promised once the switching time has been measured on the prototype, and it does not reach the 5,000 that J22 itself is designed for. Whether the plug and cable outlet at the bottom next to J14 fit into the housing (about 9 mm remain at the bottom) will be checked on the prototype.

The HallTapBoard on the rack (draft 0.6, in planning)

🧩 **Draft: do not build yet**
The HallTapBoard exists as draft 0.6 of 11 November 2026 (revised after a review on the same day), together with a housing from the 3D printer. Schematic, circuit board, manufacturing data and housing have only been checked on the computer: the KiCad rule checks report no errors, and the tool for the housing checks its dimensions. Nothing has been manufactured, printed, measured or tried out on the rack; all values are calculated or read from datasheets. **Before ordering** come points 1 to 4 of the test plan in the README: measurements on the rack at the sync controller BST-KZQ-15A-2 (voltage and level of its Hall connection, how fast the pulses come, whether it tolerates the tap, voltage spikes and lowest voltage of the supply). Whether a prototype is ordered without these measurements is decided by the user. The 10-inch firmware evaluates J22 (Chapter 8), but this too has only been built and tested on the computer.

What is it? A small circuit board (70 × 40 mm) for a box on the rack, close to the BST-KZQ-15A-2 sync controller; on the Samba Solar Rack 2-E Plus it sits in box 2 (E1b) next to the control box (E1). It “listens in” on the Hall wires of drive 1, just before its CH1 connector, without disturbing them, and passes them on via two optocouplers as two potential-free contacts (A and B) to J22 of the RearCircuitBoard_10 board. The two sides of the board are galvanically isolated: on the left is

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everything that belongs to the sync controller and the 24 V side of the drives, on the right only the two contacts to the controller. In between is a 7 mm wide strip without copper, the isolation gap (printed “TRENNUNG”). All three terminals sit on **one** edge, the bottom edge; there the three cables enter the housing through three cable glands. At the top left there are two LEDs side by side: on the left the yellow D9 printed “24V”, on the right the green D4 with “5V OK”. The yellow one lights up as soon as the 24 V arrive at the board with correct polarity. The green one only lights up when the board's voltage regulator also delivers its 5 V; only then does the tap work. If only the yellow one lights up, the 24 V are there, but something is wrong on the board (table “What the lights show” below). Since draft 0.6 both shine about equally bright (calculated; before, the yellow one was about ten times dimmer).

Safe in the event of a fault: Since draft 0.4 the supply is built so that a single defective component or an incorrectly connected power supply (up to 30 V continuously, short voltage spikes) does not overload any other component. Either the board keeps working, or the self-resetting fuse F1 switches it off. For this it has a voltage regulator that tolerates up to 40 V, a protection diode on the 5 V and two resistors in series in front of each light. Since draft 0.5 there are also two resistors each (20 kΩ in total) in front of the two protection diodes at J2: if one of these diodes fails short, the Hall signal at the sync controller is preserved instead of being short-circuited. **New in draft 0.6:** A second self-resetting fuse F2 sits between pole 3 of J2 (GND of the sync controller) and the ground of the board. Up to draft 0.5 the two were connected directly: if J1 was connected with reverse polarity while J2 was plugged in, 24 V reached GND of J2, or if the negative wire of the sync controller came loose, the current of the power supply (up to about 25 A) flowed without a fuse through the board and the thin Hall wire. Now F2 limits the current and switches off. Also new: if J2 is unplugged or a Hall wire is broken, the channel reliably reports “contact open” instead of a random value. All of this is calculated, not tested (fault table in the README).

⚠️ **Voltage between the two grounds (to be measured on the rack)**
F2 lies between the ground of the board (negative at J1, from the star point) and GND of the sync controller (pole 3 of J2). Outside the board, the two meet via the negative wires at the star point. If some voltage drops on the negative wire of the sync controller while driving, a small current therefore flows through F2: below about 0.18 V nothing noticeable, from about 0.54 V F2 switches off during travel and back on at standstill (calculated). This does no harm; the difference then lies at the Hall input, and the tap keeps working as long as it stays below about 1.7 to 2.0 V (calculated, **assumption** until measured). Therefore connect the negative wire of the cable to J1 at the star point, not behind the sync controller. This is measured on the rack: DC voltage between J2 pole 3 and J1 pole 2 while both drives are moving; target well below 1 V (test plan in the README, point 7).

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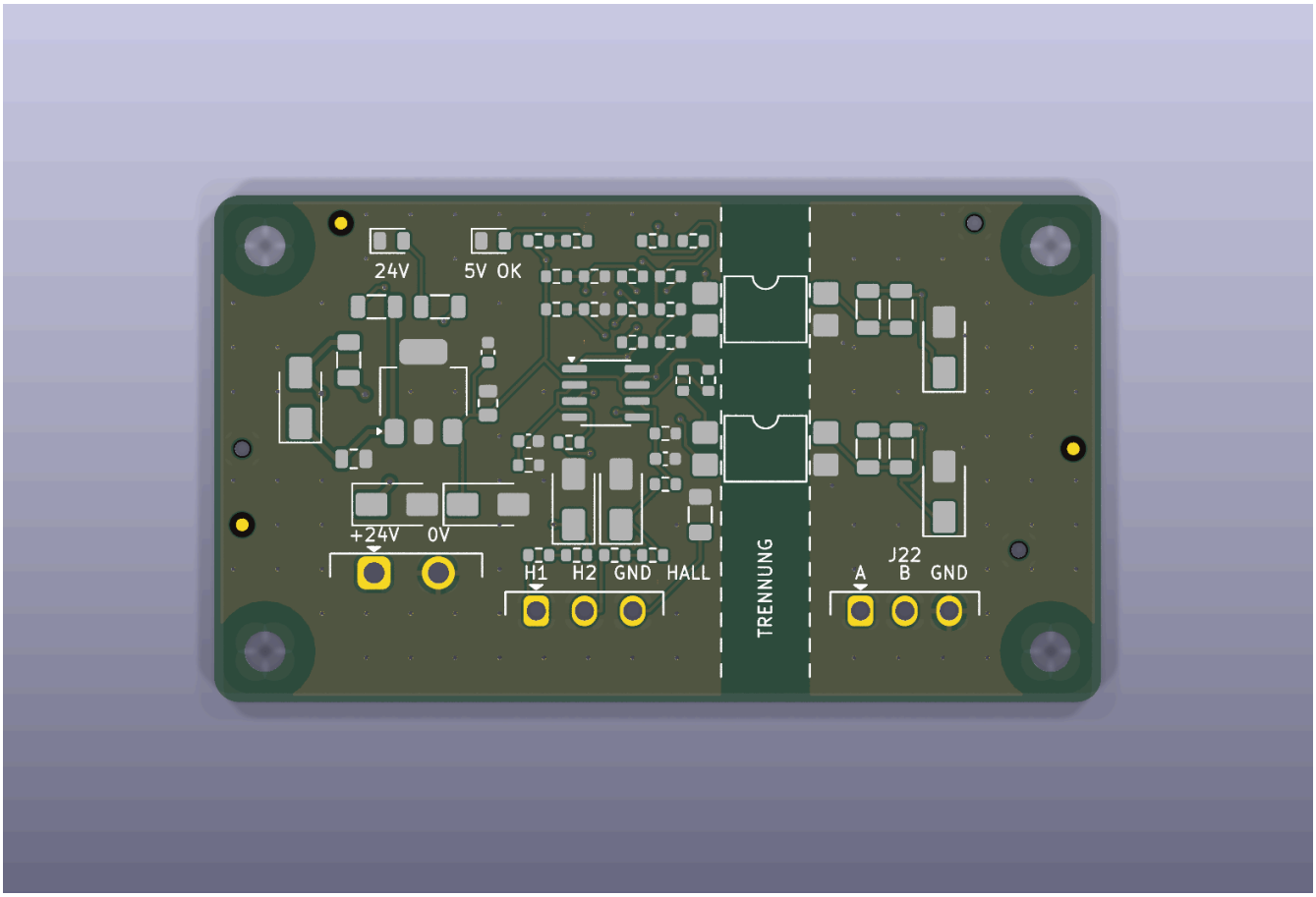


Fig. 11 – Draft 0.6 of the HallTapBoard from above (image from the KiCad project, without the bodies of the terminals). At the bottom from the left: J1 for the 24 V (+24V, 0V), J2 for the Hall signals of drive 1 (H1, H2, GND) and on the right behind the isolation gap J3 for the cable to J22 (A, B, GND). At the top left the two lights “24V” (left) and “5V OK” (right), in the middle the two optocouplers above the isolation gap.

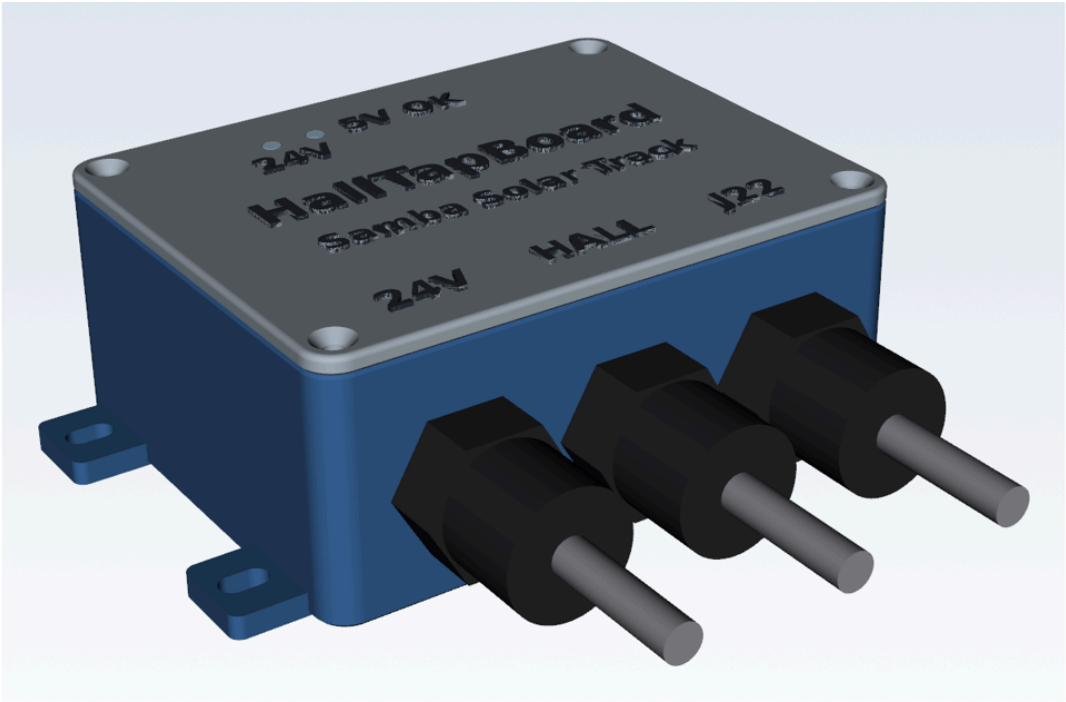
TERMINAL	POLE	PRINTING	CONNECT TO
J1 (pitch 5.08 mm)	1	+24V	+24 V via its own small cable from the power supply of the drives (on the rack HLG-600H-24, via the 1 A fuse terminal block at the star point in box 2 (E1b) – F1 on the HallTapBoard only protects the board, not the cable; in the variant with a shared power supply with its own 0.5 to 1 A fuse). **Check the polarity with the multimeter before J2 is plugged in.**
	2	0V	Minus of the power supply
J2 (pitch 3.81 mm)	1	H1	Hall1 (Ch.A), **yellow** wire, via a lever connector before the CH1 connector (drive 1)
	2	H2	Hall2, **green** wire, likewise
	3	GND	GND, **black** wire, likewise (connected to the ground on the board via the fuse F2)
J3 (pitch 3.5 mm)	1	A	Cable to J22 pole 1 of the RearCircuitBoard_10 board
	2	B	Cable to J22 pole 2
	3	GND	Cable to J22 pole 3 (common return; shield only at J22)

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The housing

The HallTapBoard gets its own small housing from the 3D printer (housing version 0.6 of 12.11.2026, changed after a test print of version 0.5; it fits the board draft 0.6 unchanged, because the holes, terminals and lights are in the same places): outside 104 × 82 × 44 mm without tabs; including the four tabs for screwing it down, the base is 128 mm wide. It has the shape of a standard industrial enclosure: on the outside a rectangle with evenly rounded edges, without attached pillars, and the lid is exactly the same size as the base. It consists of two parts, the base and the lid. Eight brass threaded inserts sit in the base: four in the domes on which the board rests, and four in the **reinforced corners** for the lid screws: inside, the corners are widely rounded, and the material between this rounding and the outer corner carries the screw. There are no pillars inside. In the front wall there are three PG 11 cable glands, one each for the 24 V cable, the Hall cable of drive 1 and the cable to the controller. Above each of the two lights the lid has a small hole with a light shaft; next to it, raised lettering on the lid says what they mean: “24V” below the left one, “5V OK” to the right of the right one. Into each shaft you insert a light guide: a piece of acrylic round rod (PMMA) with 3 mm diameter and **30 mm length**. It is flush with the lid at the top and ends about 1.5 mm above the light at the bottom; this way the light is clearly visible on top of the lid, and the shaft is closed. So with the lid closed you can see whether the board is supplied and whether its regulator is working. Between the base and the lid there is a **round cord Ø 1.5 mm** as a seal: the upper rim of the base has an inner collar with a groove into which the cord goes; the lid compresses it. The groove runs as a smooth loop with four wide corners around the interior, passing the screws on the inside, so the cord never has to go around a tight curve. At the top the groove is narrower than the cord: you press the cord in through the narrow opening, behind it the groove is wider, and the cord does not fall out again. The lid screws lie in the corners outside the seal. The lid is closed with four **countersunk screws M3 × 10**, whose heads sit flush in the four corners of the lid. The files for printing are in `hardware/HallTapBoard/gehaeuse/`: three ready-made print projects (3MF) for the Creality K2 Plus with Hyper PETG-CF – one with both parts, one with only the base and one with only the lid –, plus STL for printing on other printers and STEP for modifying.

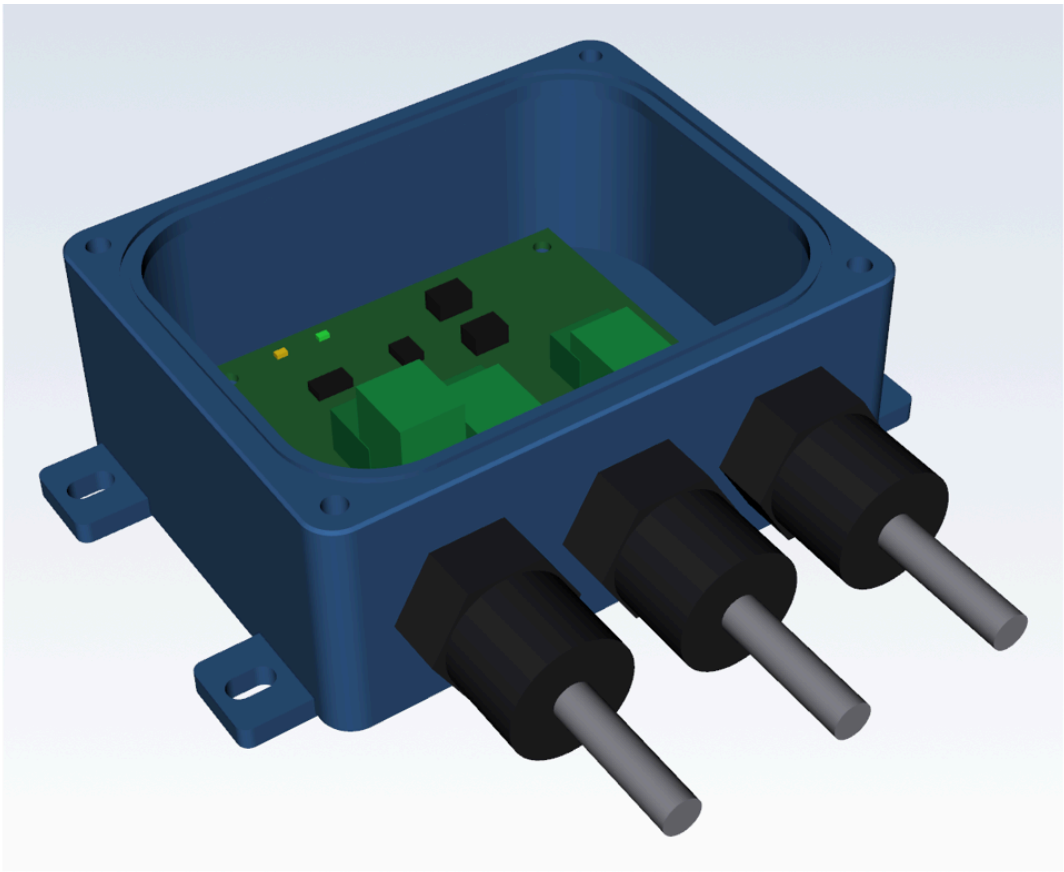
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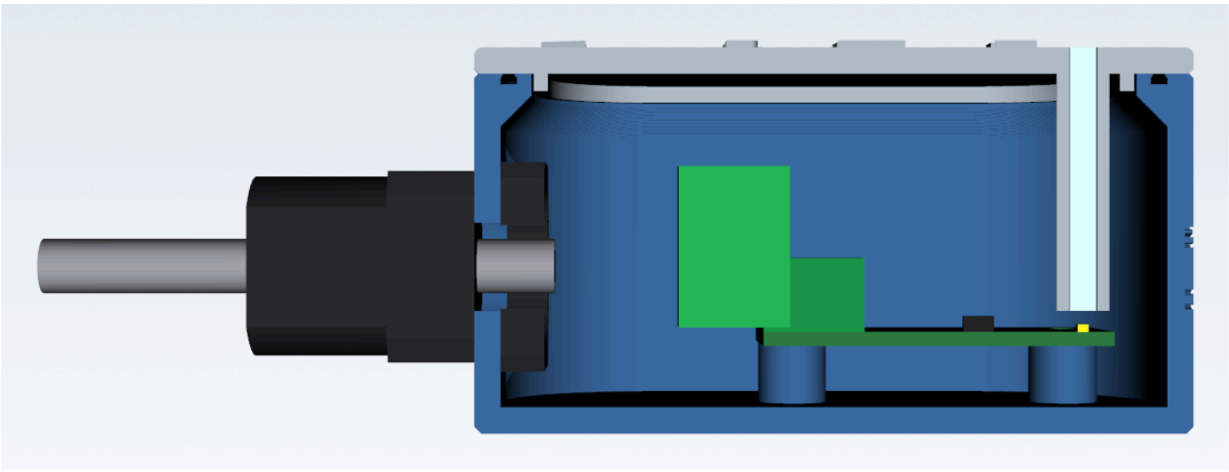
Fig. 11a – The housing closed (image from the tool `tools/halltap_gehaeuse.py`, not printed). At the front the three cable glands; on the lid above them raised lettering says “24V”, “HALL” and “J22”, at the back left the two holes for the lights with “24V” and “5V OK”. The base and the lid are the same size and have the same rounded edges all round; in the four corners of the lid the countersunk screws sit flush. Bottom left and right, the tabs with slotted holes for screwing down.

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Fig. 11b – The housing open with the board inserted. Board, plugs, components, glands and cables are only drawn as simple bodies. The plugs point to the wall with the glands; in the four reinforced corners, at the top, the holes for the lid screws. At the top of the rim on the inside the narrow collar with the groove for the sealing cord; it runs all round as a smooth loop with four wide corners, passing the screws on the inside. Nothing is in the way inside, the board can be inserted straight down.



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Fig. 11c – Section through the housing at the “24V” light (on the left the cable wall; the “5V OK” light has the same shaft 8 mm next to it). The board rests on the domes 7 mm above the bottom; from the lid, the light shaft reaches down to 2.4 mm above the board, with the light guide inside it (light blue, 30 mm) reaching to about 1.5 mm above the light. The plug of J1 (green) is the highest part; above it about 10 mm remain to the lid (calculated). At the top right and left the collar with the groove for the sealing cord (1.2 mm deep, 1.9 mm wide at the bottom, narrower than the cord at the top) and the bevel below it, so that the collar can be printed without supports.

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This is what the first printed prototype looked like (photos from 15.10.2026, black filament, still without board). It is the version before 11.11.2026: still without seal, 4 mm lower, with recessed lettering, cheese-head screws on the lid and the four pillars inside in the corners (since version 0.6 the screws sit in the reinforced corners). It has not yet been measured or tested.



Fig. 11d – Photo: on the left the base with the eight threaded inserts and the three PG 11 cable glands, on the right the tabs for screwing down; next to it the lid from the inside with the two light guides at the top right.



Fig. 11e – Photo: the base from above (old version). Inside in the corners the four pillars for the lid screws (since version 0.6 reinforced corners instead of pillars), inside the four low domes for the board, at the front the lock nuts of the glands.

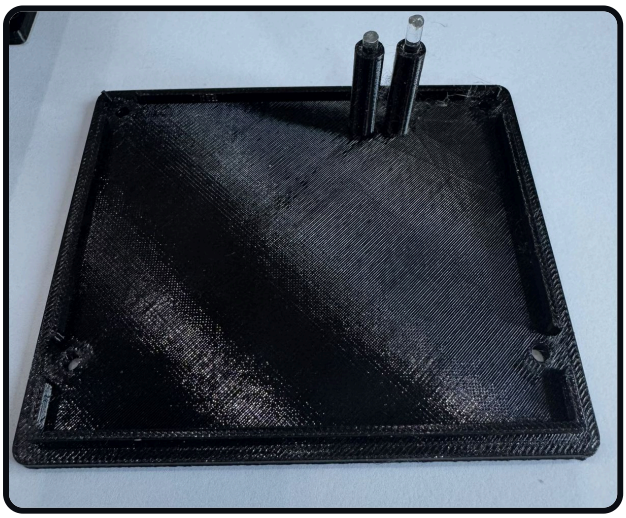


Fig. 11f – Photo: the lid from the inside. The two light shafts with the inserted acrylic rods will later reach to just above the lights of the board.



Fig. 11g – Photo: the housing closed. At the top left the light guides with “24V” and “5V OK”, in the middle “HallTapBoard” and “Samba Solar Track”, above the glands “24V”, “HALL” and “J22” (printed in one colour, without colour change).

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PART	QUANTITY	PURPOSE
Printed base and lid (Hyper PETG-CF, ASA or PETG)	1 each	Housing
Threaded insert ruthex M3 (RX-M3x5,7)	8	4 for the board, 4 for the lid
Screw M3 × 8 (cheese head)	4	Board
Countersunk screw M3 × 10 (ISO 10642, hexagon socket)	4	Lid
Round cord Ø 1.5 mm (EPDM or silicone)	about 40 cm	Seal between base and lid (about 31 cm are needed)
Washer M3 (outer 7 mm)	4	under the screws of the board
Cable gland PG 11 (Lapp SKINTOP ST, cable 4–10 mm) with lock nut	3 each	Lead in cables
Screw M4 (length according to the mounting plate)	4	Fasten the housing on the mounting plate (on the Rack 2-E Plus in box 2 (E1b))
Light guide: acrylic round rod (PMMA) Ø 3 mm, clear, 30 mm long each	2	Light of the two lights through the lid

Sources and prices are listed in Appendix G.

⚠️ **Cable not too thin**
A PG 11 cable gland seals cables with 4 to 10 mm outer diameter. It does not hold individual wires or very thin cables (for example 3 × 0.14 mm²). For the Hall cable, use a sheathed cable, for example LiYY 3 × 0.25 mm² or 3 × 0.5 mm², and measure the outer diameter before buying (the cable diameters here are **assumptions**).

ℹ️ **No substitute for the box on the rack**
The printed housing protects the board against contact, dust and splashes in the box. The sealing cord additionally keeps out moisture and dust; nevertheless, the housing has not been tested for an IP rating. It therefore belongs in a closed box (on the Rack 2-E Plus box 2 (E1b)), not outdoors.

Testing the board on the bench (incoming inspection)

Before you install a delivered board, you test it on the bench, without the rack and without the sync controller. This way a component fitted the wrong way round is noticed before it is connected to the Hall cable of the drives or to the controller. You need a multimeter, a laboratory power supply with adjustable current limiting (24 V), a second voltage source with 5 V (for example a 5 V plug-in power supply with open wires) and a 1 kΩ resistor. The expected values are calculated, not measured on a prototype.

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⛔ **Never connect J1 with reverse polarity while J2 is plugged in**

The board only tolerates reverse polarity at J1 as long as nothing is plugged into J2 and J3. If J2 is plugged into the sync controller, with J1 reversed the 24 V reach GND of the sync controller via the ground of the board; then the fuse F2 has to switch off. You therefore do the reverse polarity test only here on the bench, and before installation you always check the polarity of J1 with the multimeter before you plug in J2.

1

Inspect

Look at the board with a magnifying glass and compare the position of the polarised parts with the list in `hardware/HallTapBoard/fertigung/smd/HINWEIS.txt` (section “Vor dem Bezahlen prüfen”): D2 and D10 with the line at the cathode as in the silkscreen, U1 with the wide cooling tab towards the edge with the lights, U2 with the mark at pin 1, U3 and U4 with the notch in the middle of the top edge (pin 1 to the left of it), D4 and D9 with the cathode at the angle in the silkscreen. *Expected:* All parts sit as in the silkscreen, no solder bridges, no part missing or crooked.

2

Measure resistance without voltage

Multimeter on resistance. Measure between J1 pole 1 and J1 pole 2. *Expected:* no value near 0 Ω; the value rises slowly because a capacitor is charging. Then measure each pole of J3 against J1 pole 2 and against J2 pole 3. *Expected:* open (over 20 MΩ or “OL”), because the isolation gap lies in between. Finally J2 pole 3 against J1 pole 2. *Expected:* a few ohms (the fuse F2, according to the part description about 3.6 Ω cold; subtract the value of the test probes held together).

3

Switch on with 24 V

On the laboratory power supply set 24 V and a current limit of **50 mA**, output off. Plus to J1 pole 1, minus to J1 pole 2 (nothing plugged into J2 and J3). Switch the output on. *Expected:* Both lights are on, “24V” yellow and “5V OK” green, about equally bright. The power supply shows a current of a few mA up to about 16 mA (calculated without a Hall signal about 4 to 6 mA). Between the capacitor C2 (or the cathode of D10) and J1 pole 2 you measure 4.85 to 5.15 V. If the power supply immediately limits at 50 mA or nothing lights up: switch off, do not install the board.

4

Check channel A

The 24 V stay on. Apply the 5 V source via the 1 kΩ resistor to J2 pole 1 (H1), its minus to J2 pole 3. Set the multimeter to **diode test** (the position with the symbol of a diode), red probe to J3 pole 1, black to J3 pole 3. *Expected:* With 5 V at H1 the multimeter shows a value (the contact is closed); without the 5 V it shows “OL” (the contact is open). Which value exactly appears depends on the multimeter (**assumption**).

5

Check channel B

The same with J2 pole 2 (H2) and J3 pole 2. *Expected:* as with channel A. If the wrong channel or neither reacts, the board is assembled incorrectly: do not install it.

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6 Reverse polarity test (only here)

Output off, remove the 5 V source; nothing is plugged into J2 and J3. Swap plus and minus at J1 (plus to pole 2), current limit still 50 mA, output on. *Expected:* Both lights stay dark, the power supply shows only a few µA (almost 0 mA). Then output off and connect the right way round again. *Expected:* Both lights come on again.

Printing and preparing the housing

1 Choose the material

For the ready-made print project **Creality Hyper PETG-CF**. Carbon-fibre filament wears down the nozzle; the printer needs a hardened nozzle. Also possible: ASA, which stays dimensionally stable even at high temperatures in the box, or PETG, if it reliably stays cooler than about 60 °C in the box (**assumption**; material values not looked up). *Expected:* You have enough material for about 115 g (**estimate**).

2 Open the print project

With the Creality K2 Plus and a 0.4 mm nozzle, open the file `HallTapBoard-gehaeuse_K2Plus_HyperPETG-CF_0.4mm.3mf` in Creality Print. It contains both parts correctly oriented on one plate. If you want to print the parts separately, use `HallTapBoard-gehaeuse-unterteil_K2Plus_HyperPETG-CF_0.4mm.3mf` (only the base) and `HallTapBoard-gehaeuse-deckel_K2Plus_HyperPETG-CF_0.4mm.3mf` (only the lid); there the part lies in the middle of the plate in each case, and the settings are the same. The orientation: the base on its bottom with the opening facing up, the lid with the **outside facing up** (lettering on top). The printer, the filament Hyper PETG-CF and the process “0.20mm Standard” are set, plus on the object 5 wall lines, 6 bottom and 6 top layers, 30 % gyroid infill and tree supports: for the **base, supports also on the model** (they only form in the three large round holes of the wall, because there they stand on the lower edge of the hole), for the **lid, supports only from the build plate**. With a different printer, load the two STL files; they are already in this orientation. Then set the supports the same way: with “supports only from the build plate” for the base, the three holes get no supports. *Expected:* The preview shows supports for the base only in the three large round holes of the wall; for the lid, supports from the bed stand under the whole inside, only the two light shafts themselves stand on the bed.

3 Lettering

The lettering on the lid is bold and raised by 0.8 mm; because the lid lies with the outside facing up, it is printed freely as the topmost layers and stays sharp. For two-colour lettering, set a colour change for the lid in the slicer at the top of the cover plate (where the letters begin) and swap the filament. *Expected:* The letters stand out clearly above the surface and are readable from a distance of one metre.

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4

Underside of the lid

The inside of the lid is printed on supports. This is where the lid will later rest on the sealing cord. Remove the supports carefully and completely, and sand the surface above the groove, i.e. the rim all round, flat with sandpaper (grit 240). *Expected:* The rim is free of support residue and lies on a flat surface without a gap. Whether the surface created this way really seals has not yet been tested (see “Still open” below).

5

Print and inspect

Print both parts and let them cool. Break the supports out of the three large holes and smooth the edge with a deburring tool or knife. Put the lid on as a test without the sealing cord. *Expected:* The guide rim of the lid slides with slight play along the inside of the collar of the base, the lid and the base are flush all round, the four countersinks lie exactly above the holes in the corners. The cable gland can be inserted into its hole from outside; if it is too tight, carefully rework the hole with a step drill (nominal size 18.9 mm).

⛔ **Hot soldering iron**

To insert the threaded inserts, the soldering iron gets as hot as for soldering, and the insert stays hot for a while afterwards. Do not touch the tip or the insert with your fingers, place the base on a firm, heat-resistant surface and always put the soldering iron down in its stand.

6

Insert the threaded inserts

Set the soldering iron with an insertion tip to about the printing temperature of the material (**assumption**, no manufacturer specification looked up). Place an insert narrow side down on a hole, insert the soldering iron and let the insert sink in vertically without pressure until it is flush with the top edge. Then pull the soldering iron straight out. Do this for all eight: four in the domes at the bottom, four at the top in the reinforced corners. *Expected:* Each insert sits straight and flush; after cooling, an M3 screw can easily be screwed in by hand.

7

Fit the cable glands

Start with the **middle** gland: between two glands already fitted there is hardly any room outside for the spanner. Push the gland through its hole from outside and screw the lock nut on by hand inside. Then hold the lock nut inside and tighten the gland outside with a 22 mm spanner (Lapp specifies 4.0 Nm for this, i.e. firmly by hand, not by force). A 24 mm open-ended spanner can hardly be applied inside under the rim and next to the neighbours; then hold the lock nut with pliers or with your fingers. Then the two outer glands in the same way. Leave the dome nut outside loose for now. *Expected:* The three glands sit firmly and no longer turn.

8

Check the board before it is installed

First test every delivered board on the bench, as described above under "Testing the board on the bench". *Expected:* All test steps there have been passed.

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9

Insert the board

The terminals of the board point to the wall with the glands. Lower the board straight down from above until its four holes rest on the domes; nothing is in the way inside, the collar at the upper rim is narrow, and at the rounded corners there is a good 1 mm of clearance next to the corners of the board. Then screw it down with four M3 × 8 screws and washers (outer diameter at most 7 mm), only hand-tight. Do not put any metal standoffs in between. To remove it, the other way round: screws out, lift the board straight up and out. *Expected:* The board lies flat on the four domes and does not wobble; the two lights are at the back left.

10

Insert the light guides

Saw two pieces of 30 mm each from the acrylic round rod (Ø 3 mm) and sand both ends smooth with fine sandpaper, ideally polishing the top surface. Press each piece from above into one of the two small holes in the lid until it is flush with the top side. The hole is drawn at 3.2 mm; it usually prints somewhat smaller, so that the rod sits tightly (**assumption**). If it goes in too stiffly, rework the hole by hand with a 3.1 mm drill; if it is loose, put a drop of clear glue on the rim at the top (no superglue, it makes acrylic cloudy). *Expected:* Both rods sit firmly, flush at the top, and protrude 27 mm from the underside of the lid (that is the length of the shaft).

11

Insert the sealing cord

At the top, the groove in the collar of the base is narrower than the cord. Lay the round cord Ø 1.5 mm on the groove without tension and, with a blunt tool, for example the back of a plastic spatula (**assumption**), press it bit by bit through the narrow opening into the groove, once all round; the groove passes the screws on the inside, its four corners are widely rounded (about 17 mm), the cord follows them without kinking. Do not use anything sharp that could damage the cord. Cut the ends to length and let them butt against each other (about 31 cm are needed, a remainder is left from 40 cm sold by the metre), and glue the joint with a drop of silicone. Place the joint on a straight section, not in a corner. Printed slots often come out 0.1 to 0.2 mm narrower than drawn (**assumption**); if the cord will not go in at all or still falls out, that is a result for the test print (please report it). *Expected:* The cord lies in the groove everywhere, holds by itself (even when you turn the base upside down) and protrudes evenly about 0.3 mm above the rim.

Installing and connecting

⛔ **Switch off the power supply first**

The boxes on the rack carry the 24 V of the drives at high currents. Before you connect or re-plug anything: disconnect both 24 V power supplies from the mains (in the variant with a shared power supply, that one), which also makes the controller (RearCircuitBoard_10) dead. Work on the 230 V side of the power supply is only for a qualified electrician.

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⛔ **Never connect Hall signals directly to J22**

The Hall wires of the drives go only to J2 of the HallTapBoard, never directly to J22 of the controller. Only then do the ground of the drives and the ground of the controller stay separate.

⚠️ **Do not connect the red wire (VCC)**

The HallTapBoard does not need VCC of the sync controller; it has its own supply from the 24 V. The red wire goes only into the CH1 connector. The motor wires (blue, brown) never belong on the HallTapBoard.

This is how you install and connect the HallTapBoard once it has been manufactured. Seen from above onto the enclosure, with the cable wall at the front, the cable glands are arranged from left to right as follows: **24V** (to J1), **HALL** (to J2), **J22** (to J3); this is also printed on the lid.

1

Make everything dead

Disconnect both 24 V power supplies from the mains (in the variant with a shared power supply, that one). *Expected:* The sync controller and the READY light are off, the display is running on its battery at most; the drives cannot be moved.

2

Choose the location and fasten the housing

On the Samba Solar Rack 2-E Plus, the HallTapBoard sits in box 2 (E1b) at the top of the mounting plate, above the DIN rail with fuses and busbar terminals. Position, holes and a 1 : 1 template are given in the assembly manual Rack 2-E Plus V3.1.2, chapters 13.3 and 13.4; the Hall cable of drive 1 comes over from the control box (E1). Screw the base down with four M4 screws through the slotted holes of the tabs. On another rack, find a place from which the connecting terminals at drive 1 can be reached with a short cable and the cable to the controller can run separately from the motor cables. *Expected:* The enclosure is firmly fixed; the cable glands point towards the cables.

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3 Connect the supply

Run a separate small cable (two wires 0.5 to 1.5 mm², sheathed) from the power supply through the dome nut and the "24V" cable gland into the enclosure, let the sheath end about 2 cm behind the gland on the inside, and connect the wires to the 2-pin plug of J1: pin 1 = +24 V, pin 2 = minus. The cable comes from the star point of the power supply of the drives (on the rack in box 2 (E1b)), stays entirely inside this box and is as short as possible. Its positive wire goes to the 1 A fuse terminal block (Phoenix Contact UK 6-FSI/C-LED24 with a 1 A automotive blade fuse, assembly manual Rack 2-E Plus V3.1.2, chapter 13.3); the self-resetting fuse F1 on the HallTapBoard only protects the board, **not the cable**. Nevertheless, route the cable so that its sheath cannot be crushed anywhere or chafe on sharp edges. The negative wire goes to the star point, not to the negative terminal behind the sync controller (see "Voltage between the two grounds" above). Never hang it behind the 25 A of the drives. In the variant with a shared power supply, the positive wire has its own fuse of 0.5 to 1 A slow-blow at the power supply (**suggestion**, Chapter 6). *Expected:* With the plug still disconnected, the multimeter (resistance) shows no short circuit between the two wires; the positive wire is marked.

4 Tap the Hall wires

The BST-KZQ-15A-2 sync controller has no Hall terminal to clamp onto; the drives plug into CH1 and CH2 with a 6-pin connector. Therefore, for drive 1, the wires Ch.A (yellow), Ch.B (green) and GND (black) each run via a lever connector WAGO 221-413 (**assumption**) with a short piece of wire to the CH1 connector (wiring manual, chapter 7). Into the third opening of each lever connector goes one wire of a short sheathed cable (ideally 0.25 to 0.5 mm² per wire, outer diameter at least 4 mm). On the WAGO side, strip the fine-stranded wire 11 mm and insert it without a ferrule, then pull firmly on the wire; approval for ferrules in this connector is not confirmed (**assumption**). First measure VCC (red) against GND at the CH1 connector: the HallTapBoard is designed for 5 V; which voltage the BST-KZQ-15A-2 supplies is not confirmed (**assumption**); if it is below about 3.4 V or above about 15 V, do not connect. Run the cable through the "HALL" cable gland and connect the three wires to the 3-pin plug of J2 (pitch 3.81 mm): pin 1 = H1, pin 2 = H2, pin 3 = GND. *Expected:* Every wire is firmly seated; the wires to the CH1 connector are as firmly seated as before.

5 Cable to the controller

Run the shielded cable (for example LiYCY 3 × 0.25 mm² or 3 × 0.5 mm², up to 15 m) through the "J22" cable gland and connect it to the plug of J3 (pitch 3.5 mm): pin 1 = A, pin 2 = B, pin 3 = return. Do **not** connect the shield here; cut it short and insulate it with heat-shrink tubing; it is connected only at J22 pin 3 (see above, "Pulse input HALL"). *Expected:* same pin assignment at both ends, shield at one end only.

6 Check the polarity of J1

Before you plug in J2: leave plug J1 disconnected, briefly switch on the power supply and measure at plug J1 (multimeter on DC voltage, red probe on pin 1, black on pin 2), then switch the power supply off again. *Expected:* about +24 V. If the multimeter shows a minus sign before the value, the wires are swapped: swap them and measure again. If J2 is plugged in while J1 has reversed polarity, the fuse F2 trips (see above).

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7 Plug in and label

Power supply off. Plug each plug into its terminal at the bottom edge of the board: J1 on the left (marking +24V/0V), J2 next to it (H1/H2/GND), J3 on the right behind the isolation gap (A/B/GND). The plugs of J2 and J3 have different pitches and should not fit into the wrong terminal (to be checked on the sample); label them anyway. *Expected:* All plugs sit flush, no pin is exposed; there is room between the plug and the lock nut, the wires lie loosely. Space there is tight (calculated, probably only about 12 to 14 mm, **assumption**), and the plugs are slightly offset to the side of their gland; the wires therefore run at a slight angle.

8 Seal the cable glands

Pull each cable back a little until the wires inside only sag slightly, then tighten the dome nut on the outside by hand and finally with a 22 mm spanner (at most 2.5 Nm according to Lapp; do not overtighten, otherwise the cable sheath suffers). *Expected:* No cable can be pulled out or turned any more, and no plug is pulled out of its terminal in the process.

9 Close the lid

Check that the sealing cord still lies in the groove everywhere. Place the lid so that the lettering "24V – HALL – J22" is above the cable glands, and fasten it with four countersunk screws M3 × 10 into the four corners: crosswise, evenly, until the lid rests on the rim all round. The head then sits flush or slightly lower in the countersink (according to the standard the heads may be slightly smaller, so "flush" alone is not a reliable sign). Do not tighten any further, otherwise the threaded inserts will spin. *Expected:* The lid rests on the rim all round, the cord is not squeezed out anywhere, no cable is pinched; the two small holes marked "24V" and "5V OK" are at the rear left.

10 Switch on and check the lights

Switch on both power supplies (in the variant with a shared power supply, that one); the drives are stationary. *Expected:* You see light through both holes in the lid: on the left "24V" yellow, on the right "5V OK" green, about equally bright; the sync controller reports no error. If only one or none lights up, look it up in the table "What the lights show" below. DC voltage at J22 of the controller between pin 1 and pin 3, then pin 2 and pin 3: about 5 V with the contact open or about 1 V with it closed (depending on the position of the motor).

11 Test run

Move once over the whole travel. *Expected:* The drives move evenly as before, the sync controller shows **no error code**. If it shows one (for example E10, no Hall signal), unplug the HallTapBoard immediately (lid off, pull plug J2), note the code and look it up in the error list (wiring manual, chapter 7).

What the lights show (troubleshooting)

The two lights show how far the supply gets on the board. All cases are calculated, not tested.

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"24V" (YELLOW, LEFT)	"5V OK" (GREEN, RIGHT)	MEANING	WHAT YOU CHECK
on	on	**Everything OK:** 24 V with correct polarity, the regulator supplies 5 V, the tap is working.	Nothing.
on	off	The 24 V arrives, but the board gets **no 5 V**. The tap is not working.	Switch off the power supply, remove the lid, inspect the board: burnt or loose components, solder bridges, swarf. If that does not help, a component of the 5 V supply is defective (regulator U1, a short circuit on 5 V, for example a capacitor C2/C3, the protection diode D10 or the comparator U2, or the light D4 itself and its resistors R17/R23). If U1 has only shut down thermally, it is hot: do not touch it. Replace the board or have it checked by a qualified person.
off	on	The supply is OK; only the yellow light itself (D9 or one of its resistors R18/R24) is defective.	The tap still works. Replace it when convenient.
off	off	**No 24 V** at the board.	Power supply switched on? Fuse of the cable at the power supply? Plug J1 plugged in correctly, wires tight? **Polarity** at J1 (pin 1 = +24 V): with reversed polarity a diode blocks and both stay off. If J2 was plugged in at the time, the fuse F2 has also tripped: power supply off, connect the right way round, wait a few minutes, then switch on again. Measure at J1 with the multimeter: expected 24 to 25 V. If the 24 V is present and both are still off, the self-resetting fuse F1 has tripped because of a short circuit on the board (for example a failed component before the regulator: D1, C1 or U1): power supply off, wait a few minutes, inspect the board. If F1 trips again after switching back on, a component is defective; replace the board.
dim	bright	Rare: The regulator U1 has failed short, the protection diode D10 holds the 5 V at about 8 V, and F1 has tripped. The yellow one only glows faintly, the green one is brighter than usual.	Switch off the power supply immediately and replace the board. Nothing is overloaded (calculated), but the tap does not work reliably: the counted tilt cannot be trusted until the board is replaced.

Neither of the two lights indicates the fuse F2 at the Hall ground. If it has tripped, the pulses usually still arrive (then via the star point). Whether it has tripped you measure during a movement between J2 pin 3 and J1 pin 2: more than about 0.5 V there means F2 has tripped; more than about 1 V is too much voltage between the two grounds (power supply off, check the wiring of the negative lines: is the minus of J1 at the star point?). At standstill, F2 switches back on by itself after cooling down.

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ℹ️ **What the board withstands (calculated)**

A swapped wire with 24 V on H1 or H2 does no harm; the channel then just shows nothing meaningful. If 24 V reaches GND of J2 (pin 3), if J1 is connected with reversed polarity while J2 is plugged in, or if the negative line of the sync controller comes loose, the fuse F2 limits the current and trips; up to draft 0.5 the current of the power supply then flowed without a fuse through the board and the Hall wire. Reversed 24 V at J1 without J2 plugged in is blocked by a diode (reverse polarity protection); both lights then stay off. **Continuous 24 V on A or B** (a swapped wire in the cable to the controller), on the other hand, can destroy the optocoupler of this channel, already at about 50 °C in the control box. The board tolerates up to 30 V continuously at J1 without damage; short spikes are absorbed by the protection diode D1. If a single component fails, the fuse F1 trips or the board keeps working; no other component is overloaded as a result (calculated, not tested; details in the README). The Hall line of drive 1 is loaded with at most about 6 µA, even when the HallTapBoard has no voltage. If a protection diode at J2 fails short, the Hall signal at the sync controller only drops from 5 V to about 4.7 V; it still recognises it (**assumption** about its switching threshold and about VCC 5 V). If J2 is unplugged or a Hall wire is broken, the board reports "contact open"; the controller only notices this during a movement (no pulses arrive), not at standstill. If a protection diode at J2 fails short, this channel fails (the controller reports missing pulses). When the power supply is switched on and off, the counter can jump a few steps without anything having moved; it is correct again at every end position (to be observed on the rack). The board is not suitable for Hall signals with 3.3 V (only 0.29 V margin to the switching threshold, calculated); therefore measure the HIGH level of the Hall wires beforehand.

🧩 **Still open**

Whether the BST-KZQ-15A-2 sync controller tolerates the tap, how many pulses per second arrive and which voltage VCC carries at its Hall connection is not known (will be measured on the rack, test plan items 1 to 4 in the README). According to the calculation, the HallTapBoard reliably keeps up with a few hundred pulses per second per channel; whether it manages 1 000 depends on the switching time of the optocouplers, which the datasheet does not specify for this operating mode (**assumption**, to be measured on the sample). Also open is the voltage between the two grounds on the rack (see above). The regulator U1 (C108469) was almost sold out at JLCPCB on 11.11.2026 (40 pieces); check its stock and that of the fuse C70064 (two per board) on the day of ordering. To be checked on the test print: fit of the inserts and the cable glands, height of the lock nut (5 mm assumed), play of the lid, whether the sealing cord holds in the narrow upper part of the groove and can still be pressed in, room for the plugs and wires under the lid, how firmly the light guides sit in the 3.2 mm hole and how well both lights can be seen through them. **Seal:** the sealing surface of the lid is printed on supports, and four screws in the 3 mm thick lid probably do not compress the cord equally everywhere; only a test print will show whether the housing is watertight (not confirmed). Space in the control box and the plug-in protection between J2 and J3 are checked on the sample.

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Long cables up to 15 metres

There can be up to 15 metres between the controller and the sensors and drives. Long cables outdoors pick up interference, for example from the motors or during thunderstorms. The RearCircuitBoard_10 board therefore has a protection circuit on every connection. The right cable does the rest.

CABLE	RECOMMENDED CABLE	NOTE
Sun sensor	Shielded, 5 wires, 0.25 mm² or more	The sensor cable is 3 m long and is extended. Route it separately from all 24 V cables: J2 withstands 12 V, 24 V on a measuring wire destroys two resistors of the board (warning above). Connect the shield to GND only at the controller. The measuring resistors sit at the end of the cable on the board, which makes the signal insensitive. The supply wire is current-limited: a short circuit in the cable does not harm the controller and is reported on the display (firmware in preparation, Chapter 8).
Limit switches	Shielded or twisted in pairs	Two wires per switch. Do not run them in the same cable as the motor cables.
Switching outputs	Twisted in pairs	Two wires per contact. With variant B, the power relays are best placed close to the controller, so that only the motor cable is long.
Anemometer (RS485)	Twisted wire pair for A and B, plus supply	RS485 is made for long cables; 15 m is not critical.
Pulse input HALL (J22)	Shielded, 3 wires, 0.25 to 0.5 mm² (for example LiYCY 3 × 0.25 mm²)	Up to 15 m. Connect the shield only at J22 pin 3. Do not run it in the same cable as the motor cables. Designed for up to 200 pF per metre of cable (**assumption**).
24 V power supply → controller (J11)	Two wires, 1.5 mm² (0.75 mm² or more possible)	On the rack: short cable from the small power supply next to the tablet. Variant with a shared power supply: the power supply is located at the drives, the controller up to 15 m away; at 1 A, about 0.35 V is lost over 15 m with 1.5 mm², about 0.7 V with 0.75 mm²; separate 2 A fuse at the power supply (Chapter 6).
24 V power supply → drives / sync controller	Two wires, 4 mm² or more	Keep it short, a few metres. Up to about 24 A; 25 A fuse at the power supply. Voltage drop: table in Chapter 6.
Lights in the enclosure	Single wires or ribbon cable	Only short cables in the housing.
Motor cables	Cross-section according to current, 1.5 mm² or more	Do not run through the board. At 5 A, just under 2 V is lost over 15 m with 1.5 mm², a good 1 V with 2.5 mm².

⛔ **Never extend I2C**

The connection between the board and RearCircuitBoard_10 (I2C) is only intended for a few centimetres. Over long cables it becomes unreliable. That is why the components on the RearCircuitBoard_10 board sit directly at the display board, and only the robust signals go outside.

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Still to be checked on the installation

The cable recommendations are general guide values, not measurements. Whether the push-button input of a controller works without interference over 15 m will only be shown by trying it.

Connecting the plugs of the switching outputs (J3 to J6)

The plugs of the four switching outputs look different from the others: they have two rows of round openings, and next to each opening there is a small pusher. The plugs of UP and DOWN (J3, J4) have two openings per row, i.e. four; those of EAST and WEST (J5, J6) have three per row, i.e. six. The wires are not screwed down but simply pushed in; a spring holds them (spring connection). Only two openings are ever used, diagonally opposite each other. Which two depends on the terminal (Fig. 12).

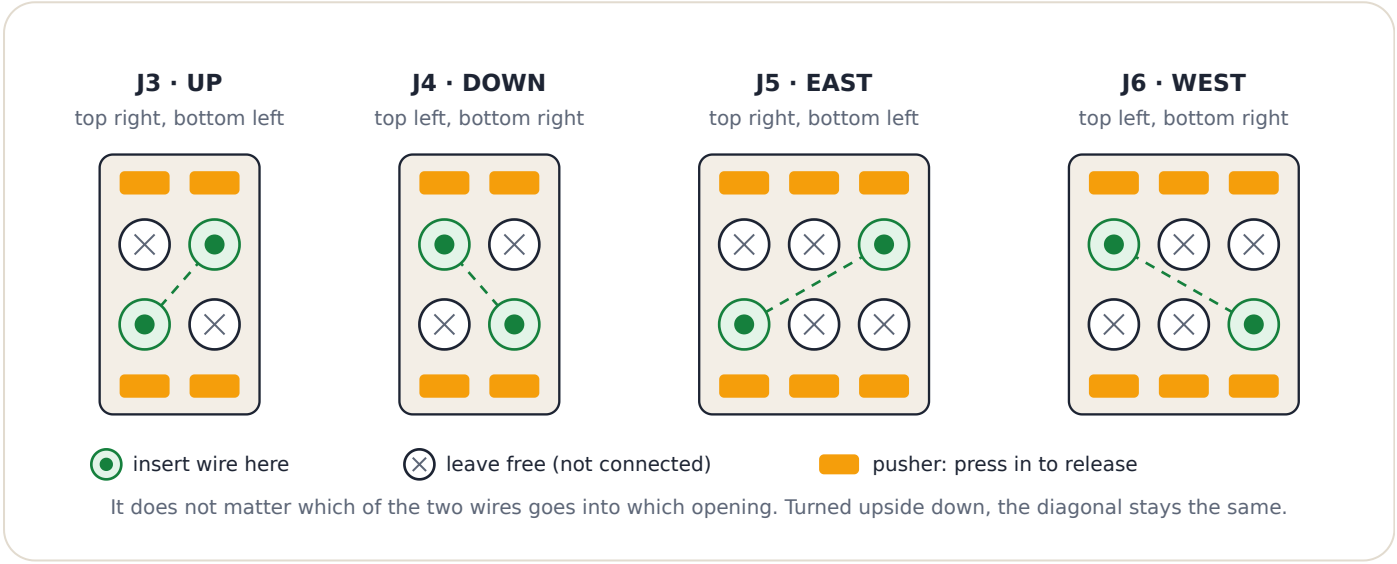


Fig. 12 – The plugs of the switching outputs, seen from the side where the wires go in (pushers at top and bottom). UP and DOWN (J3, J4) each have two openings at the top and bottom, EAST and WEST (J5, J6) three each. For UP and EAST, the two wires go in at top right and bottom left, for DOWN and WEST at top left and bottom right; for EAST and WEST these are the outer openings, the middle ones stay free. The assignment is derived from the manufacturer's drawings; for the terminal on the board it is confirmed by a photo from the manufacturer, for the plug it is an assumption until it has been checked on the sample.

1 Make the drives dead

Switch off the power supply of the drives before you pull or insert a plug. *Expected:* No drive can start while you are working.

2 Strip the wires

Strip the two wires of the cable to the drive controller by 8 to 9 mm (manufacturer's specification). Wires up to 1.5 mm² fit. A stranded wire is best fitted with a ferrule. *Expected:* The bare end is 8 to 9 mm long, no strand is nicked or sticking out.

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3

Hold the plug correctly

Hold the plug so that you are looking at the round openings and the pushers are at the top and bottom. *Expected:* On the plug for J3 or J4 you see two openings at the top and two at the bottom, on the plug for J5 or J6 three at the top and three at the bottom; the pushers sit above and below them, as in Fig. 12.

4

Choose the two correct openings

For J3 (UP) and J5 (EAST): top right and bottom left. For J4 (DOWN) and J6 (WEST): top left and bottom right. For J5 and J6 these are the outer openings on the far right and far left; the two middle ones stay free. It does not matter which wire goes into which of the two openings, because the contact is potential-free. If you turn the plug upside down, it stays the same two openings. All other openings stay empty. *Expected:* You know which two openings you are using.

5

Insert the wire

Push a stiff (solid) wire or a wire with a ferrule straight into the opening as far as it will go. For a stranded wire without a ferrule, first press in and hold the pusher next to the opening with the tip of a small screwdriver, insert the wire and then release the pusher. *Expected:* The wire is in the plug up to the insulation, no bare metal is visible.

6

Pull test

Pull firmly on each wire, but without jerking. *Expected:* It stays put. If it slips out, it was not inserted all the way: repeat step 5.

7

Label and plug in

Label the plug like its terminal, for example "J3 UP", and push it into the terminal as far as it will go. *Expected:* The plug sits firmly in the terminal and does not wobble.

8

Check

During the first commissioning in step 10: leave the plug inserted, disconnect the other end of the cable from the drive controller and measure between the two wires with the continuity tester while you hold down the button of the output. Do not measure at the pins of the terminal, otherwise you are not testing the plug as well. *Expected:* It beeps while the button is pressed, not when released. If it stays silent, a wire is in a wrong opening or the plug is on the wrong terminal.

Releasing a wire: Press in the pusher next to the opening and hold it, pull out the wire, then release the pusher. Never tear out a stuck wire without using the pusher.

⚠️ **When soldering in the terminals J3 to J6**

The pins of a terminal (four on J3 and J4, six on J5 and J6) are close together. After soldering, check with a continuity tester that no pin is connected to its neighbour. A solder bridge to one of the free pins cancels the protection against swapped plugs on that terminal.

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🧩 **To be checked on the sample**

Which two openings belong to the contact is derived from the manufacturer's drawings. For the terminal on the board, the assignment is confirmed by a photo from the manufacturer; for the plug (which opening belongs to which pin) it remains an **assumption** until it has been measured on a real part. If it is wrong, the contact does not switch and the drive does not move; this shows up during the check (last step above). The datasheet only gives the stripping length of 8 to 9 mm and wires up to 1.5 mm². Whether ferrules up to 1.5 mm² fit and how firmly stranded wires without a ferrule hold will be checked on the sample. Releasing with the pusher is usual for spring connections; the datasheet does not describe it.

7 The battery and the charge lock

The 10.1-inch board has a battery. If the 24 V at J11 fails, it keeps running on it; in the variant with a shared power supply, the drives then also stop (Chapter 6). So that the battery is not damaged, the board only lets it charge at a suitable temperature.

Waveshare supplies the board with a lithium battery (3.7 V, 4000 mAh according to the manufacturer). A charger on the board charges it from the 5 V coming from the RearCircuitBoard_10, at up to 1.33 A. The charger does not check the temperature of the battery. Lithium batteries, however, may only be charged within a limited temperature range. Nothing is changed on the board or the battery; the RearCircuitBoard_10 board takes care of the protection itself.

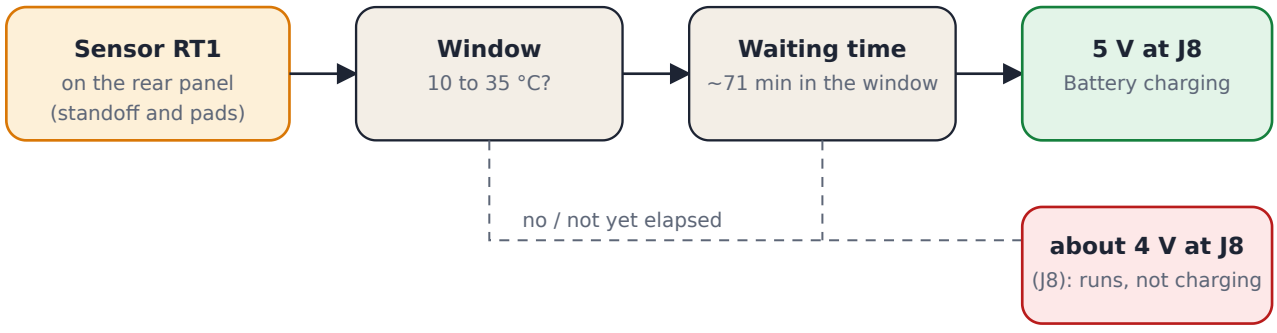


Fig. 13 – How the charge lock works. Only if the temperature at the rear panel stays in the window without interruption and the waiting time has elapsed does the board get the full 5 V. Otherwise the RearCircuitBoard_10 lowers the voltage to about 4 V; that is enough for the board, but not for charging.

ITEM	WHAT APPLIES
Window	Charging takes place only between 10 and 35 °C at the rear panel (with tolerances about 8 to 12 °C and 33 to 37 °C).
Waiting time	About 71 minutes (with tolerances 45 to 101 minutes) after the temperature has entered the window. Each time the 24 V is switched on, the waiting time starts again; so the battery does not charge during the first hour or so. This is intended.

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ITEM	WHAT APPLIES
Voltage during the lock	3.8 to 4.1 V at the J8 socket of the board. The charger only charges from about 4.5 V (manufacturer's specification without tolerance). That it really does not charge at 4.1 V will be measured over hours before ordering (**open**).
Where it is measured	The sensor sits on a small copper island at the top left of the board. A brass standoff between two thermal pads conducts the temperature of the rear panel to it; behind the rear panel is the battery (position is an **assumption**, will be measured). So the sensor measures the rear panel, not the battery itself.
Standoff and pads	M3 brass standoff, 4.7 mm across flats. Which spacer and which pads it will be is only decided during the first assembly on the manufactured circuit board (Chapter 5). At 13.9 mm distance between board and rear panel (50 mm housing, without washers on the standoffs, measured) there are two options: 12 mm spacer (C551332) with one 1.0 mm and one 1.5 mm pad, the pads are compressed by (12 + 2.5 − 13.9) / 2.5 = 24 %; or 13 mm spacer (C551333) with two 0.5 mm pads, (13 + 1.0 − 13.9) / 1.0 = 10 %. 10 to 30 % is intended. With two washers per standoff (14.9 mm), the 13 mm spacer and one 1.0 mm and one 1.5 mm pad fit (24 %). At a distance of 8.2 to 9.1 mm it is 6 mm high (then 1.5 mm pads), at 9.1 to 10.0 mm 8 mm high (then 1.0 mm pads). The pads are 6 × 6 mm, electrically insulating and flame-retardant.
Handle the island with care	The copper island hangs on a narrow finger (1.9 × 2.5 mm) and two bridges of circuit-board material (each 1.0 × 2.5 mm, without copper), so that the heat of the board hardly reaches it. When installing, hold the board by the edge, lower it straight, do not press on the island and only press the thermal pads lightly (Chapter 5). According to the calculation (**assumption**), the finger and bridges withstand the contact pressure with a large reserve; this will be checked on the sample (**open**).
Test jumper J21	2-3 = operation. 1-2 forces the lock, for testing only. If the jumper is missing, operation applies.

⛔ **Below 10 °C, no USB cable on the "UART" socket**

A PC on the "UART" socket of the board supplies the board with 5 V via USB. The charger then charges the battery despite the charge lock. So do not plug in the USB cable in cold conditions.

⚠️ **Soldering faults that can cancel the lock**

If the sensor fails or the supply of the lock is missing, it locks (fail-safe). Some rare soldering faults on the comparator U18 (an open pin or a solder bridge), however, cancel the cold or the warm limit without this being noticed. That is why commissioning checks both limits: once with freeze spray, once with heat. In operation, the firmware cross-checks the lock continuously and warns in the header when something does not add up (see below); it cannot replace the lock.

Battery level on the display (in preparation, not checked on the device)

At the very top right of the dashboard, the firmware shows how full the battery is: a battery symbol and the percentage, for example "87 %" (Fig. 13f). Below 20 % the display turns orange, below 10 % red. If it shows "--", the measurement is not providing a usable value. When the

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battery is charging, a yellow lightning bolt appears in front of the battery symbol (Fig. 13g; amber in the light designs, so that it remains readable on a light background). The display is for information only; the controller does not act on it.

How the firmware detects charging: the charger reports the charging state only to the red light on the small adapter board of the display board, not to the chip. With the RearCircuitBoard_10 board, the firmware therefore measures the voltage that the display board receives from the RearCircuitBoard_10: if it is above 4.4 V, the charger is charging (during the charge lock the board gets less). Without the RearCircuitBoard_10, for example on the display board alone with a USB cable, it checks whether the battery voltage rises over a few minutes. The lightning bolt then only appears after about 5 to 10 minutes and disappears just as late. When the battery is almost full (from 4.15 V), the lightning bolt no longer appears. All these thresholds are **assumptions** and have not yet been checked on the device.

How the firmware measures: on the display board, the battery is connected via two resistors (a voltage divider that divides the voltage to one third) to a measuring input of the chip. The firmware measures every 2 seconds, smooths the values and converts the voltage into a percentage. Nothing is changed on the board for this.

- **Too high while charging:** As long as the 24 V is present and the charger is charging, the battery voltage is higher than at rest. The display then shows more than is in the battery. Only without charging (power failure or full battery) is it reasonably accurate.
- **Without battery:** If no battery is fitted, the firmware measures the voltage of the charger (about 4.2 V) and would show "full". In that case, switch off the display in the setup under **SCREEN** with the **BATTERY FITTED** switch (Standby and motion detector). It is switched on by default, because the board is supplied with a battery.
- The relationship between voltage and charge level is a typical characteristic curve for lithium batteries (**assumption**); it has not yet been measured on the board's battery.

🧩 **Still open**
Before ordering, the delivered board will be measured over hours to see whether the charger reliably does not charge at up to 4.1 V. If this measurement turns out badly, the charge lock via the 5 V does not work, and the circuit board will not be ordered in this form. The position of the battery is also measured; after that, the position of the island is fixed. The distance to the rear panel has been measured (13.9 mm without washers); spacer and pads are chosen during the first assembly on the manufactured circuit board. How well the sensor follows the rear panel is measured at the same time: the island hangs on a narrow finger and two bridges and is largely separated from the rest of the board; according to the calculation, the sensor still picks up about 16 to 29 % of the difference between rear panel and board (**assumption**).

The firmware cross-checks the charge lock (in preparation, not tested on the device)

Since 3 Nov 2026 the firmware keeps a constant eye on the charge lock. Via the second measuring chip (ADS1115 at 0x49) it reads the temperature at sensor RT2 on the copper island, the voltage the board receives from the circuit board (the J8 socket of the board), and whether the

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lock is currently releasing; from the lamp chip it receives the signals “24 V present” and “5 V present”. One pass over all four readings takes 0.8 seconds. If something does not add up, for example full 5 V at J8 with 5 °C at the rear panel, it shows a warning in the header and in the browser after 30 seconds and writes it to the log. The warnings, their causes and what you then do are listed in Chapter 8.

The firmware can only read and report. It has no way to switch the lock itself, and it changes nothing about tracking or storm protection. It calculates the temperature from the sensor according to the manufacturer's data sheet; a warning is only raised from 2 °C outside the window (below 8 °C or above 37 °C), so that the tolerances of the circuit do not trigger false alarms. All values are calculated and not yet measured on the device (**assumption**).

⛔ **Connecting the battery: at your own risk**

Until 3 Nov 2026 the rule was: the battery stays unplugged until the firmware checks the lock. This obligation was dropped with the decision of 3 Nov 2026: with a firmware that checks the charge lock (version from 3 Nov 2026 on; the header then knows the warnings from Chapter 8), you may connect the battery. The firmware checks the lock continuously and warns; it cannot prevent charging. The lock in the hardware has not yet been measured on a manufactured device. Whoever connects the battery now bears the risk themselves. The cold and heat test during commissioning remains mandatory, and below 10 °C the USB cable stays unplugged.

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8 Software and operation (in preparation)

The firmware for the ESP32-P4-WIFI6-Touch-LCD-10.1 has been started: it can be built and was flashed onto a test board on 20.10.2026 (firmware 0.4.6): it starts, the display shows the setup, touch, sounds and Wi-Fi work. Everything else has not yet been checked on the device (in preparation). Firmware 0.4.6 still showed the user interface small (800 × 480 pixels) in the middle of the screen; since firmware 0.4.7 the dashboard uses the whole display (see below). The connection to the anemometer (via terminal J15 of the RearCircuitBoard_10 board) has also been written, but never tried with a real anemometer. In addition there are sounds from the loudspeaker, the search for new firmware on the internet, tilt on command (optionally with Hall tap), pauses for the drives (duty cycle) and repeated moves to the flat position in a storm (all below, likewise unchecked). Since firmware 0.7.0 (in preparation, only tested on the PC) the setup program for the first installation on the rack and the calculated position of the sun with location, sunrise and sunset and night position have been added. This chapter will be written as soon as the firmware has been checked on the device.

🧩 **In preparation**
The firmware has been flashed onto the test board: setup, touch, sounds and Wi-Fi work; everything else for this board has only been built and tested on the computer. Flashing, setup, operation on the display and in the browser, settings and log will be described here once this firmware is finished and checked on the device.

Flashing the firmware for the first time (flashed on the test board)

A new board still carries the Waveshare demo software. To turn it into the controller, you write the Samba Solar Track firmware into its memory (flash) once via a USB cable. This is called flashing. The file for this is called the factory image. After that, new versions arrive conveniently over the network (New firmware from the internet), without a cable.

There are three ways. For most people, **way A in the browser** is the easiest: you do not need a program, only a suitable browser. Ways B and C help if A does not work. The firmware is only available ready-compiled; its source code is not public.

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ℹ️ **Flashed on the test board**

On 20.10.2026, firmware 0.4.6 was flashed onto an ESP32-P4-WIFI6-Touch-LCD-10.1 test board using way A (ESPHome Web). It starts: the display shows the setup, touch and sounds work, and Wi-Fi works. The user interface then still had 800 × 480 pixels in the middle of the 1280 × 800 screen; since firmware 0.4.7 it uses the whole display. This also shows that the firmware Waveshare puts on the built-in ESP32-C6 radio chip at the factory fits on this board. Whether every board carries the same version from the factory is not known. Ways B and C have not been tried individually; the information comes from the Waveshare and Espressif documentation, the board's schematic and a check of the generated file.

ℹ️ **Where the file comes from**

For the first flashing you need the file `samba-solar-track-10zoll-<Version>-factory.bin` ("factory image"). Since firmware 0.4.7 it is publicly available at `https://firmware.samba-solar.de/erstinstallation/` . The start page `https://firmware.samba-solar.de/` names the current file. Next to it is the file with the licences of the third-party libraries it contains (`DRITTANBIETER.md`).

Which file for what

FILE	PURPOSE	ADDRESS IN MEMORY	SETTINGS
`…-10zoll-<Version>-factory.bin` (factory image)	**first flashing** onto a new or erased board; contains everything the board needs to start	0x0	are **erased**, the setup appears again afterwards
`…-10zoll-<Version>.bin`	Update of a board on which Samba Solar Track is already running (only ways C and D, otherwise over the network)	0x10000	are kept

The file without "factory" is not enough on its own for a new board: it lacks the start part (bootloader) and the memory layout. Conversely, the factory image always erases the stored settings, including Wi-Fi and token.

What you need

PART	NOTE
PC or laptop	Windows, macOS or Linux
USB-C cable with data lines	A charge-only cable does not work. No cable is supplied with the board.
Browser with Web Serial (ways A and B)	Chrome, Edge or Opera; Firefox from version 151 (not tried on the board there). Safari and iPhone/iPad cannot do it.
The factory image	`samba-solar-track-10zoll-<Version>-factory.bin` , see above

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Preparation

1 Preferably before installation

Flash the firmware while the board is not yet wired into the device. If the RearCircuitBoard_10 board is already plugged on, switch off the 24 V power supply first. During flashing all switching contacts stay open, the drives do not move. *Expected:* No drive is supplied with power.

2 Connect the board

Plug the USB-C cable into the socket labelled **UART**, the other end into the PC. Do not use the "USB OTG" socket. The PC supplies the board with power. *Expected:* The display lights up.

3 Check the port on the PC

The board has the CH343 USB chip. *Windows:* In Device Manager, a new COM port appears under "Ports (COM & LPT)". *macOS:* In Terminal, `ls /dev/cu.*` shows a new entry. *Linux:* `ls /dev/ttyACM* /dev/ttyUSB*`; your user must be in the `dialout` group. If nothing appears under Windows or macOS, install the driver from the manufacturer WCH (Windows: CH343SER, macOS: CH34xSER) and restart the PC. Whether Windows 10/11 and macOS recognise the chip without a driver has not been checked. *Expected:* A new port is there and disappears when you unplug the cable.

Way A: in the browser with ESPHome Web (recommended, tried on the test board)

ESPHome Web is a free website that writes firmware to ESP32 boards via the browser. Samba Solar Track is not ESPHome firmware; however, the site writes any finished file to the board. It always writes the file to the beginning of memory (0x0) and erases the whole memory beforehand. So use only the factory image here.

1 Open the site

Go to `https://web.esphome.io/` in the browser. *Expected:* The site shows the **CONNECT** button.

2 Connect

Click **CONNECT**, select the port from "Preparation", step 3, in the window, then **Connect**. *Expected:* The site reports a connected device.

3 Select the file

Click **INSTALL** (not "Prepare for first use"), then select the file `samba-solar-track-10zoll-<Version>-factory.bin` and confirm **INSTALL**. *Expected:* First the erasing runs, then a progress bar; this takes a few minutes.

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4

Wait

Do not unplug the cable until the site reports "Installation complete". *Expected:* The site reports that the installation is finished.

5

Restart

Briefly unplug the USB cable and plug it back in. *Expected:* The display shows the Samba Solar Track setup (after flashing).

Way B: in the browser with the Espressif web flasher

Espressif, the manufacturer of the chip, offers its own site at `https://espressif.github.io/esptool-js/`. Here you choose the address yourself; it does not work with Safari.

1

Connect

Open the site, leave or set "Baudrate" to 460800, click **Connect** and select the port. *Expected:* The window below shows "Chip is ESP32-P4".

2

File and address

Enter `0x0` under "Flash Address" and select the factory image with "Choose file". *Expected:* A line with 0x0 and the file name.

3

Write

Click **Program** and wait until "Leaving..." or the end is reported. Then briefly unplug the USB cable and plug it back in. *Expected:* The display shows the setup.

Way C: with esptool on the command line

esptool is the Espressif program that all the other ways use in the background. It needs Python. For the 10.1-inch board it must be **at least version 5.3**; older versions do not fully know the chip revision of this board.

1

Install esptool

Install Python 3 (`python.org`), then in the command prompt or terminal: `pip install "esptool>=5.3"` . *Expected:* `esptool version` shows 5.3 or newer.

2

Write the factory image

In the folder with the file; replace `<PORT>` with your port (for example `COM5` , `/dev/cu.wchusbserial…` or `/dev/ttyACM0`):

```
esptool --chip esp32p4 -p <PORT> -b 460800 write-flash 0x0 samba-solar-track-10zoll-<Version>-factory.bin
```

Expected: At the end "Hash of data verified" and "Hard resetting".

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3 Later: update without erasing

If Samba Solar Track is already running and the setup is to be kept, write only the file without "factory" to 0x10000:

```
esptool --chip esp32p4 -p <PORT> -b 460800 write-flash 0x10000 samba-solar-track-10zoll-<Version>.bin
```

Expected: After the restart, setup, Wi-Fi and token are still there.

Viewing the log (in case of problems)

If the board does not start properly or keeps restarting, the log helps: the messages that the firmware outputs via the USB cable.

1 Connect

Connect the board to the **UART** socket as above, open `https://web.esphome.io/` and click **CONNECT**. *Expected:* The port appears for selection.

2 Open the log

Click **LOGS**. *Expected:* The messages scroll past; after a restart, the firmware version and the reset reason are at the top.

3 Save

Copy or download the messages and send them to the project. *Expected:* A text file with the lines around the error.

If the board does not connect

Normally the PC switches the board into download mode by itself (according to the schematic via the DTR and RTS lines of the USB chip). If that does not work, switch over by hand:

1 Download mode by hand

Hold down the **BOOT** button on the board, briefly press **RESET**, then release BOOT. Then connect again in the browser or with esptool. *Expected:* The display stays dark, the connection works.

2 After flashing

Press **RESET** once or unplug and replug the cable, otherwise the board stays in download mode. *Expected:* The firmware starts.

OBSERVATION	CAUSE	REMEDY
No port is listed in the browser's selection window	Charging cable without data lines, wrong socket or missing driver	Different cable, "UART" socket, install the driver from the manufacturer WCH
The site does not offer a connection	Browser without Web Serial (Safari, iPhone, older Firefox)	Use Chrome, Edge or Opera

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OBSERVATION	CAUSE	REMEDY
"Failed to connect" / "Timed out"	Board not in download mode	Hold BOOT, press RESET, release BOOT, connect again
Port busy ("Access denied", "busy")	Another program has the port open (serial monitor, second browser tab)	Close the other program
esptool does not know the chip or hangs	esptool too old	`pip install -U "esptool>=5.3"`
The display stays dark after flashing	Board still in download mode, or wrong file or address	Press RESET; check whether the factory image was written to 0x0
Board keeps restarting, or Wi-Fi does not respond	possibly a different version of the ESP32-C6 radio chip firmware than on the test board	View the log (above, "Viewing the log") and send the messages to the project

After flashing

After the factory image has been flashed, the board starts as if new. It shows the setup, with the operating mode as the first item (seen like this on the test board on 20.10.2026, firmware 0.4.6). Since firmware 0.4.7 the user interface uses the whole display. If the RearCircuitBoard_10 board is not yet plugged on, the controller reports an error of the inputs and outputs or offers the simulation; that is correct without the RearCircuitBoard_10. After that, continue with the first commissioning of the hardware. Since firmware 0.7.0 the setup program opens after the setup; it must be run through completely once on the finished rack before the automatic works. If you will only bring the controller to the rack later, choose CONTINUE LATER.

The dashboard on the large display (generated on the computer, not checked on the device)

Since firmware 0.4.7, the dashboard uses the whole display with 1280 × 800 pixels. At the top is the header with logo, time, Wi-Fi, the **AUTOMATIC** switch, the gear wheel for the setup and, on the far right, the battery level (Chapter 7); below it the cards **WIND**, **SUN SENSOR** and **MANUAL**, and at the bottom the status bar with **PARK FLAT**. In the *VERTICAL ONLY* operating mode, the **TILT** card is added, with the buttons for the fixed angles (Tilt on command). With two axes it is omitted, and the remaining cards become wider. If an anemometer with wind direction is set up, a **compass rose** appears in the **WIND** card below the speed: north at the top, east on the right as on a map. The yellow needle points with its dot in the direction the wind is coming from; below it the same is shown as text, for example "Wind from SW (228°)". If the wind direction sensor does not respond, the needle disappears and the text reads "Wind direction --". If there is something to warn about (lamps, supply, charge lock, Hall tap, new firmware), the warning appears in yellow in place of the time (Warnings about the supply and the charge lock). Note: the crosshair of the sun sensor next to it is labelled the other way round (east on the left), because it shows the view towards the south.

Manual mode: With the **AUTOMATIC** switch off, the elevation moves as long as a button is held. **STEEPER** (arrow up) tilts the panels steeper, their edge rises; **FLATTER** (arrow down) moves towards the flat position. With two axes, **EAST** and **WEST** are added. Which output of the board switches follows the setting “Flat position is in UP direction” under SETTINGS → COMMISSIONING:

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by default STEEPER switches the output DOWN and FLATTER the output UP. The status band shows the same words (“MANUAL: MOVING STEEPER”, “AUTO: TRACKING ELEVATION FLATTER”). Up to firmware 0.5.7 the buttons were named after the outputs UP and DOWN; when upgrading, note: by default the upper button now moves in the opposite direction.

SUN SENSOR card: Since firmware 0.7.0 the button with sunrise and sunset is at the top right of the card; it opens the SUN page (Sun, location and night). The yellow dot in the circle shows where the sun is, seen from the module. In the centre the module points exactly at the sun; if the dot is at the top, the sun is higher than the module points, and the module would have to become flatter. The green ring is the threshold from which the automatic starts tracking. At the bottom of the card the same thing is shown as a number under the heading **SUN DEVIATION** (with two axes, separately for El and Az). This number is **not the position of the panels**, but the difference between the opposite sensors of the sun sensor in percent. The edge of the circle stands for 30 % (labelled “±30 %”): if the deviation is larger, the dot stays at the edge while the number keeps running. This way small deviations around the centre are clearly visible, and that is exactly where the automatic works.

Example in manual mode: The module is aligned (about −2 %). FLATTER moves to the flat position and stops there at the limit switch; the deviation then reads, for example, −14 %, because the sun is not exactly perpendicular above the module. STEEPER moves from there to the steep position; the deviation keeps growing, because the module points further and further away from the high sun. From 30 % the dot stays at the edge, the number keeps rising, for example to +80 %.

Tilt below the circle (since firmware 0.6.0): Below the circle a sketch shows the current tilt of the panels: ground, joint and the panel at the current angle, next to it the angle in degrees. Below that it says where it comes from: **MEASURED** (counted with the Hall tap), **ESTIMATED** (from the travel time; in the TILT card then with “~”) or **UNKNOWN** (“--°”, the sketch then shows no panel). The sketch is purely schematic; it does not show which side of the module goes up on the rack. It exists in both operating modes. During a reset of the synchronous control, the band showing the progress of the reset takes its place.



Fig. 13a – Vertical only: four cards, below them TILT with the angle large at the top. In the SUN SENSOR card “±30 %” at the circle, below it the sketch of the tilt and the SUN DEVIATION. In manual mode STEEPER and FLATTER.

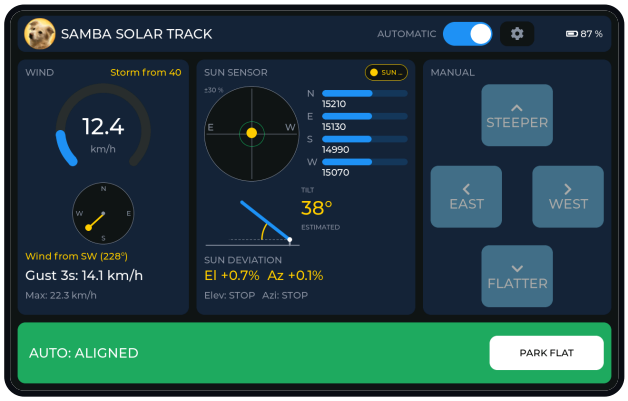


Fig. 13b – Two axes with wind direction: three wider cards, in the WIND card the compass rose (wind from SW), below the circle the estimated tilt “38°” with “ESTIMATED”, in manual mode plus EAST and WEST.

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Status

The other screens (setup, SETTINGS, network, time, remote access, design, sound, screen, firmware, calibration) also use the whole display: they are laid out for 1280 × 800 pixels, with large lettering (for example Fig. 24). The large dashboard was flashed onto the test board on 20.10.2026, and the device can then be reached on the network. How it looks there, whether fingers hit the buttons and whether it runs smoothly has not yet been confirmed; the enlarged other screens have only been built on the computer. The pictures here were generated by the screenshot tool from the real code of the user interface.

Display language (in preparation, generated on the computer, not checked on the device)

The display is available in seven languages: English, German, French, Spanish, Dutch, Portuguese and Polish. The factory default is **English**. At the very first start, the language selection therefore appears first (Fig. 13h), even before the setup. The buttons carry the name of the language in the language itself, for example DEUTSCH or FRANÇAIS, so that you can find your language even if you do not understand the currently selected one.

1. ### Select the language

   Tap your language. *Expected:* The button turns blue with a tick, and all texts switch to this language immediately.

2. ### Next

   At the first start, tap the green NEXT button (in German: WEITER). *Expected:* The setup appears in the selected language. The choice is saved, even after a restart.

3. ### Change later

   Open the setup (gear wheel in the dashboard) and tap the button with the name of the language at the top right, for example **ENGLISH ...**. *Expected:* The language selection appears; after the selection, BACK leads back to the setup. The controller keeps running without interruption, including the storm protection.

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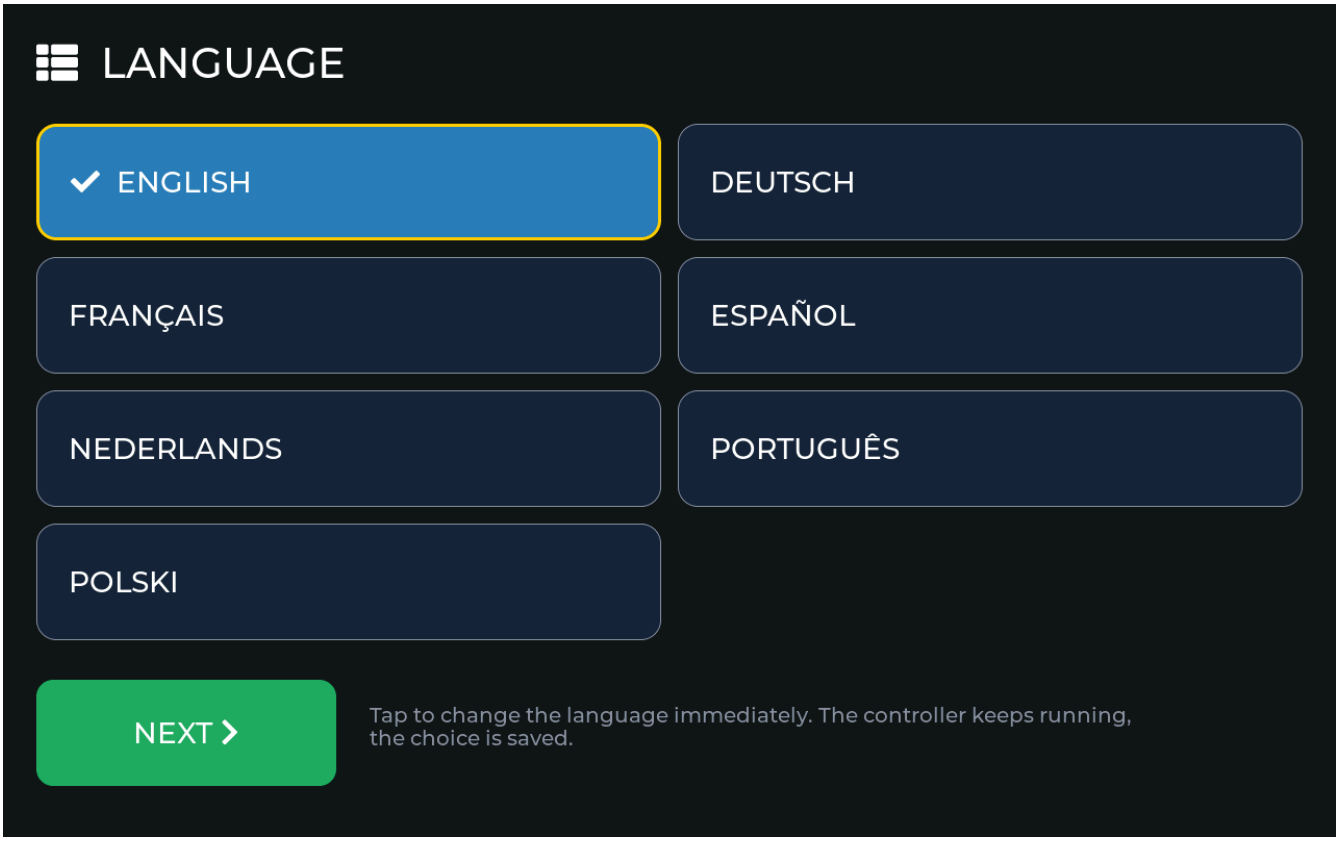


Fig. 13h – The language selection at the first start, here with English selected; the green button leads on to the setup.

Still open

The translations were created by machine by Claude on 28.10.2026; no native speaker has checked them (**assumption**). If a device already has older firmware and has been set up, it stays German after the update. The messages in the log always stay German. The web interface shows the same language as the device; you can also choose it there under SETUP (Fig. 23c). This manual is also available in all seven languages, as a page and as a PDF; you change the language at the top of the table of contents. The German version is authoritative; the other versions of the manual are also machine-translated and not checked by native speakers (**assumption**).

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The setup program (first installation) (since firmware 0.7.0, in preparation, only tested on the PC)

> 🧩 **Firmware 0.7.0: only tested on the PC**
> The setup program has been written and checked on the PC with automatic tests and in the simulation. It has never run on a real rack. It has been published since firmware 0.7.0 (11.11.2026). The screenshot tool generated the pictures from the real code of the user interface, with simulated values. There are no limits yet for travel times, Hall steps and angles: they will only be set once the drives have been delivered and measured on the rack. Until then “passed” means: end position reached or confirmed by you; the measured values are in the test report.

The setup program guides you step by step on the display through the first commissioning on the **fully assembled and wired rack**. It asks what is installed on your system and then checks every function and every travel path: whether the circuit board responds, whether the anemometer measures, whether the rack moves flat in the right direction, whether it reaches fully steep and fully flat again, whether it turns east (with two axes), whether the sun sensor is connected correctly and whether the automatic really moves to the sun. Where the device can measure by itself (travel time, response of the anemometer, brightness), it measures. Where only you can see it (“Has the rack become flatter?”), it asks you and treats NO as an error, with a hint on what to do.

🔒

Mandatory, once completely

Until all mandatory steps have been passed, the automatic is locked. This applies to every device, also after an update.

🌬️

Storm protection always works

Storm, anemometer failure and PARK FLAT have priority the whole time. A storm aborts every test move.

⏸️

Interrupting allowed

CONTINUE LATER ends every move; the progress stays saved, even through a power failure and a restart.

📄

Test report

At the end the device stores a report with all values. You view it in the web interface and save it as a PDF or as a text file.

What you need

- The rack is fully assembled, all drives, the sun sensor and the anemometer are connected (wiring manual), the hardware check in Chapter 9 has been passed.
- A day with **light wind** and **daylight** (for the sun sensor and the test run; if need be, a bright lamp helps).
- You stand at the display **and** can see the rack. If that is not possible, a second person at the rack helps.
- With one axis only: something to measure angles, for example a spirit level app on your phone (step “Move to angle”).
- Optional, but recommended: the name and password of your Wi-Fi (for the time) and the location of the rack.

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The procedure at a glance

The steps depend on the operating mode. With **two axes** (vertical + horizontal) the program additionally checks the rotation from east to west and all four sides of the sun sensor; there is no step “Move to angle” there, because tilt on command is only available with one axis. With **one axis** (vertical only) the program calibrates the travel time and moves the rack to 30° once. At the top of the screen there is always “STEP x OF y” and a green bar; without a reset of the sync controller there are 13 steps in both operating modes, with a reset 14.

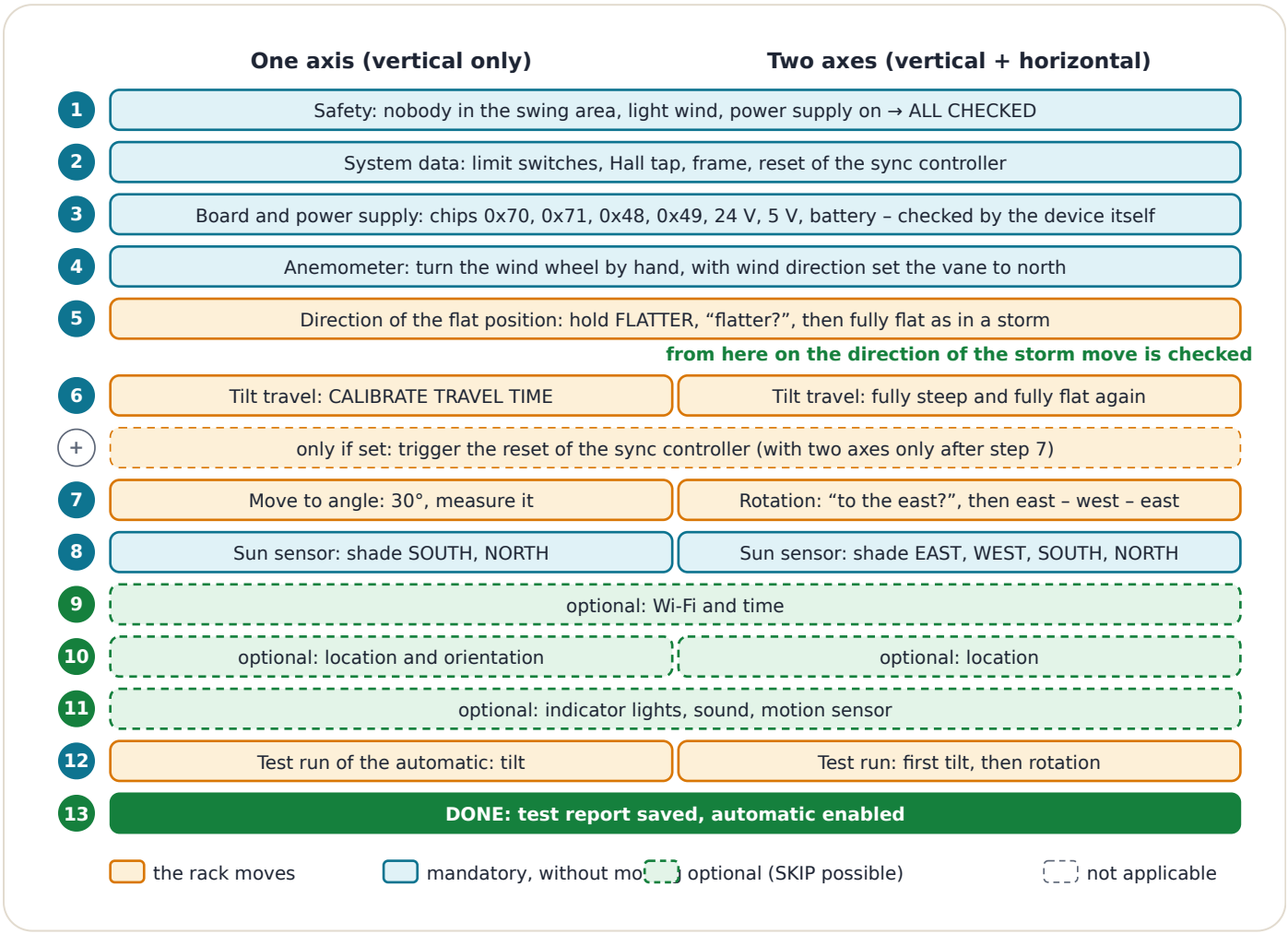


Fig. 60 – The procedure of the setup program. A row across the full width applies equally to both operating modes; where they differ, one axis is on the left and two axes on the right. The numbers on the left are the same as in the “STEP x OF 13” display. If a reset of the sync controller is set, the step “+” is added, and from there on the display counts one higher (“OF 14”). Step 7 differs depending on the operating mode.

Starting

1 Complete the setup

At the first start you choose the language, and in the setup the operating mode and the anemometer (Language, after flashing). *Expected:* After saving, the dashboard appears.

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2 The setup program opens

As long as the setup program is missing, it opens once by itself after the start. If you have closed it, the status bar at the bottom is violet: “SETUP PROGRAM MISSING: AUTOMATIC LOCKED – TAP HERE TO CONTINUE” (Fig. 61a, 61b). Tap the bar. *Expected:* The SETUP PROGRAM screen appears at the point where you stopped.

3 Read, do, confirm

On the left a sketch shows what is happening (partly animated), on the right is what you should do, below it a measured value in yellow. The large green button at the bottom right goes on (NEXT, START, YES). *Expected:* After each passed step the green bar at the top gets longer.

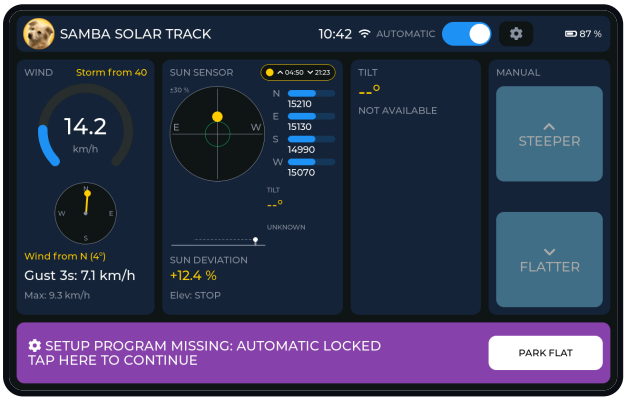


Fig. 61a – One axis: the status bar is violet, the automatic is locked. Tapping it continues the setup program.

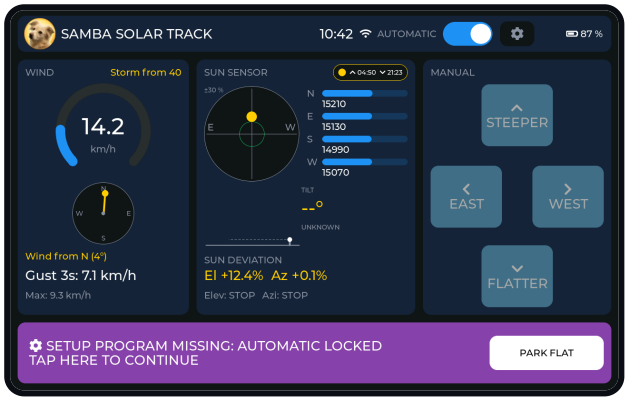


Fig. 61b – Two axes: the same bar. PARK FLAT and manual mode remain usable.

The buttons of the setup program

BUTTON	WHAT IT DOES
NEXT / START / YES (green, bottom right)	Next step, start the procedure or answer the question with yes. For some questions the button says more precisely what you confirm, for example “YES, FLATTER” or “YES, TO THE EAST”.
NO	What you saw does not match the question. The program then shows what to do and offers REPEAT.
■ **STOP** (red)	Appears during every move. Stops the rack **immediately**; the step then starts again from the beginning. STOP does not hold back a storm: in a storm the rack moves flat anyway.
FLATTER, EAST (blue, press and hold)	Short test move as long as your finger is on the button, at most 5 seconds. Releasing stops immediately.
STEP BACK	To the previous step, for example to repeat it.
SKIP	Only for the optional steps (Wi-Fi, location, lamps/sound/motion sensor). Moving and safety steps can never be skipped (“This step is mandatory.”).
CONTINUE LATER (grey, bottom left)	Back to the dashboard. Every move stops, the step reached stays saved. The automatic remains locked.

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The steps in detail

1 Safety

The rack is about to move. Check: nobody is in the swing area, no animals either; no ladder, no tools, no cable in the way; the wind is light; the power supply at the rack is switched on. Then **ALL CHECKED**. *Expected:* Step 2 appears.

2 System data

A list of what is installed on your rack. Tapping a row changes it. With one axis: limit switches of the elevation, Hall tap, frame, reset of the sync controller of the elevation. With two axes: limit switches of the elevation and the azimuth, reset of the sync controller per axis. If you do not know something, leave the default; the program checks everything shortly afterwards. For the Samba Solar Rack 2.0-E Plus: no limit switches on the tracker (tilt on command). *Expected:* Each row shows the correct value; NEXT.

3 Board and power supply

Tap **CHECK**. The device queries the chips of the RearCircuitBoard_10 board (switching outputs 0x70, indicator lights 0x71, sun sensor 0x48, supply measurement 0x49, reset contacts 0x72 only with J23) and checks 24 V, 5 V as well as battery and charge lock. *Expected:* All fields green, on the right “All OK.” (Fig. 64b). A red field names on the right what is missing (Fig. 64a); then check plugs and cables according to the wiring manual. As soon as the fault has been fixed, the field turns green by itself. Grey means: not present or not set.

4 Anemometer

START, then turn the wind wheel of the anemometer firmly by hand. *Expected:* The value at the bottom right rises by at least 2 km/h, and “Turning detected. The anemometer works.” appears. If it says “Anemometer not responding: check cable”, the connection via RS485 is not right (Anemometer on RS485). *With wind direction:* Then you set the vane so that it points north and answer whether the display shows about north (N, 0 degrees). If NO: check the mounting, the north mark and the wind direction values under SETTINGS → COMMISSIONING, then REPEAT.

5 Direction of the flat position

The most important step: here the device determines in which direction the rack becomes flat in a storm. Hold the **FLATTER** button (5 seconds at most) while watching the rack; then answer the question “Has the rack become flatter?”. *If NO*, the device switches the direction itself (“Good that we checked: the direction has been switched.”), then REPEAT; you do not need to rewire anything. *If YES, FLATTER*, the rack moves fully flat, as in a storm, until it stops by itself. *Expected:* “The flat position is checked. From now on the storm protection keeps the rack safe.” The travel time is in the report. If the rack is already flat, the program asks right away whether it is flat.

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6 Tilt travel

One axis: Tap **CALIBRATE TRAVEL TIME** and follow the steps (Calibrating the travel time); afterwards you come back here. Without a calibrated travel time you cannot continue (“The travel time has not been calibrated yet.”). *Two axes:* **START**; the rack moves fully steep to the end stop and then fully flat, and the device measures both times. Then the question: “Did the rack move fully steep and fully flat again without catching or bumping?” If NO: check mechanics and drive (assembly manual of the rack), then REPEAT. *Expected:* “Tilt travel checked.” or “Travel time calibrated.”

7 Rotation east – west (two axes) or move to angle (one axis)

Two axes – rotation east – west: Hold the **EAST** button and watch whether the rack turns east, that is, towards where the sun rises in the morning. *If NO*, the EAST and WEST contacts are swapped: switch off the power supply, swap the plugs at J5 and J6 (wiring manual), switch on, REPEAT. *If YES, TO THE EAST*, the rack turns by itself fully east, then fully west and back east, each time to the end stop. *Expected:* “Rotation checked. The rack is in the morning position.”

One axis – move to angle: **MOVE TO 30 DEGREES**; the rack moves to 30° using tilt on command. Then you lay a phone with a spirit level app flat on a panel and read the angle (Fig. 70). You type in the number and confirm with the tick. *Expected:* “Angle recorded. It is in the test report.” 0 to 90 degrees are allowed. The program still accepts a deviation (no limits, see above); it shows how accurate the estimate from the travel time is on your rack.

8 Sun sensor

START. The program names one side of the sensor, for example “Now shade the SOUTH side of the sensor”. Hold your flat hand next to the sensor so that its shadow falls on this side, **without touching it** (Fig. 71). *Expected:* After about one second of shadow the device detects the side; take your hand away, then the next side follows. With one axis these are SOUTH and NORTH, with two axes EAST, WEST, SOUTH and NORTH. If a different side becomes dark than the one named, the device assigns the inputs correctly by itself and notes this in the report; you do not need to rewire anything. If it is too dark, it reports “Too dark: repeat in daylight or with a bright lamp.”

9 Wi-Fi and time (optional)

With Wi-Fi the device gets the exact time from the internet. Then it reliably moves to the night position after sunset, and you can operate it with your phone. **NETWORK …** opens the Wi-Fi settings. *Expected:* “✓ Time is set”, then NEXT. Without Wi-Fi: SKIP; the device then detects the night from the brightness (night position).

10 Location (optional)

LOCATION … opens the LOCATION page (below). With one axis the orientation of the rack is added (below). *Expected:* “✓ Location 51.05 / 13.74” (your location), then NEXT.

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11 Lamps, sound, motion sensor (optional)

START. One after the other: are all indicator lights on the housing lit (READY and FAIL)? Did you hear the sound (ONCE MORE plays it again)? Wave in front of the display: does the line below show motion? Answer each question with YES or NO. *Expected:* The answers are in the report; a NO does not stop the program.

12 Test run of the automatic

START (it needs sun or a bright lamp in front of the sensor). The device moves the rack off a little for 4 seconds and then lets the automatic work; it must set the rack exactly towards the sun again. With two axes first the tilt, then the rotation. Question: “Has the rack turned back to the sun?” *Expected:* YES, “The test run is passed.” If the automatic moves away from the sun, the device switches the direction itself (“The automatic moved away from the sun. The direction has been switched. Please once more.”). If it has not aligned after 3 minutes, “The automatic mode did not align. Is there enough sun?” appears.

13 Done

“Done! All checks are passed. The automatic is now enabled.” **TO THE DASHBOARD**. *Expected:* The status bar shows AUTO, the rack follows the sun. The test report is saved.

ℹ️ **Additional step: reset of the sync controller (only if set)**

If a reset via J23 or J5 has been set in step 2, **TRIGGER RESET** triggers it; the sync controller fully retracts the drives (flat). A blue box shows the axis, the phase (pulse, then the run to the end position) with the remaining time and a progress bar (Fig. 68e); per axis the reset takes 5 seconds plus the axis’s maximum run time (see Resetting the sync controller). *Expected:* “Reset checked.” With two axes and two resets, both follow one after the other. With one axis the step comes after the tilt travel (before “Move to angle”), with two axes after the rotation; the display then counts up to 14.

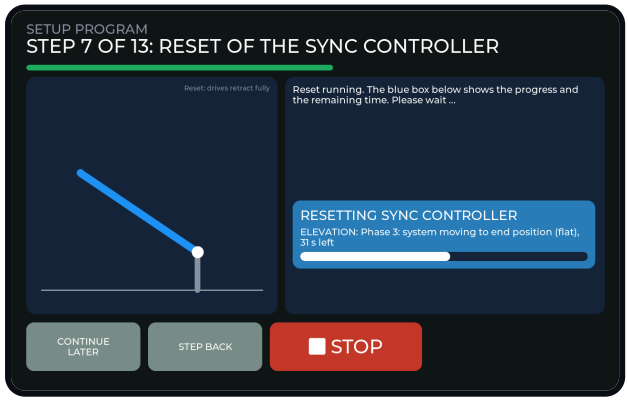


Fig. 68e – Reset of the sync controller in the setup program: the blue box shows axis, phase, remaining time and progress. When the reset is completed it turns green, if it is aborted orange.

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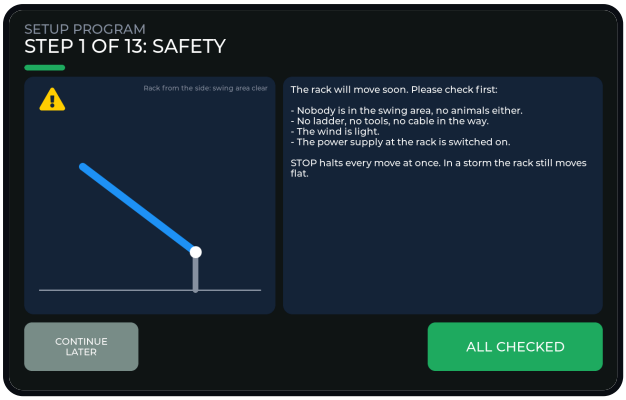


Fig. 62 – Step 1, safety (the same for both operating modes). The sketch shows the swing area that must be clear.

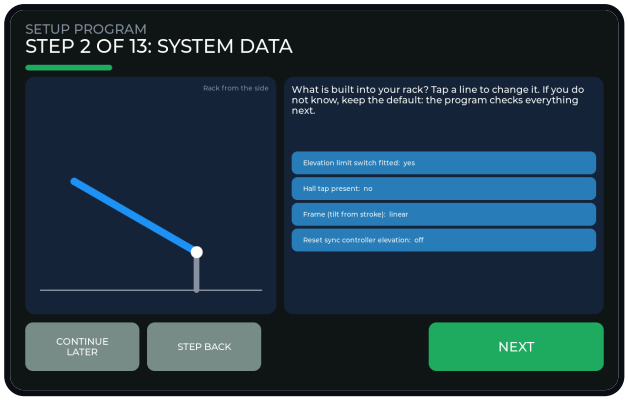


Fig. 63a – System data, one axis: limit switches, Hall tap, frame, reset.

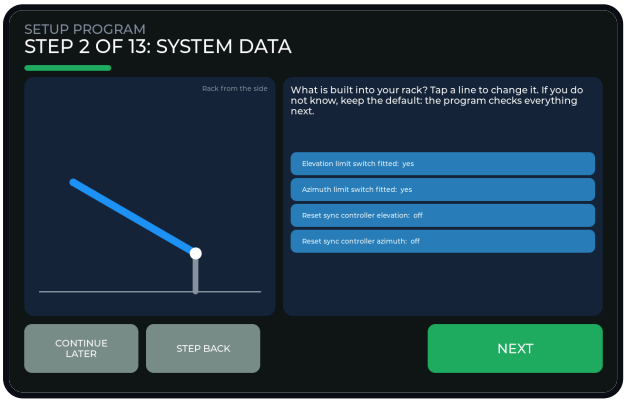


Fig. 63b – System data, two axes: limit switches and reset per axis.

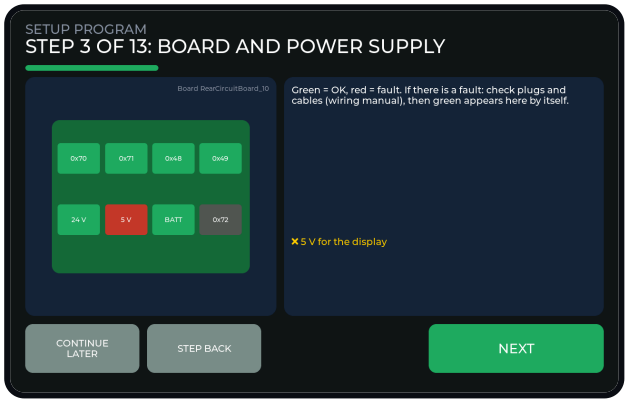


Fig. 64a – Board and power supply: the red field “5 V” reports “5 V for the display” (example).

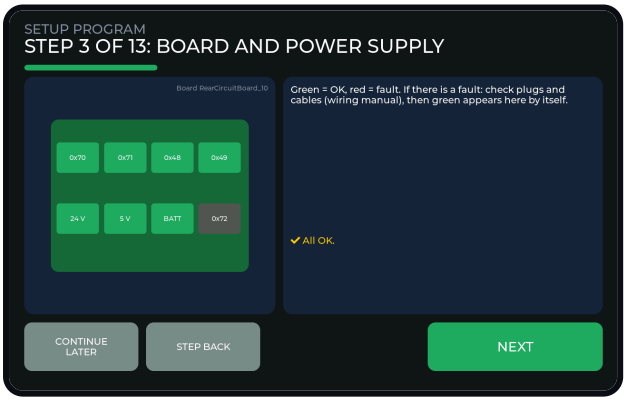


Fig. 64b – All green: “All OK.” The grey field 0x72 is not set (no reset via J23).

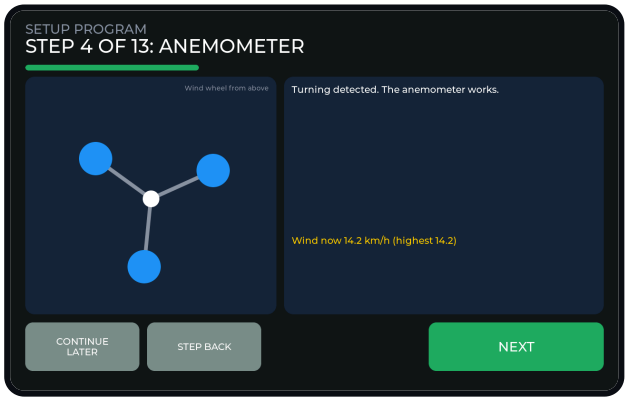


Fig. 65a – Anemometer: the wind wheel was turned, the value rose to 14.2 km/h.

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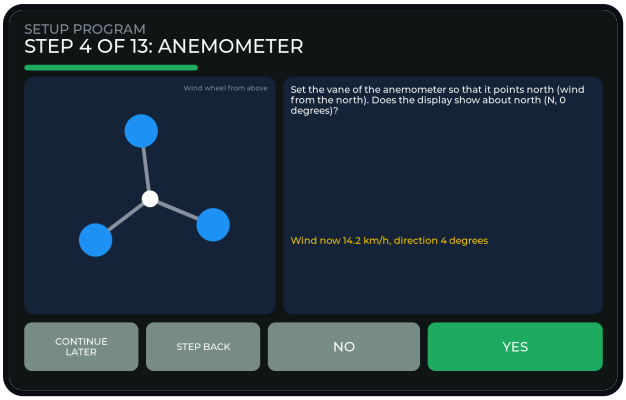


Fig. 65b – With wind direction: vane to the north, the display shows 4 degrees. YES.

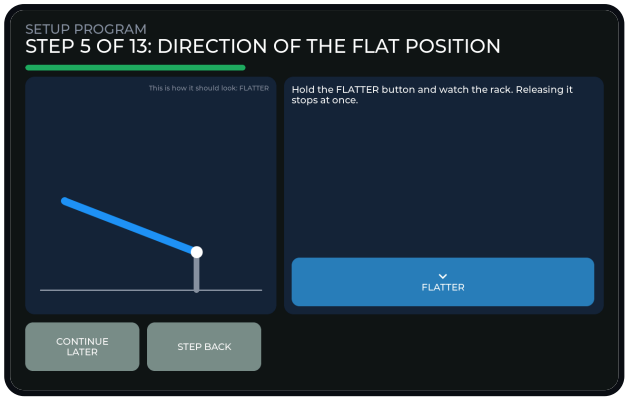


Fig. 66a – Direction of the flat position: hold FLATTER and watch the rack.

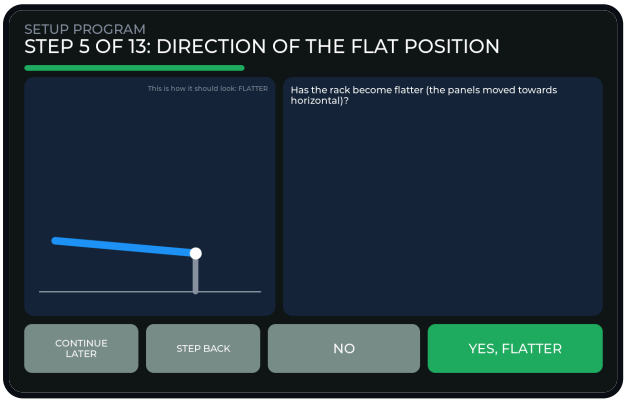


Fig. 66b – The question afterwards. NO switches the direction.

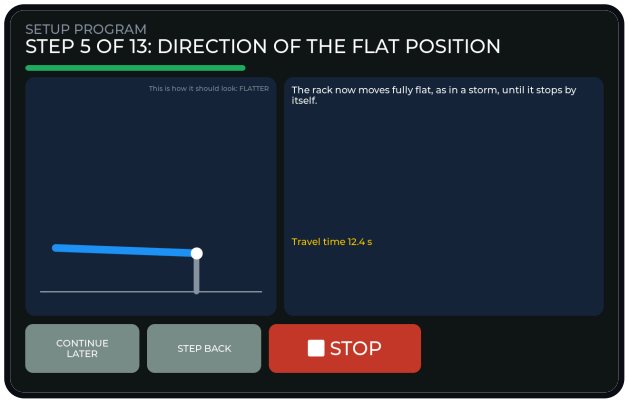


Fig. 66c – The move to the flat position as in a storm, with travel time and STOP.

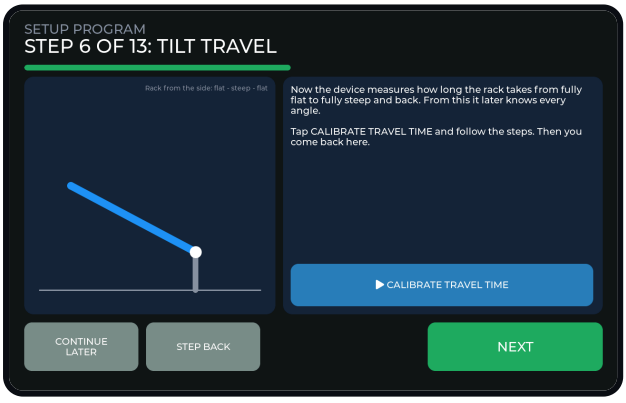


Fig. 67a – One axis: tilt travel via CALIBRATE TRAVEL TIME.



Fig. 67b – Two axes: “Moving STEEPER to the end stop ...” with the time running.

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Fig. 67c – Two axes: did the rack move fully steep and flat again without catching?

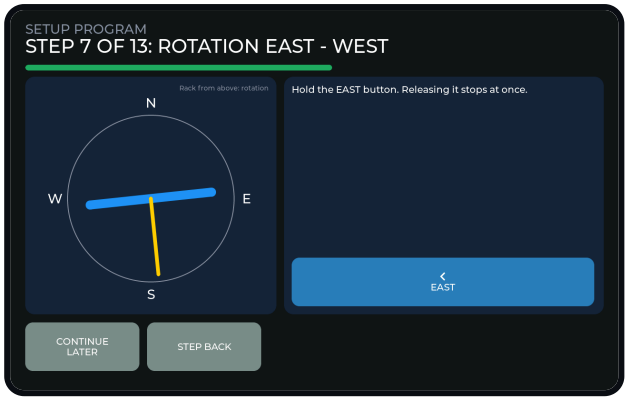


Fig. 68a – Rotation east – west: hold EAST. The sketch shows the rack from above.

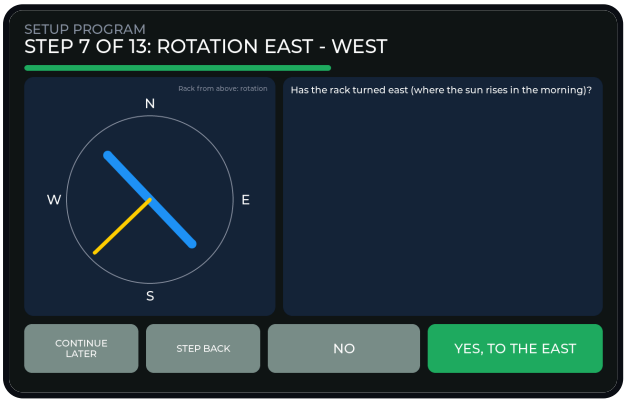


Fig. 68b – Has the rack turned east?

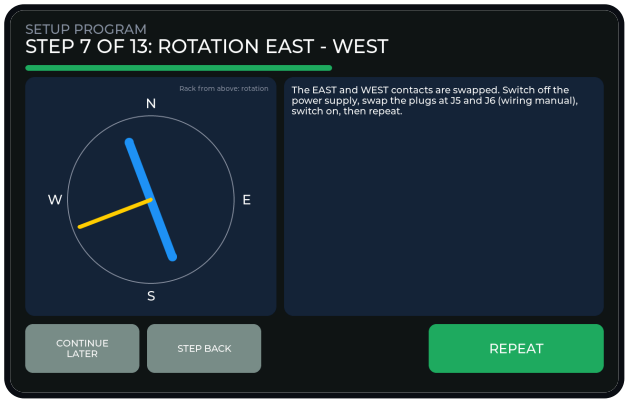


Fig. 68c – After NO: swap the plugs at J5 and J6, then REPEAT.

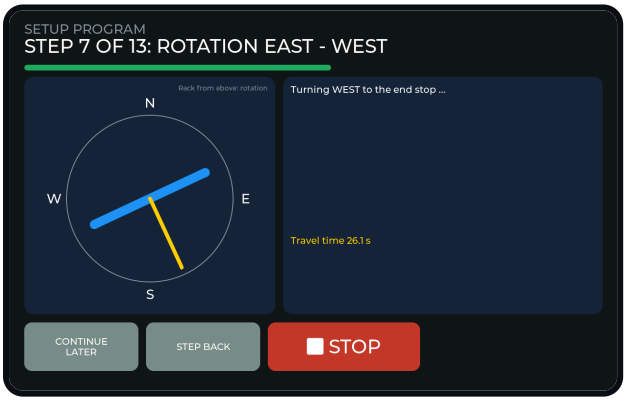


Fig. 68d – The travel east – west – east runs by itself.

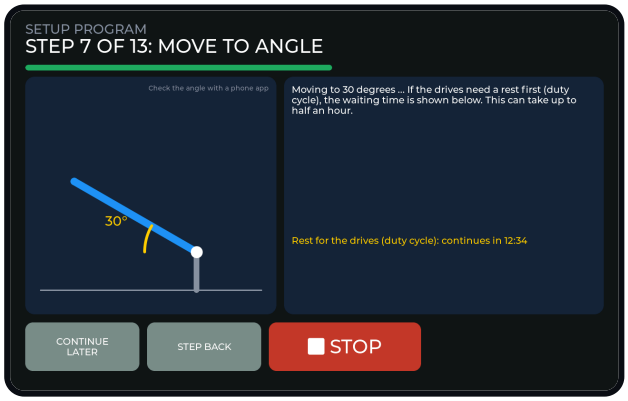


Fig. 69a – One axis: the rack moves to 30°. Here the drives first take a rest; the waiting time is shown at the bottom.

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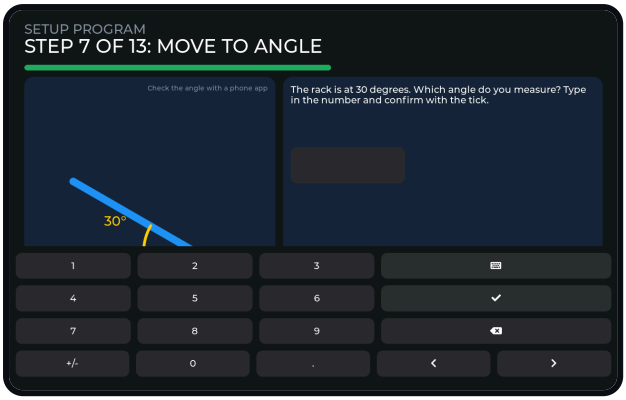
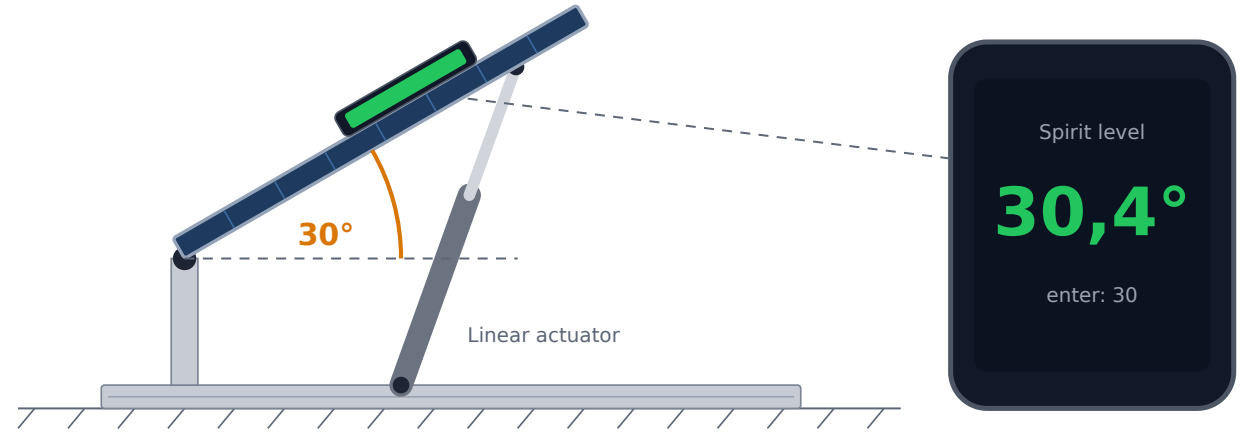


Fig. 69b – Enter the measured angle with the number keys and confirm with the tick.

Waiting for the drives

The drives may only run part of the time (duty cycle, 10 % for the Samba Solar Rack). After the drives in the previous steps and the reset of the sync controller, their stock of run time is often used up. In the steps with drives the bottom right then shows “Rest for the drives (duty cycle): continues in …” with a counting-down time. This can take up to half an hour, also in the simulation. Just wait; the setup program then continues by itself.

Measuring the angle: lay the phone flat on the panel



Lay it along the tilt (the way the panels tip), not across. A whole number is enough.

Fig. 70 – Step “Move to angle” (one axis only): the phone lies lengthwise on a panel, the app shows the angle to the horizontal. The frame is drawn schematically; your rack may look different.

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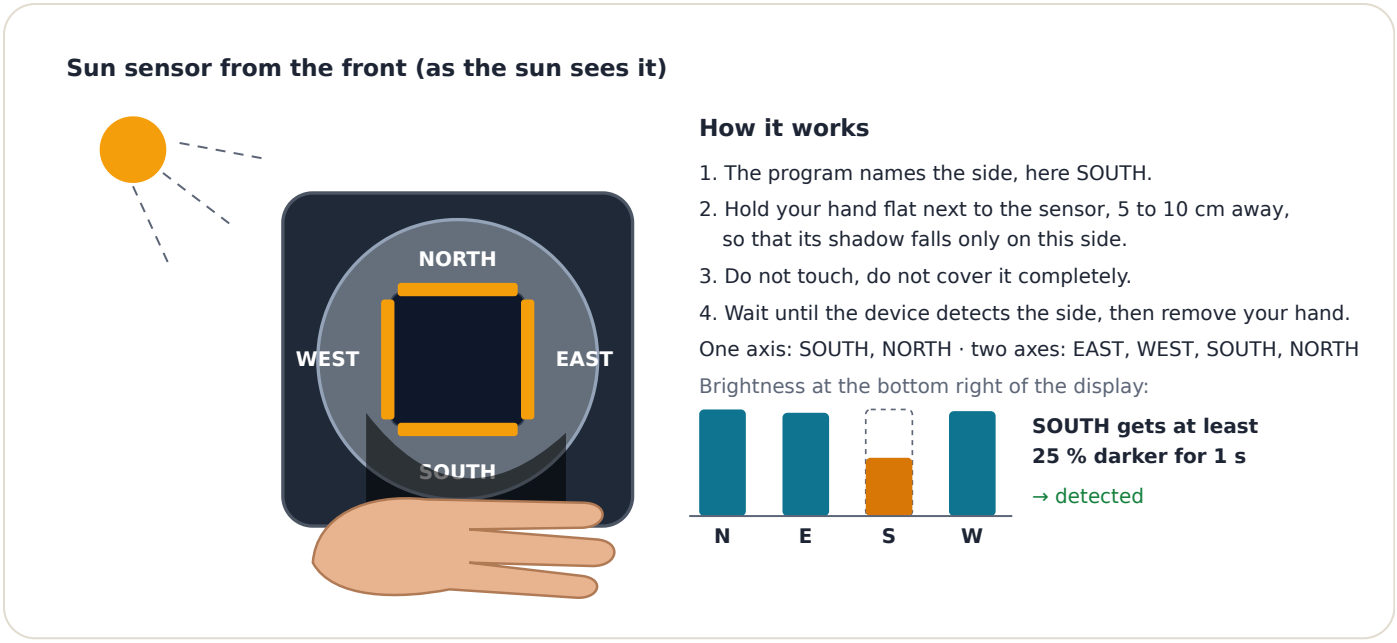


Fig. 71 – Step “Sun sensor”: the shadow of the hand falls on one side of the sensor, and its measured value drops. The sensor is drawn after the photo in Fig. 4a (dome, below it the cube with the sensors); the position of the sensors in the cube is simplified.

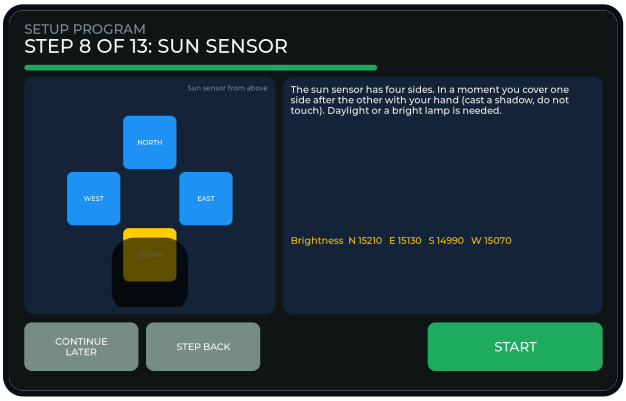


Fig. 72a – Sun sensor: explanation and brightness of the four sides.

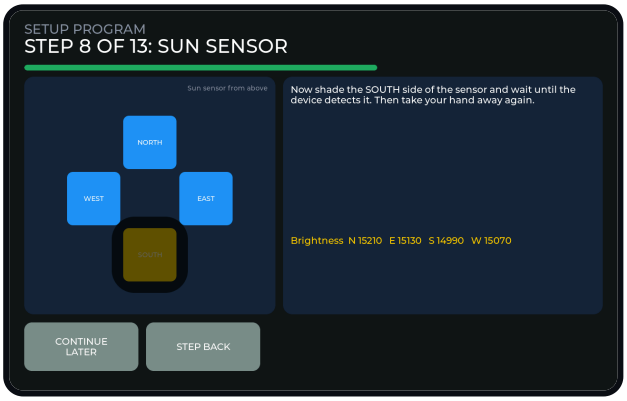


Fig. 72b – “Now shade the SOUTH side of the sensor”: the sketch shows the hand over the side.

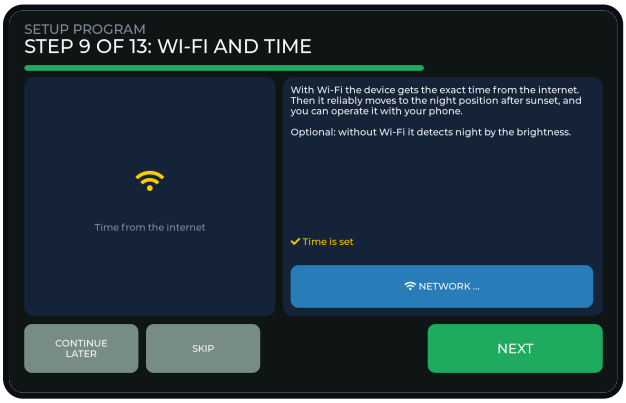


Fig. 73a – Wi-Fi and time (optional): “✓ Time is set”.

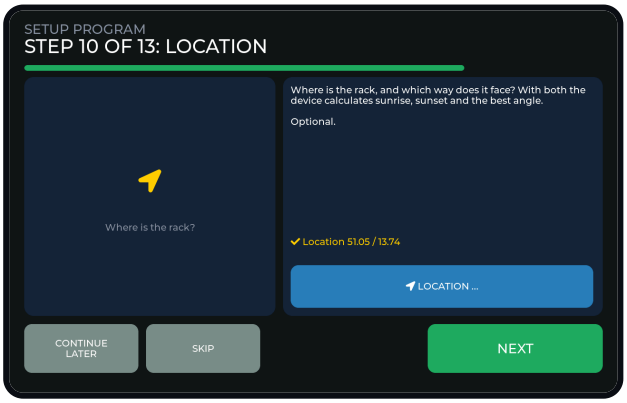


Fig. 73b – Location, one axis: location and viewing direction of the rack.

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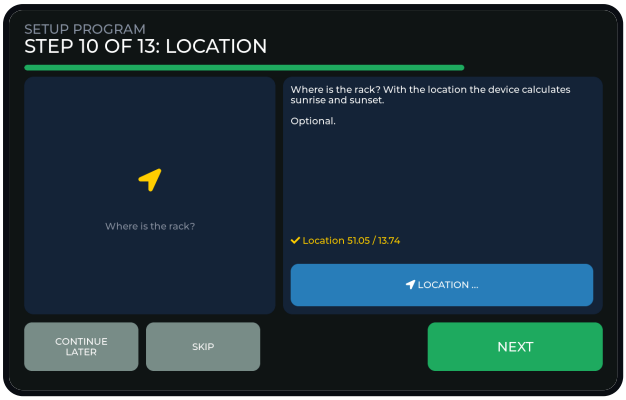


Fig. 73c – Location, two axes: only the location.

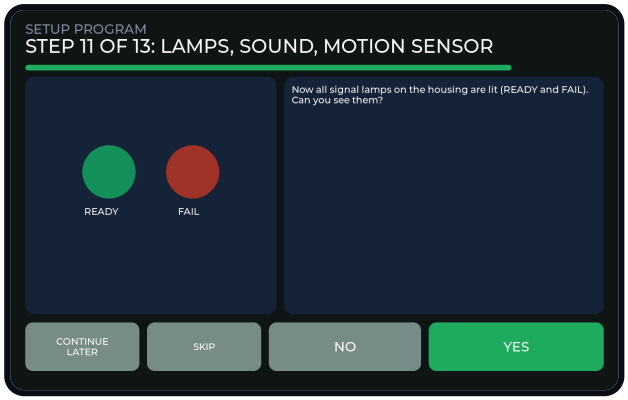


Fig. 74a – Are the READY and FAIL lights on the housing lit?

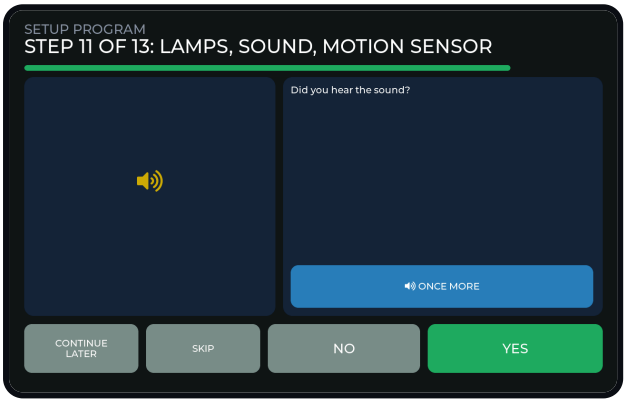


Fig. 74b – Heard the sound? ONCE MORE plays it again.

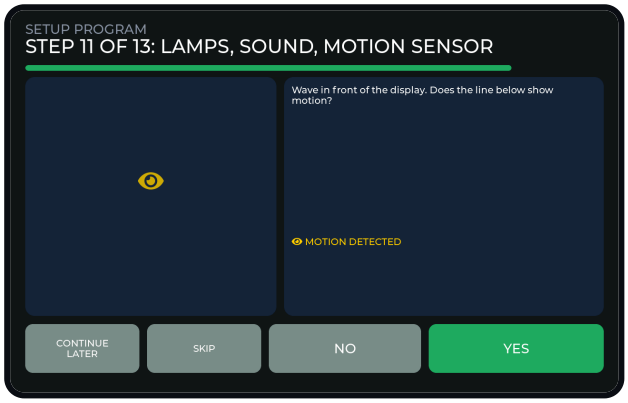


Fig. 74c – Waving in front of the display: “MOTION DETECTED”.

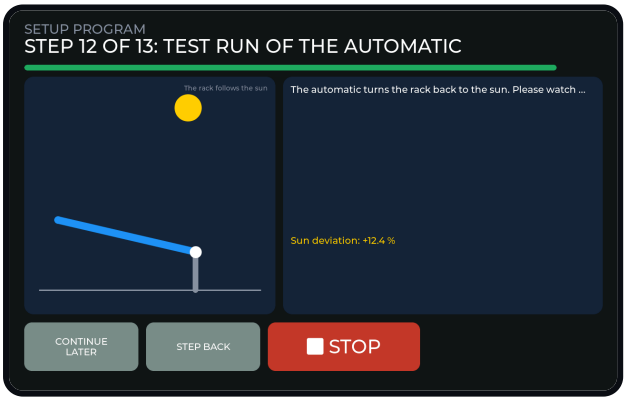


Fig. 75a – Test run: the automatic aligns the rack with the sun again.

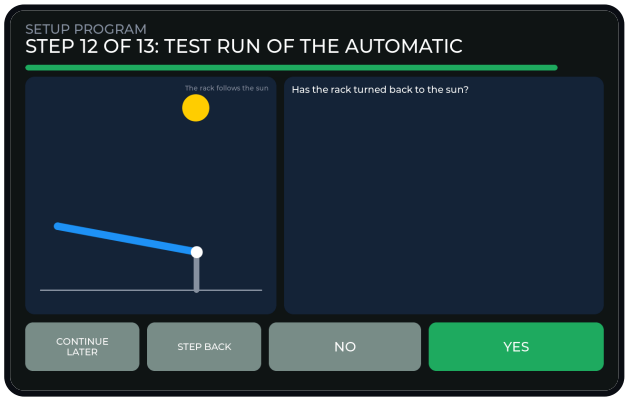


Fig. 75b – Has the rack turned back to the sun?

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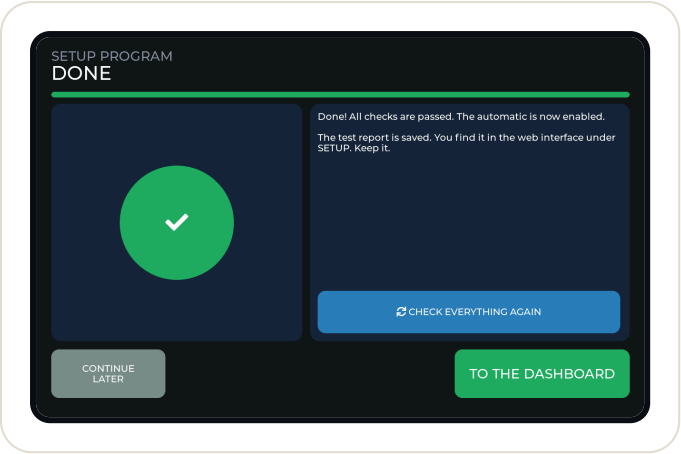


Fig. 76 – Done: the automatic is enabled, the test report saved. With **CHECK EVERYTHING AGAIN** a new check starts later (below).

If something goes wrong

If a step does not work, the right side says “This step did not work. The reason is shown below in red. Fix it, then tap REPEAT.” Below it the reason is shown in red (Fig. 76b). The table lists the reasons and what you do.

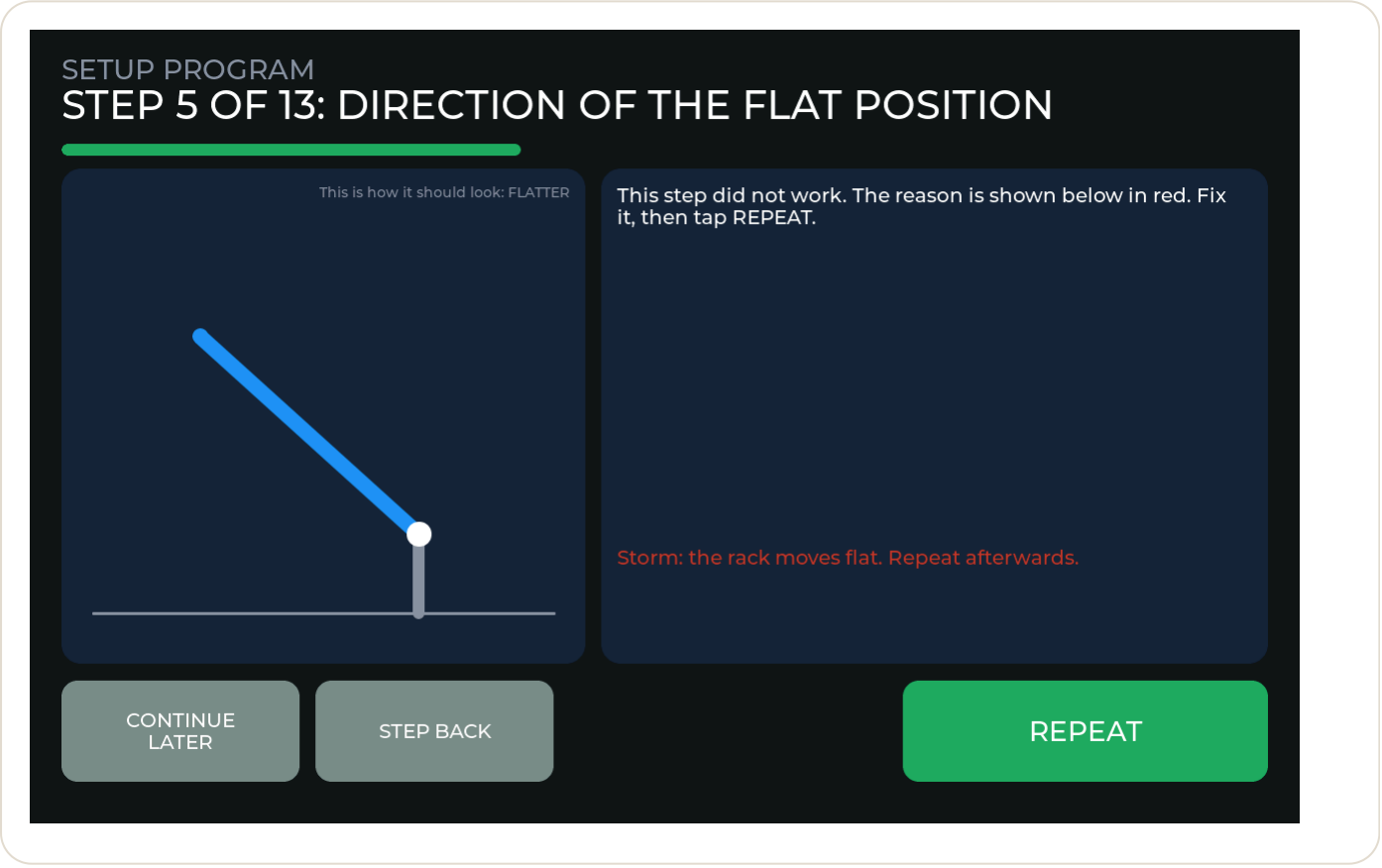


Fig. 76b – Abort due to a storm in step 5 (direction of the flat position): on the right the instructions, below them in red “Storm: the rack moves flat. Repeat afterwards.” At the bottom REPEAT, STEP BACK and CONTINUE LATER.

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MESSAGE	MEANING	WHAT YOU DO
“Storm: the rack moves flat. Repeat afterwards.”	A storm came during a test move; the storm protection has priority.	Wait until the storm is over, then REPEAT.
“Drive fault (maximum run time). Acknowledge and repeat.”	A drive did not reach its end position within the maximum run time.	Check the mechanics and the maximum run time, acknowledge the fault on the dashboard, REPEAT.
“Drive stopped without end position (Hall pickup). Check the mechanics, then repeat.”	With Hall tap: the drive stops before the end position is reached.	Is something in the way or jammed? Then REPEAT.
“First put all points in order.”	In step 3 a field is still red.	Check plugs and cables until everything is green.
“The wind wheel has not been turned yet.”	The wind changed by less than 2 km/h.	Turn it more firmly; check whether the anemometer responds.
“Angle cannot be reached (travel time or Hall pickup).”	Travel time not calibrated or Hall tap faulty.	Back to step 6 and calibrate again.
“RESET ABORTED …” / “Reset not possible …”	The reset of the sync controller did not run to the end or did not start.	Read the cause in the status bar; check the connection J23 or J5 and chip 0x72.
“Too dark …”	There is too little light for the sun sensor and the test run.	Repeat in daylight or with a bright lamp.
“This step is still missing.”	At completion a mandatory step is missing; the program jumps there.	Carry out the step.

The test report

On completion the device saves a test report: date, firmware, operating mode, PASSED, skipped or open for each step, and the measured values (travel times, wind value when turning, angle read off, assignment of the sun sensor, answers on lamps, sound and motion sensor). It is the basis for queries and for comparison if a part is replaced later. Keep it. The last three reports remain in the device. A run in the simulation is marked as “SIMULATION” in the report.

1

Open the web interface

In the browser, open the web interface of the device (address on the display under remote access) and choose **SETUP** at the bottom. *Expected:* The **SETUP PROGRAM** block shows the status (Fig. 78).

2

View

In the row **LATEST REPORT** (or REPORT 2, REPORT 3 for the older ones) tap **VIEW**. *Expected:* The report appears as an A4 sheet: the result at the top, below it the installation, the sequence, the travel times as bars, the assignment of the sun sensor and all steps with their measured values (Fig. 78a). The texts are in the language of the device.

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3

Light or dark

At the top choose **LIGHT SHEET** or **DARK SHEET**. Light is meant for the printer, dark for saving as a file (Fig. 78b). *Expected:* The sheet changes immediately; the browser remembers the choice.

4

Save as PDF

Tap **SAVE AS PDF**. The print dialog of the browser opens. Choose “Save as PDF” as the destination to get a file, or a printer to have it on paper. *Expected:* an A4 PDF without margins with two pages: on page 1 result, installation, sequence, travel times and sun sensor, on page 2 the table of all steps with measured values and the fields for signatures.

5

As a text file (optional)

In the same row tap **TEXT**. *Expected:* The browser saves the report as a plain text file, as shown below.

ℹ️ **Dark sheet in the PDF**

Chrome and Edge keep the dark areas in the PDF file. Firefox and Safari leave out backgrounds when printing, depending on the setting; there tick “Print backgrounds” (or “Background graphics”) in the print dialog. Without it, dark text comes out on white.

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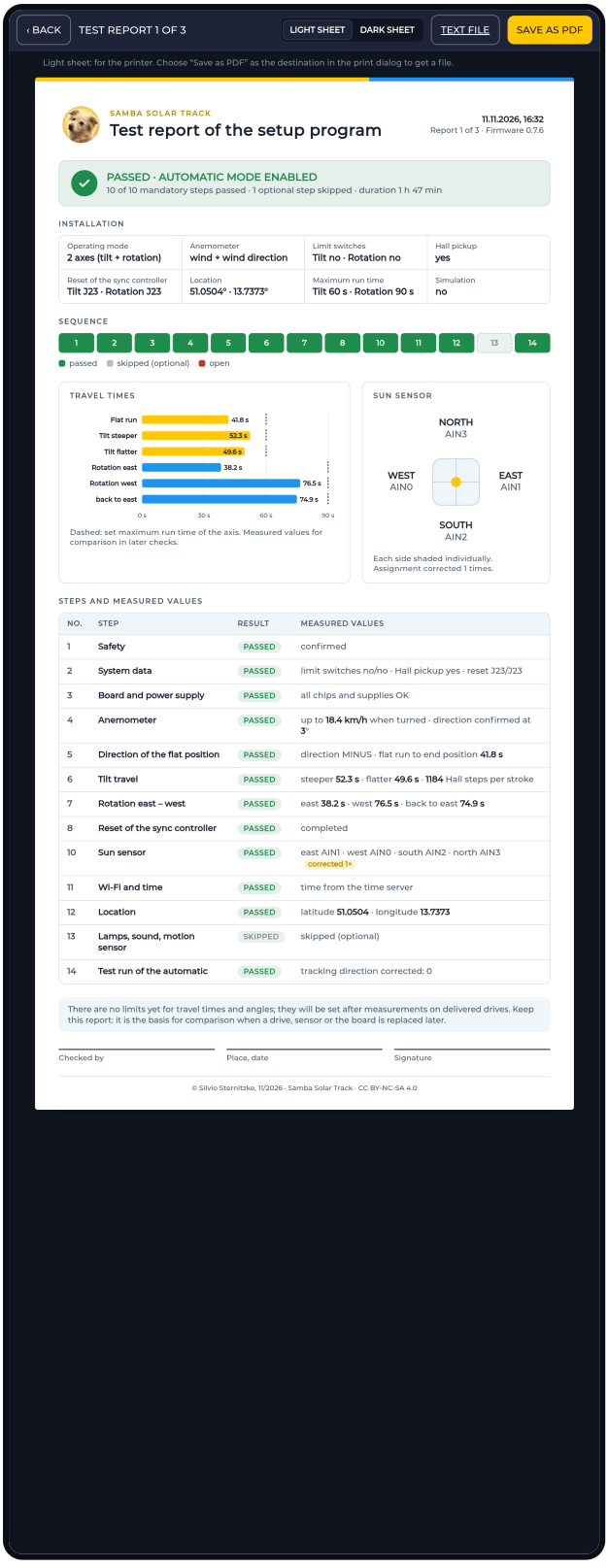


Fig. 78a – The test report in the web interface, light sheet (example values, two axes). At the top the buttons to switch the sheet, for the text file and SAVE AS PDF.

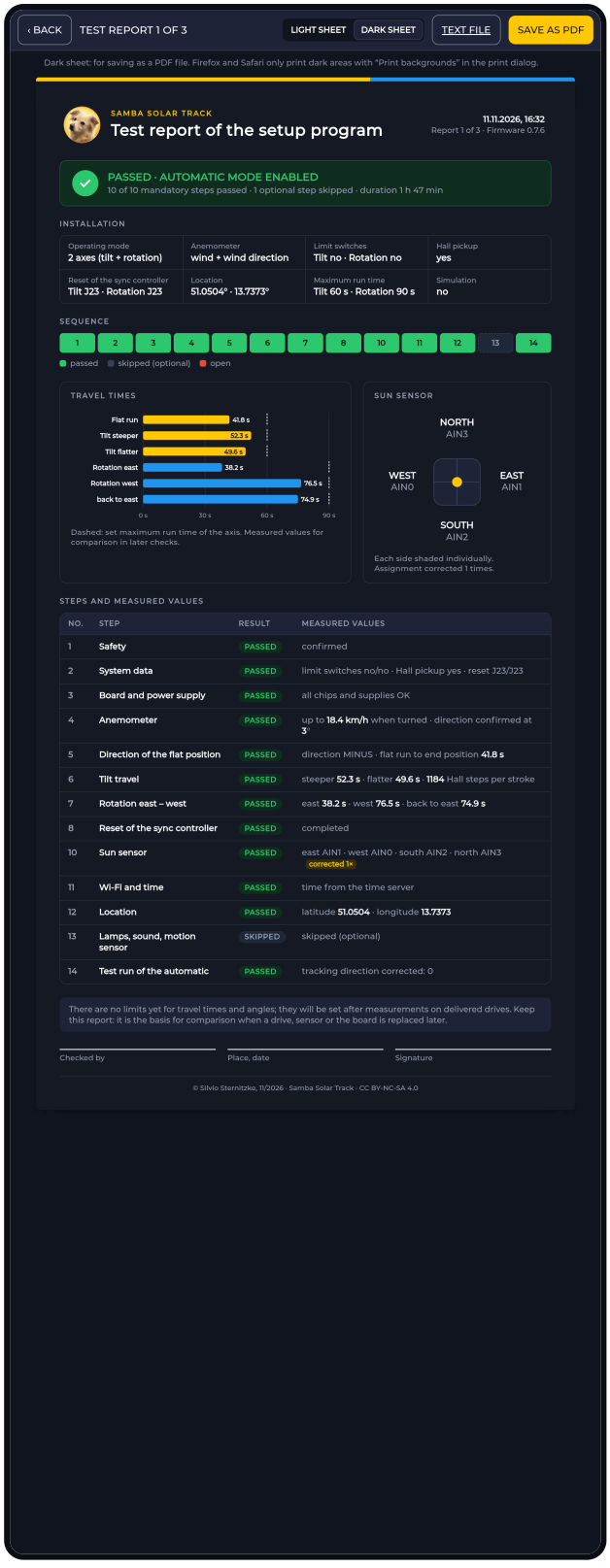


Fig. 78b – The same report as a dark sheet, for saving as a file.

Reports created with firmware before 0.7.6 exist only as a text file; VIEW then reports “This report is only available as a text file”.

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Test report – example in the firmware format (two axes), not from the real hardware

```
SAMBA SOLAR TRACK - PRUEFBERICHT DES SETUP-PROGRAMMS
====================================================
Datum:             10.11.2026 14:32:05
Firmware:          0.7.0
Betriebsart:       vertikal + horizontal (2 Achsen)
Windmesser:        Wind + Windrichtung
Stand:             ABGESCHLOSSEN, Automatik frei

 1. Sicherheit                     BESTANDEN
 2. Daten der Anlage               BESTANDEN
      Endschalter Elevation nein, Azimut nein; Hall-Abgriff nein; Reset Elevation 0, Azimut 0
 3. Platine und Versorgung         BESTANDEN
 4. Windmesser                     BESTANDEN
      beim Drehen von Hand bis 14.2 km/h, Richtung bestaetigt bei 4 Grad
 5. Richtung der Flachstellung     BESTANDEN
      Flachstellung Richtung PLUS, Flachfahrt bis zur Endlage 12.4 s
 6. Fahrweg der Elevation          BESTANDEN
      steiler 48.3 s, flacher 46.9 s
 7. Azimut                         BESTANDEN
      Ost 26.1 s, West 51.0 s, zurueck nach Ost 50.6 s
10. Sonnensensor                   BESTANDEN
      Zuordnung: Ost AIN0, West AIN1, Sued AIN2, Nord AIN3; berichtigt: 0
11. Netzwerk und Uhrzeit           BESTANDEN
12. Standort                       BESTANDEN
      Breite 51.0504, Laenge 13.7373, Ausrichtung des Racks 0 Grad
13. Leuchten, Ton, Bewegungsmelder BESTANDEN
      Leuchten ja, Ton ja, Bewegungsmelder ja
14. Probelauf der Automatik        BESTANDEN
      Richtung der Nachfuehrung berichtigt: 0

Grenzwerte fuer Fahrzeiten und Winkel gibt es noch nicht; sie werden nach der Messung an
gelieferten Antrieben festgelegt. Die Messwerte oben dienen dem Vergleich bei spaeteren
Pruefungen.
```

The report numbers all possible steps fixed from 1 to 14 and leaves out those that do not apply to your system (in the example 8, reset, and 9, move to angle). Its numbers therefore differ from the “STEP x OF 13” display from step 8 on. The numbers in the example are made up.

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Fig. 77 – Web interface while the setup program is missing: at the top the notice with the step (“STEP 8 OF 13”), the status bar violet. The setup program can only be operated at the display, because someone has to watch the rack while it runs.

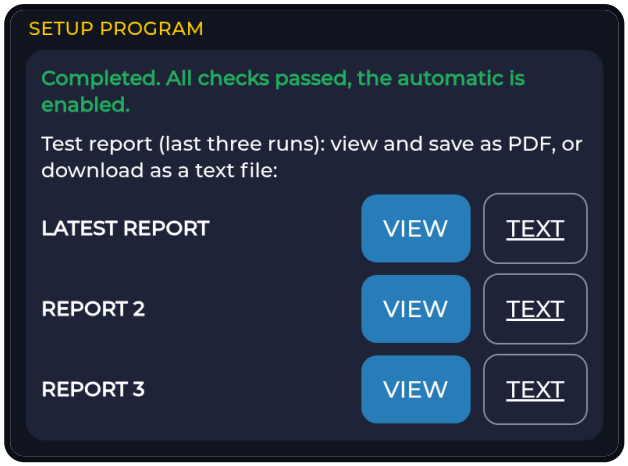


Fig. 78 – Web interface, SETUP (detail): “Completed” and the three test reports, each with VIEW and TEXT.

Checking again later

The setup program is also meant for later, for example after replacing a drive, the sun sensor, the anemometer or the circuit board. It is in the setup under **SETTINGS → COMMISSIONING** as the green row **SETUP PROGRAM (CHECK EVERYTHING)**. Once it is completed, the row opens the DONE page (Fig. 76). The individual values (limit switches, channels of the sun sensor, direction of the flat position, travel times …) remain individually adjustable there as before.

1

Start a new check

On the DONE page tap **CHECK EVERYTHING AGAIN**. *Expected:* The query “CHECK EVERYTHING AGAIN?” explains: the setup program starts from the beginning; until all steps are passed again, the automatic is locked; storm protection and parking keep working; the previous test report remains saved.

2

Confirm

Tap **START OVER** (CANCEL leaves everything as it is). *Expected:* The setup program is back at step 1, the status bar is violet, the automatic is locked.

3

Go through all steps

As the first time, up to the DONE page. *Expected:* The automatic is enabled again, a new test report is saved.

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ℹ️ **Only completed in the simulation?**
If the setup program was only completed in the simulation, nothing of it has been checked on the real rack. If the device then starts with the circuit board connected, the setup program therefore starts again by itself, and the automatic remains locked until it has been passed on the real rack.

ℹ️ **Manual steps in the simulation**
In the simulation there are no wind cups and no sensor to shade. Since firmware 0.7.2 the device then does these steps itself: it turns the wind cups, points the vane north, shades the sides of the sun sensor one after the other and makes it bright enough, even in the simulated night. You only tap the buttons as described. Up to firmware 0.7.1 the setup program therefore stopped at the wind sensor in the simulation.

🧩 **Still open in firmware 0.7.0**
Individual sections of the setup program cannot yet be repeated selectively; CHECK EVERYTHING AGAIN always starts at step 1. If after a replacement you only want to recalibrate one value, use the individual functions under SETTINGS → COMMISSIONING (for example CALIBRATE TRAVEL TIME).

Sun, location and night (since firmware 0.7.0, in preparation, only tested on the PC)

🧩 **Firmware 0.7.0: only tested on the PC**
The calculation of the position of the sun has been checked with automatic tests against values from an astronomical reference; display, map and night position have been built on the PC and tried out in the simulation, but never on a rack. The screenshot tool generated the pictures from the real code (location Dresden, example time 13:30).

Since firmware 0.7.0 the device knows where the sun is. It calculates the position of the sun from two things: the **time** (comes from the internet via Wi-Fi) and the **location** of the rack (latitude and longitude, entered on the display or in the web interface). From this it knows sunrise and sunset for every day, the sun elevation and the compass direction of the sun (sun azimuth). The calculation follows the method of the US weather agency NOAA, with its own code; the deviation is below about 0.05° and below one minute for sunrise and sunset.

What for: Above all for the **night**: after sunset the rack reliably moves to the night position, even if a street lamp or the moon shines on the sensor. With one axis the device also shows the **best tilt angle** and can move to it in cloudy weather if you wish (below). **During the day the sun sensor stays in charge**: it is more accurate than any calculation, because it sees the real sun.

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> ⚠️ **No time without Wi-Fi**
> The board has no clock of its own with a battery. The time only comes via Wi-Fi from a time server on the internet. Without Wi-Fi (or without an entered location) the controller still works: it then detects the night from the brightness of the sun sensor as before, and tracking follows the sensor as always. This is not an error; the SUN button then shows “SUN …” in yellow.

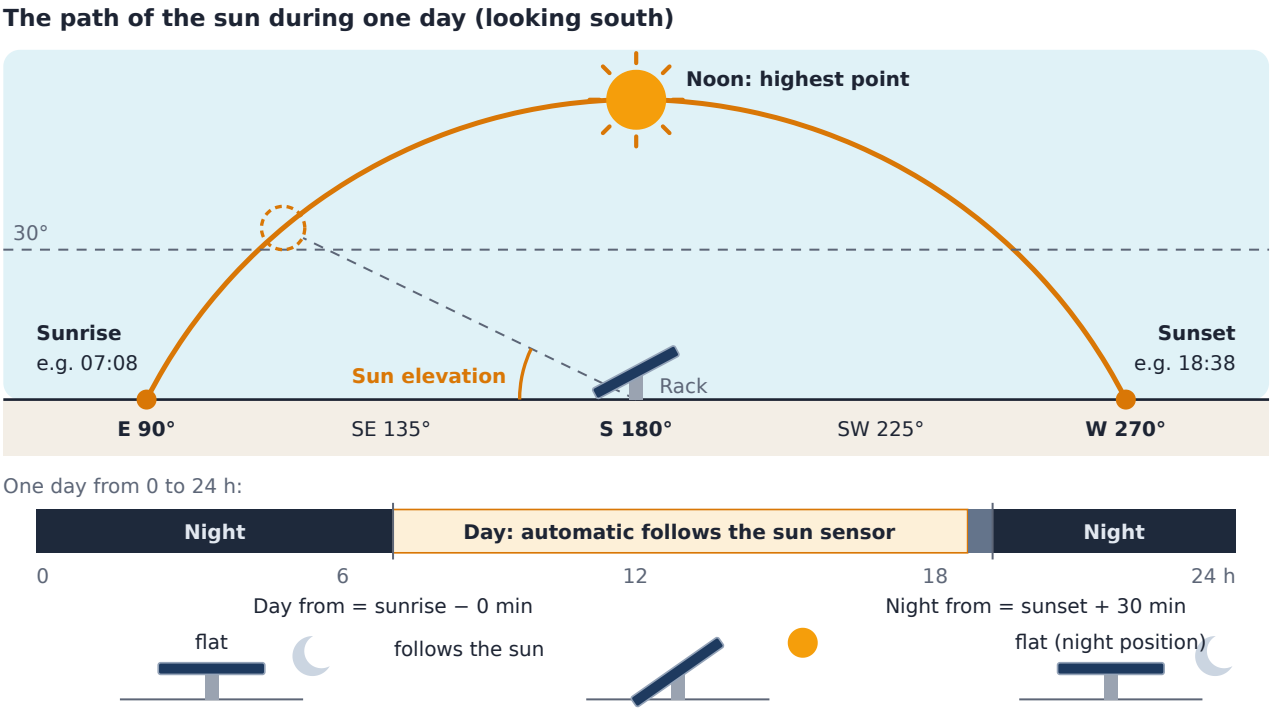


Fig. 80 – Top: the daily arc, as also shown on the SUN page: the sun rises in the east, is highest in the south at midday and sets in the west (drawn for the start of spring and autumn; in summer it rises in the north-east and sets in the north-west, in winter in the south-east and south-west). The **sun elevation** is the angle above the horizon, the number below the compass directions is the **azimuth of the sun**. Bottom: from “Night from” the rack moves to the night position, from “Day from” it follows the sun again. The times are an example.

The SUN button on the dashboard

At the top right of the SUN SENSOR card there is a small button with a yellow dot and today’s times: **▲ 04:50 ▼ 21:23** means sunrise at 4:50 and sunset at 21:23. Tapping it opens the SUN page. If the time or the location is missing, it shows **SUN …** in yellow; at the Arctic or Antarctic Circle it can also show “SUN: POLAR DAY” or “POLAR NIGHT”. The button exists in both operating modes.

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Fig. 81a – One axis: at the top right of the SUN SENSOR card, the button with sunrise and sunset.



Fig. 81b – Two axes: the same button.

The SUN page

On the left is the **daily arc**: the orange line is the path of the sun today, the yellow dot is the sun now, above it the time of noon and the highest sun elevation (“NOON 13:06 - 62° HIGH”). The compass directions at the bottom apply when looking south (north in the southern hemisphere). When the page opens, the sun runs once from sunrise to its current position (Fig. 82c). If the sun is below the horizon, it says “The sun is below the horizon.”

On the right are the numbers: TIME, SUN ELEVATION, DIRECTION (the azimuth of the sun, for example “191° (SSW)”), SUNRISE / SUNSET, NIGHT FROM and DAY FROM. With **one axis** these are followed by BEST TILT ANGLE (the angle at which the panels currently face the sun best, calculated with the orientation of the rack; at night “FLAT (NIGHT)”) and RACK FACES (“15° TO THE WEST”). With **two axes** it shows AT NIGHT: “FLAT, THEN EAST” or “FLAT”.

At the bottom: **BACK**, **LOCATION …** and, with one axis, **RACK ORIENTATION …**; with two axes the **NIGHT TO THE EAST: ON / OFF** button (night position).



Fig. 82a – SUN page, one axis: daily arc, numbers, best tilt angle 28°, the rack faces 15° to the west.



Fig. 82b – SUN page, two axes: AT NIGHT “FLAT, THEN EAST”, at the bottom the NIGHT TO THE EAST: ON button.

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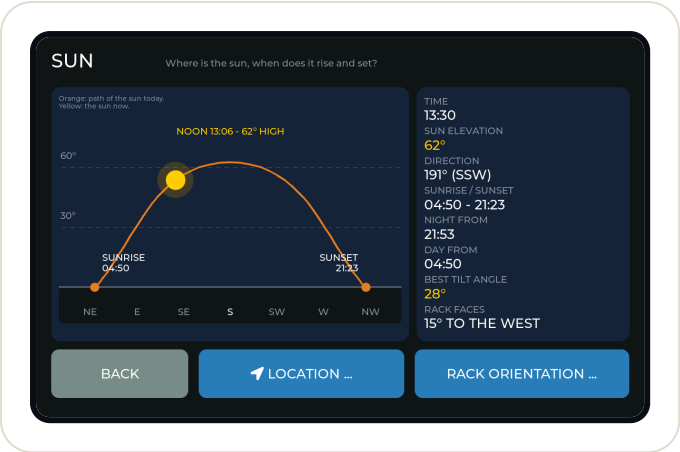


Fig. 82c – When opening: the sun runs along the arc from sunrise to its current position.

If something is missing: Without time or location, instead of the arc the page shows what is missing, with a red cross in front: “1. THE TIME … set up Wi-Fi (NETWORK button)” and “2. THE LOCATION … tap the LOCATION button”, below it the two buttons **NETWORK …** and **LOCATION …**. Once both are there, the arc appears by itself.

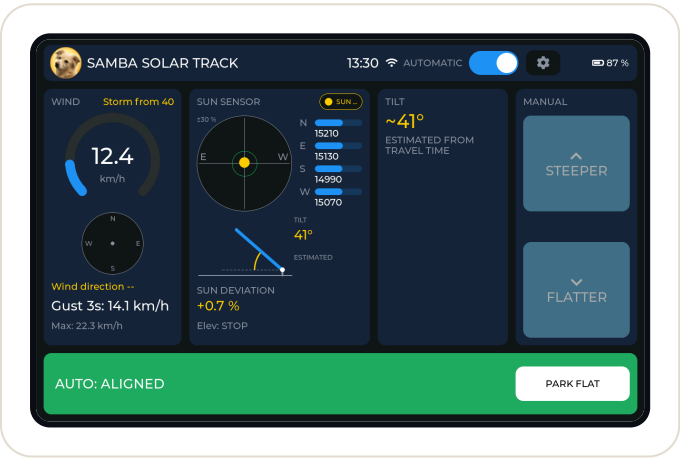


Fig. 83a – Time or location missing: the button shows “SUN …” in yellow.

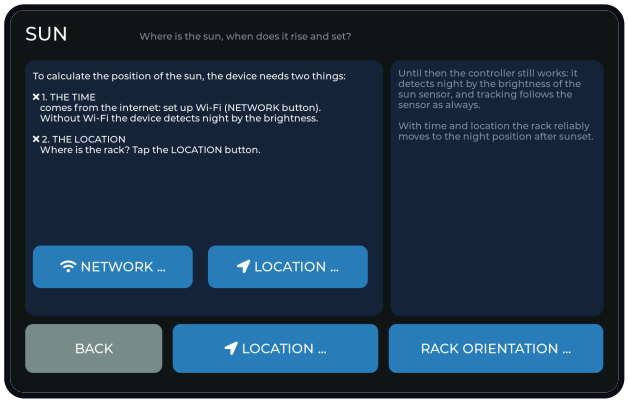


Fig. 83b – The SUN page says what is missing and takes you straight there with NETWORK … and LOCATION ….

Entering the location (LOCATION page)

You open the LOCATION page from the SUN page (**LOCATION …**) or in the setup program. It shows a map that is **stored in the device**: coasts, country borders and lakes from Natural Earth and all places with 5 000 inhabitants or more from GeoNames. So it does not need the internet. For the position of the sun an accuracy of about **10 km** is enough; the nearest larger town is quite sufficient.

1

Search for a place

Tap the **SEARCH PLACE** field and type the first letters with the keyboard, for example “Dres”. *Expected:* Matching places with their country appear below, for example “Dresden (DE)” (Fig. 84b). If nothing is found: “No place found. Try the nearest larger town or tap on the map.”

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2 Tap the place

Tap the right place. *Expected:* The map jumps there, a red dot marks it, and LATITUDE (N +) and LONGITUDE (E +) are filled in.

3 Or tap on the map

With **+** and **−** you zoom in seven steps (“ZOOM 5/7”); swiping moves the map. A tap sets the point. *Expected:* “Point set. To refine, zoom in closer and tap again.”

4 Or enter numbers

If you know the coordinates (for example from a map app on your phone), type them in directly, with a dot instead of a comma: latitude −90 to 90 (north positive), longitude −180 to 180 (east positive), for example 51.0504 and 13.7373 for Dresden. *Expected:* “Numbers taken over. Save with APPLY.”

5 Save

APPLY (green). *Expected:* The page closes, the SUN page shows the daily arc for your location, and the button on the dashboard shows the times. **BACK** leaves the page without saving; **CLEAR** deletes the location (then night is again detected by brightness).

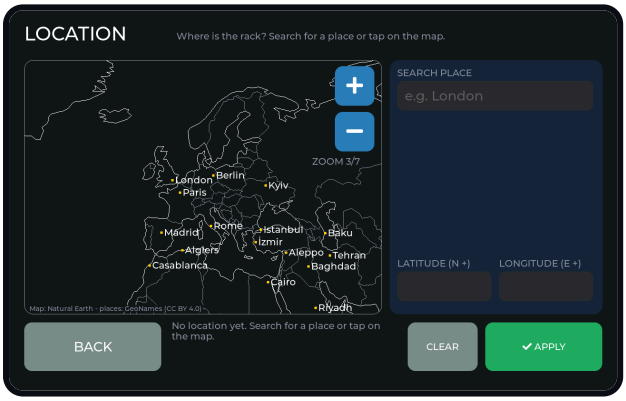


Fig. 84a – No location yet: the map shows Europe, the number fields are empty.

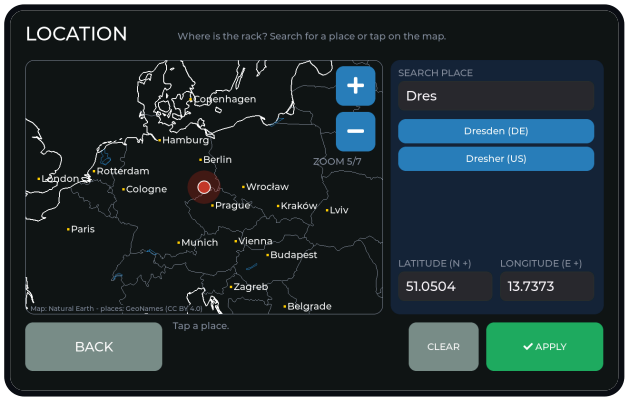


Fig. 84b – Search “Dres”: Dresden (DE) and Dresher (US) to choose from.

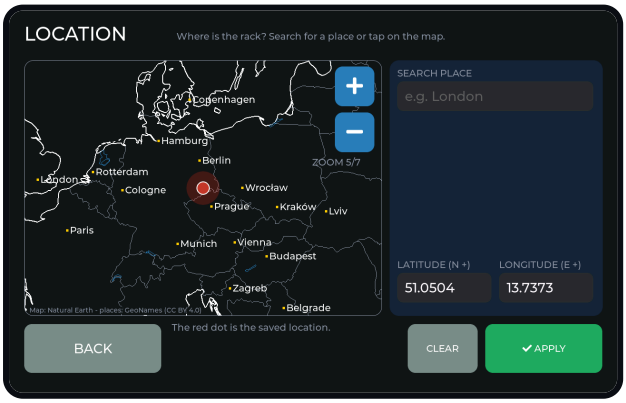


Fig. 84c – The red dot is the saved location, on the right latitude 51.0504 and longitude 13.7373. At the bottom left of the map, the sources.

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The orientation of the rack (one axis only)

A rack with one axis can only tilt the panels, not rotate them. It always faces the same compass direction, ideally exactly south. If it stands a little askew, for example towards the south-west, the best tilt angle is different from when it faces exactly south. So that the device calculates it correctly, you enter where your rack faces. The orientation only affects the best tilt angle (display and cloud cover); sunrise, sunset and the night do not depend on it. With two axes this value does not exist, because there the azimuth turns the rack towards the sun anyway.

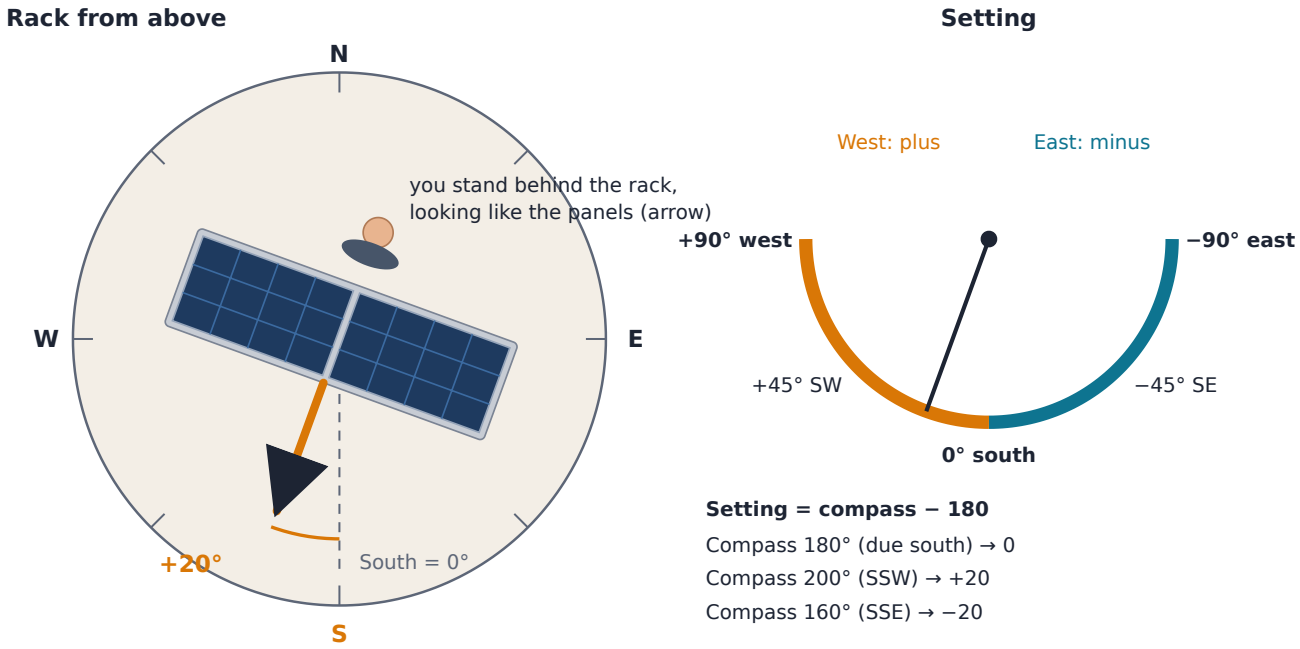


Fig. 85 – Left: the rack from above (north at the top, as on a map): the panels face in the direction of the arrow, here turned 20° from south towards west. Right: the scale of the setting: due south = 0, towards the west positive (afternoon sun), towards the east negative (morning sun), allowed −90 to +90.

1

Stand behind the rack

Stand behind the rack and look in the same direction as the panels. *Expected:* You look over the panels in the direction they point.

2

Measure with the compass

Measure with a compass where you are looking; a compass app on your phone is enough. Hold the phone level and not directly against metal of the rack. *Expected:* A number around 180° (south), for example 200°.

3

Convert

Subtract 180 from the compass reading: 200° gives +20, 160° gives −20. *Expected:* A value between −90 and +90. A few degrees of compass error do not matter.

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4

Enter

SUN page → **RACK ORIENTATION …**; set the value with **5° EAST**, **1° EAST**, **1° WEST**, **5° WEST**. *Expected:* The sketch on the left turns the rack accordingly, in large letters it shows “20° TO THE WEST (SSW)” and below it “Best angle now: …”.

5

Save

APPLY. *Expected:* The SUN page shows RACK FACES “20° TO THE WEST”. BACK discards the change.

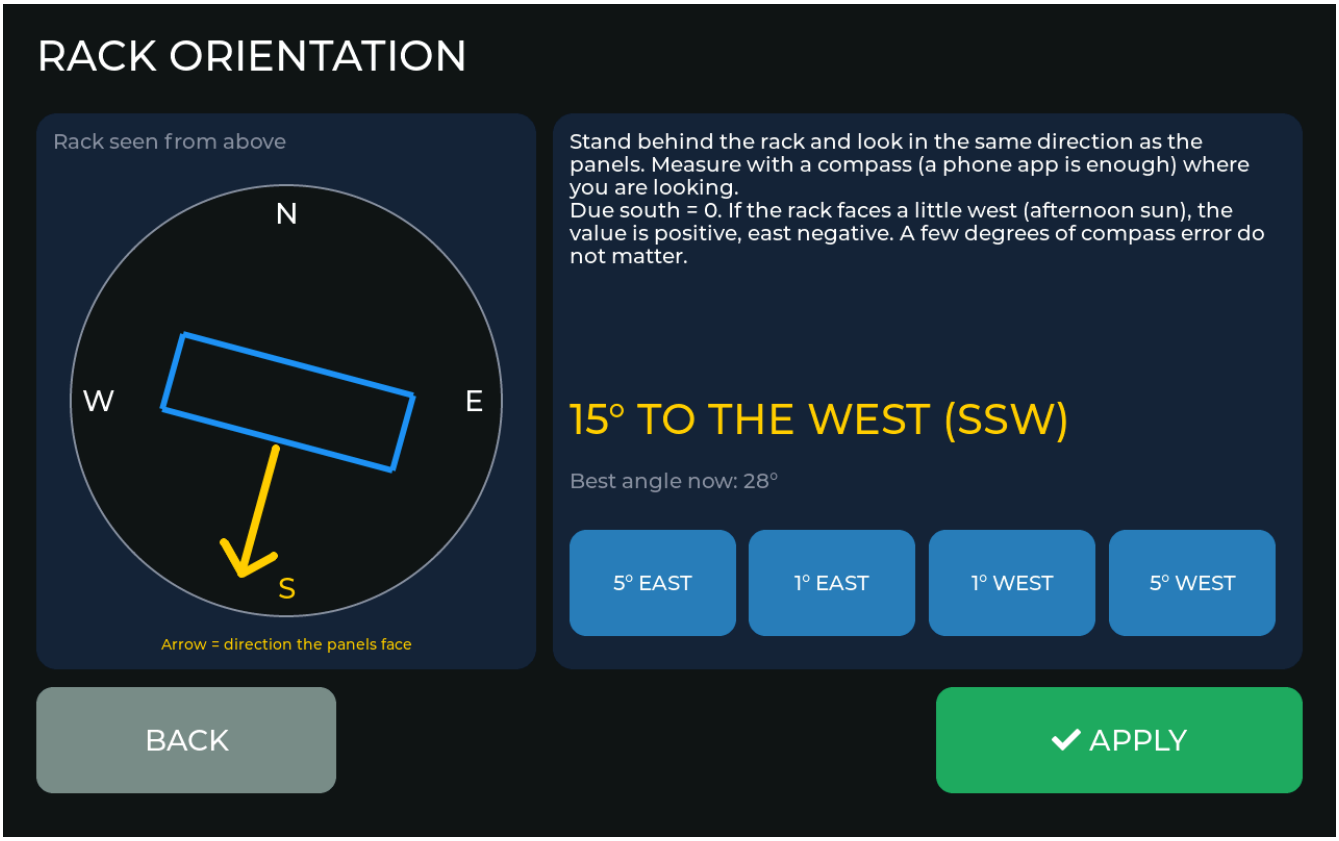


Fig. 86 – The RACK ORIENTATION page: on the left the rack from above with the yellow arrow of the viewing direction, on the right the explanation, the value (“15° TO THE WEST (SSW)”), the best angle and the four buttons.

The night position

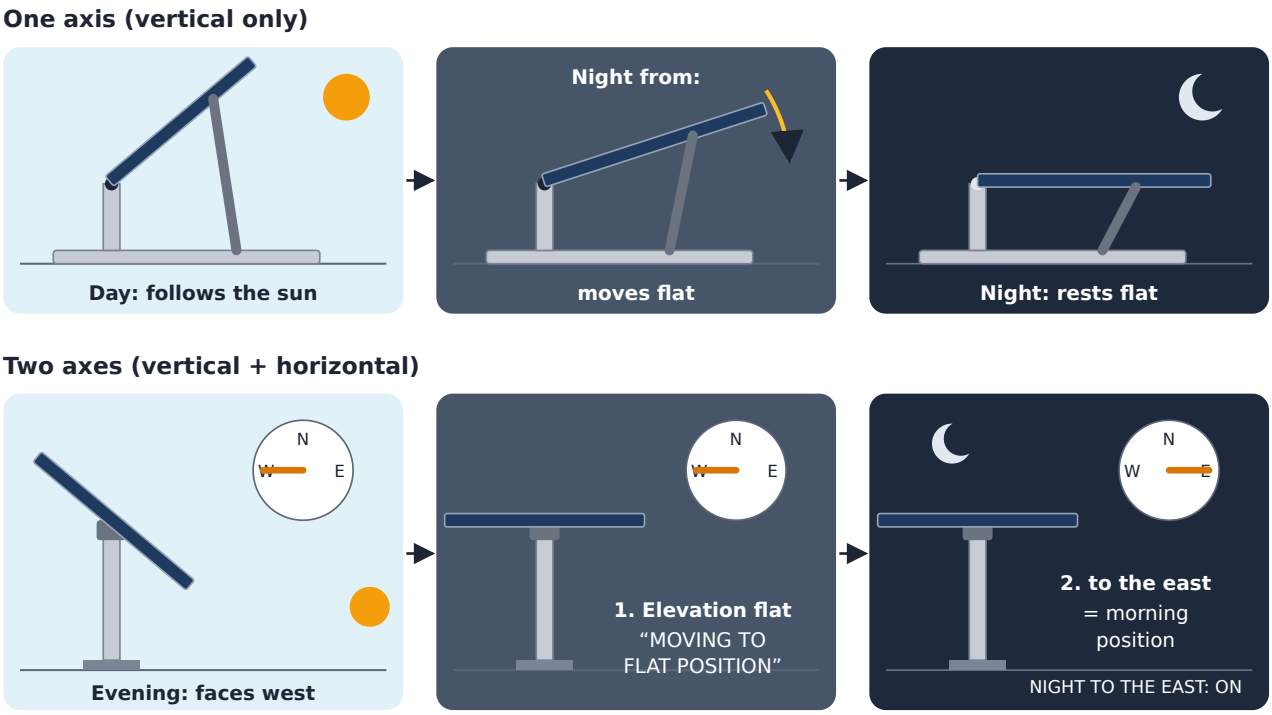
When is it night? With time and location, night begins at the “Night from” time (sunset plus 30 minutes, adjustable) and ends at the “Day from” time (sunrise minus 0 minutes, adjustable). Both times are shown on the SUN page. Without time or without location, **brightness** applies as before: if the sum of the four sensors of the sun sensor stays below 4 000 for 5 minutes, it is night; if it stays above 6 000 for 5 minutes, it is day. In the simulation, brightness always applies.

What the rack does at night:

- **One axis (vertical only):** The elevation moves to the flat position (the same direction as in a storm) and stays there until morning. Status bar: “NIGHT: MOVING TO FLAT POSITION”, then “NIGHT: RESTING IN FLAT POSITION”.

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- **Two axes (vertical + horizontal):** First the elevation moves flat. Only when it has finished does the azimuth turn east to the end stop, so that in the morning the rack already faces the rising sun (morning position). Never more than one axis moves at a time. Status bar: “NIGHT: MOVING TO FLAT POSITION”, then “NIGHT: MOVING TO MORNING POSITION (EAST)”, then “NIGHT: RESTING IN FLAT POSITION”. The turn to the east is switched on by default; you can switch it off with the **NIGHT TO THE EAST: ON / OFF** button on the SUN page (or the setting “Return east at night”). The rack then stays flat at night in the direction it had in the evening.
- **In the morning**, from “Day from” (or when it is bright enough), the automatic takes over again and follows the sun sensor.



Small circle: viewing direction of the panels from above.
If NIGHT TO THE EAST is off, step 2 is skipped; the rack rests flat in the direction of the evening.

Fig. 87 – The night position. Top, one axis: from “Night from” the tilt moves flat and stays there. Bottom, two axes: first flat, then (by default) east to the morning position; the arrow in the small circle shows the viewing direction of the panels from above. The frames are drawn schematically; your rack may look different.

> ℹ️ **The night move is a normal move**
>
> It has no priority like the storm move: limit switches, dead time, max. run time and duty cycle apply as for every move. Without limit switches, the move to the flat position waits until the duty cycle allows a complete move; this is shown in the log (“Night position: move to flat position waiting … s for the duty cycle”). Storm, drive fault, manual mode, parking, anemometer failure, tilt on command, sensor fault and a missing setup program still take precedence over the night: as long as the setup program is not completed, there is no night move, because the travel paths have not yet been checked.

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Moving to the calculated angle when cloudy (one axis only, off by default)

In heavy cloud cover the sun sensor sees no sun; the rack then stays where it is. With one axis, the device can instead move to the **calculated best angle** (from the position of the sun and the orientation of the rack), because even behind clouds most of the light comes from the direction of the sun. This is **off by default** and is switched on under SETTINGS → CONTROL with “When cloudy, move to calculated angle”.

- **Requirements:** one axis, time and location available, daytime, calibrated travel time or Hall tap (the move is carried out as with tilt on command).
- **Cloudy** means: the sum of the four sensors stays below “Cloudy below brightness (sum)” (default 20 000) for 5 minutes. It ends when it stays above 1.25 times that value (default 25 000) for 5 minutes; then the sensor takes over again.
- The device moves to at most 60° and only readjusts when the calculated angle has changed by at least 3°, so that the drives do not run constantly.
- Status bar (blue): “CLOUDY: MOVING TO CALCULATED 35°”, then “CLOUDY: CALCULATED ANGLE 35° (SUN POSITION)”. Without a calibrated travel time: “CLOUDY: TRAVEL TIME NOT CALIBRATED – NO MOVE”, with a faulty Hall tap “CLOUDY: HALL TAP FAULT – NO MOVE”.

> 🧩 **Assumption**
> The threshold of 20 000 for “cloudy” is estimated and has not yet been measured in real weather. Whether moving to the calculated angle when cloudy yields more than staying put has not yet been measured on this rack; that is why it is off by default.

New in the status bar

STATUS BAR	COLOUR	MEANING
SETUP PROGRAM MISSING: AUTOMATIC LOCKED / TAP HERE TO CONTINUE	violet	The setup program has not yet been completed. Tapping continues it. In the web interface: “… – CONTINUE AT THE DISPLAY”.
NIGHT: MOVING TO FLAT POSITION	night blue	Night: the elevation moves flat (both operating modes).
NIGHT: MOVING TO MORNING POSITION (EAST)	night blue	Two axes only: the elevation is flat, the azimuth turns east.
NIGHT: RESTING IN FLAT POSITION	night blue	Night position reached, nothing moves.
CLOUDY: MOVING TO CALCULATED …° / CLOUDY: CALCULATED ANGLE …° (SUN POSITION)	blue	One axis only, when switched on: moving to the calculated angle, or arrived there.
CLOUDY: TRAVEL TIME NOT CALIBRATED – NO MOVE / CLOUDY: HALL TAP FAULT – NO MOVE	blue	Cloudy, but the angle cannot be approached: first calibrate or check the Hall tap.

The colours in the table are approximate; on the display they depend on the selected design.

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New settings

SETTING (GROUP UNDER SETTINGS)	DEFAULT	PERMITTED	PURPOSE
Location: latitude (north positive) (COMMISSIONING)	0 = not entered	−90 to 90 degrees, 4 decimal places	Latitude of the rack. Latitude 0 and longitude 0 together mean “not entered”. Easiest via the LOCATION page.
Location: longitude (east positive) (COMMISSIONING)	0 = not entered	−180 to 180 degrees	Longitude of the rack; negative west of Greenwich.
Rack orientation (south 0, west positive) (COMMISSIONING, one axis only)	0	−90 to 90 degrees	Where the rack faces (above).
Night from sunset plus (CONTROL)	30 min	0 to 180 min	This long after sunset, night begins.
Day from sunrise minus (CONTROL)	0 min	0 to 180 min	This long before sunrise, night ends.
Return east at night (CONTROL, two axes only)	yes	yes or no	After the move to flat, turn to the morning position. The same as the NIGHT TO THE EAST button.
When cloudy, move to calculated angle (CONTROL, one axis only)	no	yes or no	See cloud cover.
Cloudy below brightness (sum) (CONTROL, one axis only)	20 000	0 to 130 000	Threshold for “cloudy” (sum of the four sensors).
Night below / day above brightness (sum), **Night or day only after** (CONTROL, as before)	4 000 / 6 000, 5 min	Day threshold greater than night threshold	Night by brightness when time or location are missing. The waiting time also applies to “cloudy”.

All new settings apply immediately. Like the other settings, they can be changed remotely in the web interface under SETTINGS.

In the web interface

On the **OVERVIEW**, below the status bar, is the **SUN** card with the same daily arc and the same numbers as on the display (Fig. 88a, 88b); if time or location are missing, it says what is missing. Under **SETUP** are the **LOCATION** block and (one axis only) the **RACK ORIENTATION** block:

- **LOCATION:** a map to tap on (Leaflet with map images from OpenStreetMap), a search field with SEARCH and the fields latitude (north +) and longitude (east +), plus APPLY (Fig. 88c). The map images and the search are loaded from the internet by **your browser**, not by the device; without internet only the number fields work.
- **RACK ORIENTATION:** the same explanation as on the display, a compass rose, the buttons −5°, −1°, +1°, +5°, a number field and APPLY (Fig. 88d).
- Only a signed-in device may operate it, and only if remote control is not locked.

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What the web interface reveals

If you type a place into the search in the web interface, your browser sends this search text to the OpenStreetMap search service Nominatim. For the map images it asks the OpenStreetMap map servers for the area shown. Like every website, both see the public IP address of your internet connection. The map on the display sends nothing to the internet.



Fig. 88a – Web interface, one axis: at the bottom the SUN card with the best tilt angle and “Rack faces 15° to the west”.



Fig. 88b – Web interface, two axes: “At night: flat, then east if wanted (morning position)”.

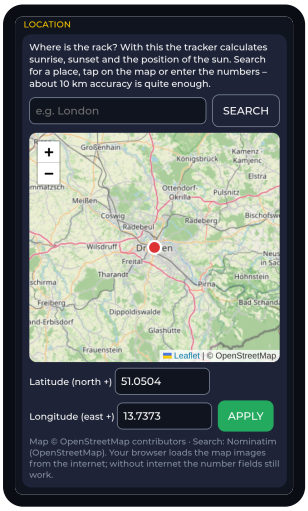


Fig. 88c – Phone, SETUP → LOCATION (detail): search, map with dot, latitude and longitude.

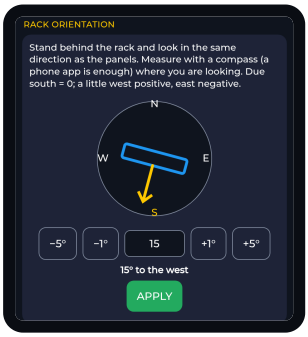


Fig. 88d – Phone, RACK ORIENTATION (detail, one axis only).

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🧩 **Open points and assumptions**

Not yet checked on the device: whether the board reliably reaches the time server (otherwise night stays detected by brightness), whether the map on the display zooms and pans smoothly, and how long the move to flat at night waits on a rack without limit switches. It is assumed that the 0 position of the night position is the same as the storm position.

Sounds from the loudspeaker (in preparation, heard on the test board)

The 10.1-inch board has a built-in loudspeaker. The firmware that has been started uses it for sounds. On the test board the sounds can be heard on 20.10.2026 (firmware 0.4.6); volume levels and alarm sounds have not yet been checked individually. What is written here describes what the firmware contains for this.

SOUND	WHEN	HOW OFTEN
Confirmation: a short, bright beep	every time you tap a button or a switch on the display	once per tap; can be switched off
Fault of an I/O chip: three short, high tones	the chip of the switching outputs on the RearCircuitBoard_10 does not respond	immediately, then every 20 seconds
Storm: three times high-low	the storm protection has laid the module flat	immediately, then every minute
Drive fault: two long tones	a drive has not reached its end position in time	immediately, then every 30 seconds
Anemometer failed: high, then low	the anemometer no longer responds, there is no storm protection	immediately, then every 2 minutes

The alarm sounds play as long as the condition persists. If several exist at the same time, you only hear the most important one (in the order of the table). An alarm sound always takes precedence over the confirmation beep. As long as the setup has not been completed, there are no alarm sounds.

Adjusting: In the setup, the **SOUND …** button leads to a separate screen. There you switch the confirmation sounds on or off and set the volume in ten levels with the − and + buttons.
Expected: After every change of volume, a short test tone sounds at the new level; the setting applies immediately and is kept after switching off. By default the confirmation sounds are on and the volume is set to level 7.

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⚠️ **Alarm sounds cannot be switched off**
So that a storm or a fault is not missed, alarm sounds play at least at level 7, even if you set the volume lower. Louder than level 10 is not possible: this limit is built in, so that the power supply and the fuse of the RearCircuitBoard_10 are not overloaded even with a full alarm sound and a charging battery. Whether the levels are clearly audible on the real device and how much current the loudspeaker really needs will be measured as soon as the board is available; until then the limit is an **assumption**.

Standby and motion detector (in preparation, not checked on the device)

The 10.1-inch board has a built-in camera. The firmware that has been started uses it as a motion detector: if nobody stands in front of the display for a while, it shows a screensaver or switches off the backlight. When someone comes to the display, it lights up again by itself. So far this has only been built and tested on the computer; whether the camera reliably detects motion on the device has not yet been checked.

ℹ️ **The camera does not record**
The firmware evaluates the camera images only in working memory to detect motion. It does not store any image, does not display any and does not send any over the network, not even to the web interface. Of each image, only the average brightness of small fields remains.

What happens:

- After the set **waiting time** without touch and without motion in front of the display, the screensaver appears (Fig. 13e): a dark area with the logo, the time, the status from the status bar in its colour and the wind. The text moves to a different place every 20 seconds, the light is dimmed. Or the display switches off completely, if you set it that way.
- **Waking up:** Motion in front of the display or a touch. The touch that wakes the display does not trigger anything: so you can tap anywhere without accidentally pressing a button. Only once you have lifted your finger does the next touch take effect.
- **In case of danger the display stays on:** In a storm, if the anemometer fails, in case of a drive fault or a fault of an I/O chip, the display lights up and stays on as long as the condition persists.
- The controller carries on working quite normally in standby, including the storm protection.
- In the dark (room without light) the camera does not detect any motion; then only a touch wakes it. If it suddenly becomes bright, for example because someone switches on the light or steps directly in front of the camera, the display also wakes up. Someone who stands completely still is no longer detected as motion after a few seconds.

Adjusting:

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1

Open the screen

Open the setup (gear wheel in the dashboard) and tap **SCREEN …** (Fig. 13d). *Expected:* The SCREEN screen appears.

2

Choose the type

Tap **ALWAYS ON**, **SCREENSAVER** or **DISPLAY OFF**. *Expected:* The selected button is blue with a yellow frame. With ALWAYS ON, the waiting time and the motion detector are irrelevant and locked.

3

Set the waiting time

With − and + you choose 1, 2, 3, 5, 10, 15, 20, 30, 45 or 60 minutes. *Expected:* The number in the middle changes.

4

Motion detector

Use the switch to turn the motion detector on or off. If it is off, only a touch wakes the display. *Expected:* Next to it it says "MOTION SENSOR ON" or "OFF", below it "Camera ready." If it says "Camera unavailable" in orange, the firmware has not found the camera; then only a touch wakes the display.

5

Check the motion detector

Below it, a line continuously shows what the camera sees: frames per second, brightness (0 to 255) and how many fields are currently moving, for example "5 fps, brightness 92, motion 412 (from 96)". Wave at a distance of about one metre in front of the display. *Expected:* The number after "motion" rises above 96, and the line turns yellow; this motion would then wake the display. If it says "too dark", the room is too dark for the camera; if it says "Camera idle", ALWAYS ON is selected or the motion detector is off. Write down the values in case waking does not work: they help with the adjustment.

6

Battery fitted

With the **BATTERY FITTED** switch you show or hide the battery level at the top right of the dashboard (Chapter 7). *Expected:* Next to it it says "BATTERY FITTED" or "NO BATTERY"; in the dashboard the indicator appears or disappears, and the gear wheel and AUTOMATIC shift accordingly.

7

Back

Tap **BACK**. *Expected:* The setup appears again. Everything applies immediately and is kept after switching off; a restart is not necessary.

By default, standby is set to **screensaver after 5 minutes** with the motion detector switched on (suggestion, not yet final). This cannot be set remotely (web interface, ioBroker).

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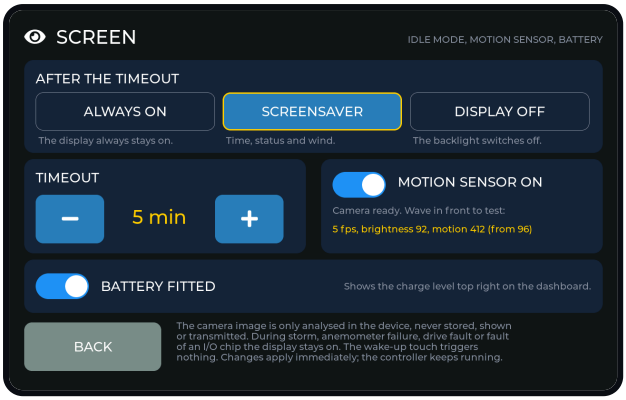


Fig. 13d – The SCREEN screen: type of standby, waiting time, motion detector and "Battery fitted".

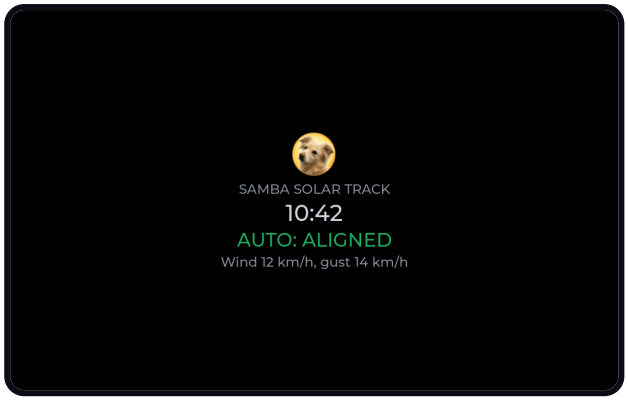


Fig. 13e – The screensaver: logo, time, status as in the status bar, and wind.

> 🧩 **Still open (to be checked on the device)**
> On the test board (21.10.2026) the screensaver appears, and a touch wakes the display; motion does not wake it yet. In addition, the device restarted immediately when new firmware was installed over the network; the next firmware therefore stops the camera before it writes. Flash this firmware once via USB (Flashing the firmware for the first time); after that it works over the network again. The next firmware also determines by itself in which part of the image data the brightness is stored, and shows the camera's measured values on the SCREEN screen (step "Check the motion detector"). Still open is whether the camera is found, how reliably it detects motion (distance, light, false alarms for example due to flickering light or animals), whether the display stays steady while the camera is running and whether the dimmed brightness of the screensaver is suitable. The detection thresholds are estimated (**assumption**) and will be adjusted after the test.

New firmware from the internet (in preparation, not checked on the device)

The firmware for the 10.1-inch board can check by itself whether there is newer firmware and install it from the internet at your request. This too has so far only been built and tested on the computer. What is written here describes what the firmware contains for this.

> 🧩 **First release, not yet checked on the device**
> For the device to accept new firmware, it must be signed with the project's key (signature). The current key has been included since firmware 0.4.7. Older firmware (up to 0.4.6) only knows an earlier key that is no longer used: a device with this firmware will never find a new version over the internet. In that case, install 0.4.7 once via the web interface (SETUP, firmware update) or via the USB cable as above; after that it continues over the internet. Whether checking and installation on the device work as described here will only be tried with versions 0.4.7 and 0.4.8. If the check reports an error (for example "no connection to release source"), the INSTALL button stays locked.

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⚠️ **Once via USB: changing to firmware 0.7.1**
Up to firmware 0.7.0 the board's flash was divided unfavourably: firmware larger than about 3.4 MB only fitted into one of the two places for the firmware. The update to 0.7.0 therefore usually fails with “Schreiben in den Flash gescheitert (Abschlussprüfung: ESP_ERR_OTA_VALIDATE_FAILED)” (writing to the flash failed, final check). Nothing is damaged; the previous firmware keeps working. Since firmware 0.7.1 the memory is divided anew. The new division only reaches the board over the USB cable: flash the factory image from 0.7.1 once, as described under Flashing the firmware for the first time. This deletes all settings: note down your values first (SETTINGS, location, Wi-Fi). Afterwards the setup appears again, followed by the setup program. All later versions come over the internet again.

What the device does by itself: When Wi-Fi is connected, it checks once a day (the first check 2 to 20 minutes after connecting). For this it downloads a small file, the manifest, over an encrypted connection (HTTPS) from the project's release source at `https://firmware.samba-solar.de/`. It never installs by itself. If there is a newer version, the header shows **NEW FIRMWARE** in small yellow letters at the top, with the number below it. There is no sound for this.

Viewing and installing:

1 **Open the screen**
Gear wheel at the top right, then the **FIRMWARE …** button in the setup. The button only exists once the setup has been saved. *Expected:* The FIRMWARE screen shows the running version with its build date, when the last check took place, and the result.

2 **Check now (optional)**
Button **CHECK NOW**. This is possible at most once per minute and only with Wi-Fi connected. *Expected:* After a few seconds it shows "UP TO DATE", "NEW FIRMWARE AVAILABLE" with number, date and short text, or "CHECK FAILED" with the reason.

3 **Install**
Button **INSTALL** (only with "NEW FIRMWARE AVAILABLE"), and confirm the prompt with INSTALL. *Expected:* "DOWNLOADING …" with a bar. The controller keeps regulating quite normally, including the storm protection. With **CANCEL** you can stop the download; then nothing changes.

4 **Flat position**
After downloading, the device checks the file. Then the elevation moves to the flat position, as when parking. *Expected:* The FIRMWARE screen shows "MOVING TO THE FLAT POSITION BEFORE WRITING", the status bar shows parking, and the module lies down flat.

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5

Writing and restart

Only when the elevation is flat does the device switch off all outputs and write the new firmware. *Expected:* Above the screen it says "FIRMWARE UPDATE – Verified firmware is being written"; then the controller restarts. The settings are kept.

6

After the restart

Do nothing and let the device run for at least two minutes. *Expected:* The new version is shown on the FIRMWARE screen, and the module stays flat for now. After two minutes the new firmware confirms itself, as soon as the display and the RearCircuitBoard_10 are working and – only if Wi-Fi is switched on – the device has connected to Wi-Fi at least once since the restart (the self-test). Then it cancels parking and moves to the previous position: the automatic tracks again, or a chosen angle is approached again.

ℹ️ **Flat first, then write**

Before every update, from the internet as well as via the web interface, the elevation moves to the flat position. This way the module lies flat if something goes wrong during the update. If the flat position is not reached, the device writes nothing and the current firmware keeps running; the reason is shown on the FIRMWARE screen and in the web interface. This happens in manual mode (switch on AUTOMATIC first), on a fault, when a storm begins, when someone cancels parking, or when the movement takes longer than the maximum run time plus about half a minute. If there was no park request before, the previous position applies again afterwards. If the device returns to the previous firmware after a failed self-test, that firmware also moves back to the previous position (**assumption**: not yet tried on the device). If you set or cancel parking yourself before that happens, your choice applies.

⚠️ **There is no storm protection while writing**

Writing takes an estimated one minute or so (**assumption**, not yet measured). During this time all drives are stopped, and the module is not laid flat in a storm. That is why the installation does not begin while the storm protection is active or a drive is moving. If a storm comes during the download or between download and writing, the device aborts and discards the downloaded file, even if the storm is quickly over again; you then have to confirm the installation again. It is best to install in calm weather.

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If the new firmware does not start properly

If the new firmware hangs or restarts within the first two minutes, the device automatically starts the previous firmware again. The same happens if it does not pass the self-test in time (for example because the display or the RearCircuitBoard_10 does not respond or because it never connects to Wi-Fi): it then first switches off all outputs and starts the previous firmware. The deadline is 12 minutes after the restart; if Wi-Fi is switched on, 30 minutes, so that a router that is briefly down does not immediately lead to a rollback. If Wi-Fi is switched off, it plays no role in the self-test. Both deadlines are suggested values, not yet tried on the device. As long as the new firmware is waiting for the self-test, an orange band is shown at the top between the product name and AUTOMATIC across all screens: the time until the check, then "FAILED", the state of the conditions (OUTPUTS and WI-FI: OK, MISSING or OFF) and the time until the rollback. Without the RearCircuitBoard_10 board it says "NO BOARD: SWITCH ON SIMULATION" (Fig. 13c). Why the self-test fails is also written in the log. After a rollback, the display reports "UPDATE ROLLED BACK"; tap OK.

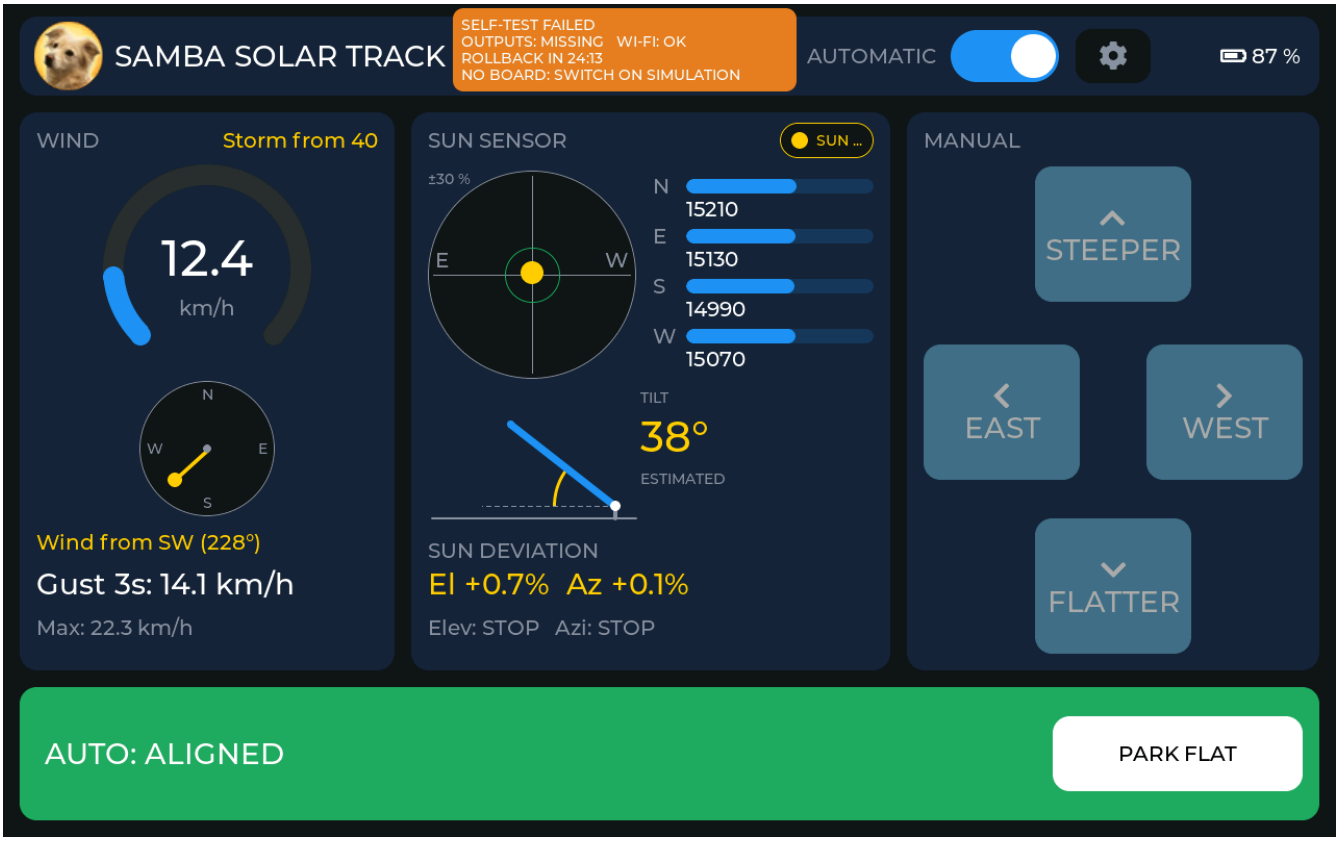


Fig. 13c – The self-test band: the switching outputs are missing (no RearCircuitBoard_10, no simulation), Wi-Fi was connected, rollback in 24:13.

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Fig. 13f – At the top right the battery level, here almost empty (8 %, red).



Fig. 13g – The battery is charging: yellow lightning bolt in front of the battery symbol (46 %).

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ℹ️ The same version is then no longer offered in the header, but can still be installed again on the FIRMWARE screen. On the 10.1-inch board this also applies to an update via the web interface.

Remotely: In the web interface, under SETUP, there is the block "Firmware from the internet" with the same information and the buttons CHECK NOW and INSTALL. Only a logged-in device (token) may operate it, and the "Remote control locked" switch on the display locks both. The daily check continues even when remote control is locked.

ℹ️ **What the device reveals in the process**
When checking, the device only sends a simple request with its firmware version; no identifier, no password, no token. Like any website, the project's site sees the public IP address of your internet connection. There is no switch to turn off the check; without Wi-Fi it does not take place.

Tilt on command: moving to a fixed angle (in preparation, not checked on the device)

Instead of tracking the sun, the elevation can move to a fixed angle on your command, for example a flat one in summer or a steep one in winter. After that, **the automatic pauses** until you cancel the request. This exists only in the 10-inch version and only in the *VERTICAL ONLY* operating mode. With two axes (vertical + horizontal), the TILT card, desired angles, travel times and frame are hidden. This function too has so far only been built and tested on the computer; the pictures were generated by the screenshot tool from the real code of the user interface.

⚠️ **Without Hall tap the position is estimated, not measured**
By default there is no sensor for the tilt. The firmware calculates the tilt from the time during which it actually switched the drive on and from the calibrated travel times. That is why the display always shows "~30°" or "UNKNOWN", never an exact number. How accurate this is is described further below; all information on this is **assumptions** until it has been measured on the rack. It gets more accurate with the Hall tap: the firmware then counts the pulses of the drive and shows the tilt without "~".

Which angles there are: always **0° (flat)**, **30°** and **60° (steep)**, plus up to two preset angles that you set under SETTINGS (whole degrees from 1 to 59, not 30, not the same one twice; 0 means off). The buttons are always arranged by angle. As long as the travel time has not been calibrated, only 0° and 60° are possible; the other angles are greyed out.

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Fig. 14 – The *TILT* card is in the dashboard between *SUN SENSOR* and *MANUAL*. "--°" and "UNKNOWN": since switching on, the firmware does not yet know the position. Without a calibrated travel time only 0° and 60° can be selected, 30° is greyed out.

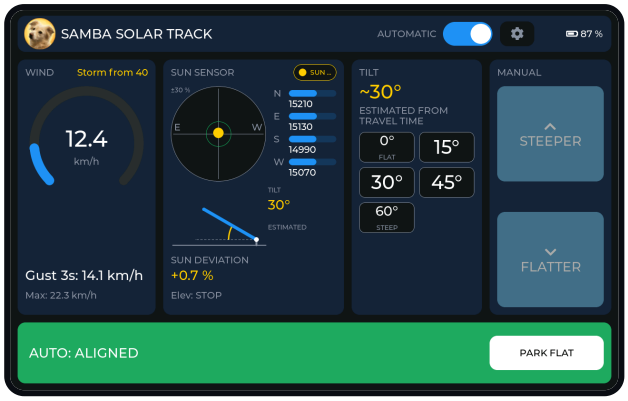


Fig. 15 – Calibrated and with the preset angles 15° and 45°: all five angles can be selected.

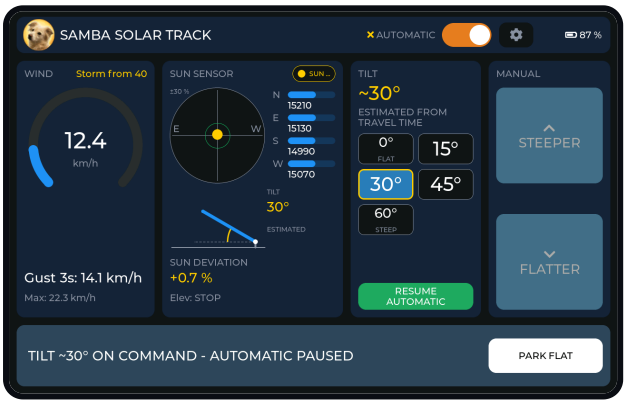


Fig. 16 – 30° has been commanded: the active angle is blue with a yellow border; *RESUME AUTOMATIC* cancels it.

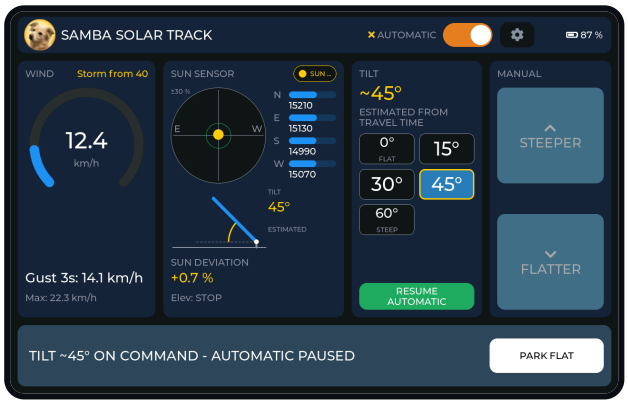


Fig. 17 – Reached: "TILT ~45° ON COMMAND - AUTOMATIC PAUSED". In the card, 45° is marked.

DISPLAY IN THE TILT CARD	MEANING
--°, below it **UNKNOWN - FIRST MOVE VIA FLAT**	The position has not been known since switching on, or the estimate is too uncertain (more than 6°).
~31°, below it **ESTIMATED FROM TRAVEL TIME**	Estimated tilt. Since firmware 0.6.0 the angle is shown large at the top of the card; up to 0.5.9 it was small in one line ("CURRENT APPROX. 31° ...").
30°, below it **MEASURED (HALL TAP)**	Tilt counted with the Hall tap, without "~" (With Hall tap).
Angle button blue with yellow border	This tilt on command is in effect, even during storm, parking or manual mode. Below it is the *RESUME AUTOMATIC* button.
AUTOMATIC switch orange, with a yellow cross in front	The automatic is switched on but paused, because a tilt on command is in effect. With *RESUME AUTOMATIC* the switch turns blue again and the cross disappears.
NOT AVAILABLE	Without a completed setup there is no tilt on command; the buttons are locked.

Moving to an angle:

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1 Read the position

Look at the **TILT** card in the dashboard (Fig. 14 or 15). *Expected:* At the top the angle is shown large, for example “~31°” with “ESTIMATED FROM TRAVEL TIME” below it, or “--°” with “UNKNOWN - FIRST MOVE VIA FLAT”.

2 Choose the angle

Tap the desired angle in the card. *Expected:* The button turns blue with a yellow border, *RESUME AUTOMATIC* appears below it, the AUTOMATIC switch at the top turns orange with a yellow cross in front of it, and the status bar shows the sequence (next step).

3 Watch

If the position is unknown or too uncertain, the elevation first moves flat to the end position, the reference move. Without limit switches it takes the full maximum run time (90 seconds on the Samba Solar Rack 2.0-E Plus), even if the frame has long since stopped. Then follows the move to the angle. *Expected:* one after the other "TILT 45°: REFERENCE MOVE TO FLAT POSITION", "TILT 45°: MOVING STEEPER, ~12°" and finally "TILT ~45° ON COMMAND - AUTOMATIC PAUSED" (Fig. 17). There is no readjustment afterwards.

4 Back to automatic

Tap **RESUME AUTOMATIC** in the card. *Expected:* The status bar shows AUTO again, the AUTOMATIC switch is blue again without a cross, and tracking starts again.



Fig. 18 – The reference move: the elevation first moves flat to the end position. Whether “flat” means UP or DOWN on the drive is determined by the setting “Flat position is in UP direction”.



Fig. 19 – The move to the angle, with the running estimate "~12°".

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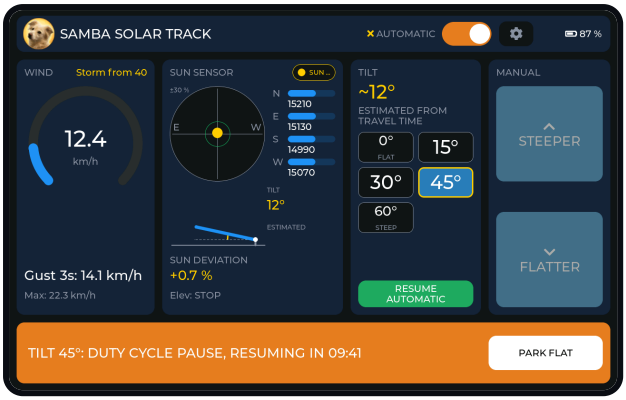


Fig. 20 – The drives need a pause (duty cycle, below). After that it continues by itself.



Fig. 21 – 30° has been requested, but the travel time has not been calibrated: nothing moves. Calibrate first (below).

What has priority (the controller's order of priority): missing setup, fault of an I/O chip, storm, drive fault, manual mode, parking on command and anemometer failure come first. Only after that comes the tilt on command, and it in turn takes precedence over sensor faults, a missing setup program, night, cloud cover and automatic. So the tilt on command also applies at night and when the sun sensor is faulty.

SITUATION	WHAT HAPPENS
Storm, anemometer fails	Take precedence: the elevation moves flat. Afterwards it moves back to the commanded angle by itself. The card says "STORM PROTECTION HAS PRIORITY - ANGLE IS APPROACHED AFTERWARDS" (Fig. 22).
PARK FLAT	Remains a separate button and takes precedence ("PARKING HAS PRIORITY ..."). After parking is cancelled, the system moves back to the angle.
Manual mode (automatic off)	The tilt on command is suspended; it takes effect as soon as the automatic is on again ("TAKES EFFECT ONCE AUTOMATIC IS ON").
Fault	Takes precedence ("FAULT HAS PRIORITY"), nothing moves.
Power failure, restart	The commanded tilt remains stored, like the parking request. After switching on, however, the position is unknown: the first move to an intermediate angle begins with a reference move.

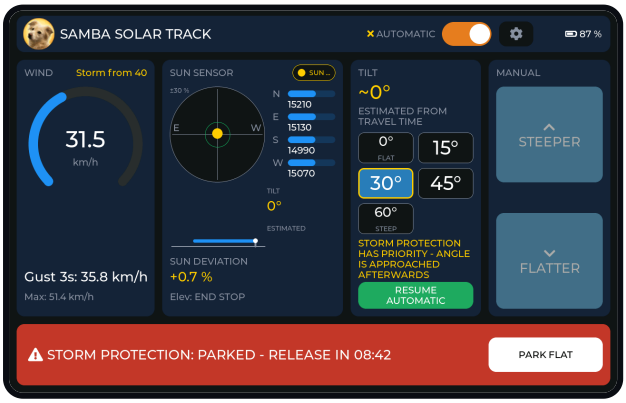


Fig. 22 – Storm while a 30° tilt is in effect: the module lies flat, 30° stays marked, and the card shows the note in yellow lettering.

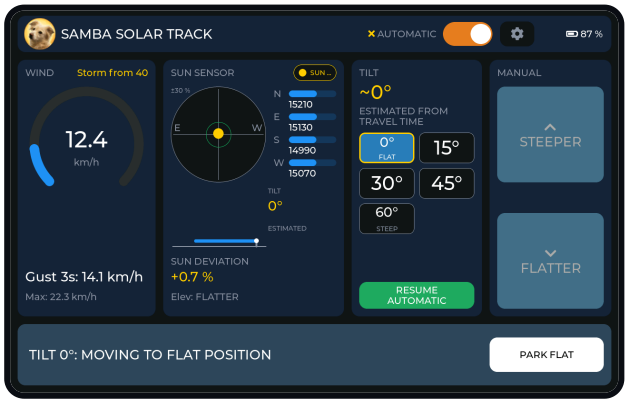


Fig. 23 – 0° selected: the elevation moves to the flat position.

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In the browser: For the 10-inch version, the web interface shows the same layout as the display (Fig. 23a and 23b): on a wide screen (PC, tablet, phone in landscape) the header, the cards WIND (with compass rose if an anemometer with direction is set up), SUN SENSOR, TILT and MANUAL in the same widths and the status bar below; on the phone one below the other. The TILT card has the same buttons, the angle large (“approx. 31°”) and below it where it comes from (“estimated from travel time”); the SUN SENSOR card shows, as on the display, “±30 %”, the sketch of the tilt and the SUN DEVIATION. As on the display, the AUTOMATIC switch is orange with a yellow cross in front of it while a commanded tilt pauses the automatic, and the battery with its charge level and, while charging, the yellow lightning bolt is shown at the top right; on a phone the battery appears small below the name (Fig. 23c). Warnings about the supply and the charge lock are shown by the page as yellow bands below the status bar (see below). Under SETUP you also choose the language: tapping changes it on the display and in the browser at the same time. Only a logged-in device may operate it, and only if remote control is not locked. Calibration is only available on the display. The pictures come from the preview tool of the web interface (`python3 tools/web_vorschau.py`) with a simulated device, not from the real tracker.



Fig. 23a – Web interface, vertical only: four cards as on the display, tilt 30° active.



Fig. 23b – Web interface, two axes: three cards, compass rose in the WIND card.

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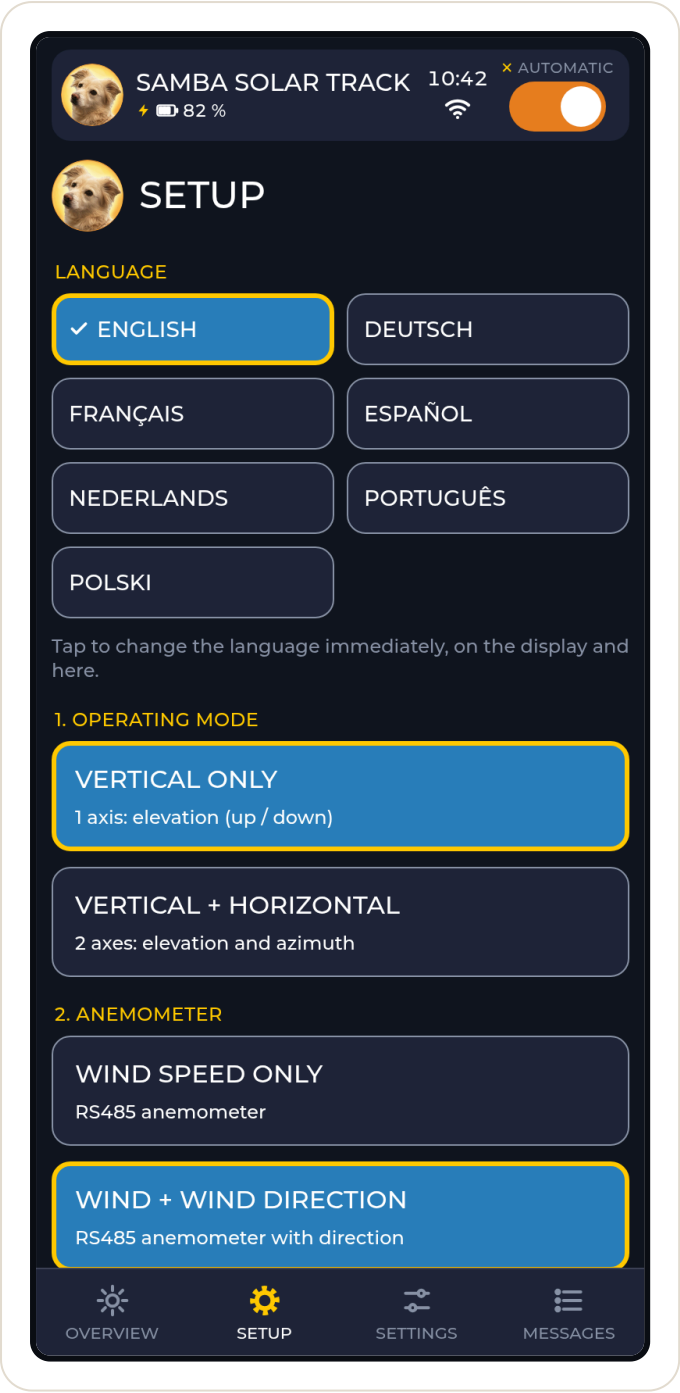


Fig. 23c – Web interface on a phone, SETUP: the language selection at the top; in the header the battery while charging and the AUTOMATIC switch showing the pause.

Setting preset angles, travel times and frame

The values are in the setup under **SETTINGS**: the two preset angles in the CONTROL group, the travel times, the frame, the start-up loss and the three values of the Hall tap in the COMMISSIONING group. The values of the COMMISSIONING group can only be changed while no drive is moving.

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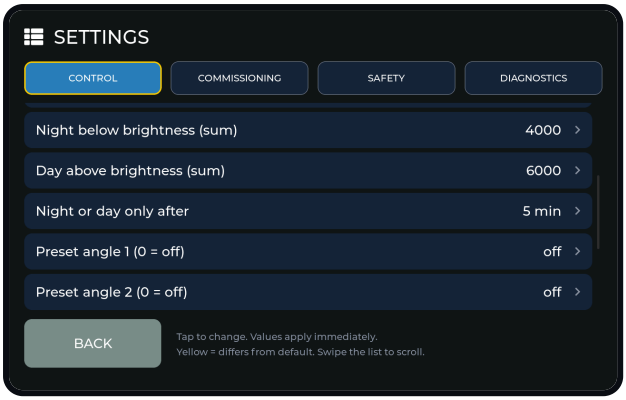


Fig. 24 – CONTROL group: preset angles 1 and 2, off by default.

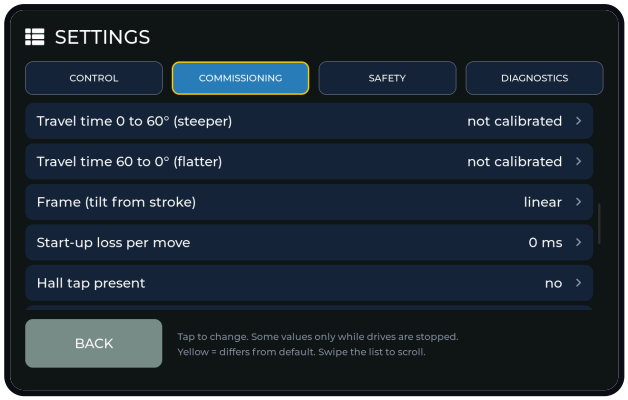


Fig. 25 – COMMISSIONING group: travel times, frame, start-up loss and "Hall tap present". Further down follow the other values of the Hall tap and the *CALIBRATE TRAVEL TIME* button (Fig. 34).

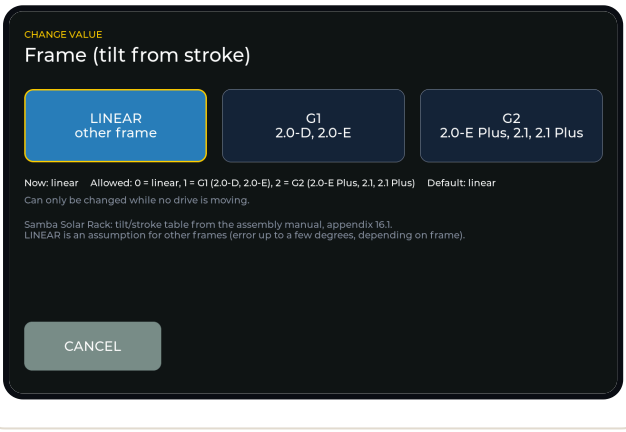


Fig. 26 – The frame: G1 or G2 for a Samba Solar Rack, LINEAR for other frames.

VALUE	DEFAULT	PERMITTED	PURPOSE
Travel time 0 to 60° (steeper)	not calibrated	10 to 300 s	Time for the whole travel from flat to steep. Best calibrated.
Travel time 60 to 0° (flatter)	not calibrated	10 to 300 s	Time for the travel back. Best calibrated.
Frame (tilt from stroke)	linear	linear, G1 (2.0-D, 2.0-E), G2 (2.0-E Plus, 2.1, 2.1 Plus)	Converts the travel of the drive into degrees (table from the assembly manual, Appendix 16.1).
Start-up loss per move	0 ms	0 to 3000 ms	Part of every move during which the frame is not yet moving.
Preset angles 1 and 2	off	1 to 59°, not 30, not twice; 0 = off	Additional buttons in the TILT card. Applies immediately.

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ℹ️ **Good to know**

The travel times only apply if they are shorter than 0.9 × the maximum run time of the elevation. If you set the maximum run time shorter, they count as not calibrated until the values fit again. "LINEAR" is an **assumption** for other frames; on a G2 frame it would be off by up to about 2.6°. If you change a preset angle that is currently commanded, or switch it off, the tilt on command remains in effect; *RESUME AUTOMATIC* cancels it. For the Samba Solar Rack 2.0-E Plus (sync controller BST-KZQ-15A-2, no limit switches on the tracker) the following also applies: "Elevation limit switch fitted" = no, max. run time elevation 90 s. Entered by hand, it would be about 75 s for the 2.0-E Plus according to the assembly manual; calibrating is more accurate.

Calibrating the travel time

Once during commissioning and, if possible, again in autumn and spring: temperature changes the speed of the drives (**assumption**, not measured). Location: **Setup → SETTINGS → COMMISSIONING → CALIBRATE TRAVEL TIME** (bottom line; scroll the list down for it, Fig. 34; with Hall tap the button is called CALIBRATE: TRAVEL TIME AND HALL STEPS). Control and storm protection keep running in the meantime.

1

Prepare

Clear the swivel range: nobody is standing at the frame, nothing is in the way. Open the CALIBRATE TRAVEL TIME screen. If the automatic is ON, tap **AUTOMATIC OFF**. *Expected:* "STEP 1 OF 4: PREPARE" (Fig. 27), after which **START** can be selected.

2

START

Tap **START**. *Expected:* "MOVING FLAT TO END POSITION" (Fig. 28). Without limit switches the contact stays closed for the full maximum run time, even if the frame has long since stopped. If the duty cycle does not quite allow this, it says "WAITING FOR DUTY CYCLE" with the remaining time.

3

MEASURE UP

Tap **MEASURE UP**. The time runs (Fig. 29). Tap **ARRIVED** as soon as the frame stops and the motor noise stops (the drive is then at its own limit switch). *Expected:* "UP measured" with the time to a tenth of a second. With limit switches on the tracker and with Hall tap, the measurement ends by itself.

4

Pause

A pause of about 15 minutes is recommended so that the drives cool down (duty cycle; Fig. 30). The pause is not enforced. *Expected:* The screen shows the recommendation and the **MEASURE DOWN** button.

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5 MEASURE DOWN

Tap **MEASURE DOWN** and again **ARRIVED** as soon as the frame is flat. *Expected:* The second time appears. If the drive stops in the middle of the measurement because of a duty cycle pause, the screen shows "DUTY CYCLE PAUSE ... NOT arrived" (Fig. 31): then do **not** tap; the measurement continues by itself afterwards.

6 Apply

The screen shows both times (Fig. 32). Tap **APPLY**. *Expected:* Under SETTINGS the times are shown instead of "not calibrated"; in the TILT card all angles can be selected.

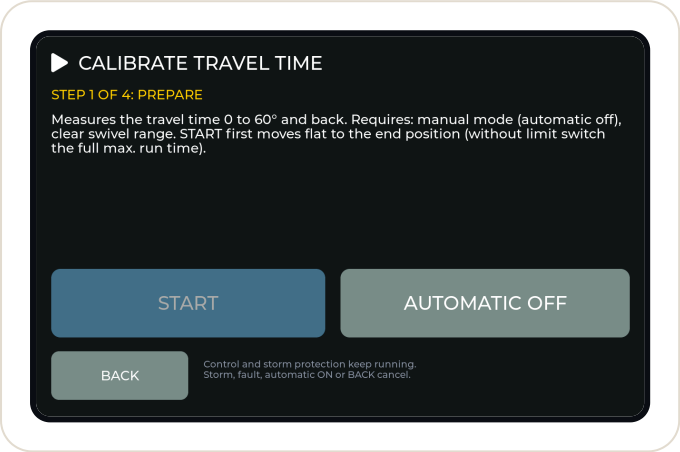


Fig. 27 – Step 1. The automatic is still on, so START is locked.

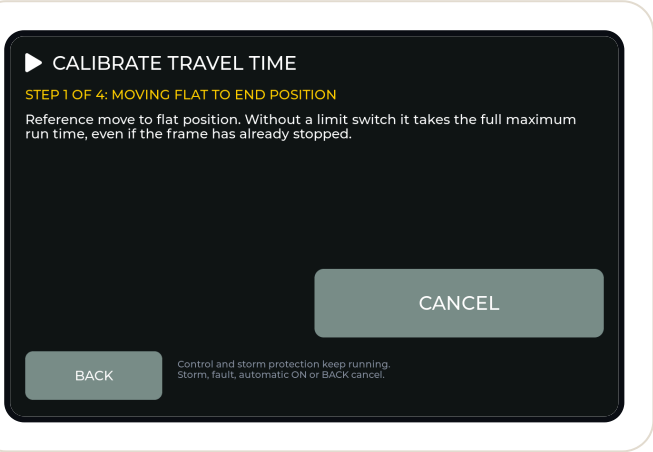


Fig. 28 – The reference move to the flat position.

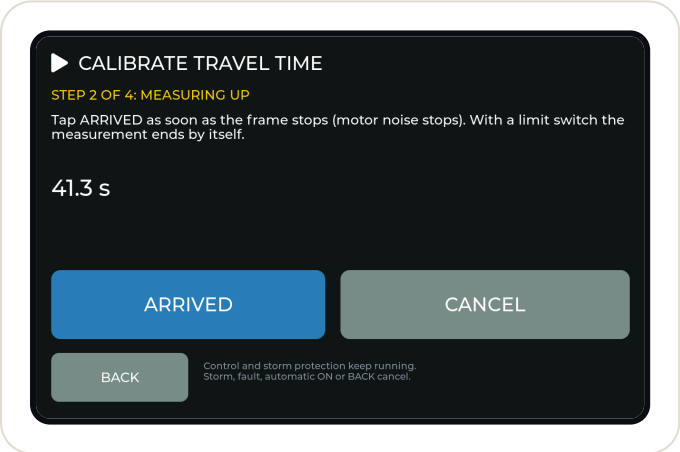


Fig. 29 – The move to steep is measured. Tap ARRIVED as soon as the frame stops.

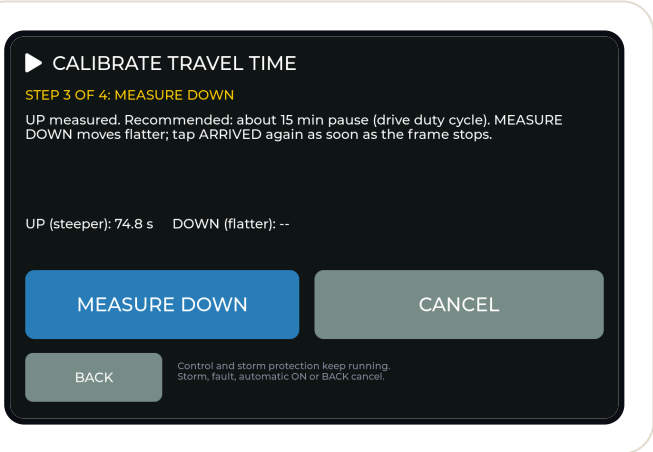


Fig. 30 – UP has been measured; a pause of about 15 minutes is recommended.

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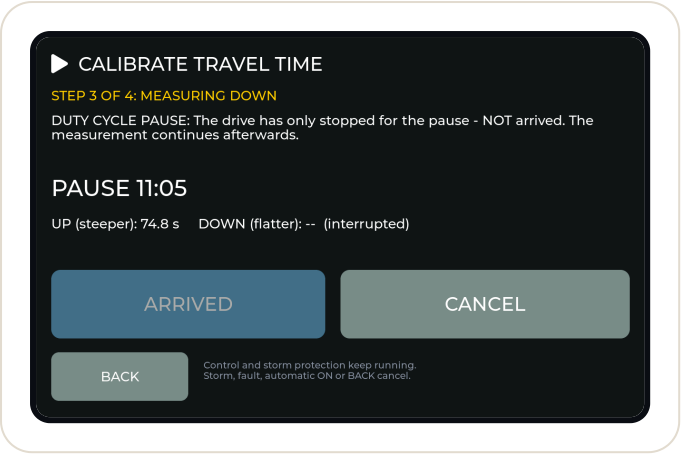


Fig. 31 – The drive has only stopped because of the duty cycle: do not tap ARRIVED.

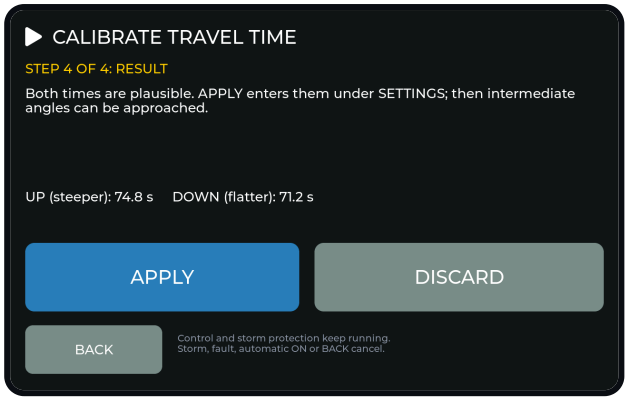


Fig. 32 – Both times are plausible. APPLY enters them under SETTINGS.

The firmware checks the measurement: each time must last at least 10 seconds and be shorter than 0.9 × the maximum run time, and UP and DOWN may differ by at most a factor of 2. Otherwise the reason appears in yellow lettering, and nothing is applied (Fig. 33). Storm, a fault, automatic ON or BACK abort the calibration. While you are calibrating, the device rejects manual moves from remote.

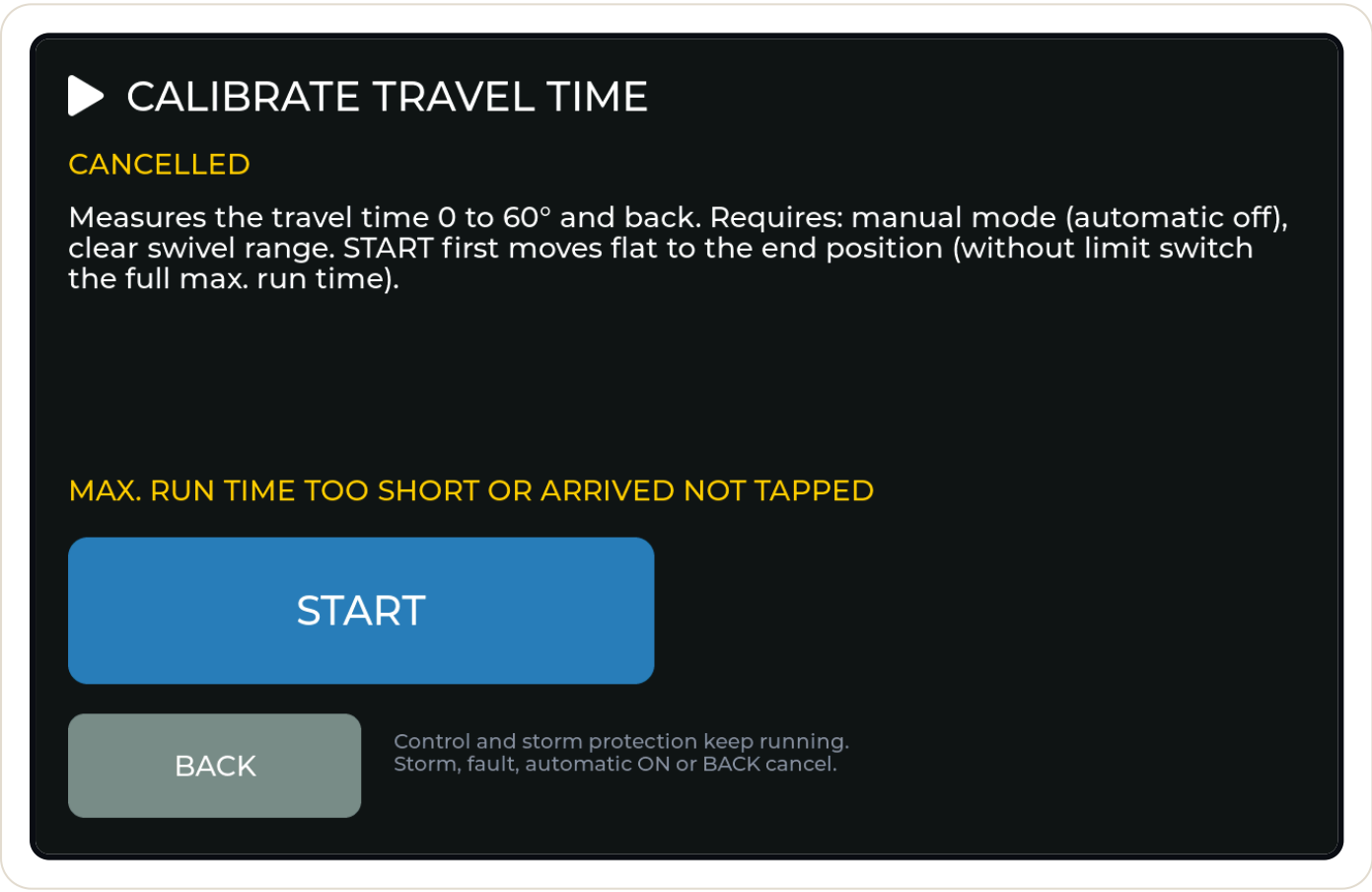


Fig. 33 – Aborted. The reason is shown in yellow lettering; START begins the calibration again.

How accurate is it?

All figures in this table are **assumptions**, not measured. They will be checked on the rack.

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SOURCE OF DEVIATION	ORDER OF MAGNITUDE
End positions of the mechanics	0° ± 1°, 60° ± 1.5° (assembly manual)
Table of frame G1/G2	below 0.1°
Speed after calibration (load, voltage)	about ± 3 to 5 % of the travel, ± 1 to 1.5° at 30°
Temperature (winter versus calibration in summer)	up to about 3 to 6° at 30°
Start-up and overrun per move	up to just under 1° per move

Expectation: a move from a reference ± 1 to ± 3°, in winter with a calibration from summer up to about ± 5°. At standstill the estimate does not change. After a day of automatic with many short tracking moves it is usually "UNKNOWN"; the first move to an intermediate angle then begins with a reference move. With every move the uncertainty grows; if it is at most 3°, the elevation moves directly, otherwise flat first. Fixed values for this (0.3° per move, plus 8 % of the degrees travelled, tolerance 1°, shortest move 0.5 s, 2° surcharge after a reset in the middle of a move) are built into the firmware and are also **assumptions**.

⛔ **Without Hall tap the firmware does not detect movement by hand**
If you move the module at the control box (rocker switch, radio remote control, buttons of the sync controller) or with the hand crank, the estimate is no longer correct. Afterwards, select **0°** or **PARK FLAT** once, so that the firmware has a reference again. With the Hall tap switched on, it counts movements via the sync controller, as long as its pulses arrive.

⚠️ **A run-time end position can be wrong**
Without limit switches on the tracker, an end position is considered reached when the contact was closed for the whole maximum run time (run-time end position). If the drives had no voltage at that time (the power supply of the drives was off, the fuse of the drives had tripped, or in the variant with a shared power supply that one was off and the board ran on the battery) or if the sync controller (for example BST-KZQ-15A-2) aborted the move without feedback, this reference is not correct, and neither is the estimate afterwards. Without Hall tap the firmware cannot detect this; it therefore never shows more than “CA.”. With Hall tap it detects it by the missing pulses.

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🧩 **Still open or to be checked on the device**

After a restart without loss of power (setup saved, update, crash), the firmware takes over the estimate from a memory that survives the restart; if a drive was moving at the moment of the reset, the estimate becomes 2° less certain. Whether this memory holds on the board is an **assumption**. If the drives have no power during a move (power supply off and board running on the battery, fuse of the drives tripped) or the sync controller aborts the move, the contacts still close; after the max. run time the firmware then remembers a wrong run-time end position and a wrong tilt. That is why it moves flat again during a storm, an anemometer failure and parking (see below). Whether the drive really moves is detected by the Hall tap, if it is fitted and switched on (also only checked on the computer). In the simulation the function runs with simulated drives; their position is never carried over into real operation.

Serial Monitor – example, not from the real hardware

```
[  120.010][I][main  ] Neigung auf Befehl: 45 Grad (vorher keine), Quelle Display
[  120.010][I][track ] Neigung 45 Grad: Bezugsfahrt noetig
[  210.020][I][elpos ] Bezug Flachstellung (Laufzeit-Endlage), Stellung war unbekannt
[  210.540][I][track ] Neigung 45 Grad: Zielfahrt STEILER von ca. 0.0 Grad, erwartet 55.1 s
[  266.160][I][track ] Neigung 45 Grad erreicht: ca. 45.0 Grad nach 55.6 s, Unsicherheit 3.9 Grad
```

The lines are an example in the firmware's format; the numbers are calculated for the Rack 2.0-E Plus with frame G2 and 75 seconds of travel time. The status line every 5 seconds then ends with „| Neigung ca. 45 Grad (+-4), Ziel 45“ (tilt approx. 45°, target 45).

With Hall tap (more accurate)

What is it? The elevation drive has a Hall sensor: with every revolution of its motor it outputs short pulses. The small HallTapBoard on the rack picks up these pulses at the sync controller (the Hall tap) and passes them on via optocouplers, galvanically isolated, to the pulse input J22 of the controller (Chapter 6). The controller counts them with its pulse counter and thus knows how far the drive has moved. It no longer has to estimate the tilt from the travel time. This exists only in the 10-inch version and only in the operating mode *VERTICAL ONLY*. Only the pulses of drive 1 are counted.

🧩 **In preparation, not tested on the device**

The evaluation has only been built and tested on the computer. The HallTapBoard is a draft (0.6) and has not been manufactured yet; whether the BST-KZQ-15A-2 sync controller tolerates the tap has not been clarified yet (Chapter 6). The screenshot tool generated the images from the real code of the user interface.

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ℹ️ **The Hall tap is off by default**
Without it, tilt on command works as described above: estimated from the travel time, with “APPROX.” in front of the number. You do not need the Hall tap to use tilt on command.

Switching on – step by step:

1 **Connect**
Install and connect the HallTapBoard (Chapter 6) and plug its cable into J22 of the RearCircuitBoard_10 board (pulse input HALL). *Expected:* Nothing is visible on the display; on the HallTapBoard “24V” (yellow) and “5V OK” (green) light up.

2 **Switch on**
On the display: gear → SETTINGS → COMMISSIONING, swipe the list all the way down, tap “Hall tap present” and select YES (Fig. 34). This is only possible while no drive is moving. *Expected:* The line shows “yes” in yellow, because the value differs from the default. The bottom button is now called **CALIBRATE: TRAVEL TIME AND HALL STEPS**.

3 **Calibrate**
Automatic off (manual mode), then open **CALIBRATE: TRAVEL TIME AND HALL STEPS**. The screen is still called CALIBRATE TRAVEL TIME. Proceed as for calibrating the travel time above: START, MEASURE UP, pause, MEASURE DOWN. The measuring moves end **by themselves** as soon as the frame stops (Fig. 35). You only need ARRIVED if the frame has visibly stopped and the time keeps running anyway. *Expected:* The result shows a time and a number of Hall steps for UP and for DOWN each, the two numbers roughly the same (Fig. 36).

4 **Apply**
Tap **APPLY**. *Expected:* Under SETTINGS, “Hall steps per stroke” shows a number instead of “not calibrated” (the average of UP and DOWN), and the travel times show the measured times.

5 **If it does not work**
If “HALL COUNTS BACKWARDS - INVERT COUNT DIRECTION (SETTINGS)” appears during calibration, or “HALL TAP FAULT” in the header: SETTINGS → “Invert Hall count direction” → YES, then calibrate again. *Expected:* The message no longer appears at the next calibration. If it says “TOO FEW HALL STEPS - CHECK HALL TAP”, hardly any pulses are arriving: check the cable at J22 and the HallTapBoard (both lamps must be lit, Chapter 6).

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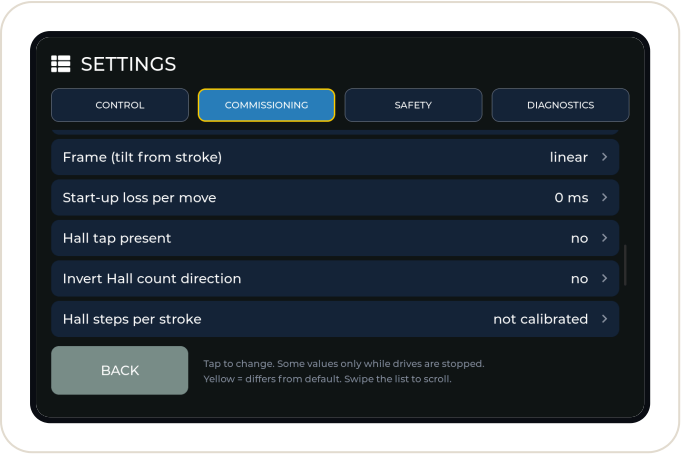


Fig. 34 – End of the COMMISSIONING list: “Hall tap present”, “Invert Hall count direction” and “Hall steps per stroke”, below them the button for calibration. Still at the default here: Hall tap “no”, so the button is called CALIBRATE TRAVEL TIME.

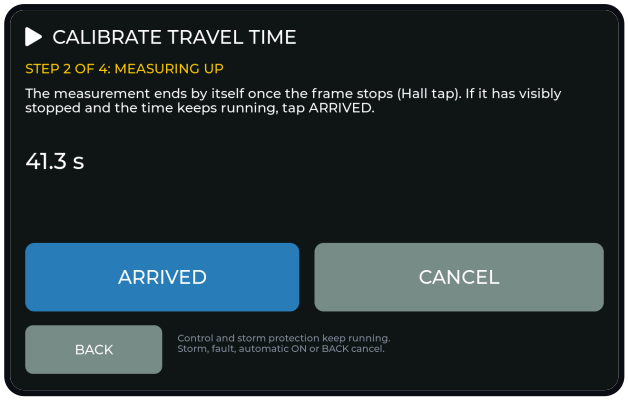


Fig. 35 – With the Hall tap, the measurement ends by itself as soon as the frame stops.

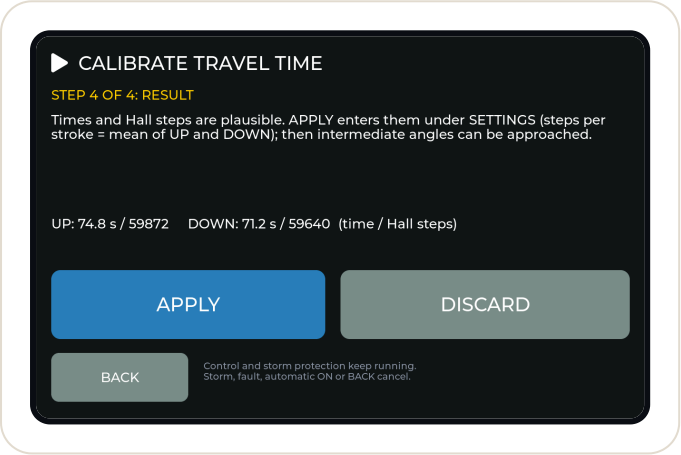


Fig. 36 – Result: time and Hall steps for each direction. APPLY enters the times and the average of the steps.

The firmware checks the result: the times must pass the check described above, UP and DOWN must each count at least 1,000 Hall steps and may differ by at most 3 % (suggested value). Otherwise the reason is shown in yellow text and nothing is applied.

In operation:

- The **TILT** card shows the counted tilt **without “~”**, for example large “30°” and below it “MEASURED (HALL TAP)” (Fig. 37); the sketch below the circle says “MEASURED”. How accurate this is will be measured on the rack; 0.2° is assumed (**assumption**).
- After switching on, the tilt is unknown until the elevation has reached an end position once. The first move to an intermediate angle therefore begins with a move to the flat position (reference move). With the Hall tap it ends as soon as the drive stops, not only after the max. run time (standstill end position). After that, the elevation moves directly to the angle and switches off earlier by the learned overrun.
- If the drive stops in the middle of its travel (for example because the drives have no power or the sync controller has aborted), the controller ends the move; the log reports „Antrieb steht ohne Endlage“ (drive stopped without end position). The tilt remains known.
- If you move the module at the control box (buttons of the sync controller), the controller counts along; the tilt stays correct.

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- During a storm, an anemometer failure and PARK FLAT, the elevation moves all the way to the flat position as always. The Hall tap never aborts this move in the middle of its travel. If the elevation is verifiably already flat, the move ends after about 1.5 seconds; if the tilt is unknown, it takes the full max. run time, just as without the Hall tap.
- As long as “Hall steps per stroke” is not calibrated, the display keeps showing the estimate with “APPROX.”; ending a move on standstill, however, already works.



Fig. 37 – With the Hall tap, the card shows the counted tilt without “~”: “30°”, below it “MEASURED (HALL TAP)”.

> ⚠️ **Warning “HALL TAP FAULT / TILT UNKNOWN” in the header**
>
> The controller is not counting matching pulses: none arrive although the drive should be moving, they count against the direction of travel, or far too many arrive (interference pulses). Check the cable at J22, the HallTapBoard (lamps “24V” and “5V OK”), the power supply of the drives and the count direction. As long as the warning is shown, only 0° and 60° can be selected; intermediate angles show “HALL TAP FAULT - NO MOVE” (Fig. 38). Storm protection, parking, automatic and manual mode continue to work as without the Hall tap. As soon as a move delivers matching pulses again, the warning disappears by itself; the tilt is known again from the next end position. The same warning appears in the web interface.



Fig. 38 – Hall tap fault: warning in the header, 45° is requested, nothing moves. In the card only 0° and 60° can be selected.

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VALUE (SETTINGS → COMMISSIONING)	DEFAULT	PERMITTED	PURPOSE
Hall tap present	no	yes or no	Switches on the evaluation of J22. Only switch it on if the HallTapBoard is connected.
Invert Hall count direction	no	yes or no	In case the two channels A and B arrive the other way round; the controller then counts backwards.
Hall steps per stroke	not calibrated	1,000 to 2,000,000; 0 = not calibrated	How many Hall steps the whole travel from 0° to 60° has. Entered by the calibration.

All three values can only be changed while no drive is moving.

Serial Monitor – example, not from the real hardware

```
[  120.010][I][main  ] Neigung auf Befehl: 30 Grad (vorher keine), Quelle Display
[  120.010][I][track ] Neigung 30 Grad: Bezugsfahrt noetig
[  168.420][I][motor ] Elevation: Endlage PLUS am Stillstand erkannt (Hall-Abgriff) nach 48410 ms
[  168.420][I][main  ] Ausgang Elevation: RAUF -> STOP (Hall-Abgriff: Endlage am Stillstand)
[  168.420][I][hall  ] Bezug Flachstellung (Stillstand, Stellung war unbekannt (Annahme: eigener Endschalter des
Antriebs)), Stellung war unbekannt
```

The lines are an example in the firmware's format. With the Hall tap, the status line every 5 seconds ends with „| Neigung 30.2 Grad (Hall), Ziel 30“ (tilt 30.2°, target 30).

🧩 **Assumptions (not yet checked on the device)**
Start-up time 1.5 seconds and standstill time 1 second; accuracy 0.2°; interference filter 5 µs at no more than about 1,000 pulses per second and channel; the count direction of the pulse counter; end position window 3 % of the travel; tolerance during calibration 3 %; that the BST-KZQ-15A-2 sync controller can share the Hall line of drive 1 with the tap; that the memory which survives a restart holds on the board. If the elevation stands still while the tilt is unknown, the firmware assumes that the drive is at its own limit switch.

Pauses for the drives: the duty cycle (in preparation, not tested on the device)

Electric linear actuators must not run continuously, otherwise they get too hot. For the drives of the Samba Solar Rack 2.0-E Plus, the dealer HAKIWO specifies a duty cycle of 10 % and at most 2 minutes in one go (assembly manual, chapters 3.6 and 14.4). The 10-inch firmware enforces this for each drive by itself. This, too, has only been built and tested on the computer.

1

Travel time account

The firmware keeps an account for each drive. Every move is deducted from it, every second of standstill refills it (with one ninth of the time). When full, the account allows 120 seconds of travel in one go. *Expected:* In normal operation with short tracking moves you will not notice any of this.

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2

Pause

When the account is empty, a pause begins. *Expected:* A move in progress stops. The status bar turns orange and shows “DUTY CYCLE PAUSE: DRIVE AVAILABLE AGAIN IN 13:30” (in manual mode, when “aligned” and during the rest time) or, with tilt on command, “TILT 45°: DUTY CYCLE PAUSE, RESUMING IN …” (Fig. 20). The remaining time counts down.

3

Available again

Once 90 seconds of travel time have been saved up, the drive is available again. After a completely emptied account this takes about 13.5 minutes. *Expected:* The status bar shows the normal state again; a waiting move (tilt, calibration) continues by itself.

💡 **Storm protection never waits**

Even during the pause, the elevation moves flat when a storm comes, the anemometer fails or you select **PARK FLAT**. These flat moves have priority. Everything else waits: manual moves, tracking and the move to an angle. The automatic ends a tracking move in progress when the pause begins and does not start a new one during the pause. The device accepts commands from the browser, but the drive waits for the end of the pause.

ℹ️ **Long moves to the end position wait for a full account**

Without limit switches, a move only reaches the end position if it runs the whole max. run time in one go. A reference move, the move to 0° or 60° and the reference move during calibration therefore only begin when the account allows a full move. Until then the display shows the waiting time. So do not change angles frequently.

🧩 **Assumption**

The 90 seconds until release are an **assumption** (one full move from 0° to 60° with a reserve) and will be checked on the rack. After a restart the account starts full again; the travel time before the restart is not counted.

Serial Monitor – example, not from the real hardware

```
[  930.100][W][motor ] Elevation: Einschaltdauer erreicht -> Pause, Fahrt wieder in 810 s (Flachfahrt bei Sturm,
Windmesser-Ausfall und Parken bleibt frei)
[  930.100][I][motor ] Elevation: Fahrt PLUS wegen Einschaltdauer beendet
[  930.100][I][main  ] Ausgang Elevation: RAUF -> STOP (Zustand MANUAL)
[ 1740.110][I][motor ] Elevation: Pause der Einschaltdauer beendet, Antrieb wieder frei
```

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Moving flat again during a storm (in preparation, not tested on the device)

Without limit switches on the tracker, the firmware remembers after a move over the whole max. run time: “This is the end position” (run-time end position). It then does not move any further in this direction. That is correct as long as the drives really moved. But if the **power supply** fails (the board then keeps running on the battery), the **fuse of the drives** trips, or the sync controller (for example BST-KZQ-15A-2) aborts a move, the contacts of the board still switch, only nothing moves, and the firmware wrongly believes the module is flat.

That is why the 10-inch firmware cancels a run-time end position in the “flat” direction and moves flat again:

1. ### At the start
   As soon as a storm begins, the anemometer fails or you select PARK FLAT (also right after switching on in one of these states). *Expected:* The elevation moves flat, even if according to the firmware it is already flat. Without limit switches, the contact stays closed for the full max. run time.

2. ### Every 30 minutes
   As long as the storm or the anemometer failure lasts, this is repeated every 30 minutes (not when parking). *Expected:* About every 30 minutes the contact for “flat” closes once more for the max. run time.

Each of these moves still ends at the latest after the max. run time. It has priority and also runs during a pause of the duty cycle. With limit switches fitted there is no run-time end position; then nothing changes. With the Hall tap switched on, the move ends after about 1.5 seconds if the elevation is verifiably already flat; if the tilt is unknown or the drive stops in the middle of its travel, it runs to the max. run time as without the Hall tap.

> 🧩 **Open: to be checked on the rack**
> It has not been clarified whether the BST-KZQ-15A-2 sync controller accepts that the controller keeps pressing “flat” in the end position until the max. run time without reporting an error code; it will be checked on the rack. Until then this is an **open point**. With the Hall tap, a move into the end position ends about 1 second after no more pulses arrive, instead of only after the max. run time; the question then remains open for flat moves with unknown tilt and for operation without the Hall tap.

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⚠️ **Without feedback a gap remains**
Without Hall tap, the firmware does not see whether the drives are moving. Moving flat again only helps in a storm, on anemometer failure and when parking. With the Hall tap switched on, missing pulses stand out (warning “HALL TAP FAULT” or “Antrieb steht ohne Endlage” in the log); only drive 1 is monitored. The board's input “24 V present” only reports the voltage at J11: on the rack therefore only the power supply of the controller, in the variant with a shared power supply also its failure, but never a tripped fuse of the drives.

Serial Monitor – example, not from the real hardware

```
[ 3600.000][I][track ] Zustand IDLE -> STORM (Auto ein, Boe 44.8 km/h, El +3, Az +0 Promille)
[ 3600.000][I][motor ] Elevation: Laufzeit-Endlage PLUS aufgehoben, neue Flachfahrt erlaubt
[ 3600.000][I][main  ] Ausgang Elevation: STOP -> RAUF (Zustand STORM)
[ 3690.010][I][motor ] Elevation: Laufzeit-Endlage PLUS erreicht
   … 30 Minuten später, der Sturm hält an …
[ 5400.000][D][track ] Zustand STORM haelt an: Flachfahrt wird erneut erlaubt
```

Whether the log shows „PLUS“ or „MINUS“ depends on the setting “Flat position is in UP direction”; by default flat = UP = PLUS.

⚠️ **Until then**
The RearCircuitBoard_10 board can only be checked on the hardware (Chapter 9). Without firmware, no contact switches and no lamp lights. Only connect the battery once the board has passed the cold and heat test and a firmware is running that checks the charge lock (Chapter 7, at your own risk).

Resetting the sync controller (in preparation, not checked on the device)

The sync controller of the Samba Solar Racks (BST-KZQ-15A-2) keeps the two drives in sync. If the power fails while it is moving, it locks itself (error E16) and does not move any more until it is reset. Until then, Samba Solar Track cannot move the system flat either, not even in a storm. Since firmware 0.5.4, the controller triggers this reset itself: a contact on the board connects the RST and COM terminals of the sync controller, just like the Reset button on its control panel. After that, the sync controller fully retracts the drives; on the rack, this is the flat position.

⛔ **The reset moves the system**
After the reset, the system moves completely flat by itself. Check beforehand: the swivel range is clear, nobody is standing at the system, nothing is lying on the modules. That is why the display asks for confirmation before every reset.

Setting up: Specify once which contact the reset uses: on the display via the gear → *SETTINGS* → *COMMISSIONING*. By default, both are set to OFF; then there is no reset. It can only be changed

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while no drive is moving. How J23 or J5 is connected is described in Chapter 6.

VALUE UNDER SETTINGS → COMMISSIONING	SELECTION	NOTE
Reset sync controller elevation	OFF (no reset) · J23 (terminal RST E) · J5 (only with VERTICAL ONLY)	J23 gibt es ab Platine Entwurf 0.12. J5 nur, wenn kein Azimut angeschlossen ist.
Reset sync controller azimuth	OFF (no reset) · J23 (terminal RST A)	Only with two axes (vertical + horizontal).

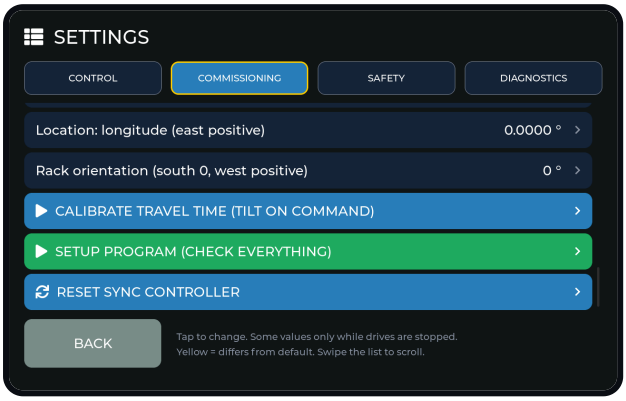


Fig. 46 – Under *SETTINGS → COMMISSIONING*: the setting “Reset sync controller elevation” (here J23) and below it the line *RESET SYNC CONTROLLER*.

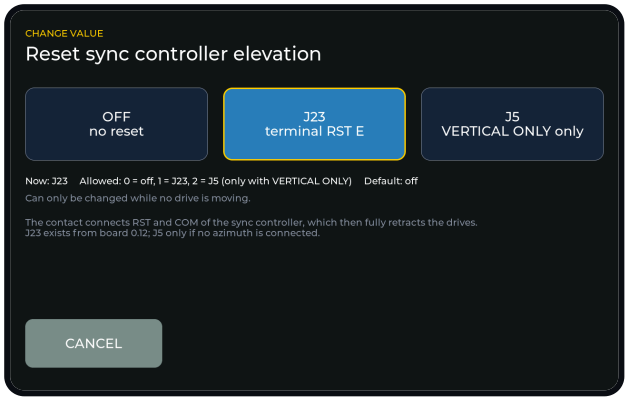


Fig. 47 – Tapping the setting opens the selection: *OFF*, *J23* (terminal RST E) or *J5* (only with VERTICAL ONLY).

Triggering a reset (for example when the control panel of the sync controller shows E16 after a power failure):

1. ## Look at the system

   Swivel range clear, nobody at the system. *Expected:* The system may now move completely flat.

2. ## Tap the line

   Gear → *SETTINGS* → *COMMISSIONING* → *RESET SYNC CONTROLLER*. *Expected:* The prompt “RESET SYNC CONTROLLER? The system moves to its end position (rack: flat). Swing area clear, nobody at the system?” appears (Fig. 48). With two axes, it asks which axis: *ELEVATION*, *AZIMUTH* or *CANCEL* (Fig. 49).

3. ## Confirm

   Tap *RESET* (with two axes, the axis). *Expected:* The display jumps straight to the start screen. The status bar at the bottom is blue: “SYNC CONTROLLER RESET - SYSTEM MOVING FLAT”; in the *SUN SENSOR* card there is the band “RESETTING SYNC CONTROLLER – Phase 1: drives off, dead time” (Fig. 50). The setup is closed in the process, without saving anything and without a restart.

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4 Watch

Phase 2: the RST contact is closed for 5 seconds (Fig. 51). Phase 3: the sync controller retracts the system; the controller waits as long as the maximum run time of the axis (90 seconds on the rack) and accepts no move command during that time (Fig. 52). *Expected:* The system moves flat. In the band the remaining time counts down and the bar grows. The move buttons of the axis are greyed out, the *TILT* card shows “RESET RUNNING - MOVING FLAT”; a selected angle is only approached after the reset.

5 End

Do nothing. *Expected:* The band turns green: “RESET FINISHED – System flat” (Fig. 53). After 10 seconds it disappears; the status bar shows the normal state again. The controller now assumes that the elevation is flat.

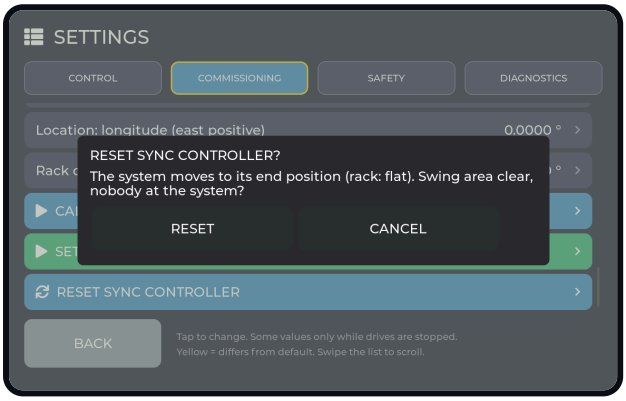


Fig. 48 – The prompt before every reset. *CANCEL* changes nothing.

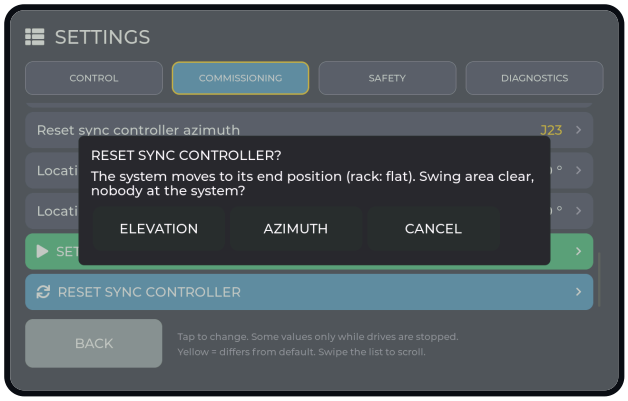


Fig. 49 – With two axes, you select the axis in the prompt.

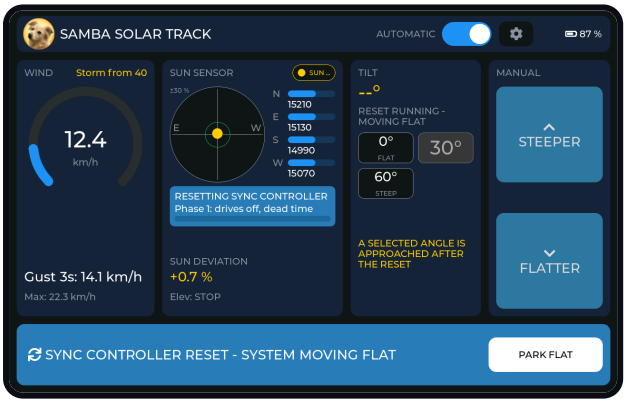


Fig. 50 – Directly after confirming: start screen, phase 1, blue status bar, *TILT* card “RESET RUNNING - MOVING FLAT”.

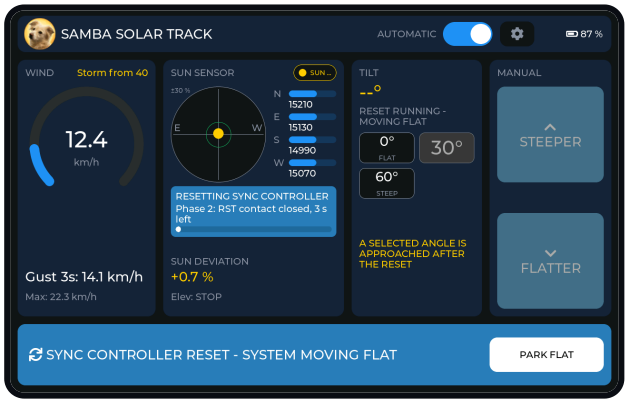


Fig. 51 – Phase 2: the RST contact is closed, 3 seconds left.

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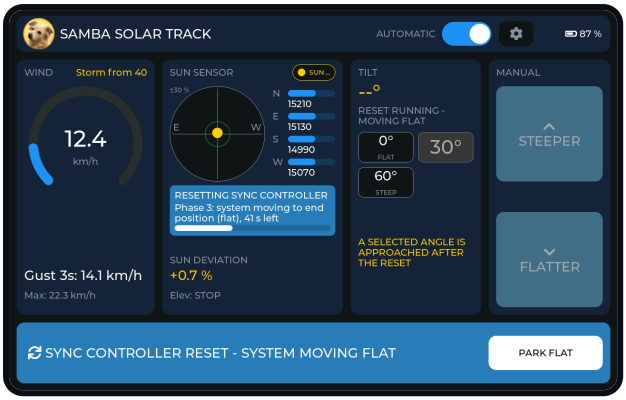


Fig. 52 – Phase 3: the system moves to its end position (flat), 41 seconds left. The move buttons are greyed out.



Fig. 53 – Finished: green band “RESET FINISHED – System flat”, for 10 seconds.

> ⚠️ **Unsaved settings in the setup**
>
> Because the reset closes the setup without saving, an operating mode or anemometer selection that has just been changed but not yet saved is lost. In this case the prompt says so (Fig. 54). If you want to keep the selection, tap *CANCEL*, save first and trigger the reset afterwards.

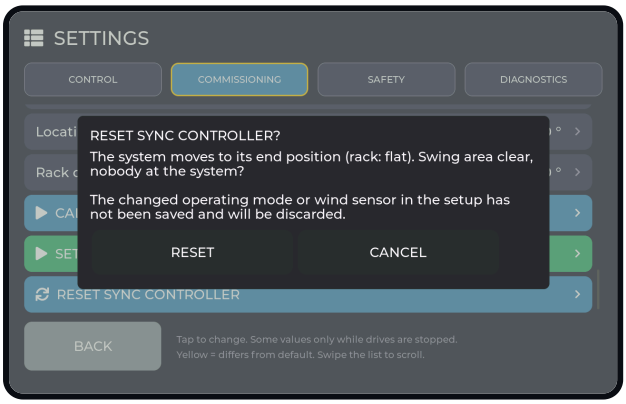


Fig. 54 – Prompt with the note that the changed operating mode or the changed anemometer has not been saved and will be discarded.

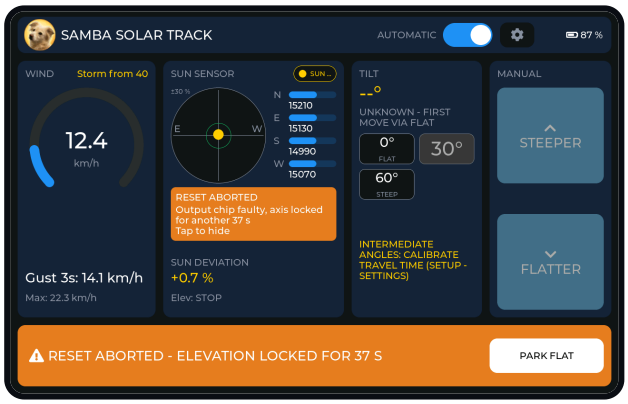


Fig. 55 – Aborted: orange band with the reason. If the contact had already closed, the axis stays locked for the remaining time; the status bar shows “RESET ABORTED - ELEVATION LOCKED FOR 37 S”.

If the reset is aborted: The band turns orange, “RESET ABORTED”, with the reason below: the chip of the switching outputs is faulty, chip 0x72 of terminal J23 no longer responds, or all drives were stopped before a restart or update. If the contact had already closed, the axis still stays locked until the end of phase 3, because the sync controller may already be moving. The band stays until you tap it; it does not lock any operation.

If the reset is refused: The display stays under *SETTINGS*, and the reason appears at the top right (Fig. 56). There is no reset

- as long as the setup is not completed,
- if the chip of the switching outputs is faulty (I/O error),

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- during a manual move (on the display or remotely), while calibrating the travel time and during a firmware update,
- after a run time error of the axis, until it has been acknowledged,
- during a duty cycle pause; for the elevation it is still possible in a storm, on anemometer failure and when parking,
- if a reset is already running,
- if no connection is set, or J23 is selected and chip 0x72 does not respond (board before draft 0.12); the line then shows “not available” (Fig. 57).

In a storm the reset is allowed, because it moves in the safe direction. The status bar then keeps showing the storm, and the band in the *SUN SENSOR* card shows the sequence (Fig. 58). With two axes, the reset of the azimuth works the same way; the band names the axis, and the firmware does not set a position for the azimuth (Fig. 59).

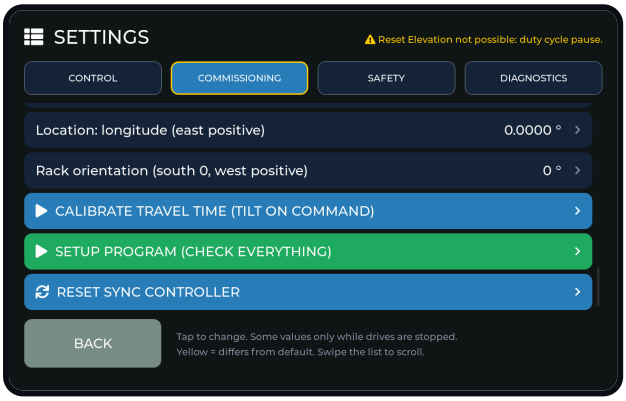


Fig. 56 – Refused: at the top right “Reset Elevation not possible: duty cycle pause.”

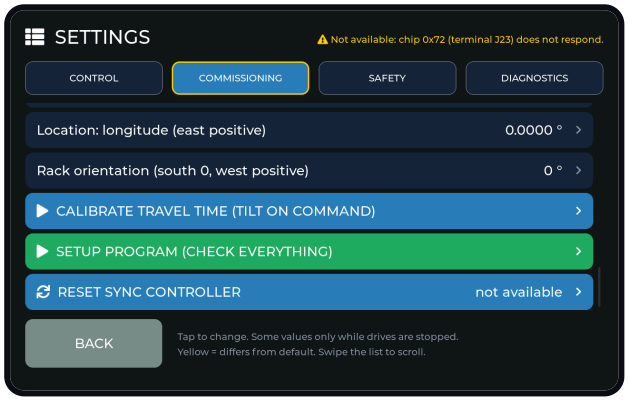


Fig. 57 – Not available: the board has no chip 0x72 (before draft 0.12) or it does not respond.

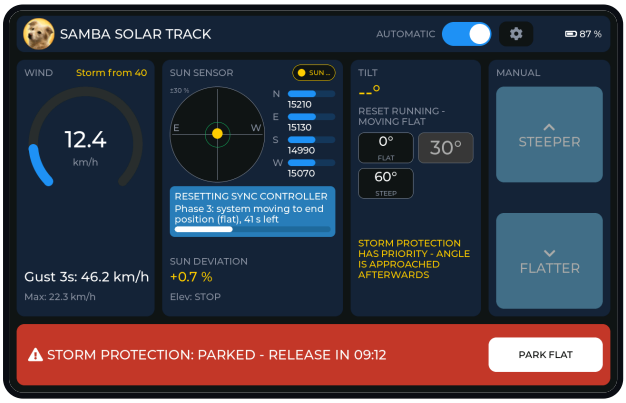


Fig. 58 – In a storm: the status bar shows the storm protection, the band shows the reset.

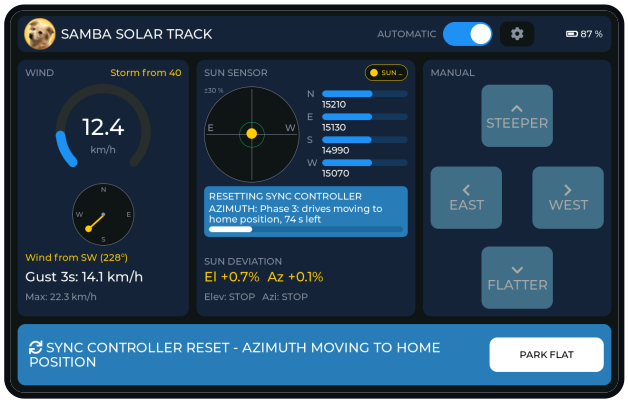


Fig. 59 – Two axes: reset of the azimuth, “AZIMUTH MOVING TO HOME POSITION”.

In the web interface, the *SETTINGS* page has the *RESET SYNC CONTROLLER* card with a *RESET* button for each configured axis. The browser asks for confirmation like the display; after that, the web interface switches to the main page and shows the band and the status bar there. This only works after logging in with the token and not if “Remote control locked” is switched on at the display. The same caution applies to remote operation: you cannot see the system.

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🧩 **Assumptions, to be checked on the rack**

The duration of the contact (5 seconds) is an **assumption**; the manufacturer gives no value. Also assumed: that the sync controller only starts moving after the contact opens, and, for the azimuth, in which direction the home position lies. The controller cannot see whether the sync controller is really moving; after phase 3 it assumes that the elevation is flat. Firmware 0.5.7 has only been built and tested on the computer.

Warnings about the supply and the charge lock in the header (in preparation, not tested on the device)

Since 3 Nov 2026 the firmware monitors the power supply of the circuit board and the charge lock and shows inconsistencies as a yellow warning in the header, in place of the time (just like “LAMPS FAILED” and “HALL TAP FAULT”). For this it reads the board's inputs “24 V present” and “5 V present” (every 100 ms) and the four readings of the chip 0x49 (temperature at the rear panel, half of 3.3 V, voltage at the J8 socket of the board, release of the lock; one pass every 0.8 seconds). All warnings are only hints: tracking, storm protection, parking and manual mode continue, and the firmware switches nothing at the lock.

WARNING IN THE HEADER	WHAT THE FIRMWARE SEES	PROBABLE CAUSE	WHAT YOU DO
24 V MISSING / CHECK POWER SUPPLY (Fig. 39)	The “24 V present” input reports: no 24 V at J11. The board is running on the battery.	Power supply off or defective, cable to the controller interrupted, fuse F1 has tripped (Chapter 6).	Check the power supply of the controller, cable and fuse. In the variant with a shared power supply the drives are down as well; in a storm the module then cannot move flat.
5 V MISSING / CONVERTER OFF (Fig. 40)	The “5 V present” input reports: the converter of the circuit board delivers no voltage although 24 V are present. The board is running on the battery.	The electronic fuse U16 or the self-latching has switched off (overload, overtemperature, overvoltage), or the converter is defective.	Switch the 24 V off for at least 30 seconds and on again. If the warning remains, check the circuit board. If it stays for more than 30 seconds, it turns into “CONVERTER OFF” (below).
CHARGE LOCK / NOT WORKING (Fig. 41)	The temperature at the rear panel is below 8 °C or above 37 °C, and yet there are more than 4.4 V at the J8 socket of the board: the battery could be charged outside the permitted window.	Soldering fault or defect at the lock (Chapter 7), sensor or copper island damaged, jumper J21 wrong, or a PC at the “UART” socket is powering the board with 5 V via USB.	Unplug the USB cable. If the warning remains: unplug the battery and check the lock according to Chapter 9. The controller keeps working.
CHARGE LOCK / STUCK (Fig. 42)	The lock reports “charging allowed”, but there are less than 4.4 V at the J8 socket of the board.	The lock releases, yet the full 5 V do not arrive: fault in the circuit of the lock or at the converter.	Harmless for the battery (it is not being charged), but it will never get full this way. Check the circuit board.

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WARNING IN THE HEADER	WHAT THE FIRMWARE SEES	PROBABLE CAUSE	WHAT YOU DO
CONVERTER OFF / SWITCH 24 V OFF 30 S (Fig. 43)	For at least 30 seconds: 24 V present, 5 V missing.	Like “5 V MISSING”, only permanent: usually the self-latching has switched off after an overvoltage and keeps the input off until the 24 V have been gone.	Switch the 24 V off for at least 30 seconds, then on again (Chapter 6). If the circuit board switches off again, it is defective.
REVERSE BLOCK OR / 24 V DETECTION FAULTY (Fig. 44)	For at least 30 seconds: 24 V missing, 5 V nevertheless present.	This cannot happen normally: either the 24 V detection is defective, or current is flowing backwards into the converter (reverse block of fuse U15 defective).	Check the circuit board; until then do not rely on a power supply failure being reported.
CHIP 0x49 MISSING / LOCK UNCHECKED (Fig. 45)	The second measuring chip does not respond (after three errors in a row). The firmware cannot check the lock; the charging indicator then estimates from the course of the battery voltage.	Normal without the circuit board. With the circuit board: the chip hangs on the board's own 3.3 V supply and is not reachable without its 5 V, i.e. in battery operation; otherwise a solder joint or the chip is defective.	If 24 V or 5 V are missing at the same time, the header shows that cause and not the chip. Otherwise check the circuit board. The controller keeps working.

When a warning comes and goes: “24 V MISSING” and “5 V MISSING” appear immediately with the input and disappear immediately when the voltage is back. The four warnings of the charge lock (Fig. 41 to 44) must be present for 30 seconds without interruption before they appear, and absent for 10 seconds before they disappear; this way short fluctuations do not trigger an alarm. If a reading is missing because chip 0x49 or the lamp chip does not respond, the corresponding warning does not come up at all. If 24 V and 5 V are missing at the same time, the board runs on the battery; that is not a fault, the header shows only “24 V MISSING”.

Order of precedence: The header always shows only one warning, the most important one: first “LAMPS FAILED” (then the inputs for 24 V and 5 V are unknown as well), then a warning of the charge lock in the order “NOT WORKING”, “REVERSE BLOCK OR 24 V DETECTION FAULTY”, “CONVERTER OFF”, “STUCK”, then “24 V MISSING”, then “5 V MISSING”, then “CHIP 0x49 MISSING”, only after that “HALL TAP FAULT” and the hint about a new firmware. In the browser the web interface shows the same warnings as yellow bands below the status bar, there all of them at once that are currently pending. Every appearance and disappearance is written to the log.

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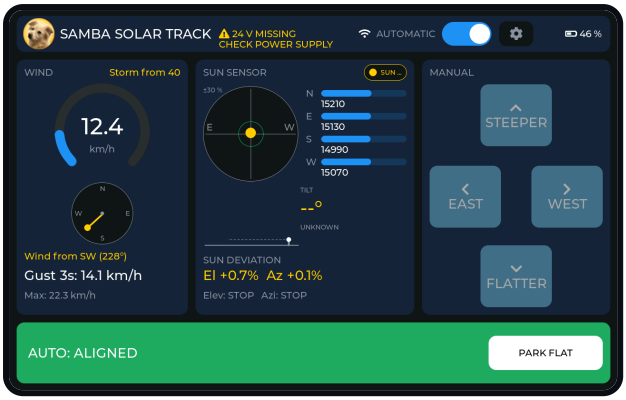


Fig. 39 – “24 V MISSING / CHECK POWER SUPPLY”: The board runs on the battery; the automatic keeps working.



Fig. 40 – “5 V MISSING / CONVERTER OFF”: 24 V are present, the converter of the circuit board delivers nothing.

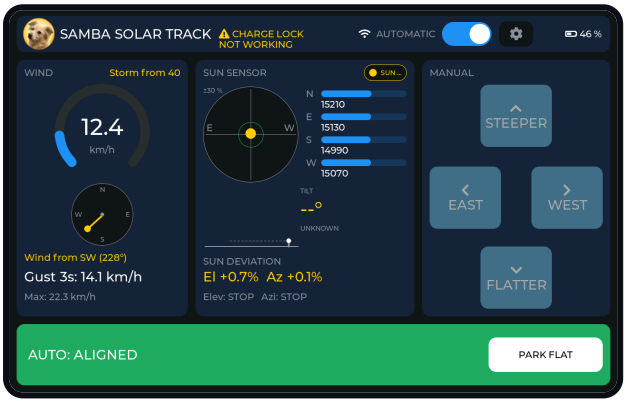


Fig. 41 – “CHARGE LOCK / NOT WORKING”: too cold or too warm, but full 5 V at the board (J8 socket).



Fig. 42 – “CHARGE LOCK / STUCK”: release detected, but no full 5 V at the board (J8 socket).



Fig. 43 – “CONVERTER OFF / SWITCH 24 V OFF 30 S”: 24 V without 5 V for 30 seconds.

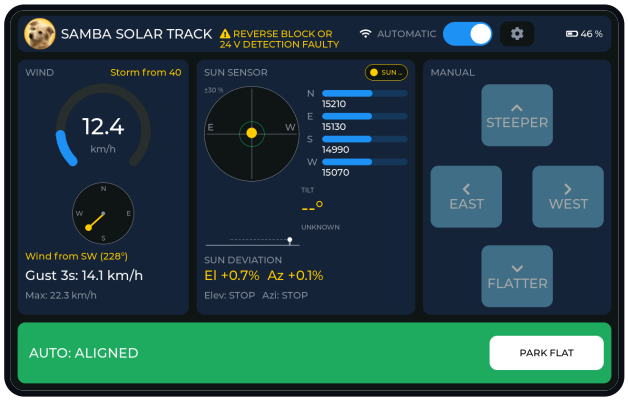


Fig. 44 – “REVERSE BLOCK OR / 24 V DETECTION FAULTY”: 5 V without 24 V for 30 seconds.

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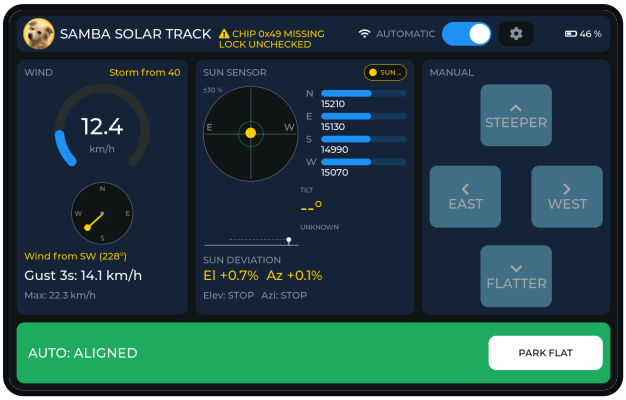


Fig. 45 – “CHIP 0x49 MISSING / LOCK UNCHECKED”: The second measuring chip does not respond.

> **Status**
> The pictures were generated by the screenshot tool from the real code of the user interface (dashboard with two axes, battery 46 %). The check is built and tested on the computer; nothing has been measured on a manufactured circuit board. Whether the thresholds (8 and 37 °C, 4.4 V at the J8 socket of the board) and the detection of the 5 V at the board's input are right in all situations will only be shown by measurement on the device (**assumption**).

9 Initial commissioning of the hardware

These steps check the board without firmware. Each step has a clear result. Only continue when it is correct. You carry out steps 4 to 7 with the board on the table, not yet on the display board. After that, on the fully assembled rack, the setup program guides you through checking all functions and travel paths.

1 Unplug the battery

For this check the battery connector on the board stays unplugged; it is only connected in the last step. *Expected:* The battery is not connected to anything.

2 Visual inspection and solder joints

Look at all solder joints of the terminals. Use the continuity tester on J3 to J6 to check that no pin is connected to its neighbour. Set jumper J16 according to the section “Anemometer”, J21 to 2-3. *Expected:* No solder bridge, both jumpers fitted.

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3

Connector check

Before switching on for the first time, compare every connector with the terminal table and the picture of the board (Fig. 5): correct connector on the correct terminal, labelling matches. Then look at J7, J10, J14 and J22 from the side. *Expected:* Every connector sits flush on its header, no pin of the header is exposed, no connector protrudes beyond the end of a terminal. As soon as there is firmware for the 10.1-inch board, this also includes: at start-up no message about limit switches or lamps appears.

4

Force the lock and measure

Connect the board on its own (not yet on the display board, J22 unplugged) to 24 V, jumper J21 to 1-2. Measure with the multimeter at the solder joints of J1: pin 2 against pin 6. *Expected:* 3.8 to 4.1 V. If the value is higher, the lock is faulty: do not use the board.

5

Operation and waiting time

24 V off, J21 to 2-3, 24 V on again (the room should be 15 to 30 °C warm). Watch the voltage at J1 pin 2 against pin 6. *Expected:* First about 4 V; after 45 to 101 minutes (usually about 71 minutes) it jumps to 4.9 to 5.2 V.

6

Cold test

At 4.9 to 5.2 V, briefly spray freeze spray next to the copper island at the top left, not directly onto components. *Expected:* The voltage immediately drops to about 4 V. After warming up it initially stays at 4 V; only after the waiting time does it rise to about 5 V again.

7

Heat test

Carefully warm the copper island above 37 °C with a hot-air tool on a low setting; check the temperature with a thermometer at the island, not above 60 °C. *Expected:* The voltage drops to about 4 V. After cooling down it returns to about 5 V after the waiting time. This test finds soldering faults that the freeze spray does not find.

8

Install

Fit the board onto the display board with spacers and pads according to Chapter 5. *Expected:* Spacers and pads sit under the island and do not slip; the board lies flat on all four spacers.

9

Switch on the board

Slide switch of the display board to ON, switch on 24 V. *Expected:* The board starts (the display shows the picture of the factory software or stays dark without firmware). The board's charging lamp, if present, stays off: without a battery there is no charging, so it does not light up anyway (**assumption**).

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10

Switching outputs and limit switches

Only with the firmware for the 10.1-inch board (in preparation): hold each button on the display and measure with the continuity tester through the plugged-in connector between the two wires (connector of the switching outputs); press each limit switch. *Expected:* Exactly the corresponding contact closes, exactly the pressed limit switch is reported.

11

Check the reset contact (from draft 0.12 or with reset via J5)

Disconnect the plug of J23 (or J5) at the sync controller or undo the cable there, place the continuity tester between the RST and COM wires and trigger “Reset sync controller” on the display (Chapter 8). *Expected:* After the dead time it beeps for about 5 seconds, then the contact is open again. After switching on and after every restart it is open.

12

Connect the battery (at your own risk)

Only once the cold and heat test have been passed and a firmware is running that checks the charge lock (Chapter 7): 24 V off, slide switch of the board to OFF, plug the battery connector into J13 of the board, board on, 24 V on. *Expected:* The header shows no warning about the charge lock (Chapter 8), the battery level is shown at the top right. If “CHARGE LOCK NOT WORKING” appears, unplug the battery again. The lock in the hardware has not yet been measured on the device; the risk is yours (decision of 3 Nov 2026).

ℹ️ **Overvoltage shutdown**

Whether the overvoltage shutdown (Chapter 5) works can only be checked with a laboratory power supply on a sample; the firmware cannot check it later. This test is part of the measurement plan before ordering (Appendix 11 E).

⚠️ **Use freeze spray and hot air carefully**

Use freeze spray only briefly and from a distance of 10 to 20 cm, hot air only on a low setting. Do not touch anything at the 24 V terminals while voltage is applied.

💡 **And then: the setup program (since firmware 0.7.0)**

Once the hardware has been checked and the rack is fully wired, the last step of commissioning follows: the setup program on the display. It checks the anemometer, the direction of the flat position, the travel paths, the rotation (with two axes), the sun sensor and the automatic, and then enables the automatic. Expected: at the end it says “Done! All checks are passed.”, and the test report is available in the web interface. Firmware 0.7.0 has only been tested on the PC.

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10 Terms simply explained

End cap	A plastic cover that closes the open end of an aluminium profile. The sun sensor holder replaces such a cap and also carries the sensor.
Stripping	Removing the plastic sheath (insulation) at the end of a wire over a certain length so that the bare metal makes contact in the terminal. Best done with wire strippers; the strands underneath must not be nicked.
Termination resistor	A resistor of 120 Ω between wires A and B of an RS485 line, at both of its ends. It prevents signals from being reflected at the end of the line.
Standoff	A small metal pillar with a thread. With an external and an internal thread it replaces a screw and at the same time holds a second board at the right distance. The HallTapBoard does not need any: it is screwed directly onto plastic bosses in its housing, so that no conductive connection is created through the mounting.
Buck converter	A circuit that makes a lower DC voltage out of a higher one, here the 5 V for the board out of 24 V. It loses little energy in the process and hardly gets warm.
Ferrule	A small metal tube that is crimped with pliers onto the stripped end of a fine-stranded wire. It holds the many thin strands together so that none sticks out and the wire sits firmly in the terminal.
ADS1115	A measuring chip. It converts the voltage of the light sensors into numbers the computer can work with.
Aluminium profile (slot profile)	A bar of aluminium with a constant cross-section, here 40 × 40 mm, with a slot on each side (slot 8) and a hole in the middle. The Samba Solar Rack, for example, is built from such profiles.
Battery (lithium battery)	A rechargeable battery. The battery of the 10.1-inch board has 3.7 V. Lithium batteries may only be charged within a limited temperature range, otherwise they age quickly or become dangerous.
Start-up loss	The part of every move in which the frame does not move yet, for example because relays pull in or the drive controller starts slowly. For tilt on command it can be entered under SETTINGS.
Anode / cathode	The two terminals of a diode. A diode lets current through in one direction only, from the anode to the cathode. On a light-emitting diode the anode is therefore plus and the cathode minus. On the board, a line or angle in the silkscreen usually marks the cathode; on the large diode D5 the cathode is the cooling tab.

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ASA / PETG	Two plastics for the 3D printer. ASA withstands sun and heat up to about 100 °C and is therefore suitable for the housing of the HallTapBoard in the box on the rack; PETG is easier to print but already gets soft from about 80 °C (values are guidelines, not looked up).
Flashing	Writing the firmware into the memory of the board, via a USB cable or over the network.
Rack orientation	The compass direction in which the panels of a one-axis rack face, as the deviation from south in degrees: due south = 0, towards the west positive, towards the east negative (compass reading minus 180). It is measured with a compass and entered on the RACK ORIENTATION page.
Azimuth	The rotation of the module from east to west.
Azimuth of the sun	The compass direction in which the sun currently stands, in degrees clockwise from north: north 0°, east 90°, south 180°, west 270°. The SUN page shows it as DIRECTION, for example “191° (SSW)”. Not to be confused with the azimuth of the rack, the rotation axis.
Assembly preview	A picture of the fully assembled board that JLCPCB shows when ordering, before payment. In it you check whether every component is placed the right way round. For polarised components a red dot marks the side JLCPCB uses as reference; this is not always pin 1.
Operating mode	Whether the controller moves one axis (vertical only: tilt) or two (vertical + horizontal: tilt and rotation from east to west). It is chosen in the setup; the screens, the steps of the setup program and the night position depend on it.
Motion sensor	Detects whether something is moving in front of a device. Here the built-in camera of the display does this: the firmware compares each image with the previous ones without storing it.
Cloudy (calculated angle)	State in which the sun sensor sees too little light to track. With one axis, the device can then, if wanted, move to the best angle calculated from the position of the sun (off by default).
Reference, reference move	A reference is an end position at which the position is reliably known. The reference move goes there (usually flat) before tilt on command moves to an intermediate angle. After a reference the uncertainty is zero.
Screensaver	A dark, calm display that the screen shows when nobody has been in front of it for a while. It saves power and does not dazzle. A touch or a movement brings back the normal display.
Gust	A short, strong blast of wind. The controller uses the average over three seconds.

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Bootloader	A small start program at the beginning of the memory. When switched on, it starts the actual firmware. It is contained in the factory image.
Latitude / longitude	The two numbers that define a place on Earth. The latitude says how far north (positive) or south (negative) of the equator the place lies, the longitude how far east (positive) or west (negative) of Greenwich. Dresden: latitude 51.05, longitude 13.74. On the device, enter with a dot: 51.0504.
Socket header / pin header	Two counterparts for plugging boards together: the pin header has pins, the socket header matching openings. The RearCircuitBoard_10 board plugs with its pin header J1 into the socket header J8 of the display board.
Codec (audio codec)	A chip that converts numbers from the computer into an audio voltage. On the 10.1-inch board it is called ES8311; a small amplifier (NS4150B) uses it to drive the speaker.
dBFS	Measure of the volume of a digital sound: 0 dBFS is the loudest sound the codec can output, smaller (negative) values are quieter. 3 dB less means half the power.
Spacer	A small brass pillar with an internal thread at both ends. Here one lies without a screw between two thermal pads and conducts the temperature of the rear panel to the sensor on the board.
Download mode	A special start state of the chip in which it accepts new firmware via USB instead of starting it. The PC normally switches it on itself; by hand: hold BOOT, press RESET, release BOOT.
Continuity tester	A simple measuring device, or a setting on the multimeter, that beeps when two points are electrically connected. With it you check, for example, whether a contact is closed.
Solid / fine-stranded	A solid wire consists of a single, stiff wire. A fine-stranded wire (litz) consists of many thin strands and is flexible. Fine-stranded wires should preferably get a ferrule before being inserted into a spring terminal.
Calibration	The guided measurement of the two travel times on the display (Setup → SETTINGS → COMMISSIONING → CALIBRATE TRAVEL TIME). Without calibration, tilt on command only knows 0° and 60°.
Duty cycle	The share of time a drive may run without getting too hot. Here it is 10 % and at most 2 minutes in one go; after that the drive needs a pause. The 10-inch firmware enforces this by itself; only the flat move during a storm, an anemometer failure and parking also runs during the pause.

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Elephant foot	In 3D printing, the first layer is pressed onto the print bed and becomes slightly wider. As a result, fine recesses on the underside of a part can close up. Many slicer programs have a compensation setting for this.
qualified electrician	A person with electrical training and experience who is allowed to work on mains voltage (230 V). Connecting power supplies, installing sockets or laying cables on the mains belongs in their hands.
Electronic fuse	A small chip that protects like a fuse but does not blow. If too much current flows, it limits it or switches off and tries again by itself after a short time. On the RearCircuitBoard_10 board, two of them protect the 24 V input (U16) and the 5 V to the display board (U15).
Elevation	The tilt of the module, from flat to steep.
Limit switches	A switch at the end of the travel. It reports: it cannot go any further.
esptool	The program from the chip manufacturer Espressif that writes firmware onto ESP32 boards. The browser pages use it in the background.
Extended / Basic	The two classes of the JLCPCB parts library. For every component type of the “Extended” class, JLCPCB charges a setup fee per order; for “Basic” it does not.
Travel time	The time the drive needs for the whole travel from 0° to 60° (steeper) or back (flatter). From it the firmware estimates the tilt.
Spring terminal (push-in)	A terminal without a screw: a spring in the connector clamps the wire. You simply push a stiff wire or a wire with a ferrule in as far as it goes. To release it, you press the small lever next to the opening and pull the wire out.
Window comparator	A circuit that checks whether a measured value lies between two limits, here whether the temperature is between 10 and 35 °C.
Firmware	The software that runs on the board.
Flat position (0 position)	The position in which the panels lie (almost) horizontal: in a storm, when parking, before an update and at night. The setup program checks in which direction the drive moves for this in the step “Direction of the flat position”.
Flash	The memory of the board that holds the firmware and the saved settings. It keeps its content even without power.

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Sense line (remote sense)	Two thin additional wires on some power supplies, through which the power supply measures the voltage at a remote point and compensates for the loss on the line. With Samba Solar Track they stay at the power supply: if they were run to the drives, the voltage at the controller would be too high.
Infill	How densely the inside of a printed part is filled, in percent. 100 % means completely full; this makes a part heavy and strong, but uses a lot of filament.
Galvanically isolated	Two circuits without a conductive connection between them. A signal still gets across, for example with light in an optocoupler. This way the two sides can have different grounds and power supplies without interfering with each other.
Lock nut	A nut that holds a cable gland in the housing wall from the inside. The gland is pushed through the hole from the outside, the lock nut goes onto its thread on the inside; then both are tightened against each other.
Frame G1 / G2	The design of the Samba Solar Rack (position of the pivot axis). It determines which travel of the drive corresponds to which tilt. G1: Rack 2.0-D and 2.0-E; G2: 2.0-E Plus, 2.1 and 2.1 Plus. For other frames there is “linear” (an assumption).
Threaded insert	A small brass sleeve with an internal thread that is melted into a hole of a printed part with a hot soldering iron (also “heat-set insert”). After that, a metal screw can often be screwed in and out there without the thread in the plastic tearing out. The housing of the HallTapBoard has eight ruthex M3 inserts, the sun sensor holder two ruthex M4 inserts.
Threaded rod	A rod with a thread over its whole length, here M8 stainless steel. In the pendulum holder it is the axis around which the pendulum swings.
Glitch filter (interference filter)	A filter in the pulse counter that does not count very short pulses. At the pulse input J22, pulses shorter than 5 µs are treated as interference.
GND	Ground, the common negative pole of the circuit.
Hall tap	Picking up the pulses of the Hall sensors at the sync controller without disturbing them. The HallTapBoard does this. If “Hall tap present” is switched on under SETTINGS, the 10-inch firmware counts the pulses and thus knows the tilt (Chapter 8).
Hall step	Every change between LOW and HIGH (edge) on one of the two channels A and B. There are 4 Hall steps per pulse. “Hall steps per stroke” is the number of steps for the whole travel from 0° to 60°.

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Hall sensor	A small sensor that detects a magnetic field. Many linear actuators have one on the motor: with every revolution it emits pulses. Whoever counts the pulses knows how far the drive has moved. With two channels (A and B) the direction can also be detected.
HallTapBoard	A separate small board for a box on the rack (draft 0.6 with a 3D-printed housing, not manufactured yet). It picks up the pulses of the Hall sensors of drive 1 at the BST-KZQ-15A-2 sync controller and passes them on via optocouplers, galvanically isolated, as two potential-free contacts to the pulse input J22 of the RearCircuitBoard_10 board. It is powered from the 24 V of the drives (Chapter 6).
HTTPS	The encrypted form of the connection to a website. The device uses a built-in list of trusted authorities (certificates) to check that it really is talking to the project's site.
Hiccup	Named after the hiccup: in the event of a short circuit, the converter switches off, waits briefly and tries again, over and over, until the fault has been cleared. This way nothing gets too hot. Some power supplies behave the same way under overload; this is unsuitable for the shared power supply.
HIGH / LOW	The two states of a digital signal: voltage present (HIGH) or no voltage (LOW).
Max. run time	This is how long a drive may run in one go. With limit switches, a longer move is a fault; without limit switches, the end position is then considered reached (run-time end position).
Hysteresis	The gap between the switch-on and switch-off threshold. At the pulse input J22 the comparator switches in one direction at about 2.05 V and only at about 2.36 V in the other. Small interference on the line therefore does not create false pulses.
I2C	A connection of two lines (SDA and SCL) to which several chips are attached. Each has its own address.
Cable shield	A braid or foil around the wires of a cable. Connected to GND at one end, it keeps interference away from the wires.
Cable gland (PG gland)	A feed-through for a cable through a housing wall. When the cap nut is tightened, a sealing ring presses around the cable sheath: the cable is then sealed and secured against being pulled out (strain relief). “PG 11” is the size of the thread (steel conduit thread); each size only seals cables in a certain range of outer diameter, for the Lapp PG 11 4 to 10 mm.
Freeze spray	A spray from electronics shops that cools a spot strongly within seconds. With it you check whether a temperature sensor reacts correctly.

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Tap drill hole	The hole into which a thread is cut. For M8 it is 6.8 mm; the central hole of many 40 × 40 aluminium profiles has exactly this size.
Comparator	A chip that compares two voltages and reports at its output which one is higher. On the RearCircuitBoard_10 board two of them are at work: in the charge lock (U18) and at the pulse input J22 (U22).
Compass	Shows the compass direction; north is 0°, south 180°. A compass app on your phone is enough to measure the orientation of the rack. Nearby metal deflects it; so keep one step away from the rack.
Creepage distance / clearance	The shortest path between two conductors across the surface of the board (creepage distance) or through the air (clearance). The longer it is, the harder it is for a leakage current, for example via moisture and dirt, to find a path.
Ball bearing	A ring with balls between an inner and an outer ring, so that something turns almost without friction. The 608 bearing (8 × 22 × 7 mm) is very common, for example in inline skates.
Cooling tab	A large metal surface on the package of a component that lies flat on the board and is soldered to it. Through it and the copper of the board, the component gives off its heat. On diode D5 the cooling tab is also a terminal, the cathode.
Charger	The circuit on the board that charges the battery from the 5 V. It only charges when its input voltage is high enough (about 4.5 V).
Charge lock	The circuit on the RearCircuitBoard_10 board that prevents the battery from being charged when it is too cold or too warm: it then lowers the 5 V for the board to about 4 V.
Run-time end position	Without limit switches on the tracker, an end position is considered reached when the contact was closed for the whole max. run time; the drive then stands at its own built-in limit switch. It can be wrong if the drives had no power at the time or their controller aborted the move. With the Hall tap there is the more accurate standstill end position instead.
Derating	In heat, a power supply delivers less than its rated power. How much is shown by a curve in the data sheet. That is why the power supply should be mounted in the shade.
Light-emitting diode (LED)	A small component that lights up when current flows through it in the right direction. It needs a series resistor that limits the current.

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Light guide	A rod of clear plastic (here acrylic glass, PMMA) that passes light along like a pipe: what shines in at the bottom comes out at the top end. This way you can see a lamp in the housing from outside, and the hole in the lid is still closed.
Light shaft	A short tube from the lid of a housing to just above a lamp on the board. It guides the light to the hole in the lid, so that you can see the lamp from outside without opening the lid.
Linear actuator	A motor that extends and retracts a rod.
Solder bridge	Solder that unintentionally connects two neighbouring pins during soldering. What should be separate is then electrically connected. You find it with the continuity tester: it beeps between the two pins. To remove it, draw off the excess solder with the hot soldering iron and then check again.
Manifest	A small text file on the project's site. It names the latest firmware: version number, date, size, a checksum of the file and the signature. From this the device recognises whether there is newer firmware and whether the file is genuine and complete.
Modbus RTU	A widespread “language” for measuring devices. The board asks, the device answers with the measured value and a checksum.
Morning position	With two axes: panels flat and turned east, so that in the morning the rack already faces the rising sun. The device moves there at night if NIGHT TO THE EAST is switched on (default).
Overrun	This is how far the drive keeps turning after being switched off. With the Hall tap, the controller measures the overrun for each direction and, when moving to an angle, switches off that much earlier.
Night position	The position of the rack at night: with one axis flat, with two axes flat and, if wanted, then turned east (morning position). It begins at the “Night from” time (sunset plus 30 minutes) or, without time and location, when it is dark.
NC / NO	Normally closed and normally open. Describes whether a contact is closed or open in its rest state.
Tilt on command	The elevation moves to a selected angle and stays there; the automatic pauses until you resume it. In the log the state is called POSITION. 10-inch version only, single axis only.
NTC	A resistor whose value decreases as the temperature rises. The board uses it to measure the temperature.

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Optocoupler	A component that transmits a signal with light: a small light-emitting diode on one side switches a transistor on the other. Input and output are electrically isolated. The output acts like a potential-free contact.
OpenStreetMap, Nominatim, Leaflet	OpenStreetMap is a free world map to which many volunteers contribute; Nominatim is its search service for places. Leaflet is a small program that displays such maps in the browser. The web interface uses all three for the map under SETUP → LOCATION; the map on the display works without them.
PCA9538	A chip with eight additional inputs and outputs, controlled via I2C.
Pendulum	A body that is suspended so it can swing and, by its own weight, aligns itself vertically downwards. The pendulum holder uses this so that the anemometer always stands vertically.
PhotoMOS	An electronic switch without moving parts. A small light-emitting diode inside switches the contact; the two sides are electrically isolated.
Plausibility check	A check whether readings fit together. This is how the firmware compares the temperature at the rear panel, the voltage at the J8 socket of the board and the release of the charge lock: full 5 V in the cold cannot be right. It only reports this; it cannot switch anything at the lock (Chapter 8).
Polarity reversal	Swapping plus and minus at the motor so that it runs backwards.
Polymer capacitor	A capacitor with solid plastic instead of liquid content. It lasts a long time and dampens voltage fluctuations well.
Potential-free	A contact that is not connected to any voltage of the controller. It behaves like a simple push-button with two terminals.
Test report	The text file that the setup program saves at the end: date, firmware, all steps with their result and the measured values. The last three are kept in the device and can be downloaded in the web interface under SETUP.
Pull-up / pull-down	A resistor that pulls a line to a fixed state as long as nothing is driving it.
Pulse counter (PCNT)	A part of the ESP32-P4 processor that counts pulses on a pin by itself, even very fast ones, without burdening the program. The 10-inch firmware uses it to count the pulses from input J22 when the Hall tap is switched on.
Quadrature	Two signals (channel A and B) that are offset from each other by a quarter period. From the order in which they change, the controller recognises the direction of movement.

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Pitch	The centre-to-centre distance between two neighbouring terminals of a terminal block or pin header, for example 3.5 mm. Connector and terminal must have the same pitch.
Reset (sync controller)	Resetting the sync controller, for example after a power failure during a move (error E16). The BST-KZQ-15A-2 sync controller then fully retracts the drives. It is triggered on the control panel of the sync controller or by Samba Solar Track via a contact between RST and COM.
RearCircuitBoard_10	The custom board for the 10.1-inch board ESP32-P4-WIFI6-Touch-LCD-10.1. It sits on the outside of the device's rear panel and plugs with the pin header J1 into its socket header J8 (in preparation).
Relay	An electrically controlled switch. A small current switches a large one.
RGB lamp	Three light-emitting diodes in one package: red, green and blue. Red and green together make yellow.
RS485	A robust data connection over two lines (A and B), suitable for long cables outdoors.
Key (signing key)	A pair of numbers that belong together. With the secret part, the developer signs a new firmware; every firmware carries the public part and uses it to check the signature. Only the developer has the secret part.
Degree of protection (IP65, IP67)	States how well a device is protected against dust and water. The first digit stands for dust (6 = dust-tight), the second for water (5 = water jets, 7 = brief immersion). A power supply outdoors needs a high degree of protection.
Safety extra-low voltage (SELV)	Short for “safety extra-low voltage”: a voltage so low that it is not dangerous to touch, and which is generated safely isolated from the mains, for example by a power supply approved for this. With Samba Solar Track, at most 30 V DC is present on the board.
Self-latching	A circuit that latches itself after being triggered. Here it keeps the 24 V input off until the 24 V is switched off.
Self-test	A short check that a freshly flashed firmware performs on itself: is the display running, and does the switching-output chip on the board answer (in the simulation the display is enough)? If Wi-Fi is switched on, the device must also have connected to the Wi-Fi at least once since the restart. Only then is the new firmware considered good. If it fails the check, the device returns to the previous firmware.

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Setup program	The guided first installation on the display (since firmware 0.7.0): it asks what is installed on the rack and checks every function and every travel path. Until it has been passed completely once, the automatic remains locked.
Signature (digital signature)	A sequence of numbers that can only be generated with the project's secret key. If anyone changes even a single character of the signed file, it no longer matches. This way the device only accepts firmware that really comes from the project.
Fuse, self-resetting	A fuse that becomes hot and high-resistance when there is too much current and closes again by itself after cooling down. It does not have to be replaced.
Simulation	A test mode without the RearCircuitBoard_10 board: the firmware only simulates tracker, sensors and drives and switches nothing.
SMD	Short for “surface-mounted device”: components that are soldered flat onto the surface of the board, without holes. JLCPCB places them by machine. The counterpart are through-hole parts such as terminals and pin headers, whose pins are inserted through holes in the board and then soldered.
Soft termination	Capacitors with flexible terminations. If the board bends, they crack less often.
Sunrise / sunset	The moment at which the upper edge of the sun appears at or disappears below the horizon, as used by weather services (sun elevation −0.833° including refraction). The device calculates it for every day from location and date.
Sun elevation	The angle of the sun above the horizon: 0° at the horizon, 90° straight above you. In Central Europe at midday about 15° in winter, about 62° in summer.
Sun position	Where the sun currently is in the sky, given by sun elevation and azimuth of the sun. The device calculates it from time and location.
Voltage regulator	A component that makes a fixed, lower voltage out of a higher, fluctuating one, for example exactly 5 V out of 24 V. A simple regulator (linear regulator) such as the TPS7B69 on the HallTapBoard converts the rest of the voltage into heat.
Voltage divider	Two resistors in series that divide a voltage in a fixed ratio. On the 10.1-inch board, R92 and R93 divide the battery voltage to one third so that the chip's measuring input can measure it.

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Voltage loss (voltage drop)	Every line has a small resistance. When current flows, part of the voltage is lost on it: the longer and thinner the line and the greater the current, the more. The calculation includes the outgoing and return wire.
Location	Latitude and longitude of the place where the rack stands. An accuracy of about 10 km is enough. Entered on the display on the LOCATION page or in the web interface; latitude 0 and longitude 0 mean “not entered”.
Pluggable terminal	A terminal in two parts: the header is soldered onto the board, the wires go into the loose plug. Usually a small screw holds them, at the switching outputs of the RearCircuitBoard_10 board a spring terminal. This way the board can be replaced without undoing the wires.
Jumper	A small plug that connects two pins lying next to each other. By moving it, a connection is switched on or off.
Shaft collar	A metal ring with a clamping screw that is screwed tight on a tube or shaft. It prevents a part from slipping or falling off.
Star wiring	Every load gets its own line directly from the terminals of the power supply, like the rays of a star. This way the large current of the motors does not flow over the controller's line, and each line has its own suitable fuse.
Standstill end position	An end position that the controller with the Hall tap recognises by the fact that no more Hall steps arrive although the output is switched on: after a start-up of 1.5 seconds, none for 1 second (**assumptions**). The move then ends immediately, not only after the max. run time. If, on the other hand, the drive stops in the middle of its travel, the controller ends the move without an end position („Antrieb steht ohne Endlage“).
End wall	The closed wall at the end of the header of a pluggable terminal. It holds the plug at the side so that it cannot be plugged in offset.
Current limiting (CV+CC)	A power supply with constant voltage (CV) and current limiting (CC) holds the voltage until the maximum current is reached; beyond that it limits the current and keeps delivering instead of switching off. The counterpart is a power supply with *hiccup*.
Sync controller	A control unit that makes two drives run at the same speed so that the frame does not twist, for example the BST-KZQ-15A-2 on the Samba Solar Rack 2.0-E Plus (until 9.11.2026 the HAKIWO SCN4 was planned there). The board only switches its push-button input and, from draft 0.12, its reset input.
Daily arc	The path of the sun across the sky in one day, from sunrise in the east via the highest point in the south (noon) to sunset in the west. The SUN page shows it as an orange line.

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Dead time	The pause between stopping and moving in the opposite direction.
Driver	A small program for the PC so that it recognises a connected device, here the board's USB chip CH343, as a port.
Isolation gap	A strip of the board without copper between two parts of the circuit that must not be conductively connected. On the HallTapBoard it is 7 mm wide; only the optocouplers bridge it, with light instead of current.
Uncertainty	How far the displayed tilt can be off at most, estimated. It grows with every move and is zero after a reference. From 6° the display shows “UNKNOWN”.
Reverse polarity protection	A diode in the supply input that only lets current through in the right direction. If plus and minus are connected the wrong way round, it blocks, and the circuit behind it gets no voltage and is not damaged. On the HallTapBoard both lamps then stay dark. There this only applies to J1 alone: if J2 is plugged into the sync controller at the same time, current flows through the ground of the board until the fuse F2 trips; therefore check the polarity of J1 before J2 is plugged in.
Via	A small plated-through hole that connects the copper layers of a board. On the RearCircuitBoard_10 board all vias are filled and capped so that no solder runs into them during soldering.
Thermal pad	A soft, electrically insulating cushion that conducts heat well and fills gaps between two parts.
Web Serial	A capability of some browsers (Chrome, Edge, Opera, newer Firefox) to talk to a device on the USB port. With it, firmware can be flashed without a separate program.
Factory image (factory.bin)	The file with everything an empty board needs to start: bootloader, memory partitioning and firmware. It is written to the beginning of the memory (0x0) and erases the saved settings.
Preset angle	A self-defined angle from 1 to 59° for tilt on command, in addition to 0°, 30° and 60°.
Time server	A computer on the internet that distributes the exact time (default pool.ntp.org). The device sets its clock with it as soon as it is connected to Wi-Fi. Without Wi-Fi it does not know the time, because the board has no clock with its own battery.

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Two-row pluggable terminal	A pluggable terminal with two rows of terminals one above the other. At the switching outputs of the RearCircuitBoard_10 board, only two diagonally opposite pins are used: for UP and DOWN (J3, J4) two of four, for EAST and WEST (J5, J6) the two outer ones of six. Their plug does not fit any single-row terminal and not J11. A plug from UP or DOWN in EAST or WEST, or vice versa, even plugged in offset, touches at most one contact pin and switches nothing; a foreign plug therefore cannot bridge the contact.

11 Appendix

Bill of materials, sources, costs, the pin header assignment, the measurement plan before ordering and the HallTapBoard (G). Status: draft 0.12 of 9 November 2026 (draft 0.11 with the reset contacts J23); in addition draft 0.9 of 8 October 2026, extended by the bridges at the sensor island and the pulse input J22 to draft 0.10 of 11 October 2026, plus, as draft 0.11, the supply of the wind sensors at J15.

A. Bill of materials (assembly variant smd)

Draft 0.12: 15 more SMD parts (U24, C78, K5, K6, D49, D50, R142–R150), all with part numbers that are already on the board: 316 parts, 94 part numbers, 44 Extended with setup fee; in the assembly variant “voll” 332 parts and 104 part numbers (the only new one is the header J23, C441173). **Check date:** C441173 and C440849 looked up in the parts library on 9.11.2026 at 17:59, prices and minimum quantities at LCSC on 9.11.2026 at 20:05; the other numbers as below. The PCA9538PW (C201530) is now needed three times per board: check the stock before ordering. Below is the status of draft 0.11:

For draft 0.11, JLCPCB assembled all 301 SMD parts (94 part numbers, of which 44 of the class “Extended” with setup fee, D15 and D47/D48 as “Preferred Extended” without setup fee and 48 “Basic”; R92 and C38 stay unpopulated). In the assembly variant “voll” (full), JLCPCB additionally solders in the 15 terminals and pin headers (316 parts, 103 part numbers). The review of 5.11.2026 adds two Zener diodes (D47/D48, 12 V to GND) at the pulse input and one new Extended type (C34, 2.2 µF in X7R). The pulse input J22 (11.10.2026) adds 20 SMD parts and no new Extended type, the supply of the wind sensors (03.11.2026) 10 SMD parts with four new Extended types (U23, D18, D19, C73), and capacitor C77 in the termination (03.11.2026) one more. The protection against 24 V at the limit switches, lights and sun sensor (03.11.2026) adds 56 SMD parts with two new Extended types (BCX56, 100 Ω 1210); in return the six ESD devices TPD4E1U06 are dropped. The designators are also printed on the board's silkscreen.

Layout (status 5.11.2026, after the review): The tracks are laid by Freerouting 2.4.1 (run 25 on the finished board 106 × 110 mm). A few short connections that Freerouting had enclosed in earlier runs are fixed beforehand as stubs. The four connections that Freerouting left open and one cut-off ground area are closed by a follow-up step with fixed tracks from the project's own grid router. The KiCad rule checks report no error (ERC 0, DRC 0) and no open connection. Since the review of 5.11.2026 a large copper area on the converter input lies on the bottom side under the converter, the input capacitors and the inductor L1 (heat from Q2), the electronic fuse U15 is

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connected at both ends of its power pads, and the 24 V path from U16 to the converter is wider and has 6 instead of 2 vias. Every line from a terminal runs first into its protection device: limit switches J7–J10 3.3 mm each, sun sensor J2 and lights J14 3.6 mm each, pulse input J22 4.0 mm, wind sensor J15.3/J15.4 12.6 and 12.8 mm, 12 V supply J15.1 25.4 mm. The lines of the azimuth limit switches from R15/R16 to their TVS diodes are 27.6 mm (EAST) and 25.1 mm (WEST) long; until October they each ran about 160 mm around the board. The tracks are only checked with the rule check, not assessed electrically.

COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR	PURPOSE
Measuring chip	ADS1115IDGS (VSSOP-10)	C37593	Extended	2	U1, U20	Sun sensor (0x48); monitoring (0x49)
I/O chip	PCA9538PW, TI (TSSOP-16)	C201530	Extended	3	U2, U8, U24	Switching outputs/limit switches (0x70); lamps, fault inputs, IO6/IO7 (0x71); reset contacts J23 (0x72, from 0.12)
Switching contact	PhotoMOS AQY212GS (SOP-4)	C415471	Extended	6	K1–K6	Switching outputs, 60 V / 1 A; K5/K6 reset contacts (from 0.12)
Current limiter	TPS2553DBV (SOT-23-6)	C55266	Extended	2	U5, U9	Sensor supply, lamps
TVS diode	SMAJ33CA (SMA), bidirectional	C115251	Extended	10	D1–D4, D11, D12, D16, D17, D49, D50	across the contacts (D49/D50 from 0.12); RS485 A/B; pulses A/B (J22)
RS485-Transceiver	THVD2410DR, TI (SOIC-8), ±70 V	C1849398	Extended	1	U11	Anemometer
Voltage regulator 3.3 V	AP7361C-33Y5-13 (SOT-89-5)	C460397	Extended	1	U12	V3V3_AUX
5 V converter	LM5146QRGYTQ1, TI (VQFN-20; since 3.11.2026 instead of LM5146RGYR)	C2876733	Extended	1	U7	24 V → 5 V
MOSFET	CSD18543Q3A, TI (SON 3.3 × 3.3)	C840100	Extended	2	Q2, Q3	Converter switch
Power inductor	Würth WE-LHMI 74437368100, 10 µH	C1519857	Extended	1	L1	Converter
Shunt	Panasonic ERJ-8BWFR010V, 10 mΩ, 1206, 1 W	C2076185	Extended	1	R53	Current limit of the converter

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COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR	PURPOSE
eFuse 5 V	TPS259470ARPWR, TI (VQFN-HR-10)	C3662799	Extended	1	U15	Fuse and reverse blocking +5V
eFuse 24 V	TPS26600PWPR, TI (HTSSOP-16)	C544399	Extended	1	U16	Fuse at the 24 V input
Temperature switch	TMP390A2DRLR, TI (SOT-563)	C5219772	Extended	1	U17	switches off U16 above 110 °C
Comparator	LM2903BIDR, TI (SOIC-8)	C2869804	Extended	2	U18, U22	Window of the charge lock; pulse input (J22)
Counter	74HC4060D (SOIC-16)	C5649	Extended	1	U19	Waiting time of the charge lock
Reference	TLV431BIDBZR, TI (SOT-23)	C398375	Extended	1	U21	Overvoltage shutdown
Fuse	BSMD2920-200-60V, self-resetting	C49233174	Extended	1	F1	24 V input
Schottky diode	SS10P6-M3/86A, Vishay (TO-277A)	C2927414	Extended	1	D5	Reverse polarity protection
TVS diode	SMBJ30A (SMB)	C113998	Extended	1	D6	24 V input
Regulator module	TPSM365R6RDNR, TI (QFN-FCMOD-11, inductor built in)	C20346008	Extended	1	U23	Supply of the wind sensors at J15.1
Schottky diode	MBR1H100SFT3G, onsemi (SOD-123FL), 100 V / 1 A	C235757	Extended	2	D18, D37	Reverse blocking J15.1; plus of the lights (J14.1) against 24 V from outside (D37 since 5.11.2026 instead of B5819W)
TVS diode	SMF30A, MDD (SOD-123FL), unidirectional	C364281	Extended	15	D19, D20–D23, D32–D36, D38–D42	J15.1; limit switches J7–J10; lights J14.1–J14.5; sun sensor J2.1–J2.5 (24 V protection)
Transistor NPN	BCX56, CJ (SOT-89), 80 V / 1 A / 0.5 W	C24280	Extended	4	Q17–Q20	Current sinks of the lights (24 V protection)
Capacitor 100 nF 100 V	Murata GRM155R62A104KE14J (0402 X5R)	C3869216	Extended	1	C73	HF input U23
Capacitor 10 nF 100 V	Yageo CC0402KRX7R0BB103 (0402 X7R)	C527014	Extended	1	C77	Termination in series with R39
Zener diode	BZT52C5V1 (SOD-123)	C173407	Extended	1	D14	TMP_VDD

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COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR	PURPOSE
Zener diode	BZT52C5V6 (SOD-123), “Preferred”	C19077402	Extended	1	D15	Clamping V5_LOG (E36); check rotation and polarity in the assembly preview (**assumption**)
Diode	1N4148WS (SOD-323)	C2128	Basic	13	D13, D24–D31, D43–D46	Timing element U19; pull-up and clamp diodes of the limit switches, clamp diodes of the sun sensor (24 V protection)
Zener diode	BZT52C12S, hongjiacheng (SOD-323), 12 V / 200 mW	C19077447	Preferred Extended	2	D47, D48	Clamping at the pulse input to GND (since 5.11.2026)
P-channel MOSFET	SSM3J351R, Toshiba (SOT-23F), −60 V	C150147	Extended	1	Q21	Reverse blocking of the sensor supply at J2.1 (24 V protection)
Resistor 100 Ω 1210	UNI-ROYAL 1210W2F1000T5E, 1 %, 0.5 W	C620661	Extended	8	R1–R4, R130–R133	Load 2 × 100 Ω per wire of the sun sensor (withstands 12 V, destroyed at 24 V)
NTC 10 kΩ	Murata NCU18XH103F60RB (0603)	C440252	Extended	2	RT1, RT2	Sensor of the charge lock; measurement by the firmware
Transistor NPN	MMBT3904 (SOT-23)	C20526	Basic	3	Q1, Q11, Q12	24 V detection; self-latching
Transistor PNP	SS8550 (SOT-23)	C8542	Basic	6	Q10, Q13–Q16, Q22	Latch; inverting stages of the current sinks; reverse blocking at J2.1
MOSFET small	2N7002 (SOT-23)	C8545	Basic	6	Q4–Q9	Charge lock, enable at U20
LED green	KT-0805G (0805)	C2297	Basic	1	D9	READY
LED RGB	S4-3528RGBTA-A, gemeinsame Anode	C2827321	Extended	1	D10	FAIL
Electrolytic capacitor 100 µF 63 V	Ymin VKMD1001J101MV (8 × 10)	C487384	Extended	1	C14	24 V input

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COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR	PURPOSE
Capacitor 2.2 µF 100 V	Samsung CL31B225KCHSFNE (1206, flexible terminations)	C16195971	Extended	9	C15, C16, C56–C61, C72	Input bank (series pairs)
Capacitor 100 nF 100 V	Murata GCE21BR72A104KA01L (0805, flexible terminations)	C3011698	Extended	3	C17, C39, C41	HF at VIN_BUCK, input U16
Capacitor 47 µF 10 V	Murata GRJ32ER71A476KE11L (1210)	C7198516	Extended	4	C20, C21, C65, C66	Output bank
Polymer 220 µF 16 V	Chemi-Con APXG160ARA221MF80G	C709481	Extended	1	C22	Output bank
Capacitor 10 µF X7R	Murata GRM188Z71A106KA73D (0603)	C2959727	Extended	1	C31	Output U15
Capacitor 22 µF 25 V	1206 X5R	C12891	Basic	3	C28, C75, C76	Output U12; output U23
Capacitor 10 µF	0603 X5R	C19702	Basic	4	C1, C13, C26, C27	Buffer
Capacitor 4.7 nF 50 V	0603 X7R (0603B472K500NT)	C53987	Basic	8	C4–C11	Filter of the measuring inputs and the limit switches (previously 100 nF)
Capacitor 100 nF	0603 X7R	C14663	Basic	22	C2, C3, C12, C18, C24, C25, C29, C30, C32, C42–C46, C48–C53, C71, C78	Interference suppression, decoupling capa-citors (C71 at U22, C78 at U24)
Capacitor 2.2 µF 25 V X7R	Samsung CL21B225KAFNNNE, 0805 (since 5.11.2026 instead of 1 µF 50 V)	C19110	Extended	1	C34	VCC of the converter
Capacitor 1 µF	0603 X5R	C15849	Basic	2	C68, C74	Filter V5_LOG; VCC of U23
Capacitor 220 nF	0603 X7R	C21120	Basic	2	C47, C67	Timer U19
Capacitor 47 nF	0603	C1622	Basic	2	C33, C40	Soft start U7, dVdT U16
Capacitor 22 nF	0603	C21122	Basic	1	C35	Compensation
Capacitor 10 nF	0603	C57112	Basic	3	C54, C62, C64	Switch-off time U15, SHDN U16, self-latching
Capacitor 2.2 nF	0603	C1604	Basic	1	C55	U15

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COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR	PURPOSE
Capacitor 1 nF	0603	C1588	Basic	4	C37, C63, C69, C70	Compensation; filter OV_REF; pulse filter (J22)
Capacitor 22 pF	0603 C0G	C1653	Basic	1	C36	Compensation
Resistor 79.6 kΩ 0.1 %	Yageo RT0603BRD0779K6L	C861560	Extended	1	R24	Output voltage
Resistor 15 kΩ 0.1 %	Yageo RT0603BRD0715KL	C326733	Extended	1	R25	Output voltage
Resistor 324 kΩ 0.1 %	Vishay TNPW0603324KBEEA	C3999394	Extended	1	R61	Thresholds U15
Resistor 64.9 kΩ 0.1 %	Yageo	C861510	Extended	1	R62	Thresholds U15
Resistor 100 kΩ 0.1 %	Yageo	C122538	Extended	1	R63	Thresholds U15
Resistor 332 kΩ	0603, 1 %	C23139	Extended	1	R67	Threshold 110 °C at U17
Resistor 4.7 kΩ 1206 (250 mW)	UNI-ROYAL 1206W4F4701T5E	C17936	Basic	2	R102, R103	Pull-up of the pulse contacts (24 V on the wire)
Resistors 1 % (0603, except R39 0805)	1 kΩ C21190 (4) · 2.2 kΩ C4190 (10, plus R137–R140) · 270 Ω C22966 (6, R17–R20, since 5.11.2026 instead of 330 Ω; R142/R143 from 0.12) · 75 Ω C4275 (4, R126–R129) · 1.5 kΩ C22843 (4, R122–R125) · 4.7 kΩ C23162 (1, R141) · 10 kΩ C25804 (26, from 0.12 with R144–R150) · 200 kΩ C25811 (2) · 120 Ω C17437 (1, 0805) · 680 Ω C23228 (2) · 100 kΩ C25803 (17, plus R134) · 12 kΩ C22790 (3) · 300 kΩ C23024 (1) · 1 MΩ C22935 (2) · 33 kΩ C4216 (4) · 9.1 kΩ C23260 (2) · 0 Ω C21189 (1) · 150 kΩ C22807 (1) · 15 kΩ C22809 (4) · 6.8 kΩ C23212 (2) · 56 kΩ C23206 (1) · 20 kΩ C4184 (1) · 13 kΩ C22797 (2) · 47 kΩ C25819 (6) · 3 kΩ C4211 (1) · 2 MΩ C22976 (2) · 470 kΩ C23178 (2, plus R135) · 8.2 kΩ C25981 (1) · 22 kΩ C31850 (10, since 3.11.2026 also R5–R8, R13–R16, R136) · 100 Ω C22775 (3, R101, R114, R115)	see type	Basic	117	remaining R (plus R104–R117, R134–R136)	Measurement, pull-ups, series resistors, dividers, timer, blocking resistors R85/R93/R94 (provisional, E29); filter, threshold, positive feedback, pull-up and series resis-tor of the pulse input; divider of U23

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COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR	PURPOSE
not populated (DNP)	R92 2.2 Ω (C25369), C38 1 nF 100 V (C43847)	–	–	0	R92, C38	Space for a snubber at the switching node

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Bottlenecks and check date

Bottlenecks (stock at JLCPCB, snapshot): L1 C1519857 289 pcs (second source noted per E32: Changjiang FXL1040-100-M C475915 or Sunlord MWSA1004S-100MT C408487, both only after checking the dimension drawing, land pattern and keep-out area), U15 C3662799 1,061, D5 C2927414 1,733, C7198516 2,019 (4 per board), C3011698 2,442 (3 per board), C201530 2,743 (2 per board), U7 C3188679 3,583, R61 C861321 3,778, C709481 3,960, C5649 4,032, C544399 4,923; connector KF2EDGKS-3.5-2*3P C2923216 only 90 pcs at LCSC. **Status 3.11.2026, 17:36:** since 3.11.2026 U7 is the LM5146QRGYTQ1 C2876733 (stock 63, minimum quantity 2, about 6.48 USD; the LM5146RGYR C3188679 is not available). U12 C460397 had a stock of 0 at 17:36 (still 2 at 14:13): obtain it via JLCPCB Parts Pre-order / Global Sourcing before ordering. R61 has been replaced by C3999394 (stock 4,428). **Check date:** numbers, classes and stock of the SMD parts at JLCPCB looked up on 7.10.2026 at 11:16–11:46 and 12:36–13:57 CEST, all 40 new or changed numbers again on 8.10.2026 at 09:57:56–09:58:04 CEST (manufacturer and type confirmed), the whole bill of materials with prices and stock on 8.10.2026 at 09:53–09:55 CEST (cost check); R80 C23178 (Basic) on 7.10.2026 at 15:15; D15 C19077402 and R101 C22775 after the rework E36 on 8.10.2026 at 10:49–10:51 CEST, again at 11:22 CEST (D15 Extended, 2,966,774 pcs; R101 Basic, 10,999,003 pcs). The numbers of the pulse input J22 looked up on **11.10.2026 at 14:02:05–14:02:21 and 14:23:18–14:23:21 CEST** (stock at 14:23 CEST: C115251 204,621, C2869804 119,902, C17936 3,149,389, C25804 33,719,722, C22790 1,090,057, C25819 2,212,225, C22775 10,430,569, C1588 7,097,647, C14663 56,390,393, header C441172 2,162, plug C440848 1,778). The numbers of the supply of the wind sensors looked up on **03.11.2026 at 09:33 and 10:25 CET** (stock at 10:25 CET: C20346008 4,106, C235757 20,047, C364281 118,769, C16195971 6,291, C15849 11,768,761, C12891 1,675,078, C25803 17,057,136, C25804 45,688,588, header C441174 3,168, plug C440850 1,332), again at **11:03 CET** after checking the capacitors and switching J15 to 3.81 mm (C162178 327,379, C23260 594,086, C12891 1,669,209, header C441185 5,408, plug C440861 5,122). The numbers of the 24 V protection (limit switches, lights, sun sensor) looked up on **03.11.2026 at 19:15–20:09 CET** and again at **20:53 CET** with prices: C620661 52,708 (1210, Extended, new), C15127 1,677,237 (AO3401A, Basic, new), C24280 270,003 (BCX56, Extended, new), C8598 939,627 (B5819W, Basic, new), C364281 118,769, C2128 2,897,460, C8542 914,469, C53987 826,146, C31850 2,814,522, C23178 989,522, C25803 17,019,787, C23025 2,902,637; the dropped TPD4E1U06 C124691 had 20,981. Check again before ordering.

Status 4.11.2026, 15:42 (whole parts list of both assembly variants queried again): All part numbers are available. U12 C460397 has only 2 pieces in stock with a minimum quantity of 5: obtain it via JLCPCB Parts Pre-order / Global Sourcing before ordering. Further bottlenecks: U7 C2876733 58 pieces, C7198516 83 (4 per board), L1 C1519857 219, C3011698 713. Since 04.11.2026 C73 is C3869216 (Murata GRM155R62A104KE14J, the same type as the phased-out C162178 with a different packaging suffix; Extended, stock 24,741 on 04.11.2026 at 17:22 CET).

Status 5.11.2026, 05:25 (whole parts list of both assembly variants queried again): All part numbers are available. U12 C460397 has only 2 pieces in stock with a minimum quantity of 5: obtain it via JLCPCB Parts Pre-order / Global Sourcing before ordering. Further bottlenecks: U7 C2876733 58 pieces, C7198516 83 (4 per board), L1 C1519857 219, C3011698 713.

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Status 5.11.2026, 12:18 (whole parts list queried again after the review of 5.11.2026; new C19110, C22966, plus C235757 and C2128 in larger quantity): All part numbers are available. **L1 C1519857 has only 14 in stock** (219 in the morning): tight for five assembled boards, check before ordering. Further U12 C460397 stock 2 with a minimum quantity of 5 (Parts Pre-order / Global Sourcing), U7 C2876733 52, C7198516 73, C3011698 713. At 14:58 (after changing D47/D48 to BZT52C12S C19077447, Preferred Extended, stock 1 012 564): L1 14, U7 42, C7198516 59.

B. Hand-assembled parts and sources

JLCPCB does not ship loose parts. You order plugs, jumpers, the pin header J1 and the mechanical parts from LCSC (can be put in one package with the JLCPCB order) or from a dealer that delivers to private customers. In the assembly variant “smd” the headers are added; you solder them in yourself. The liability notice in Chapter 2 applies to this. All parts that JLCPCB does not assemble in the “smd” variant are listed with quantity, LCSC number, source, price and note in the order list `hardware/RearCircuitBoard_10/fertigung/smd/RearCircuitBoard_10-selbst-bestellen.csv` (plus `.md`), which the tool creates with every export.

PART	TYPE (KEFA OR MANUFACTURER)	JLCPCB/LCSC	QTY	ASSEMBLY VARIANT	ALTERNATIVE WITH DELIVERY TO PRIVATE CUSTOMERS (GROSS PRICE PER PIECE, RETRIEVED 7.10.2026)
J7–J10 header	KF2EDGR-3.5-2P	C441171	4	smd yourself, voll JLCPCB	Reichelt MC 2 G 3,5 (Phoenix 1844210), 0.58 €
J23 header (from 0.12, only with reset via J23)	KF2EDGR-3.5-4P	C441173	1	ditto	Reichelt MC 4 G 3,5 (Phoenix 1844236), 0.99 € (9.11.2026); at LCSC from 5 pieces
J22 Grundleiste	KF2EDGR-3.5-3P	C441172	1	ditto	Reichelt MC 3 G 3,5 (Phoenix 1844223), 0.87 € (price from 7.10.2026, not retrieved again for J22)
J14 Grundleiste	KF2EDGR-3.5-5P	C441174	1	ditto	Reichelt MC 5 G 3,5 (Phoenix 1844249), 1.25 €
J15 header (pitch 3.81 mm)	KF2EDGR-3.81-5P	C441185	1	ditto	Reichelt MC 5 G 3,81 (Phoenix MC 1,5/ 5-G-3,81), price not retrieved
J2 Grundleiste	KF2EDGR-3.5-6P	C441175	1	ditto	Reichelt MC 6 G 3,5 (Phoenix 1844252), 1.50 €
J11 Grundleiste	KF2EDGRC-5.08-2P (end walls)	C441236	1	ditto	Reichelt MSTBA 2-G 5,08 (Phoenix 1757242), 0.60 €; fits the same footprint (E35)

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J3/J4 Grundleiste	KF2EDGSR-3.5-2*2P	C2923260	2	ditto	DigiKey DMC 1,5/2-G1-3,5 P20THR (Phoenix 1786837), 1.23 € net
J5/J6 Grundleiste	KF2EDGSR-3.5-2*3P	C2923261	2	ditto	DigiKey DMC 1,5/3-G1-3,5 P20THR (Phoenix 1786840), 1.91 € (**assumption** net); Bürklin 62P6910, 2.38 €
J16/J21 Stiftleiste 1 × 3	PZ254V-11-03P	C2937625	2	ditto	Reichelt MPE 087-1-003, 0.09 €
Plug J7–J10	KF2EDGK-3.5-2P	C440847	4	both	Reichelt MC 2 ST 3,5 (Phoenix 1840366), 1.40 €
Plug J22	KF2EDGK-3.5-3P	C440848	1	both	Reichelt MC 3 ST 3,5 (Phoenix 1840379), 1.99 € (ditto)
Plug J23 (from 0.12, only with reset via J23)	KF2EDGK-3.5-4P	C440849	1	both	Reichelt MC 4 ST 3,5 (Phoenix 1840382), 2.65 € (9.11.2026)
Plug J14	KF2EDGK-3.5-5P	C440850	1	both	Reichelt MC 5 ST 3,5 (Phoenix 1840395), 3.45 €
Plug J15 (pitch 3.81 mm)	KF2EDGK-3.81-5P	C440861	1	both	Reichelt MC 5 ST 3,81 (Phoenix MC 1,5/ 5-ST-3,81), price not retrieved
Plug J2	KF2EDGK-3.5-6P	C440851	1	both	Reichelt MC 6 ST 3,5 (Phoenix 1840405), 3.99 €
Plug J11	KF2EDGK-5.08-2P-AA	C440880	1	both	Reichelt MSTB 2-ST 5,08 (Phoenix 1757019), 1.40 €
Plug J3/J4	KF2EDGKS-3.5-2*2P (spring terminal)	C2923215	2	both	DigiKey DFMC 1,5/2-ST-3,5 (Phoenix 1790108), 2.68 € (**assumption** net); Bürklin 62P6731, 3.43 €
Plug J5/J6	KF2EDGKS-3.5-2*3P (spring terminal)	C2923216	2	both	DigiKey DFMC 1,5/3-ST-3,5 (Phoenix 1790111), 4.18 € (**assumption** net); Bürklin 62P6743, 5.13 €
Jumper 2.54 mm	–	C5305	2	both	Reichelt JUMPER 2,54 SW, 0.02 €

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J1 pin header 1 × 40, long pins (cut to 2 × 1 × 20)	BOOMELE 21 mm (C51353); insertion depth without washers on the standoffs 3.0–3.2 mm, with one washer 2.5–2.7 mm (decision during the first assembly); alternatives 15 mm (C52461, for 9 mm spacing in the 40 mm housing), 17 mm (C44526)	C51353	1	both	Reichelt ECON SL40G1F (17.8 mm overall, dimension above the board not documented), 0.85 €
Brass spacer M3, AF 4.7, 13 mm (**option**, decision during the first assembly)	Shuntian M3*13: at 13.9 mm gap with two 0.5 mm pads, at 14.9 mm with 1.0 + 1.5 mm pads; 6 mm / 8 mm for 8.2–10 mm	C551333; C551326 / C551328	1	both	LCSC 0.0479 USD each from 10 pieces (06.11.2026); Reichelt ECON D3X08I5MT (8 mm, AF 5 instead of 4.7; no substitute documented for 13 mm), 0.11 €
Brass spacer M3, 12 mm (**option**, decision during the first assembly)	Shuntian M3*12, same series as C551333 (4.7 mm across flats **assumption**): at 13.9 mm gap with 1.0 + 1.5 mm pads	C551332 (LCSC only)	1	both	LCSC 0.0447 USD each from 10 pieces (06.11.2026, stock 370); Reichelt ECON D3X12I5MT (12 mm, female/female, AF 5 instead of 4.7), 0.20 € (06.11.2026)
Thermal pad 6 × 6 × 0.5 mm (cut to size, **option**)	Thermal Grizzly Minus Pad 8, manufacturer no. TG-MP8-30-30-05-1R	– (not a JLCPCB part, E31)	2	both	Reichelt item no. TG-MP8-30-05-1 (30 × 30 × 0.5 mm, enough for all boards), 3.50 €, in stock (06.11.2026)
Thermal pad 6 × 6 × 1.0 mm (cut to size, **option**)	Thermal Grizzly Minus Pad 8, manufacturer no. TG-MP8-30-30-10-1R; Arctic TP-3	– (not a JLCPCB part, E31)	1	both	Reichelt item no. TG-MP8-30-1-1 (30 × 30 × 1.0 mm, enough for all boards), 3.35 €, in stock (06.11.2026); Alternate 2.90 €
Thermal pad 6 × 6 × 1.5 mm (cut to size, **option**)	Thermal Grizzly Minus Pad 8, manufacturer no. TG-MP8-30-30-15-1R	– (not a JLCPCB part, E31)	1	both	Reichelt item no. TG-MP8-30-15-1 (30 × 30 × 1.5 mm, enough for all boards), 3.45 €, delivery time 3–5 working days (06.11.2026)

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PART	TYPE (KEFA OR MANUFACTURER)	JLCPCB/LCSC	QTY	ASSEMBLY VARIANT	ALTERNATIVE WITH DELIVERY TO PRIVATE CUSTOMERS (GROSS PRICE PER PIECE, RETRIEVED 7.10.2026)
Standoff M2.5 male/female thread	XHHD DTTZ4309, brass, 12 + 6 mm (with nut, without washer, screwed in fully: 13.9 mm measured; with one or two washers on top 14.4 or 14.9 mm respectively; depth of the female thread **not documented**)	C50377852	4	both	Amazon.de B0C1FY87PY (brass assortment, according to the seller with M2.5 × 12 + 6; contents **not checked**), 14.99 €; Reichelt RPI MOUNTINGKIT1 (standoffs 10 mm, too short for 15 mm), 3.50 € per set
Screw M2.5 × 6	Shuntian PM2.5X6nie, pan head, head 4.5 mm; M2.5 × 5 if the thread of the standoffs is too shallow (number **open**, Chapter 5)	C357395 (LCSC only)	4	both	LCSC 0.0022 USD each (05.11.2026, stock 12 100, minimum quantity 100); alternative Kessler SZKM2,5X6MM (bag of 100), 4.29 €
Washer M2.5 under the screw head	DIN 433 (ISO 7092), 2.7 × 5 × 0.5 mm	**open**	4	both	For private customers **open** (no source found with evidence); for business customers Bürklin 16H912 (DIN 433, zinc-plated steel), 0.0202 € net each (05.11.2026)
Washers M2.5 between standoff and circuit board (**only if needed**)	DIN 433, 2.7 × 5 × 0.5 mm, none, one or two per standoff (check during the first assembly, Chapter 5)	**open**	up to 8	both	as the washer under the screw head
Nut M2.5 on the male thread of the standoff (mounting on the board)	DIN 934 M2.5 × 0.45, nickel-plated steel, 2 mm high, 5 mm across flats; the nut shortens the effective length of the male thread, because the standoff cannot be screwed in any further; it belongs to the mounting on the board, not to the PCB	C357431	4 (1 per standoff)	both	LCSC/JLCPCB 0.0022 USD each (queried on 05.11.2026 08:17; stock 10,061, minimum quantity 2); alternative with delivery to private customers **open**

Kefa plugs are not available from dealers in the EU (Findchips, Reichelt; 7.10.2026). Whether Phoenix plugs fit the Kefa headers is **not documented** (assumption); in the assembly variant voll, Kefa headers sit on the board, Phoenix plugs only with this assumption. If you buy from Reichelt/DigiKey, it is therefore best to take header and plug from the same manufacturer (assembly variant smd). According to its help pages, LCSC also supplies private individuals (no restriction found, **assumption**).

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C. Costs

Valid status: draft 0.12 (9.11.2026), estimate. The basis is the totals “gross, hand parts from LCSC” from release v0.5.3 (draft 0.11, everything queried again on 5.11.2026 between 16:45 and 16:53, exchange rate 1 EUR = 1.1224 USD, PCB 106 × 110 mm 66.31 USD, hand parts with the minimum quantities of LCSC, shipping about 30 USD and customs duty 0 € as an **estimate**, 19 % import VAT). Added to this are the additional costs of the reset contacts J23 (**estimate** of 9.11.2026).

ITEM	SMD · 2	SMD · 5	VOLL · 2	VOLL · 5
Basis: draft 0.11, release v0.5.3 (5.11.2026, exchange rate 1.1224)	433 €	593 €	465 €	629 €
Additional costs draft 0.12 (J23, U24, K5/K6, D49/D50, R142–R150, C78)	+7 €	+17 €	+10 €	+21 €
Total gross draft 0.12, hand parts from LCSC (estimate)	**≈ 440 €**	**≈ 610 €**	**≈ 475 €**	**≈ 650 €**
per assembled board	≈ 220 €	≈ 122 €	≈ 238 €	≈ 130 €

If you use the reset via J5 (vertical only), you do not need the header and plug J23. Hand parts from Reichelt/DigiKey have not been recalculated for 0.12 (release v0.5.3: 533 / 839 / 546 / 818 €).

⚠️ **Superseded: the following list of 5.11.2026, 14:58**
It still calculates with the PCB for 100 × 102 mm (41.98 instead of 66.31 USD), without the minimum quantities of LCSC for the hand parts and with the exchange rate 1.1179. Release v0.5.3 therefore states 32 / 26 / 28 / 25 € more than its totals. It only remains here so that the individual items can be traced.

Basis: manufacturing data draft 0.11 of 5.11.2026 (both assembly variants, after the review of 5.11.2026). **Queried again on 5.11.2026 at 14:58** in the JLCPCB parts library: class, stock, price, minimum quantity and attrition of every part number (hand-fitted parts at 05:26; since the change to two washers per standoff, the M2.5 × 6 screw C357395 at LCSC at 17:22 and the 1.5 mm pad at Reichelt at 17:27); exchange rate 1 EUR = 1.1179 USD (ECB reference rate of 3.11.2026). **Not queried again** (status 8.10.2026, therefore an **estimate**): price of the circuit board (queried for 100 × 102 mm; whether 106 × 110 mm costs more is open), basic fee, fee per Extended type (3.07 USD) and per solder joint, hand soldering, X-ray, shipping. 5 circuit boards are always manufactured, 2 or 5 assembled (Economic PCBA, 2–50 pieces). Net unless stated otherwise. A setup fee applies to 44 (smd) or 53 (voll) Extended types; D15 is “Preferred Extended” and costs none. SMD solder joints: 796 per board. In every column 5 circuit boards are manufactured, of which 2 or 5 are assembled.

ITEM	SMD · 2	SMD · 5	VOLL · 2	VOLL · 5
Circuit board, 5 pieces (4 layers, ENIG, vias filled; price of 8.10.2026 for 100 × 102 mm, **estimate**)	41.98 USD	41.98 USD	41.98 USD	41.98 USD
Basic assembly fee and stencil	9.71 USD	9.71 USD	9.71 USD	9.71 USD
Fee for Extended types (3.07 USD × 44 or 53)	135.08 USD	135.08 USD	162.71 USD	162.71 USD
SMD solder joints (796 per board)	2.55 USD	6.37 USD	2.55 USD	6.37 USD

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ITEM	SMD · 2	SMD · 5	VOLL · 2	VOLL · 5
Hand soldering of through-hole parts at JLCPCB	–	–	5.22 USD	7.68 USD
X-ray (U7, U15, Q2, Q3; **assumption** whether charged)	13.12 USD	16.40 USD	13.12 USD	16.40 USD
Parts at JLCPCB (prices of 5.11.2026, with attrition and minimum quantities)	112.53 USD	235.81 USD	120.00 USD	251.61 USD
JLCPCB order (with 0.45 USD placement check)	**315.42 USD**	**445.80 USD**	**355.74 USD**	**496.91 USD**
Hand-fitted parts (plugs, jumpers, J1, standoffs, screws; with smd also the headers; prices of 5.11.2026)	22.96 USD	47.67 USD	15.66 USD	32.04 USD
Shipping JLCPCB to Germany (**estimate**) + LCSC in the same package (**estimate**)	30 USD	30 USD	30 USD	30 USD
Total gross route A (hand parts via LCSC, thermal pads 1.0 and 1.5 mm from Reichelt, 19 % import VAT, customs duty 0 € **estimate**)	**405 €**	**570 €**	**440 €**	**608 €**
per assembled board, route A	202 €	114 €	220 €	122 €
Total gross route B (hand-fitted parts from Reichelt/DigiKey; surcharge as on 8.10.2026, **estimate**)	498 €	797 €	516 €	780 €

- **Corrected on 22.10.2026, customs:** not included in the totals are the parcel service's fee for advancing customs duty and import VAT (DHL Germany 2 %, at least 15 €) and the standard customs duty for assembled PCBs (rate not checked), together **estimated** at least 15 €.
- The setup fee for Extended parts is the largest single item; it is charged per order. That is why 5 boards cost only a little over half as much per piece as 2. The assembly variant voll (full) costs 40–51 USD more at JLCPCB (9 further Extended types, hand soldering), but saves soldering in 15 headers.
- Compared with the status of 8.10.2026 (draft 0.9) about 46–92 € more (route A gross): 7 or 8 further Extended types with setup fee (supply of the wind sensors, termination C77, protection against 24 V, Q21, C34), parts for 24–60 USD more (mainly U23 and the TVS diodes) and 148 more SMD solder joints per board.
- **Estimates:** shipping (amount only visible in the shopping cart), customs duty (above 150 € goods value the standard tariff applies, rate for assembled PCBs not checked), X-ray item. Not included: vouchers, first-order discounts, the announced EU handling fee from November 2026, the up to twelve M2.5 washers per board (price open) and the M2.5 nuts of the standoffs (C357431, 4 × 0.0022 USD per board). Since 05.11.2026 the second thermal pad (1.5 mm, 3.45 €) is included in the totals; LCSC sells the M2.5 × 6 screw only from 100 pieces (0.22 USD, not in the totals). Since the test fit of 06.11.2026, the M3 × 12 spacer (LCSC 10 pieces 0.45 USD, Reichelt 0.20 €) and the 0.5 mm pad (3.50 €) are added as options; they are not in the totals because, depending on the decision during the first assembly, other parts are dropped (**estimate:** anyone who buys all options pays about 4 € more). Whether the stack-up JLC04161H-7628 counts as standard with Economic is **not documented** (the price calculator charges no surcharge); check when ordering. Surcharges that are only fixed after uploading the Gerber files (drill density, slots of the sensor island) are not known to the query.

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D. Assignment of the pin header J1

Numbering as in the planning, pin 1 = 3.3 V; what counts is the position on the board. Waveshare's schematic and wiki number differently: if you connect according to them, you can put 5 V onto the board's 3.3 V. Confirmed on the device on 5.11.2026: the board's silkscreen (V2) shows IO48/GND at the top and 5V/3V3 at the bottom and matches the table pin by pin.

PIN	SIGNAL	PIN	SIGNAL
1, 17	3.3 V from the board	2, 4	5 V to the board
3	SDA (GPIO 7)	5	SCL (GPIO 8)
16	RS485 TX → DI (GPIO 21)	18	RS485 RX ← RO (GPIO 22)
15	RS485 DE/RE (GPIO 28)	6, 9, 14, 20, 25, 30, 34, 39	GND
19	Pulse A from J22 (GPIO 29)	21	Pulse B from J22 (GPIO 30)

All other pins remain unconnected.

E. Measurement plan before ordering (excerpt)

1. **Charging current versus voltage (release):** on the delivered board, measure the battery current at 3.8 to 4.4 V at the J8 socket of the board, each for at least 2 hours, also with a cooled cell. Up to 4.1 V no measurable charging current may flow; otherwise the board will not be ordered in this form.
2. **Position of the battery and distance to the rear panel:** The position of the battery determines the position of the sensor island. The distance to the rear panel has been measured (13.9 mm without washers), the position of J1 above J8 checked with the test fit (06.11.2026); spacer and pads are chosen during the first assembly.
3. **Heat:** transistor Q2 at 4 A and 30 V; if its top is more than 50 K above ambient, the switching frequency is lowered.
4. **Switching node:** measure overshoot and ripple, fit the prepared snubber if needed.
5. **Overvoltage shutdown:** tripping at 6.0 to 6.5 V, latching until the 24 V is switched off, required off time (specified: at least 30 seconds).
6. **Current consumption of the board** in all operating modes, connector check step on a sample.

During the first assembly of draft 0.12: the rise time of the I2C lines with a third PCA9538 on the bus (oscilloscope at J1, at 400 kHz at most 300 ns) and the duration of the RST contact (5 seconds), including the question of whether the sync controller resets with it.

The complete measurement plan is in the project's README (section on the RearCircuitBoard_10 board, “Offen, Messplan und Restrisiken” – open points, measurement plan and residual risks).

F. Documents

Schematic, drawing and 3D views of the board are in the project under `hardware/RearCircuitBoard_10/vorschau/` and the production data under `hardware/RearCircuitBoard_10/fertigung/`. For a test fit in the rear housing there is a printable model

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of the board with the tall components and plugged-in connectors under `hardware/RearCircuitBoard_10/passprobe/` (**assumption** regarding position and connector dimensions) and a printed plate that is screwed onto the standoffs instead of the board and shows the position of J1 above J8 (printed and checked on 06.11.2026, fits).

G. HallTapBoard: bill of materials, sources and costs (draft 0.6)

The bills of materials of the manufacturing data are authoritative (`hardware/HallTapBoard/fertigung/smd/HallTapBoard-bom.csv` and `fertigung/voll/`); the table reproduces them in full. The JLCPCB numbers were looked up in the parts library: the carried-over ones on 12.10.2026 between 08:34 and 08:51, those of draft 0.4 (U1, D1/D5–D8, D2, D10, C2 and the resistors 510 Ω and 5.1 kΩ) on 13.10.2026 between 18:02 and 18:12. **New in draft 0.6** are C7250 (10 MΩ), C4190 (2.2 kΩ), C17936 (4.7 kΩ 1206) and C17944 (2 kΩ 1206): looked up on 11.11.2026 at 16:40 via jlcsearch.tscircuit.com (type, value, package, class Basic); their stock and price have **not been looked up (open, check at JLCPCB before ordering)**. Dropped are C4410 (1 kΩ 1206) and C21190 (1 kΩ 0603). Class, stock and prices of the other numbers date from 13.10.2026; at the check on 11.11.2026, U1 (C108469) had only **40 pieces** left (minimum order 25). On the day of ordering, check the stock of C108469 and of C70064 (now two per board, F1 and F2). JLCPCB assembles 50 SMD parts (22 numbers, 5 of them "Extended"; for the full variant plus the three terminals: 53 parts, 25 numbers). The rotation angles for assembly are an **assumption**: check them in the JLCPCB assembly preview and choose the placement confirmation by JLCPCB (stated in `HINWEIS.txt`). Documents: `hardware/HallTapBoard/` (preview in `vorschau/`, manufacturing data in `fertigung/`), tool `tools/halltap_erzeugen.py`.

COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR
Optocoupler	Lite-On LTV-817S-TA1-C	C109227	Basic	2	U3, U4
Comparator	TI LM2903BIDR	C2869804	Extended	1	U2
Voltage regulator 5 V (up to 40 V)	TI TPS7B6950QDCYRQ1 (SOT-223); substitute NCV4264-2ST50T3G (C73115) only with a modified C2, because it requires at least 10 µF at the output	C108469	Extended	1	U1
Fuse, self-resetting	TECHFUSE nSMD005 (50 mA); F1 in the supply, F2 (since 0.6) at the Hall ground	C70064	Extended	2	F1, F2
Protection diode (both directions)	MDD SMAJ30CA	C110043	Extended	5	D1, D5–D8
Protection diode at 5 V	MDD SMAJ6.5A	C123817	Extended	1	D10
Diode (reverse polarity protection)	MDD M7 (1 A, 1000 V)	C95872	Basic	1	D2
LED green “5V OK”	KENTO KT-0805G (with R17 + R23, 2 × 2.2 kΩ: 0.45 to 0.65 mA)	C2297	Basic	1	D4
LED yellow “24V”	KENTO KT-0805Y (with R18 + R24, 2 × 4.7 kΩ 1206: 2.0 to 2.9 mA)	C2296	Basic	1	D9

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COMPONENT	TYPE	JLCPCB NO.	CLASS	QTY	DESIGNATOR
Resistor 2 kΩ (1206)	2 kΩ, 1206 (two in parallel each = 1 kΩ)	C17944	Basic	4	R15, R16, R31, R32
Resistor 4.7 kΩ (1206)	UNI-ROYAL 1206W4F4701T5E	C17936	Basic	2	R18, R24
Resistors (0603)	1 MΩ C22935 (2) · 10 MΩ C7250 (2, R29/R30) · 10 kΩ C25804 (8) · 47 kΩ C25819 (2) · 5.1 kΩ C23186 (4) · 510 Ω C23193 (4) · 2.2 kΩ C4190 (2, R17/R23)	see type	Basic	24	R3–R14, R17, R19–R23, R25–R30
Capacitors	1 µF 50 V C28323 (1) · 10 µF 25 V C15850 (1) · 100 nF C14663 (1) · 10 pF C1634 (2)	see type	Basic	5	C1–C5

BY HAND	LCSC	ALTERNATIVE FOR PRIVATE CUSTOMERS (REICHELT, GROSS)
J1 Grundleiste KF2EDGRC-5.08-2P / Stecker KF2EDGK-5.08-2P-AA	C441236 / C440880	Phoenix MSTBA 2,5/2-G-5,08 0.60 € / MSTB 2,5/2-ST-5,08 1.40 €
J2 Grundleiste KF2EDGR-3.81-3P / Stecker KF2EDGK-3.81-3P	C441183 / C440859	Phoenix MC 1,5/3-G-3,81 0.87 € / MC 1,5/3-ST-3,81 1.99 €
J3 Grundleiste KF2EDGR-3.5-3P / Stecker KF2EDGK-3.5-3P	C441172 / C440848	Phoenix MC 1,5/3-G-3,5 0.87 € / MC 1,5/3-ST-3,5 1.99 €

Order list of the plugs: which plugs belong to which terminal, the minimum quantities you have to order at LCSC and what you should not mix is listed in `hardware/HallTapBoard/STECKER-BESTELLLISTE.md` (for the RearCircuitBoard_10 board correspondingly `hardware/RearCircuitBoard_10/STECKER-BESTELLLISTE.md`). Plug and header from the same manufacturer fit most reliably.

Costs of the board – as of 13.10.2026, draft 0.5. For draft 0.6 the costs have not been recalculated yet (**open**): it has five more parts but no additional Extended type; above all, the prices of the four new numbers and of the regulator U1 need to be checked again, of which JLCPCB required at least 25 pieces on 11.11.2026 (about 9.29 USD per order). The table shows the old values (prices from 13.10.2026, parts and PCB at 18:46; exchange rate 1 EUR = 1.1188 USD; PCB 70 × 40 mm, 5 pieces 3.10 USD, of which 2 or 5 assembled; shipping 30 USD is an **estimate**, 19 % import VAT included; **corrected on 22.10.2026:** not included are 3 € customs duty per tariff number, charged since 1.7.2026 on consignments to private customers up to 150 €, and the parcel service's fee (DHL at least 15 €), together **estimated** at 18 € or more):

ITEM	SMD · 2	SMD · 5	VOLL · 2	VOLL · 5
JLCPCB order net (PCB 3.10 USD, fees, parts)	35.90 USD	39.25 USD	49.49 USD	53.84 USD
Hand parts via LCSC (plugs, with smd also headers)	5.08 USD	5.08 USD	3.41 USD	3.41 USD
Total gross, hand parts from LCSC	**75 €**	**79 €**	**88 €**	**93 €**
per assembled board	38 €	16 €	44 €	19 €
Total gross, hand parts from Reichelt	86 €	113 €	96 €	117 €

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For context (draft 0.5): the four resistors of draft 0.5 cost 0.07 to 0.17 USD more per order; the rounded euro totals stay the same. The safe power supply of draft 0.4 costs about 4 to 5 USD more per order than draft 0.3, mainly for an additional Extended number (3.07 USD) and the regulator (0.33 USD per piece).

Housing (gross prices from 13.10.2026, dealers delivering to private customers; shipping not included):

PART	QUANTITY PER HOUSING	SOURCE	PRICE
Threaded insert ruthex RX-M3x5,7 (GE-M3x57-001)	8	ruthex.de	8.99 € per 100
Screw M3 × 8 cheese head DIN 84 (Würth)	4	Kessler-Electronic SZKM3X8MM	3.79 € per 100
Countersunk screw M3 × 10 ISO 10642 (hexagon socket)	4	DIY store or screw dealer (not looked up)	open
Round cord Ø 1.5 mm, EPDM or silicone	about 40 cm (about 31 cm needed)	Seal supplier, sold by the metre (not looked up)	open
Washer M3, outer diameter 7 mm	4	Reichelt 433596	2.19 € per 100
Cable gland Lapp SKINTOP ST PG 11, 4–10 mm, black (53015220)	3	Reichelt 158578	0,95 €
Lock nut Lapp SKINTOP GMP-GL PG 11, black (53019220)	3	Reichelt 285444	0,31 €
Screw M4 × 10 cheese head DIN 84 (length depending on mounting plate)	4	Kessler-Electronic SZKM4X10MM	4.99 € per 100
Filament Creality Hyper PETG-CF (or ASA, PETG), about 115 g (**estimate**)	–	any	not looked up (**open**)
Light guide: acrylic glass round rod (PMMA) Ø 3 mm, clear	2 × 30 mm	Craft or model-making shop (sold by the metre)	not looked up (**open**)

For one housing, glands and lock nuts cost 3.78 €. The full packs of the small parts came to a one-off of about 24 € up to the version of 15.10.2026 and are enough for many housings; countersunk screws, round cord and filament are still without a price (**open**).

If you order the HallTapBoard together with the RearCircuitBoard_10 board, shipping is charged only once. Details and reasoning are in the README (section “Platine HallTapBoard”).

H. Maps, place names and web interface map: sources

WHAT	SOURCE	LICENCE	WHERE
Coasts, country borders, lakes of the map on the display	Natural Earth (naturalearthdata.com)	public domain	in the firmware, LOCATION page

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WHAT	SOURCE	LICENCE	WHERE
Places with 5 000 inhabitants or more for the map and search on the display	GeoNames (geonames.org)	CC BY 4.0, attribution “GeoNames”	in the firmware, LOCATION page (note at the bottom left of the map)
Map program of the web interface	Leaflet 1.9.4 (leafletjs.com)	BSD-2-Clause	in the firmware, delivered by the device to the browser
Map images of the web interface	OpenStreetMap (openstreetmap.org), “© OpenStreetMap contributors”	Data ODbL; use according to the rules of the OpenStreetMap tile servers	loaded by your browser from the internet, not in the device
Place search of the web interface	Nominatim (OpenStreetMap)	Nominatim usage policy	queried by your browser on the internet, not in the device

The licence texts of the third-party libraries in the firmware are in the file `DRITTANBIETER.md` next to the firmware files. Since firmware 0.7.0 it also lists Leaflet, Natural Earth and GeoNames.

12 Licence and copyright

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The licence text in the file `LICENSE` in the project folder is binding. A summary in German is in `LIZENZ.md`. It also states what is not covered by this licence: the Waveshare documents, third-party software that is downloaded during the build, and, in the firmware for the 10.1-inch board, the table for switching on the display taken over from Waveshare (Apache License 2.0, licence text in `LICENSES/Apache-2.0.txt`). Since firmware 0.7.0 the firmware also contains map data from Natural Earth (public domain), place names from GeoNames (CC BY 4.0) and the map program Leaflet (BSD-2-Clause); the map images of the web interface come from OpenStreetMap (“©

OpenStreetMap contributors”) and are loaded by the browser (Appendix 11 H). For commercial use you need the author's written permission.

The prohibition of commercial use applies to everyone who uses the project under this licence. The rights holder, Silvio Sternitzke, is not bound by it: he can sell the board himself or, by way of exception, allow others commercial use.

🧩 **Still open for a sale: not checked, no legal advice**
Today the project serves building it for private purposes. A commercial sale of the board, a kit or a device requires a separate review, for example: CE marking and EMC conformity (electromagnetic compatibility), registration under the German Electrical and Electronic Equipment Act (ElektroG, WEEE), RoHS (restriction of hazardous substances), product safety under the EU regulation GPSR and the German Packaging Act. None of these points has been checked or fulfilled.

ℹ️ **No warranty**
The documents are provided without warranty. Building and operation are at your own risk. No liability is accepted for damage caused by improper soldering, reversed polarity or incorrect connection of external components, to the extent permitted by law (Chapter 2).

13 Change history

What has changed in this manual, newest change first. The history starts with the edition of 10 November 2026.

DATE	FIRMWARE	CHANGE
13.11.2026	0.7.8 (only tested on the PC)	Alignment with the assembly manual Rack 2-E Plus V3.1.2: the cable to the HallTapBoard (L2) is now protected at the star point by a fuse terminal block Phoenix Contact UK 6-FSI/C-LED24 with a 1 A automotive blade fuse (previously: without its own fuse, miniature fuse open). With two drives, the star point, fuses and HallTapBoard sit in box 2 (E1b), the sync controller in the control box (E1); the HLG-600H-24 power supply lies on the inside of the carriers of both boxes.
12.11.2026	0.7.8 (only tested on the PC)	Housing of the HallTapBoard version 0.6 (after a test print of version 0.5): shape of an industrial enclosure – on the outside a rectangle with evenly rounded edges, lid exactly the same size as the base, the lid screws sit in the reinforced corners within the contour instead of in pillars (outside now 104 × 82 × 44 mm instead of 98 × 82 × 44 mm, 128 mm wide with tabs); the groove for the sealing cord runs as a smooth loop with four wide corners past the screws on the inside and is narrower than the cord at the top so that it does not fall out (step “Insert the sealing cord” new, about 31 cm needed instead of 36 cm); the board is now inserted straight down, the offset positioning is no longer needed; lettering on the rear wall without “draft” and within the flat surface; additionally one print project each for only the base and only the lid; Fig. 11a–11c new.

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DATE	FIRMWARE	CHANGE
11.11.2026	0.7.8 (only tested on the PC)	HallTapBoard draft 0.6 (after a review): new fuse F2 at the Hall ground – up to draft 0.5, with J1 reversed and J2 plugged in, the current of the power supply flowed through the board without a fuse; the earlier statement "does no harm" was wrong in this case; open Hall input now reliably "contact open"; lights "24V" and "5V OK" about equally bright; new resistors and JLCPCB numbers in Appendix G, costs still open. New: Testing the board on the bench with a reverse polarity test only without J2 and J3, measure the polarity of J1 before plugging in J2, insert the board under the rim of the pillars, hold the lock nuts without a 24 mm spanner. Corrected: F1 does not protect the 24 V cable to the HallTapBoard (miniature fuse at the star point recommended), supports when printing the housing (base also on the model, lid only from the print bed), 30 % infill, cavities in the base filled; limits on pulse rate, seal and plug space marked as open; chapter 8 names draft 0.6.
11.11.2026	0.7.8 (only tested on the PC)	Housing of the HallTapBoard: seal with round cord Ø 1.5 mm in a groove, countersunk screws M3 × 10 on the lid, lettering raised and larger, lid is printed with the outside facing up, print project (3MF) for Creality K2 Plus with Hyper PETG-CF; housing 4 mm higher, light guides 30 mm; Fig. 11a–11c new.
11.11.2026	0.7.8 (only tested on the PC)	Test report as PDF: the table “Steps and measured values” always starts on page 2 and is no longer split on page 1.
11.11.2026	0.7.7 (only tested on the PC)	Setup program, reset of the sync controller: blue box with phase, remaining time and progress bar as on the dashboard and in the web interface; new Fig. 68e.
11.11.2026	0.7.6 (only tested on the PC)	Test report as a designed page in the web interface (The test report): VIEW, light or dark sheet, SAVE AS PDF; new Fig. 78a and 78b, Fig. 78 renewed. The text file remains.
11.11.2026	0.7.5 (only tested on the PC)	Setup program, step “Move to angle”: reaching 30 degrees was not recognised, the step stayed at “Moving to 30 degrees …”. Fixed; during the drive the current tilt is shown at the bottom right.
11.11.2026	0.7.4 (only tested on the PC)	Setup program: if a drive waits for the rest of the drives (duty cycle), the waiting time is now shown at the bottom right; during the reset of the sync controller it shows the axis, phase and remaining time. New note “Waiting for the drives”, Fig. 69a renewed.
11.11.2026	0.7.3 (only tested on the PC)	Setup program: after passing the wind sensor step the NEXT button was missing; the program could only be left. Fixed, as was the text after the signals step. Pictures of the FIRMWARE page with 0.7.3.
11.11.2026	0.7.2 (only tested on the PC)	New note “Manual steps in the simulation” in the setup program: in the simulation the device turns the wind cups, points the vane north and shades the sun sensor itself.
11.11.2026	0.7.1 (only tested on the PC)	New warning in New firmware from the internet: changing to 0.7.1 once via USB, because the memory is divided anew (the update to 0.7.0 failed with ESP_ERR_OTA_VALIDATE_FAILED). Note in the setup program: firmware 0.7.0 is published.

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DATE	FIRMWARE	CHANGE
10.11.2026	0.7.0 (only tested on the PC)	New in Chapter 8: The setup program (first installation) with all steps for one and two axes, buttons, error messages, test report and web interface; Sun, location and night with the SUN button, the SUN page, LOCATION (map, search, numbers), RACK ORIENTATION (one axis), night position for one and two axes (NIGHT TO THE EAST), cloud cover, new texts in the status bar and new settings. New drawings Fig. 60 (sequence), 70 (re-measuring the angle), 71 (shading the sun sensor), 80 (daily arc), 85 (orientation from above) and 87 (night position), images Fig. 61–88, including Fig. 76b (cancellation due to storm). On the DONE page, CHECK EVERYTHING AGAIN starts a new check; a setup program completed only in the simulation starts again by itself on the real rack. References in Chapters 8 and 9, priority of tilt on command added, glossary extended by sun position, azimuth of the sun, sun elevation, setup program, test report, night position, morning position and other terms, Appendix 11 H with the sources of the map and place names. Firmware 0.7.0 has not been released and has not been checked on any rack.
10.11.2026	0.6.0	SUN SENSOR card explained (chapter 8): meaning of the percentage, which is now called “SUN DEVIATION”, edge of the circle at ±30 %, example from manual mode. New below the circle: the sketch of the tilt with the angle in degrees; the TILT card shows the angle large. Table, steps and Fig. 13a/b, 14, 23a/b and 37 updated.
10.11.2026	0.5.9	In French, Spanish and Portuguese the buttons STEEPER and FLATTER are shown on two lines (with two axes they were cut off in 0.5.8); images of these languages regenerated. Own paragraph on manual mode in chapter 8; Fig. 18 and the log example name the setting as on the display, “Flat position is in UP direction”. The change history now also appears in the printed table of contents.
10.11.2026	0.5.8	The buttons in manual mode are called STEEPER and FLATTER instead of UP and DOWN, as is the movement in the status bar (“MANUAL: MOVING STEEPER”, “AUTO: TRACKING ELEVATION FLATTER”), in the axis state (“Elev: STEEPER”) and in the web interface (chapter 8, fig. 13a). Which output that is follows the setting “Flat position is in UP direction”; by factory default, STEEPER is the output DOWN. Terminals, contacts, settings and the log lines “Ausgang Elevation” are still called RAUF and RUNTER (UP and DOWN). Fig. 2 and the table on the sun sensor (chapter 4) name the movement of the panels. Screenshots of the user interface regenerated.

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Samba Solar Track · Manual of the 10-inch version: Waveshare ESP32-P4-WIFI6-Touch-LCD-10.1 with the RearCircuitBoard_10 board (in preparation). The images of the user interface are generated from the program code with `python3 tools/screenshots.py`, the PDF with `python3 tools/handbuch_pdf.py`, the logo with `python3 tools/logo.py`.